

No-clustering bootstrap screen $\text{Sharpe} > 0$ (OOS 2023+)

Directed factor-mimicking portfolio catalogue

Abstract

This catalogue constructs one FMP for each configured field or composite and compares all factors on one factor- and control-complete monthly window. Factors are indexed by reported raw Sharpe, highest first. Direction is frozen upstream; this report performs no return-based sign selection.

Methodology

Signals are observed at month-end, earn the next calendar month's simple return, and refresh monthly. Every factor must independently pass breadth, history, coverage, and non-degeneracy gates; none is silently dropped. One factor- and control-complete window fixes the reported Sharpe order and feeds all reported analyses. Displayed signal signs define the pre-direction composite; the reported Direction then orients the whole FMP. Risk Loadings include HAC t-statistics; Return Contributions are loading times annualized control mean, plus alpha and realized total. Rank diagnostics use 0/1/3/6/12/36-month horizons; available-history statistics are diagnostic only. Return Uncertainty uses paired circular stationary-bootstrap draws of the common monthly panel. It measures sampling and path uncertainty, not out-of-sample generalisation.

Construction Configuration

Setting	Effective value
Factors	100
Universe	shared security screen; refreshed monthly
Formation filters	price $\geq$ None; 21d dollar volume $\geq$ none
Market-cap screen	top 90.0% of names monthly
Signal preparation	within-month winsorize, normalize, then demean
Composite	mean available normalized components; factor-specific coverage floor
Construction	one blanket profile
Rebalance / return	month-end / next calendar month simple return
Costs	effective configured costs; default 0 bps
Minimum sample	20 stocks/month; 24 common months; 90% factor coverage
Comparison window	2023-01-01 to 2026-05-01; factor- and control-complete
Winsor settings	1%-99%
Controls	MKT, SMB, VAL, MOM, BAB, QMJ
Return uncertainty	paired stationary bootstrap; 2000 draws; expected block 12 months; seed 0
Catalogue note	Each of the 100 admitted signals is one standalone factor. No clustering, PCA, varimax, or post-screen factor pruning is applied. Admission requires at least 70% of fixed-orientation 2012–2022 stationary-bootstrap Sharpes to exceed 0.

Returns PCA

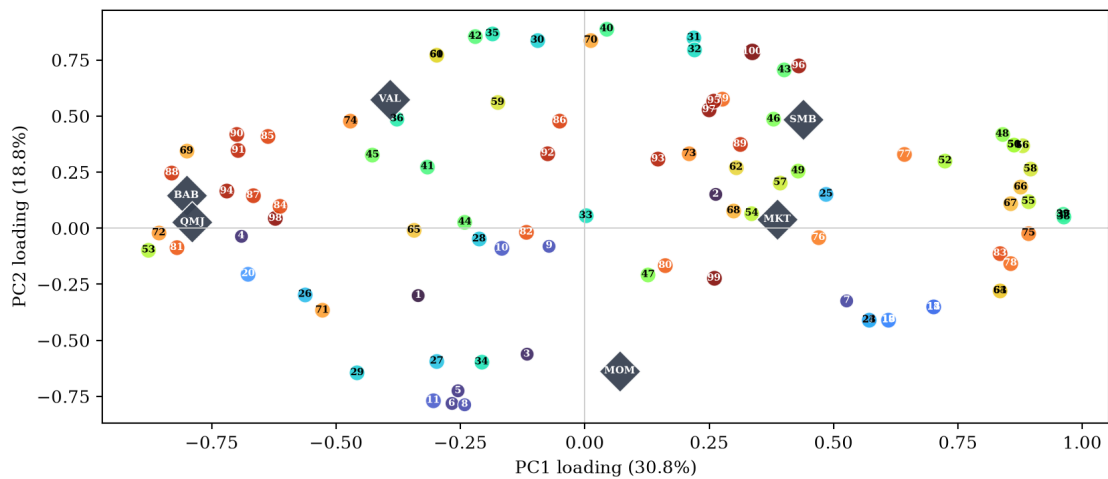


Figure: catalogue factors are numbered by reported-Sharpe index; attribution controls are named; $N=38$ common months.

Index

Idx	Factor	Dir.	N
1	share_turnover_21d	+1	38
2	accruals_noa_proxy	+1	38
3	revenue_growth_yoy	+1	38
4	sga_to_revenue	+1	38
5	momentum_21_126	+1	38
6	momentum_21_252_vol_63d	+1	38
7	rd_to_opex	+1	38
8	resid_momentum_21_252	+1	38
9	ret_zscore_63d	+1	38
10	dio	+1	38
11	momentum_21_252	+1	38
12	bab_252v_1260c_SPY	+1	38
13	bab_shrunk_252v_1260c_SPY	+1	38
14	beta_252v_1260c_SPY	+1	38
15	beta_shrunk_252v_1260c_SPY	+1	38
16	bab_252v_1260c_VTI	+1	38
17	bab_shrunk_252v_1260c_VTI	+1	38
18	beta_252v_1260c_VTI	+1	38
19	beta_shrunk_252v_1260c_VTI	+1	38
20	fcf_growth_yoy_ttm	+1	38
21	bab_252v_1260c_FRENCH_CRSP	+1	38
22	bab_shrunk_252v_1260c_FRENCH_CRSP	+1	38
23	beta_252v_1260c_FRENCH_CRSP	+1	38
24	beta_shrunk_252v_1260c_FRENCH_CRSP	+1	38
25	gross_profit_margin_ttm	+1	38
26	ebit_growth_yoy_ttm	+1	38
27	eps_growth_yoy	+1	38
28	resid_ret_252v_1260c_SPY	+1	38
29	net_income_growth_yoy	+1	38
30	evs_ttm	+1	38
31	ev_to_assets	+1	38
32	pb	+1	38
33	accruals_wc_proxy	+1	38
34	roic_nopat	+1	38
35	ps_ttm	+1	38
36	dpo	+1	38
37	corr_1260d_FRENCH_CRSP	+1	38
38	corr_1260d_VTI	+1	38
39	corr_1260d_SPY	+1	38
40	assets_to_mc	+1	38
41	dso	+1	38
42	sales_to_price	+1	38
43	altman_z	+1	38
44	capital_intensity	+1	38
45	asset_turnover	+1	38
46	equity_growth_yoy	+1	38
47	debt_issuance_yoy	+1	38
48	dollar_volume_21d	+1	38
49	growth_gmar_5y	+1	38
50	log_market_cap	+1	38
51	market_cap	+1	38
52	surprise	+1	38
53	fcf_margin	+1	38
54	accruals	+1	38
55	ebit_margin_ttm	+1	38
56	amihud_illiq_21d	+1	38
57	growth_gpoa_5y	+1	38
58	ev	+1	38
59	ret_volume_corr_63d	+1	38
60	gp_ev	+1	38
61	gross_profit_to_ev	+1	38
62	stoch_k_14d	+1	38
63	earnings_yield	+1	38
64	ebit_to_ev	+1	38
65	is_positive_revenue	+1	38
66	net_income_margin_ttm	+1	38
67	roe_volatility	+1	38
68	accruals_total	+1	38
69	fcf_to_market_cap	+1	38
70	pe_ttm	+1	38
71	surprisePercentage	+1	38
72	fcf_to_equity	+1	38
73	bb_pctb_20d	+1	38
74	accruals_percent	+1	38
75	net_operating_assets_proxy	+1	38
76	net_income_growth_yoy_sign_flip	+1	38
77	interest_to_debt	+1	38
78	net_income_to_market_cap	+1	38
79	asset_growth_yoy	+1	38
80	working_capital_proxy	+1	38
81	lottery_max_5_21d	+1	38
82	growth_acc_5y	+1	38
83	cash_earnings_ttm	+1	38
84	gpoa	+1	38
85	debt_to_assets	+1	38
86	skew_63d	+1	38
87	gp_to_equity	+1	38
88	equity_issuance_yoy	+1	38
89	rsi_14d	+1	38
90	ncav_to_mc	+1	38
91	debt_to_equity	+1	38
92	capex_to_revenue	+1	38
93	momentum_0_21	+1	38
94	cash_ratio	+1	38
95	price_zscore_63d	+1	38
96	momentum_21_756	+1	38
97	ma_dist_63d	+1	38
98	current_ratio	+1	38
99	is_negative_invested_capital	+1	38
100	value_proxy_1260d	+1	38

Returns

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
share_turnover_21d	14.68%	5.19%	2.83	5.22	-1.41%	0.329	13.00%	5.19%
accruals_noa_proxy	4.78%	3.69%	1.30	2.41	-2.67%	0.067	5.69%	3.69%
revenue_growth_yoy	6.13%	5.34%	1.15	1.88	-5.88%	0.393	5.06%	5.34%
sga_to_revenue	6.89%	6.03%	1.14	2.02	-6.45%	0.425	4.14%	6.03%
momentum_21_126	10.68%	9.44%	1.13	2.83	-4.34%	0.622	6.32%	9.44%
momentum_21_252_vol_63d	9.96%	9.17%	1.09	2.55	-5.09%	0.735	6.87%	9.17%
rd_to_opex	12.65%	11.73%	1.08	1.58	-13.92%	0.441	13.91%	11.73%
resid_momentum_21_252	9.50%	9.23%	1.03	2.41	-4.97%	0.703	7.17%	9.23%
ret_zscore_63d	5.46%	5.32%	1.03	1.84	-4.88%	0.029	6.19%	5.32%
dio	4.55%	4.47%	1.02	1.55	-9.00%	0.239	4.73%	4.47%
momentum_21_252	9.33%	9.62%	0.97	2.45	-5.50%	0.721	6.85%	9.62%
bab_252v_1260c_SPY	7.24%	7.50%	0.97	2.20	-7.32%	0.651	3.78%	7.50%
bab_shrunk_252v_1260c_SPY	7.24%	7.50%	0.97	2.20	-7.32%	0.651	3.78%	7.50%
beta_252v_1260c_SPY	7.24%	7.50%	0.97	2.20	-7.32%	0.651	3.78%	7.50%
beta_shrunk_252v_1260c_SPY	7.24%	7.50%	0.97	2.20	-7.32%	0.651	3.78%	7.50%
bab_252v_1260c_VTI	6.77%	7.17%	0.94	2.19	-6.82%	0.630	3.04%	7.17%
bab_shrunk_252v_1260c_VTI	6.77%	7.17%	0.94	2.19	-6.82%	0.630	3.04%	7.17%
beta_252v_1260c_VTI	6.77%	7.17%	0.94	2.19	-6.82%	0.630	3.04%	7.17%
beta_shrunk_252v_1260c_VTI	6.77%	7.17%	0.94	2.19	-6.82%	0.630	3.04%	7.17%
fcf_growth_yoy_ttm	4.71%	5.00%	0.94	1.41	-10.05%	0.584	5.18%	5.00%
bab_252v_1260c_FRENCH_CRSP	6.41%	6.94%	0.92	2.14	-6.38%	0.598	2.70%	6.94%
bab_shrunk_252v_1260c_FRENCH_CRSP	6.41%	6.94%	0.92	2.14	-6.38%	0.598	2.70%	6.94%
beta_252v_1260c_FRENCH_CRSP	6.41%	6.94%	0.92	2.14	-6.38%	0.598	2.70%	6.94%
beta_shrunk_252v_1260c_FRENCH_CRSP	6.41%	6.94%	0.92	2.14	-6.38%	0.598	2.70%	6.94%
gross_profit_margin_ttm	6.04%	6.65%	0.91	1.42	-8.08%	0.452	6.15%	6.65%
ebit_growth_yoy_ttm	4.27%	4.75%	0.90	1.30	-8.69%	0.515	5.23%	4.75%
eps_growth_yoy	4.01%	4.82%	0.83	1.35	-5.32%	0.470	2.36%	4.82%
resid_ret_252v_1260c_SPY	5.40%	6.59%	0.82	1.39	-6.09%	0.068	7.63%	6.59%
net_income_growth_yoy	3.09%	4.00%	0.77	1.40	-6.18%	0.558	1.84%	4.00%
evs_ttm	5.93%	7.68%	0.77	1.88	-6.79%	0.436	7.10%	7.68%
ev_to_assets	6.22%	8.17%	0.76	1.64	-7.99%	0.610	5.62%	8.17%
pb	5.32%	7.89%	0.67	1.26	-12.69%	0.600	5.07%	7.89%
accruals_wc_proxy	2.03%	3.30%	0.61	1.65	-2.79%	0.176	3.87%	3.30%
roic_nopat	2.29%	3.95%	0.58	0.90	-7.46%	0.375	1.84%	3.95%
ps_ttm	4.95%	8.82%	0.56	1.34	-8.46%	0.532	5.56%	8.82%
dpo	2.53%	4.64%	0.55	0.94	-7.00%	0.367	5.59%	4.64%
corr_1260d_FRENCH_CRSP	6.17%	11.47%	0.54	0.91	-16.62%	0.645	6.67%	11.47%
corr_1260d_VTI	6.13%	11.51%	0.53	0.90	-16.68%	0.651	6.63%	11.51%
corr_1260d_SPY	5.90%	11.60%	0.51	0.86	-16.98%	0.660	6.59%	11.60%
assets_to_mc	4.27%	8.56%	0.50	1.12	-10.09%	0.611	3.61%	8.56%
dso	1.98%	4.26%	0.47	1.14	-4.35%	0.116	1.86%	4.26%
sales_to_price	4.20%	9.04%	0.47	1.08	-9.80%	0.544	4.58%	9.04%
altman_z	2.46%	6.23%	0.40	0.81	-6.97%	0.444	1.32%	6.23%
capital_intensity	1.93%	4.91%	0.39	0.58	-7.78%	0.178	2.10%	4.91%
asset_turnover	2.25%	6.12%	0.37	0.64	-11.44%	0.417	3.19%	6.12%
equity_growth_yoy	1.64%	4.78%	0.34	0.59	-8.43%	0.247	3.06%	4.78%
debt_issuance_yoy	1.00%	3.12%	0.32	0.45	-6.33%	0.110	0.86%	3.12%
dollar_volume_21d	2.86%	9.11%	0.31	0.60	-12.89%	0.596	5.33%	9.11%
growth_gmar_5y	1.90%	6.16%	0.31	0.43	-13.71%	0.360	0.77%	6.16%
log_market_cap	2.16%	10.30%	0.21	0.39	-14.34%	0.591	5.39%	10.30%
market_cap	2.16%	10.30%	0.21	0.39	-14.34%	0.591	5.39%	10.30%
surprise	0.89%	4.26%	0.21	0.43	-4.52%	0.450	2.56%	4.26%
fcf_margin	1.50%	7.71%	0.20	0.36	-12.26%	0.675	2.16%	7.71%
accruals	0.90%	4.87%	0.19	0.34	-9.30%	0.293	2.13%	4.87%
ebit_margin_ttm	1.54%	8.34%	0.19	0.31	-12.85%	0.682	1.97%	8.34%
amihud_illiq_21d	1.71%	9.82%	0.17	0.32	-16.04%	0.608	4.34%	9.82%
growth_gpoa_5y	0.91%	5.90%	0.15	0.20	-15.79%	0.416	-0.58%	5.90%
ev	1.61%	10.51%	0.15	0.28	-16.05%	0.611	4.95%	10.51%
ret_volume_corr_63d	0.49%	3.97%	0.12	0.28	-6.74%	0.391	3.22%	3.97%
gp_ev	0.88%	8.21%	0.11	0.24	-12.90%	0.510	1.74%	8.21%
gross_profit_to_ev	0.88%	8.21%	0.11	0.24	-12.90%	0.510	1.74%	8.21%
stoch_k_14d	0.37%	6.27%	0.06	0.11	-10.54%	0.151	0.52%	6.27%
earnings_yield	0.32%	7.68%	0.04	0.07	-12.88%	0.683	-0.98%	7.68%
ebit_to_ev	0.32%	7.68%	0.04	0.07	-12.88%	0.683	-0.98%	7.68%
is_positive_revenue	0.14%	6.27%	0.02	0.04	-9.08%	0.340	0.74%	6.27%
net_income_margin_ttm	0.08%	7.95%	0.01	0.02	-13.82%	0.667	0.48%	7.95%
roe_volatility	0.05%	6.29%	0.01	0.01	-11.23%	0.555	-0.11%	6.29%
accruals_total	-0.05%	4.80%	-0.01	-0.02	-10.15%	0.326	1.46%	4.80%
fcf_to_market_cap	-0.10%	8.11%	-0.01	-0.03	-13.79%	0.673	1.24%	8.11%
pe_ttm	-0.21%	7.78%	-0.03	-0.06	-9.74%	0.583	2.84%	7.78%
surprisePercentage	-0.19%	3.74%	-0.05	-0.09	-7.08%	0.384	-1.11%	3.74%
fcf_to_equity	-0.68%	8.43%	-0.08	-0.13	-20.17%	0.759	-0.08%	8.43%
bb_pctb_20d	-0.51%	6.20%	-0.08	-0.19	-11.64%	0.129	-0.26%	6.20%
accruals_percent	-0.56%	5.75%	-0.10	-0.21	-11.11%	0.409	-0.31%	5.75%
net_operating_assets_proxy	-1.05%	10.54%	-0.10	-0.18	-18.37%	0.661	2.36%	10.54%
net_income_growth_yoy_sign_flip	-0.56%	4.58%	-0.12	-0.17	-12.19%	0.204	-0.92%	4.58%
interest_to_debt	-0.55%	4.47%	-0.12	-0.24	-8.83%	0.330	2.17%	4.47%
net_income_to_market_cap	-0.84%	6.62%	-0.13	-0.22	-11.29%	0.647	-1.76%	6.62%
asset_growth_yoy	-0.65%	5.01%	-0.13	-0.24	-9.21%	0.222	0.38%	5.01%
working_capital_proxy	-1.24%	4.66%	-0.27	-0.36	-12.01%	0.466	-1.30%	4.66%
lottery_max_5_21d	-2.71%	9.52%	-0.28	-0.51	-20.92%	0.566	-4.93%	9.52%
growth_acc_5y	-1.19%	3.85%	-0.31	-0.63	-9.16%	0.280	-2.59%	3.85%
cash_earnings_ttm	-2.14%	6.49%	-0.33	-0.58	-19.13%	0.677	0.42%	6.49%
gpoa	-2.89%	7.47%	-0.39	-0.62	-20.80%	0.525	-2.51%	7.47%
debt_to_assets	-2.48%	5.72%	-0.43	-1.04	-12.08%	0.574	-3.24%	5.72%
skew_63d	-1.56%	3.54%	-0.44	-0.91	-8.51%	0.400	0.55%	3.54%
gp_to_equity	-3.47%	7.65%	-0.45	-0.73	-22.38%	0.554	-3.54%	7.65%
equity_issuance_yoy	-3.23%	6.70%	-0.48	-0.90	-15.73%	0.720	-1.74%	6.70%
rsi_14d	-3.30%	6.80%	-0.49	-1.01	-17.48%	0.193	-3.29%	6.80%
ncav_to_mc	-3.42%	6.84%	-0.50	-1.03	-15.26%	0.575	-4.12%	6.84%
debt_to_equity	-2.89%	5.39%	-0.54	-1.04	-14.16%	0.529	-4.46%	5.39%
capex_to_revenue	-2.56%	4.58%	-0.56	-0.73	-14.04%	0.284	-0.93%	4.58%
momentum_0_21	-4.26%	6.90%	-0.62	-1.20	-22.62%	0.109	-4.30%	6.90%
cash_ratio	-4.09%	5.83%	-0.70	-1.02	-18.91%	0.541	-4.93%	5.83%
price_zscore_63d	-5.10%	6.67%	-0.77	-1.57	-22.49%	0.469	-5.36%	6.67%
momentum_21_756	-7.18%	8.50%	-0.85	-1.99	-27.38%	0.459	-7.13%	8.50%
ma_dist_63d	-6.13%	7.22%	-0.85	-1.60	-25.66%	0.449	-5.75%	7.22%
current_ratio	-5.96%	6.57%	-0.91	-1.23	-22.93%	0.573	-8.03%	6.57%
is_negative_invested_capital	-11.88%	11.66%	-1.02	-1.56	-42.91%	0.454	-18.46%	11.66%
value_proxy_1260d	-8.88%	8.62%	-1.03	-2.18	-30.47%	0.602	-9.34%	8.62%

Return Uncertainty



Figure: point estimates and paired stationary-bootstrap 95% intervals.

Idx	Factor	Ann. ret. [95%]	Sharpe [95%]	Pr(S<=0)	Skew [95%]	Max DD [95%]	ES 5% [95%]
1	share_turnover_21d	14.7% [10.7%, 18.6%]	2.83 [2.33, 3.45]	0.0%	0.50 [0.07, 1.18]	-1.4% [-1.9%, -0.7%]	-1.4% [-1.4%, -0.6%]
2	accruals_noa_proxy	4.8% [2.4%, 7.2%]	1.30 [0.64, 2.08]	0.1%	0.28 [-0.04, 0.66]	-2.7% [-3.9%, -1.4%]	-1.5% [-1.8%, -1.0%]
3	revenue_growth_yoy	6.1% [0.2%, 12.3%]	1.15 [0.04, 2.30]	2.1%	0.54 [-0.16, 1.11]	-5.9% [-9.2%, -0.9%]	-2.3% [-2.4%, -0.6%]
4	sga_to_revenue	6.9% [2.4%, 10.9%]	1.14 [0.39, 2.20]	0.1%	-0.36 [-0.78, 0.46]	-6.4% [-9.8%, -3.1%]	-3.1% [-3.3%, -1.4%]
5	momentum_21_126	10.7% [4.3%, 16.8%]	1.13 [0.44, 1.91]	0.2%	-0.26 [-0.68, 0.27]	-4.3% [-11.7%, -2.9%]	-5.1% [-6.2%, -2.9%]
6	momentum_21_252_vol_60d	6.0% [4.2%, 15.4%]	1.09 [0.43, 1.85]	0.2%	-0.35 [-0.93, 0.47]	-5.1% [-12.7%, -3.2%]	-5.1% [-7.0%, -2.0%]
7	rd_to_opex	12.6% [28.6%]	1.08 [-0.10, 2.20]	3.6%	0.84 [0.30, 1.29]	-13.9% [-19.9%, -4.8%]	-3.3% [-3.4%, -2.8%]
8	resid_momentum_21_252	9.5% [3.0%, 15.9%]	1.03 [0.31, 1.88]	0.2%	-0.30 [-0.93, 0.65]	-5.0% [-13.0%, -3.4%]	-5.3% [-7.2%, -1.9%]
9	ret_zscore_63d	5.5% [0.4%, 10.5%]	1.03 [0.07, 2.13]	1.8%	-0.24 [-0.66, 0.32]	-4.9% [-9.2%, -0.8%]	-3.0% [-3.5%, -1.2%]
10	dio	4.6% [-2.3%, 10.8%]	1.02 [-0.54, 2.92]	8.4%	0.12 [-0.14, 0.65]	-9.0% [-14.1%, -0.8%]	-2.0% [-2.1%, -0.7%]
11	momentum_21_252	9.3% [3.6%, 14.8%]	0.97 [0.35, 1.71]	0.2%	-0.50 [-1.12, 0.54]	-5.5% [-13.9%, -3.4%]	-5.6% [-7.9%, -2.1%]
12	bab_252v_1260c_SPY	7.2% [3.0%, 11.7%]	0.97 [0.38, 1.62]	0.2%	0.29 [-0.18, 1.04]	-7.3% [-10.5%, -2.4%]	-3.5% [-4.5%, -1.9%]
13	bab_shrunk_252v_1260c_SPY	7.2% [3.0%, 11.7%]	0.97 [0.38, 1.62]	0.2%	0.29 [-0.18, 1.04]	-7.3% [-10.5%, -2.4%]	-3.5% [-4.5%, -1.9%]
14	beta_252v_1260c_SPY	7.2% [3.0%, 11.7%]	0.97 [0.38, 1.62]	0.2%	0.29 [-0.18, 1.04]	-7.3% [-10.5%, -2.4%]	-3.5% [-4.5%, -1.9%]
15	beta_shrunk_252v_1260c_SPY	7.2% [3.0%, 11.7%]	0.97 [0.38, 1.62]	0.2%	0.29 [-0.18, 1.04]	-7.3% [-10.5%, -2.4%]	-3.5% [-4.5%, -1.9%]
16	bab_252v_1260c_VTI	6.8% [2.5%, 11.2%]	0.94 [0.33, 1.68]	0.3%	0.24 [-0.30, 1.08]	-6.8% [-9.8%, -2.7%]	-3.6% [-4.3%, -1.7%]
17	bab_shrunk_252v_1260c_VTI	6.8% [2.5%, 11.2%]	0.94 [0.33, 1.68]	0.3%	0.24 [-0.30, 1.08]	-6.8% [-9.8%, -2.7%]	-3.6% [-4.3%, -1.7%]
18	beta_252v_1260c_VTI	6.8% [2.5%, 11.2%]	0.94 [0.33, 1.68]	0.3%	0.24 [-0.30, 1.08]	-6.8% [-9.8%, -2.7%]	-3.6% [-4.3%, -1.7%]
19	beta_shrunk_252v_1260c_VTI	6.8% [2.5%, 11.2%]	0.94 [0.33, 1.68]	0.3%	0.24 [-0.30, 1.08]	-6.8% [-9.8%, -2.7%]	-3.6% [-4.3%, -1.7%]
20	fcf_growth_yoy_ttm	4.7% [-2.7%, 11.7%]	0.94 [-0.54, 2.94]	10.3%	-0.12 [-0.60, 0.38]	-10.0% [-16.3%, -1.6%]	-2.2% [-2.4%, -1.2%]
21	bab_252v_1260c_FRENCH_CRS	4.4% [0.3%, 11.2%]	0.92 [0.27, 1.72]	0.4%	0.27 [-0.30, 1.11]	-6.4% [-9.4%, -2.9%]	-3.6% [-4.1%, -1.6%]
22	bab_shrunk_252v_1260c_FRENCH_CRS	4.4% [0.3%, 11.2%]	0.92 [0.27, 1.72]	0.4%	0.27 [-0.30, 1.11]	-6.4% [-9.4%, -2.9%]	-3.6% [-4.1%, -1.6%]
23	beta_252v_1260c_FRENCH_CRS	4.4% [0.3%, 11.2%]	0.92 [0.27, 1.72]	0.4%	0.27 [-0.30, 1.11]	-6.4% [-9.4%, -2.9%]	-3.6% [-4.1%, -1.6%]
24	beta_shrunk_252v_1260c_FRENCH_CRS	4.4% [0.3%, 11.2%]	0.92 [0.27, 1.72]	0.4%	0.27 [-0.30, 1.11]	-6.4% [-9.4%, -2.9%]	-3.6% [-4.1%, -1.6%]
25	gross_profit_margin_ttm	6.0% [-1.8%, 13.9%]	0.91 [-0.31, 1.97]	7.4%	0.50 [-0.07, 0.89]	-8.1% [-13.8%, -3.6%]	-2.5% [-3.0%, -1.8%]
26	ebit_growth_yoy_ttm	4.3% [-2.9%, 11.3%]	0.90 [-0.71, 2.61]	12.6%	-0.24 [-0.80, 0.20]	-8.7% [-14.6%, -2.1%]	-2.4% [-2.5%, -2.0%]
27	eps_growth_yoy	4.0% [-1.3%, 9.9%]	0.83 [-0.29, 2.06]	9.0%	0.15 [-0.29, 0.57]	-5.3% [-9.3%, -1.3%]	-2.4% [-2.8%, -1.8%]
28	resid_ret_252v_1260c_SPY	4.4% [-1.1%, 11.9%]	0.82 [-0.18, 1.94]	5.7%	-0.34 [-0.73, 0.12]	-6.1% [-13.2%, -3.2%]	-4.0% [-4.6%, -2.0%]
29	net_income_growth_yoy	3.1% [-1.5%, 7.5%]	0.77 [-0.36, 2.11]	8.7%	-0.13 [-0.37, 0.16]	-6.2% [-9.3%, -3.9%]	-1.8% [-2.0%, -1.4%]
30	evs_ttm	5.9% [1.2%, 10.4%]	0.77 [0.14, 1.66]	0.4%	-0.43 [-1.08, 0.19]	-6.8% [-10.4%, -3.9%]	-4.7% [-5.4%, -2.4%]
31	ev_to_assets	6.2% [0.4%, 11.8%]	0.76 [0.05, 1.49]	1.9%	0.13 [-0.21, 0.48]	-8.0% [-13.1%, -4.8%]	-4.3% [-4.8%, -2.6%]
32	pb	5.3% [-2.2%, 12.4%]	0.67 [-0.27, 1.77]	8.9%	-0.07 [-0.30, 0.23]	-12.7% [-18.8%, -5.3%]	-3.6% [-4.1%, -3.1%]
33	accruals_wc_proxy	2.0% [0.3%, 3.8%]	0.61 [0.09, 1.11]	0.9%	0.47 [-0.04, 0.81]	-2.8% [-4.4%, -1.3%]	-1.6% [-1.8%, -1.1%]
34	roic_nopat	2.3% [-2.4%, 6.7%]	0.58 [-0.58, 2.16]	15.7%	-0.06 [-0.75, 0.52]	-7.5% [-13.6%, -1.3%]	-2.2% [-2.4%, -1.1%]
35	ps_ttm	5.0% [-0.4%, 10.3%]	0.56 [-0.04, 1.35]	3.4%	-0.43 [-0.94, 0.17]	-8.5% [-13.2%, -5.1%]	-5.2% [-6.2%, -2.9%]
36	dpo	2.5% [-2.4%, 7.9%]	0.55 [-0.54, 1.92]	16.8%	-0.01 [-0.46, 0.48]	-7.0% [-10.5%, -3.2%]	-2.3% [-2.7%, -1.5%]
37	corr_1260d_FRENCH_CRS	6.2% [-4.4%, 17.8%]	0.54 [-0.42, 1.52]	13.2%	0.03 [-0.47, 0.51]	-16.6% [-23.9%, -8.3%]	-5.4% [-5.9%, -3.9%]
38	corr_1260d_VTI	6.1% [-4.4%, 17.8%]	0.53 [-0.42, 1.52]	13.2%	0.04 [-0.46, 0.52]	-16.7% [-24.0%, -8.5%]	-5.3% [-5.9%, -3.9%]
39	corr_1260d_SPY	5.9% [-4.7%, 17.6%]	0.51 [-0.44, 1.49]	14.2%	0.02 [-0.49, 0.50]	-17.0% [-24.3%, -8.7%]	-5.3% [-6.0%, -4.1%]
40	assets_to_mc	4.3% [-1.9%, 10.1%]	0.50 [-0.21, 1.26]	8.8%	-0.06 [-0.39, 0.34]	-10.1% [-16.0%, -5.7%]	-4.9% [-5.8%, -2.6%]
41	dso	2.0% [-1.2%, 5.1%]	0.47 [-0.26, 1.36]	10.9%	-0.31 [-0.97, 0.32]	-4.3% [-7.6%, -5.1%]	-2.5% [-2.6%, -1.7%]
42	sales_to_price	4.2% [-1.4%, 9.9%]	0.47 [-0.15, 1.30]	7.6%	-0.46 [-0.99, 0.18]	-9.8% [-15.3%, -5.2%]	-5.6% [-6.6%, -2.7%]
43	altman_z	2.5% [-1.4%, 6.5%]	0.40 [-0.24, 1.07]	10.3%	0.38 [-0.08, 0.82]	-7.0% [-12.0%, -3.8%]	-2.8% [-2.8%, -2.1%]
44	capital_intensity	1.9% [-3.8%, 8.3%]	0.39 [-0.94, 1.61]	27.4%	0.38 [-0.08, 0.90]	-7.8% [-14.0%, -3.1%]	-2.3% [-2.4%, -1.6%]
45	asset_turnover	2.2% [-5.0%, 8.9%]	0.37 [-0.70, 1.94]	24.0%	-0.55 [-0.81, 0.35]	-11.4% [-19.7%, -2.1%]	-3.9% [-3.9%, -1.5%]
46	equity_growth_yoy	1.6% [-3.1%, 6.3%]	0.34 [-0.67, 1.36]	23.9%	0.23 [-0.17, 0.70]	-8.4% [-13.2%, -2.5%]	-2.2% [-2.3%, -1.7%]
47	debt_issuance_yoy	1.0% [-2.7%, 4.8%]	0.32 [-0.93, 1.63]	29.6%	-0.21 [-0.58, 0.09]	-6.3% [-10.3%, -3.3%]	-1.7% [-1.8%, -1.3%]
48	dollar_volume_21d	2.9% [-4.0%, 10.1%]	0.31 [-0.51, 1.10]	20.2%	0.15 [-0.32, 0.60]	-12.9% [-19.8%, -6.2%]	-5.0% [-6.2%, -3.2%]
49	growth_gmar_5y	1.9% [-6.2%, 10.6%]	0.31 [-1.52, 1.50]	31.4%	0.84 [-1.09, 1.60]	-13.7% [-19.2%, -4.4%]	-3.3% [-4.1%, -1.6%]
50	log_market_cap	2.2% [-4.7%, 9.5%]	0.21 [-0.49, 0.91]	25.8%	-0.11 [-0.50, 0.26]	-14.3% [-22.4%, -9.2%]	-5.8% [-6.7%, -4.2%]
51	market_cap	2.2% [-4.7%, 9.5%]	0.21 [-0.49, 0.91]	25.8%	-0.11 [-0.50, 0.26]	-14.3% [-22.4%, -9.2%]	-5.8% [-6.7%, -4.2%]
52	surprise	0.9% [-1.9%, 3.8%]	0.21 [-0.46, 0.90]	23.8%	0.04 [-0.39, 0.46]	-4.5% [-8.9%, -2.8%]	-2.3% [-2.6%, -1.6%]
53	fcf_margin	1.5% [-5.2%, 8.0%]	0.20 [-0.68, 1.15]	30.4%	-0.27 [-0.75, 0.18]	-12.3% [-22.4%, -5.6%]	-3.6% [-3.8%, -3.0%]
54	accruals	0.9% [-3.1%, 4.8%]	0.19 [-0.64, 1.02]	31.5%	0.15 [-0.38, 0.81]	-9.3% [-14.0%, -2.5%]	-2.8% [-3.1%, -1.7%]
55	ebit_margin_ttm	1.5% [-4.5%, 7.9%]	0.19 [-0.58, 0.96]	30.3%	0.02 [-0.49, 0.59]	-12.9% [-19.8%, -7.7%]	-4.2% [-4.8%, -3.1%]
56	amihud_illiq_21d	1.7% [-5.6%, 9.7%]	0.17 [-0.65, 0.98]	31.2%	0.07 [-0.39, 0.55]	-16.0% [-24.0%, -7.4%]	-5.3% [-6.5%, -3.9%]
57	growth_gpoa_5y	0.9% [-7.1%, 9.8%]	0.15 [-1.71, 1.51]	39.1%	0.32 [-0.98, 0.85]	-15.8% [-21.6%, -5.0%]	-3.3% [-3.9%, -2.3%]
58	ev	1.6% [-5.8%, 9.4%]	0.15 [-0.56, 0.90]	32.1%	-0.21 [-0.61, 0.15]	-16.0% [-24.6%, -9.5%]	-5.8% [-6.3%, -4.4%]
59	ret_volume_corr_63d	0.5% [-2.6%, 3.3%]	0.12 [-0.74, 0.78]	36.1%	-0.23 [-0.57, 0.05]	-6.7% [-10.5%, -2.8%]	-2.1% [-2.2%, -1.7%]
60	gp_ev	0.9% [-4.9%, 6.6%]	0.11 [-0.51, 1.10]	35.5%	-0.55 [-1.27, 0.03]	-12.9% [-21.3%, -4.4%]	-5.6% [-5.7%, -2.5%]
61	gross_profit_to_ev	0.9% [-4.9%, 6.6%]	0.11 [-0.51, 1.10]	35.5%	-0.55 [-1.27, 0.03]	-12.9% [-21.3%, -4.4%]	-5.6% [-5.7%, -2.5%]
62	stoch_k_14d	0.4% [-4.6%, 5.1%]	0.06 [-0.68, 0.90]	41.1%	-0.27 [-0.75, 0.54]	-10.5% [-18.1%, -4.2%]	-3.7% [-4.1%, -2.0%]
63	earnings_yield	0.3% [-6.4%, 6.8%]	0.04 [-0.91, 0.86]	44.2%	-0.00 [-0.43, 0.47]	-12.9% [-23.1%, -7.5%]	-4.1% [-4.7%, -3.2%]
64	ebit_to_ev	0.3% [-6.4%, 6.8%]	0.04 [-0.91, 0.86]	44.2%	-0.00 [-0.43, 0.47]	-12.9% [-23.1%, -7.5%]	-4.1% [-4.7%, -3.2%]
65	is_positive_revenue	0.1% [-5.2%, 5.4%]	0.02 [-0.89, 0.85]	46.2%	-0.15 [-0.48, 0.21]	-9.1% [-18.2%, -6.0%]	-3.8% [-3.8%, -2.9%]
66	net_income_margin_ttm	0.1% [-5.5%, 5.8%]	0.01 [-0.73, 0.73]	48.4%	0.05 [-0.43, 0.58]	-13.8% [-20.8%, -7.6%]	-4.2% [-4.7%, -3.2%]

Idx	Factor	Ann. ret. [95%]	Sharpe [95%]	Pr(S<=0)	Skew [95%]	Max DD [95%]	ES 5% [95%]
67	roe_volatility	0.1% [-5.2%, 6.0%]	0.01 [-0.85, 0.97]	48.4%	-0.42 [-0.78, -0.05]	-11.2% [-19.6%, -3.8%]	-3.8% [-3.9%, -2.5%]
68	accruals_total	-0.1% [-4.8%, 4.5%]	-0.01 [-0.90, 1.12]	52.6%	-0.57 [-1.09, 0.56]	-10.1% [-17.3%, -2.5%]	-3.5% [-3.8%, -1.5%]
69	fcf_to_market_cap	-0.1% [-6.7%, 6.7%]	-0.01 [-0.76, 1.00]	53.8%	-0.12 [-0.65, 0.35]	-13.8% [-24.1%, -5.4%]	-4.2% [-4.9%, -2.8%]
70	pe_ttm	-0.2% [-5.0%, 4.5%]	-0.03 [-0.67, 0.58]	55.8%	0.26 [-0.29, 0.67]	-9.7% [-18.5%, -0.9%]	-4.0% [-4.4%, -3.3%]
71	surprisePercentage	-0.2% [-3.9%, 3.7%]	-0.05 [-1.16, 1.02]	55.5%	0.04 [-0.27, 0.34]	-7.1% [-13.7%, -0.5%]	-1.9% [-1.9%, -1.6%]
72	fcf_to_equity	-0.7% [-11.7%, -0.08%]	-1.35, 1.35]	56.5%	-0.28 [-0.72, 0.19]	-11.6% [-35.2%, -4.2%]	-4.5% [-4.8%, -3.0%]
73	bb_pctb_20d	-0.5% [-4.8%, 4.1%]	-0.08 [-0.75, 0.73]	61.9%	-0.16 [-0.88, 0.61]	-11.1% [-17.2%, -4.9%]	-3.6% [-4.7%, -2.4%]
74	accruals_percent	-0.6% [-5.5%, 3.8%]	-0.10 [-0.79, 0.97]	61.3%	-1.00 [-1.34, -0.16]	-11.1% [-20.4%, -4.2%]	-4.3% [-5.1%, -2.0%]
75	net_operating_assets_proxy	-0.1% [-8.0%, 6.7%]	-0.10 [-0.80, 0.63]	65.0%	-0.19 [-0.67, 0.29]	-18.4% [-28.6%, -11.2%]	-5.8% [-5.8%, -4.2%]
76	net_income_growth_yoy	-0.6% [-11.6%, 4.9%]	-0.12 [-1.56, 1.14]	59.2%	0.20 [-0.49, 0.83]	-12.2% [-21.0%, -3.4%]	-2.6% [-3.0%, -1.7%]
77	interest_to_debt	-0.5% [-4.0%, 3.4%]	-0.12 [-1.05, 0.76]	65.0%	-0.12 [-0.65, 0.91]	-8.8% [-13.0%, -3.9%]	-2.8% [-3.7%, -1.7%]
78	net_income_to_market_cap	-0.8% [-6.1%, 4.2%]	-0.13 [-1.00, 0.61]	65.2%	0.04 [-0.43, 0.60]	-11.3% [-20.6%, -6.3%]	-3.8% [-4.2%, -2.5%]
79	asset_growth_yoy	-0.7% [-4.0%, 2.9%]	-0.13 [-0.82, 0.58]	68.5%	-0.22 [-0.68, 0.27]	-9.2% [-14.4%, -4.2%]	-3.0% [-3.6%, -2.1%]
80	working_capital_proxy	-1.2% [-7.3%, 5.2%]	-0.27 [-1.49, 1.42]	65.1%	-0.66 [-1.03, -0.03]	-12.0% [-23.3%, -3.7%]	-3.2% [-4.1%, -1.8%]
81	lottery_max_5_21d	-2.7% [-5.8%, 0.4%]	-1.30, 0.65]	73.1%	0.17 [-0.28, 0.65]	-20.9% [-35.4%, -8.4%]	-4.4% [-4.5%, -3.9%]
82	growth_acc_5y	-1.2% [-4.2%, 1.8%]	-0.31 [-1.07, 0.50]	79.1%	-0.71 [-1.22, -0.21]	-9.2% [-13.8%, -3.9%]	-2.7% [-3.1%, -2.0%]
83	cash_earnings_ttm	-2.1% [-8.1%, 4.3%]	-0.33 [-1.35, 0.67]	76.8%	-0.16 [-0.61, 0.32]	-19.1% [-25.6%, -7.1%]	-3.4% [-3.4%, -2.8%]
84	gpoa	-2.9% [-5.2%, 0.2%]	-1.42, 0.88]	73.8%	-0.39 [-0.69, 0.14]	-20.8% [-34.6%, -3.8%]	-4.7% [-5.5%, -2.4%]
85	debt_to_assets	-2.5% [-6.1%, 0.7%]	-0.43 [-0.94, 0.15]	94.0%	-0.88 [-1.26, -0.45]	-12.1% [-20.5%, -4.7%]	-4.1% [-4.6%, -3.2%]
86	skew_63d	-1.6% [-4.2%, 1.2%]	-0.44 [-1.32, 0.34]	87.5%	-0.54 [-0.93, -0.01]	-8.5% [-13.3%, -4.0%]	-2.4% [-2.7%, -1.8%]
87	gp_to_equity	-3.5% [-5.4%, 0.4%]	-1.62, 0.89]	75.8%	-0.34 [-0.69, 0.23]	-22.4% [-40.9%, -4.0%]	-4.8% [-5.4%, -2.9%]
88	equity_issuance_yoy	-3.2% [-8.2%, 1.4%]	-0.48 [-1.20, 0.23]	91.5%	-0.40 [-0.75, 0.02]	-15.7% [-25.4%, -6.9%]	-4.3% [-5.1%, -3.2%]
89	rsi_14d	-3.3% [-8.7%, 2.1%]	-0.49 [-1.30, 0.34]	87.8%	-0.32 [-0.72, 0.22]	-17.5% [-25.8%, -6.7%]	-4.3% [-5.4%, -2.9%]
90	ncav_to_mc	-3.4% [-9.4%, 2.0%]	-0.50 [-1.22, 0.33]	89.0%	-0.91 [-1.19, -0.59]	-15.3% [-28.9%, -5.8%]	-5.0% [-5.7%, -3.6%]
91	debt_to_equity	-2.9% [-8.8%, 2.1%]	-0.54 [-1.49, 0.48]	85.0%	-0.51 [-1.10, 0.19]	-14.2% [-26.1%, -2.8%]	-4.0% [-4.2%, -2.1%]
92	capex_to_revenue	-2.6% [-8.4%, 2.7%]	-0.56 [-1.61, 0.86]	78.8%	-0.71 [-1.17, 0.23]	-14.0% [-25.5%, -2.6%]	-3.6% [-3.8%, -1.3%]
93	momentum_0_21	-4.3% [-3.2%, 0.4%]	-1.44, 0.46]	86.9%	0.12 [-0.17, 0.37]	-22.6% [-31.5%, -6.8%]	-4.1% [-4.2%, -3.2%]
94	cash_ratio	-4.1% [-3.5%, 0.4%]	-1.30, 0.68]	82.0%	-0.35 [-1.00, 0.57]	-18.9% [-34.4%, -3.8%]	-4.5% [-5.0%, -2.2%]
95	price_zscore_63d	-5.1% [-1.1%, 0.4%]	-2.08, 0.15]	95.3%	0.31 [-0.60, 1.12]	-22.5% [-30.8%, -8.3%]	-4.4% [-5.6%, -2.6%]
96	momentum_21_756	-7.2% [-11.5%, -2.6%]	-1.46, -0.29]	99.7%	0.53 [-0.33, 1.14]	-27.4% [-32.0%, -15.4%]	-5.0% [-5.6%, -4.0%]
97	ma_dist_63d	-6.1% [-1.2%, 0.4%]	-2.20, 0.14]	95.5%	0.30 [-0.44, 0.96]	-25.7% [-34.9%, -9.6%]	-4.7% [-5.9%, -3.2%]
98	current_ratio	-6.0% [-2.7%, 0.4%]	-2.51, 0.49]	86.6%	-0.44 [-0.73, -0.09]	-22.9% [-42.1%, -5.3%]	-4.7% [-5.0%, -2.7%]
99	is_negative_invested_capital	-8.9% [-13.9%, -4.3%]	-1.02 [-2.15, 0.01]	97.4%	-0.04 [-0.86, 0.42]	-42.9% [-55.3%, -18.2%]	-7.7% [-8.3%, -5.8%]
100	value_proxy_1260d	-8.9% [-13.9%, -4.3%]	-1.03 [-1.79, -0.48]	99.9%	0.52 [-0.53, 1.00]	-30.5% [-36.7%, -17.1%]	-5.9% [-6.4%, -4.1%]

Risk Loadings

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
share_turnover_21d	-0.02 (-0.21)	-0.23 (-2.46)	-0.16 (-1.58)	-0.04 (-0.73)	0.08 (0.55)	0.05 (0.63)	13.83% (4.24)	0.337
accruals_noa_proxy	-0.03 (-0.42)	0.09 (1.66)	-0.02 (-0.33)	0.05 (0.85)	0.06 (0.46)	-0.03 (-0.30)	5.43% (1.95)	0.069
revenue_growth_yoy	-0.07 (-0.76)	-0.05 (-0.42)	-0.32 (-2.36)	0.01 (0.08)	0.35 (2.05)	-0.16 (-1.31)	4.59% (1.44)	0.396
sga_to_revenue	0.08 (1.10)	-0.28 (-3.72)	0.11 (1.27)	-0.00 (-0.07)	0.16 (1.15)	0.06 (0.58)	5.40% (1.52)	0.437
momentum_21_126	-0.03 (-0.26)	-0.14 (-1.35)	-0.12 (-0.77)	0.42 (4.22)	0.43 (1.84)	-0.06 (-0.35)	6.16% (1.85)	0.622
momentum_21_252_vol_63d	-0.15 (-2.32)	-0.12 (-1.10)	-0.30 (-3.19)	0.37 (6.82)	0.63 (4.38)	-0.19 (-1.62)	6.86% (1.84)	0.735
rd_to_opex	-0.53 (-4.64)	0.12 (0.66)	0.25 (1.24)	0.37 (3.56)	-0.29 (-0.84)	-0.68 (-3.61)	13.54% (1.88)	0.441
resid_momentum_21_252	-0.21 (-3.06)	-0.10 (-0.94)	-0.32 (-3.62)	0.36 (6.99)	0.62 (3.69)	-0.22 (-1.57)	7.23% (2.37)	0.703
ret_zscore_63d	-0.02 (-0.19)	0.07 (0.45)	0.04 (0.34)	0.02 (0.20)	0.03 (0.15)	-0.02 (-0.09)	6.46% (1.57)	0.030
dio	-0.09 (-0.88)	-0.08 (-1.03)	-0.23 (-3.02)	-0.08 (-1.57)	0.12 (1.01)	-0.05 (-0.38)	5.14% (2.44)	0.243
momentum_21_252	-0.14 (-1.86)	-0.08 (-0.72)	-0.31 (-3.07)	0.39 (6.98)	0.58 (3.68)	-0.07 (-0.57)	6.90% (1.94)	0.721
bab_252v_1260c_SPY	-0.16 (-2.05)	-0.29 (-2.96)	-0.18 (-1.98)	-0.05 (-0.74)	-0.34 (-1.81)	-0.32 (-2.40)	3.27% (1.48)	0.653
bab_shrunk_252v_1260c_SPY	-0.16 (-2.05)	-0.29 (-2.96)	-0.18 (-1.98)	-0.05 (-0.74)	-0.34 (-1.81)	-0.32 (-2.40)	3.27% (1.48)	0.653
beta_252v_1260c_SPY	-0.16 (-2.05)	-0.29 (-2.96)	-0.18 (-1.98)	-0.05 (-0.74)	-0.34 (-1.81)	-0.32 (-2.40)	3.27% (1.48)	0.653
beta_shrunk_252v_1260c_SPY	-0.16 (-2.05)	-0.29 (-2.96)	-0.18 (-1.98)	-0.05 (-0.74)	-0.34 (-1.81)	-0.32 (-2.40)	3.27% (1.48)	0.653
bab_252v_1260c_VTI	-0.15 (-2.03)	-0.35 (-3.67)	-0.15 (-1.80)	-0.04 (-0.54)	-0.34 (-1.91)	-0.26 (-2.14)	2.65% (1.26)	0.631
bab_shrunk_252v_1260c_VTI	-0.15 (-2.03)	-0.35 (-3.67)	-0.15 (-1.80)	-0.04 (-0.54)	-0.34 (-1.91)	-0.26 (-2.14)	2.65% (1.26)	0.631
beta_252v_1260c_VTI	-0.15 (-2.03)	-0.35 (-3.67)	-0.15 (-1.80)	-0.04 (-0.54)	-0.34 (-1.91)	-0.26 (-2.14)	2.65% (1.26)	0.631
beta_shrunk_252v_1260c_VTI	-0.15 (-2.03)	-0.35 (-3.67)	-0.15 (-1.80)	-0.04 (-0.54)	-0.34 (-1.91)	-0.26 (-2.14)	2.65% (1.26)	0.631
fcf_growth_yoy_ttm	0.09 (1.59)	0.01 (0.18)	-0.30 (-4.35)	-0.09 (-1.72)	0.43 (3.68)	0.16 (1.68)	5.39% (2.38)	0.584
bab_252v_1260c_FRENCH_CRSP	-0.14 (-1.87)	-0.35 (-3.76)	-0.13 (-1.52)	-0.02 (-0.36)	-0.33 (-1.83)	-0.24 (-1.99)	2.34% (1.06)	0.599
bab_shrunk_252v_1260c_FRENCH_CRSP	-0.14 (-1.87)	-0.35 (-3.76)	-0.13 (-1.52)	-0.02 (-0.36)	-0.33 (-1.83)	-0.24 (-1.99)	2.34% (1.06)	0.599
beta_252v_1260c_FRENCH_CRSP	-0.14 (-1.87)	-0.35 (-3.76)	-0.13 (-1.52)	-0.02 (-0.36)	-0.33 (-1.83)	-0.24 (-1.99)	2.34% (1.06)	0.599
beta_shrunk_252v_1260c_FRENCH_CRSP	-0.14 (-1.87)	-0.35 (-3.76)	-0.13 (-1.52)	-0.02 (-0.36)	-0.33 (-1.83)	-0.24 (-1.99)	2.34% (1.06)	0.599
gross_profit_margin_ttm	0.01 (0.09)	0.22 (1.80)	0.37 (3.52)	0.23 (2.96)	-0.36 (-2.02)	-0.07 (-0.60)	5.10% (1.55)	0.461
ebit_growth_yoy_ttm	-0.08 (-1.38)	-0.01 (-0.08)	-0.32 (-4.10)	-0.06 (-0.94)	0.51 (4.03)	-0.07 (-0.71)	5.72% (2.17)	0.519
eps_growth_yoy	-0.06 (-0.81)	-0.13 (-1.70)	-0.25 (-2.67)	0.03 (0.47)	0.40 (2.72)	-0.15 (-1.20)	2.40% (1.02)	0.470
resid_ret_252v_1260c_SPY	-0.08 (-0.49)	0.16 (0.88)	0.00 (0.01)	-0.01 (-0.11)	0.10 (0.37)	0.02 (0.06)	8.74% (1.95)	0.083
net_income_growth_yoy	-0.05 (-0.77)	-0.16 (-2.27)	-0.24 (-3.45)	-0.02 (-0.34)	0.33 (2.70)	-0.09 (-0.74)	2.07% (1.31)	0.559
evs_ttm	0.29 (2.40)	0.20 (1.43)	0.23 (1.68)	-0.14 (-1.66)	0.01 (0.02)	0.24 (1.30)	6.83% (2.28)	0.437
ev_to_assets	0.22 (1.93)	0.11 (0.86)	0.38 (3.54)	-0.18 (-3.59)	-0.01 (-0.03)	-0.13 (-0.87)	5.14% (1.61)	0.612
pb	0.19 (2.02)	0.12 (1.05)	0.47 (4.61)	-0.09 (-2.19)	-0.18 (-0.90)	-0.07 (-0.47)	4.70% (1.42)	0.601
accruals_wc_proxy	0.01 (0.16)	0.16 (2.05)	-0.08 (-1.41)	-0.01 (-0.15)	0.01 (0.10)	0.10 (1.00)	3.65% (1.98)	0.178
roic_nopat	-0.09 (-1.05)	-0.14 (-1.86)	-0.18 (-2.75)	-0.03 (-0.58)	0.06 (0.60)	-0.01 (-0.08)	2.39% (1.00)	0.382
ps_ttm	0.39 (2.83)	0.19 (1.14)	0.33 (2.09)	-0.17 (-1.59)	0.04 (0.13)	0.30 (1.37)	5.39% (1.65)	0.532
dpo	0.06 (1.00)	0.22 (4.36)	-0.11 (-1.42)	-0.12 (-2.37)	0.13 (0.89)	0.18 (2.53)	5.84% (2.45)	0.368
corr_1260d_FRENCH_CRSP	-0.30 (-1.90)	0.22 (2.54)	0.03 (0.25)	0.05 (0.64)	-0.50 (-2.04)	-0.56 (-2.18)	5.30% (1.02)	0.650
corr_1260d_VTI	-0.30 (-1.95)	0.22 (2.55)	0.01 (0.14)	0.05 (0.60)	-0.51 (-2.12)	-0.56 (-2.19)	5.27% (1.00)	0.656
corr_1260d_SPY	-0.29 (-1.94)	0.25 (2.83)	0.01 (0.05)	0.04 (0.56)	-0.51 (-2.14)	-0.55 (-2.18)	5.22% (0.98)	0.665
assets_to_mc	0.28 (2.48)	0.11 (0.76)	0.44 (3.51)	-0.16 (-2.94)	0.03 (0.12)	-0.04 (-0.23)	3.28% (0.99)	0.612
dso	0.04 (0.64)	0.01 (0.09)	0.03 (0.30)	-0.04 (-0.63)	0.16 (0.98)	-0.03 (-0.27)	2.07% (0.94)	0.118
sales_to_price	0.43 (3.31)	0.17 (1.07)	0.33 (2.15)	-0.18 (-1.95)	0.02 (0.06)	0.36 (1.77)	4.43% (1.29)	0.545
altman_z	0.11 (1.17)	-0.05 (-0.51)	0.09 (0.83)	-0.24 (-3.28)	0.03 (0.17)	-0.22 (-1.86)	0.62% (0.18)	0.450
capital_intensity	0.01 (0.10)	0.05 (0.54)	0.18 (1.69)	0.11 (1.53)	-0.05 (-0.28)	0.10 (0.70)	2.95% (0.95)	0.190
asset_turnover	0.29 (4.75)	0.20 (1.97)	-0.10 (-0.85)	-0.02 (-0.29)	0.05 (0.24)	0.50 (5.23)	2.86% (0.92)	0.418
equity_growth_yoy	-0.00 (-0.00)	0.11 (1.42)	0.17 (1.85)	-0.01 (-0.06)	-0.24 (-1.47)	0.04 (0.33)	3.02% (0.99)	0.247
debt_issuance_yoy	-0.05 (-1.11)	-0.02 (-0.21)	-0.08 (-1.34)	0.02 (0.38)	0.07 (0.53)	-0.08 (-0.90)	0.63% (0.25)	0.113
dollar_volume_21d	-0.11 (-0.63)	0.34 (2.89)	0.02 (0.17)	-0.09 (-0.82)	-0.27 (-0.92)	-0.32 (-1.30)	3.99% (1.18)	0.603
growth_gmar_5y	0.00 (0.02)	0.07 (0.58)	0.36 (2.47)	0.22 (1.79)	-0.25 (-1.49)	-0.12 (-0.80)	-0.05% (-0.02)	0.367
log_market_cap	-0.15 (-0.84)	0.42 (3.61)	0.04 (0.27)	-0.05 (-0.52)	-0.36 (-1.17)	-0.33 (-1.37)	3.92% (0.88)	0.597
market_cap	-0.15 (-0.84)	0.42 (3.61)	0.04 (0.27)	-0.05 (-0.52)	-0.36 (-1.17)	-0.33 (-1.37)	3.92% (0.88)	0.597
surprise	-0.05 (-0.55)	0.14 (3.68)	-0.02 (-0.21)	-0.05 (-0.97)	-0.18 (-2.01)	-0.06 (-0.62)	1.96% (1.05)	0.456
fcf_margin	0.13 (2.53)	-0.19 (-2.04)	-0.22 (-3.15)	-0.13 (-1.77)	0.33 (2.75)	0.40 (4.84)	3.13% (1.01)	0.681
accruals	0.04 (0.38)	0.10 (1.06)	-0.17 (-1.56)	-0.04 (-0.52)	-0.17 (-1.13)	0.15 (1.18)	1.22% (0.38)	0.304
ebit_margin_ttm	-0.14 (-1.30)	0.29 (3.49)	0.03 (0.36)	0.08 (1.21)	-0.31 (-1.86)	-0.35 (-3.07)	0.71% (0.17)	0.689

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
amihud_illiq_21d	-0.13 (-0.75)	0.37 (3.40)	0.04 (0.35)	-0.06 (-0.58)	-0.38 (-1.28)	-0.31 (-1.21)	2.95% (0.78)	0.614
growth_gpoa_5y	-0.08 (-0.63)	-0.04 (-0.39)	0.38 (3.29)	0.19 (1.96)	-0.22 (-1.60)	-0.24 (-1.72)	-0.79% (-0.25)	0.417
ev	-0.24 (-1.49)	0.42 (3.90)	-0.02 (-0.16)	-0.01 (-0.15)	-0.32 (-1.16)	-0.42 (-1.91)	3.56% (0.78)	0.616
ret_volume_corr_63d	-0.05 (-0.52)	0.16 (3.19)	-0.02 (-0.42)	-0.11 (-4.10)	0.14 (1.62)	-0.05 (-0.60)	3.52% (1.38)	0.394
gp_ev	0.40 (4.60)	0.13 (1.17)	0.08 (0.59)	-0.26 (-2.67)	0.14 (0.47)	0.39 (2.67)	1.72% (0.47)	0.510
gross_profit_to_ev	0.40 (4.60)	0.13 (1.17)	0.08 (0.59)	-0.26 (-2.67)	0.14 (0.47)	0.39 (2.67)	1.72% (0.47)	0.510
stoch_k_14d	0.14 (1.48)	0.04 (0.41)	0.02 (0.16)	-0.04 (-1.02)	-0.25 (-1.21)	0.07 (0.50)	-1.86% (-0.46)	0.201
earnings_yield	-0.25 (-2.91)	0.05 (0.52)	-0.13 (-1.62)	0.09 (2.11)	-0.18 (-1.51)	-0.47 (-6.12)	-1.83% (-0.56)	0.688
ebit_to_ev	-0.25 (-2.91)	0.05 (0.52)	-0.13 (-1.62)	0.09 (2.11)	-0.18 (-1.51)	-0.47 (-6.12)	-1.83% (-0.56)	0.688
is_positive_revenue	-0.18 (-2.03)	0.02 (0.18)	0.26 (1.92)	0.16 (1.24)	0.13 (0.67)	-0.21 (-1.41)	2.01% (0.56)	0.356
net_income_margin_ttm	-0.09 (-0.88)	0.30 (3.57)	0.10 (1.10)	0.09 (1.50)	-0.33 (-1.99)	-0.29 (-2.43)	-0.82% (-0.21)	0.676
roe_volatility	-0.14 (-1.48)	0.04 (0.59)	-0.06 (-0.91)	-0.05 (-0.96)	-0.17 (-0.90)	-0.34 (-2.32)	-1.01% (-0.29)	0.563
accruals_total	-0.01 (-0.06)	0.09 (0.86)	-0.23 (-2.61)	-0.10 (-1.32)	-0.08 (-0.50)	0.08 (0.58)	0.89% (0.32)	0.331
fcf_to_market_cap	0.33 (4.98)	-0.03 (-0.34)	0.01 (0.08)	-0.15 (-2.51)	0.15 (1.05)	0.62 (8.04)	1.98% (0.83)	0.677
pe_ttm	0.10 (0.98)	0.12 (0.82)	0.27 (2.66)	-0.25 (-3.27)	-0.15 (-0.65)	0.09 (0.56)	3.24% (1.09)	0.585
surprisePercentage	0.03 (0.39)	-0.05 (-1.07)	-0.12 (-1.90)	0.04 (0.91)	0.21 (2.23)	0.05 (0.51)	-0.97% (-0.53)	0.384
fcf_to_equity	0.29 (4.19)	-0.14 (-2.01)	-0.27 (-4.87)	-0.16 (-2.61)	0.28 (1.87)	0.64 (7.91)	0.45% (0.17)	0.760
bb_pctb_20d	0.12 (1.24)	0.04 (0.39)	0.04 (0.37)	-0.06 (-1.30)	-0.18 (-0.92)	0.01 (0.09)	-2.34% (-0.74)	0.168
accruals_percent	0.31 (2.95)	0.10 (1.13)	0.02 (0.21)	-0.06 (-0.62)	-0.03 (-0.18)	0.47 (3.55)	-0.36% (-0.13)	0.409
net_operating_assets_proxy	-0.38 (-2.61)	0.38 (3.55)	-0.23 (-2.01)	0.06 (0.77)	-0.29 (-1.20)	-0.44 (-2.04)	1.31% (0.30)	0.664
net_income_growth_yoy_sign_flip	0.05 (0.73)	0.01 (0.11)	0.06 (0.72)	0.06 (1.22)	-0.32 (-1.90)	0.09 (0.67)	-2.01% (-0.72)	0.226
interest_to_debt	-0.12 (-1.04)	0.20 (2.42)	-0.00 (-0.07)	-0.01 (-0.39)	-0.09 (-0.66)	-0.10 (-0.66)	1.98% (0.89)	0.331
net_income_to_market_cap	-0.18 (-2.05)	0.09 (1.01)	-0.05 (-0.61)	0.09 (1.90)	-0.15 (-1.41)	-0.39 (-4.78)	-2.69% (-0.91)	0.654
asset_growth_yoy	0.02 (0.26)	0.11 (1.72)	0.13 (1.29)	-0.04 (-0.49)	0.00 (0.00)	-0.06 (-0.55)	0.22% (0.06)	0.223
working_capital_proxy	-0.15 (-2.69)	-0.18 (-3.14)	-0.27 (-2.71)	-0.14 (-2.13)	0.06 (0.42)	-0.14 (-1.35)	-0.89% (-0.34)	0.469
lottery_max_5_21d	0.11 (0.95)	-0.41 (-4.57)	-0.07 (-0.66)	-0.10 (-1.54)	0.40 (1.56)	0.23 (1.47)	-3.10% (-0.53)	0.577
growth_acc_5y	-0.04 (-0.49)	-0.09 (-1.69)	0.12 (1.85)	0.06 (0.90)	0.17 (1.92)	-0.20 (-2.16)	-2.37% (-1.15)	0.282
cash_earnings_ttm	-0.26 (-3.53)	0.32 (5.98)	-0.03 (-0.49)	0.15 (3.96)	-0.19 (-2.20)	-0.28 (-3.06)	-0.15% (-0.06)	0.679
gpoa	0.26 (3.86)	-0.01 (-0.11)	-0.33 (-3.16)	-0.18 (-1.98)	0.30 (1.23)	0.47 (5.00)	-2.19% (-0.67)	0.526
debt_to_assets	0.26 (3.59)	-0.01 (-0.10)	0.08 (1.01)	-0.02 (-0.25)	0.20 (1.52)	0.32 (2.37)	-2.94% (-1.38)	0.575
skew_63d	-0.06 (-0.86)	0.11 (2.72)	-0.03 (-0.78)	-0.11 (-4.42)	0.19 (1.81)	-0.12 (-1.28)	0.82% (0.39)	0.403
gp_to_equity	0.28 (4.38)	-0.02 (-0.30)	-0.32 (-3.06)	-0.17 (-1.99)	0.42 (1.90)	0.43 (5.55)	-3.23% (-1.02)	0.555
equity_issuance_yoy	0.19 (3.02)	-0.05 (-0.52)	0.19 (2.96)	0.00 (0.01)	0.00 (0.02)	0.48 (4.09)	-0.96% (-0.39)	0.725
rsi_14d	0.14 (1.58)	0.05 (0.41)	0.06 (0.53)	-0.09 (-1.67)	-0.18 (-0.83)	-0.03 (-0.23)	-5.53% (-1.41)	0.232
ncav_to_mc	0.28 (2.94)	-0.12 (-1.04)	0.09 (0.89)	-0.10 (-1.32)	0.13 (0.71)	0.40 (2.49)	-3.41% (-1.38)	0.579
debt_to_equity	0.21 (3.37)	-0.11 (-1.18)	-0.06 (-0.57)	-0.10 (-1.59)	0.35 (2.49)	0.17 (1.40)	-4.20% (-1.99)	0.530
capex_to_revenue	0.06 (0.65)	0.09 (0.95)	-0.22 (-2.54)	-0.17 (-2.33)	0.15 (0.90)	0.10 (0.73)	-0.99% (-0.25)	0.284
momentum_0_21	0.12 (1.08)	0.01 (0.08)	-0.01 (-0.07)	-0.11 (-1.10)	-0.03 (-0.11)	-0.08 (-0.68)	-6.62% (-1.33)	0.151
cash_ratio	0.26 (3.44)	-0.05 (-0.64)	-0.03 (-0.32)	-0.04 (-0.63)	0.12 (0.85)	0.42 (3.03)	-4.57% (-1.65)	0.542
price_zscore_63d	0.16 (1.67)	0.02 (0.21)	0.03 (0.24)	-0.27 (-4.95)	-0.18 (-0.82)	-0.04 (-0.41)	-6.29% (-2.07)	0.476
momentum_21_756	0.17 (1.72)	0.10 (0.93)	0.25 (1.79)	-0.23 (-3.28)	-0.21 (-1.04)	-0.10 (-1.03)	-8.11% (-1.64)	0.464
ma_dist_63d	0.16 (1.44)	0.08 (0.69)	-0.02 (-0.18)	-0.28 (-5.13)	-0.16 (-0.73)	-0.03 (-0.27)	-6.92% (-1.93)	0.459
current_ratio	0.21 (2.49)	-0.29 (-3.39)	-0.16 (-1.33)	-0.15 (-2.25)	0.08 (0.46)	0.36 (2.77)	-7.41% (-2.72)	0.576
is_negative_invested_capital	0.06 (0.25)	-0.62 (-3.48)	-0.32 (-1.50)	-0.31 (-2.61)	-0.38 (-1.22)	-0.17 (-0.72)	-19.47% (-2.54)	0.458
value_proxy_1260d	0.12 (1.04)	-0.08 (-0.61)	0.17 (1.63)	-0.40 (-6.58)	-0.14 (-0.51)	-0.22 (-0.99)	-9.65% (-2.32)	0.603

Return Contributions

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
share_turnover_21d	-0.36%	1.55%	0.97%	-0.41%	-0.29%	-0.60%	13.83%	14.68%
accruals_noa_proxy	-0.73%	-0.64%	0.12%	0.50%	-0.22%	0.31%	5.43%	4.78%
revenue_growth_yoy	-1.44%	0.35%	2.01%	0.08%	-1.29%	1.83%	4.59%	6.13%
sga_to_revenue	1.58%	1.90%	-0.68%	-0.05%	-0.58%	-0.69%	5.40%	6.89%
momentum_21_126	-0.66%	0.97%	0.74%	4.36%	-1.60%	0.71%	6.16%	10.68%
momentum_21_252_vol_63d	-3.19%	0.81%	1.88%	3.85%	-2.34%	2.09%	6.86%	9.96%
rd_to_opex	-11.12%	-0.83%	-1.55%	3.90%	1.07%	7.64%	13.54%	12.65%
resid_momentum_21_252	-4.38%	0.71%	2.02%	3.76%	-2.30%	2.46%	7.23%	9.50%
ret_zscore_63d	-0.49%	-0.50%	-0.27%	0.19%	-0.13%	0.20%	6.46%	5.46%
dio	-1.79%	0.52%	1.46%	-0.84%	-0.45%	0.51%	5.14%	4.55%
momentum_21_252	-2.83%	0.56%	1.91%	4.13%	-2.15%	0.80%	6.90%	9.33%
bab_252v_1260c_SPY	-3.39%	1.99%	1.10%	-0.53%	1.27%	3.54%	3.27%	7.24%
bab_shrunk_252v_1260c_SPY	-3.39%	1.99%	1.10%	-0.53%	1.27%	3.54%	3.27%	7.24%
beta_252v_1260c_SPY	-3.39%	1.99%	1.10%	-0.53%	1.27%	3.54%	3.27%	7.24%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_shrunk_252v_1260c_SPY	-3.39%	1.99%	1.10%	-0.53%	1.27%	3.54%	3.27%	7.24%
bab_252v_1260c_VTI	-3.02%	2.37%	0.95%	-0.38%	1.27%	2.93%	2.65%	6.77%
bab_shrunk_252v_1260c_VTI	-3.02%	2.37%	0.95%	-0.38%	1.27%	2.93%	2.65%	6.77%
beta_252v_1260c_VTI	-3.02%	2.37%	0.95%	-0.38%	1.27%	2.93%	2.65%	6.77%
beta_shrunk_252v_1260c_VTI	-3.02%	2.37%	0.95%	-0.38%	1.27%	2.93%	2.65%	6.77%
fcf_growth_yoy_ttm	1.88%	-0.10%	1.89%	-0.93%	-1.59%	-1.83%	5.39%	4.71%
bab_252v_1260c_FRENCH_CRSP	-2.82%	2.41%	0.81%	-0.25%	1.23%	2.69%	2.34%	6.41%
bab_shrunk_252v_1260c_FRENCH_CRSP	-2.82%	2.41%	0.81%	-0.25%	1.23%	2.69%	2.34%	6.41%
beta_252v_1260c_FRENCH_CRSP	-2.82%	2.41%	0.81%	-0.25%	1.23%	2.69%	2.34%	6.41%
beta_shrunk_252v_1260c_FRENCH_CRSP	-2.82%	2.41%	0.81%	-0.25%	1.23%	2.69%	2.34%	6.41%
gross_profit_margin_ttm	0.21%	-1.48%	-2.31%	2.46%	1.33%	0.73%	5.10%	6.04%
ebit_growth_yoy_ttm	-1.74%	0.05%	1.96%	-0.58%	-1.90%	0.75%	5.72%	4.27%
eps_growth_yoy	-1.32%	0.89%	1.56%	0.28%	-1.50%	1.70%	2.40%	4.01%
resid_ret_252v_1260c_SPY	-1.58%	-1.06%	-0.01%	-0.13%	-0.38%	-0.17%	8.74%	5.40%
net_income_growth_yoy	-1.12%	1.07%	1.48%	-0.21%	-1.22%	1.01%	2.07%	3.09%
evs_ttm	6.11%	-1.39%	-1.46%	-1.47%	-0.02%	-2.67%	6.83%	5.93%
ev_to_assets	4.59%	-0.77%	-2.39%	-1.87%	0.02%	1.50%	5.14%	6.22%
pb	3.89%	-0.83%	-2.94%	-0.97%	0.68%	0.78%	4.70%	5.32%
accruals_wc_proxy	0.20%	-1.08%	0.48%	-0.10%	-0.03%	-1.09%	3.65%	2.03%
roic_nopat	-1.80%	0.94%	1.14%	-0.29%	-0.22%	0.13%	2.39%	2.29%
ps_ttm	8.09%	-1.28%	-2.03%	-1.75%	-0.15%	-3.32%	5.39%	4.95%
dpo	1.31%	-1.48%	0.70%	-1.29%	-0.49%	-2.06%	5.84%	2.53%
corr_1260d_FRENCH_CRSP	-6.17%	-1.50%	-0.17%	0.53%	1.87%	6.31%	5.30%	6.17%
corr_1260d_VTI	-6.19%	-1.50%	-0.09%	0.49%	1.90%	6.26%	5.27%	6.13%
corr_1260d_SPY	-6.11%	-1.69%	-0.03%	0.44%	1.90%	6.17%	5.22%	5.90%
assets_to_mc	5.85%	-0.74%	-2.71%	-1.73%	-0.10%	0.42%	3.28%	4.27%
dso	0.83%	-0.05%	-0.16%	-0.46%	-0.60%	0.34%	2.07%	1.98%
sales_to_price	9.03%	-1.17%	-2.03%	-1.91%	-0.06%	-4.08%	4.43%	4.20%
altman_z	2.22%	0.35%	-0.54%	-2.52%	-0.10%	2.45%	0.62%	2.46%
capital_intensity	0.15%	-0.31%	-1.12%	1.17%	0.17%	-1.08%	2.95%	1.93%
asset_turnover	6.14%	-1.34%	0.59%	-0.23%	-0.19%	-5.60%	2.86%	2.25%
equity_growth_yoy	-0.00%	-0.75%	-1.08%	-0.05%	0.91%	-0.39%	3.02%	1.64%
debt_issuance_yoy	-1.03%	0.12%	0.50%	0.16%	-0.25%	0.87%	0.63%	1.00%
dollar_volume_21d	-2.29%	-2.35%	-0.15%	-0.92%	1.01%	3.58%	3.99%	2.86%
growth_gmar_5y	0.07%	-0.48%	-2.23%	2.28%	0.95%	1.36%	-0.05%	1.90%
log_market_cap	-3.22%	-2.85%	-0.22%	-0.52%	1.34%	3.71%	3.92%	2.16%
market_cap	-3.22%	-2.85%	-0.22%	-0.52%	1.34%	3.71%	3.92%	2.16%
surprise	-1.09%	-0.97%	0.10%	-0.51%	0.68%	0.72%	1.96%	0.89%
fcf_margin	2.81%	1.28%	1.34%	-1.33%	-1.23%	-4.51%	3.13%	1.50%
accruals	0.76%	-0.70%	1.05%	-0.40%	0.64%	-1.67%	1.22%	0.90%
ebit_margin_ttm	-2.99%	-1.96%	-0.19%	0.87%	1.15%	3.96%	0.71%	1.54%
amihud_illiq_21d	-2.71%	-2.52%	-0.28%	-0.64%	1.43%	3.48%	2.95%	1.71%
growth_gpoa_5y	-1.76%	0.28%	-2.34%	2.04%	0.82%	2.68%	-0.79%	0.91%
ev	-4.93%	-2.88%	0.12%	-0.13%	1.19%	4.68%	3.56%	1.61%
ret_volume_corr_63d	-0.95%	-1.13%	0.14%	-1.15%	-0.54%	0.59%	3.52%	0.49%
gp_ev	8.27%	-0.91%	-0.50%	-2.75%	-0.53%	-4.41%	1.72%	0.88%
gross_profit_to_ev	8.27%	-0.91%	-0.50%	-2.75%	-0.53%	-4.41%	1.72%	0.88%
stoch_k_14d	2.98%	-0.30%	-0.13%	-0.45%	0.92%	-0.79%	-1.86%	0.37%
earnings_yield	-5.13%	-0.33%	0.78%	0.91%	0.67%	5.23%	-1.83%	0.32%
ebit_to_ev	-5.13%	-0.33%	0.78%	0.91%	0.67%	5.23%	-1.83%	0.32%
is_positive_revenue	-3.70%	-0.11%	-1.65%	1.70%	-0.47%	2.36%	2.01%	0.14%
net_income_margin_ttm	-1.96%	-2.02%	-0.61%	0.98%	1.22%	3.30%	-0.82%	0.08%
roe_volatility	-2.90%	-0.29%	0.37%	-0.58%	0.63%	3.83%	-1.01%	0.05%
accruals_total	-0.11%	-0.61%	1.44%	-1.04%	0.28%	-0.89%	0.89%	-0.05%
fcf_to_market_cap	6.79%	0.24%	-0.04%	-1.61%	-0.55%	-6.91%	1.98%	-0.10%
pe_ttm	2.16%	-0.84%	-1.70%	-2.67%	0.56%	-0.96%	3.24%	-0.21%
surprisePercentage	0.65%	0.36%	0.75%	0.39%	-0.80%	-0.58%	-0.97%	-0.19%
fcf_to_equity	6.10%	0.98%	1.65%	-1.68%	-1.04%	-7.15%	0.45%	-0.68%
bb_pctb_20d	2.46%	-0.28%	-0.24%	-0.64%	0.68%	-0.15%	-2.34%	-0.51%
accruals_percent	6.36%	-0.66%	-0.15%	-0.59%	0.12%	-5.28%	-0.36%	-0.56%
net_operating_assets_proxy	-7.88%	-2.62%	1.44%	0.68%	1.09%	4.93%	1.31%	-1.05%
net_income_growth_yoy_sign_flip	1.02%	-0.07%	-0.35%	0.66%	1.17%	-0.99%	-2.01%	-0.56%
interest_to_debt	-2.48%	-1.33%	0.03%	-0.14%	0.32%	1.08%	1.98%	-0.55%
net_income_to_market_cap	-3.78%	-0.61%	0.31%	0.96%	0.57%	4.38%	-2.69%	-0.84%
asset_growth_yoy	0.50%	-0.74%	-0.84%	-0.42%	-0.00%	0.63%	0.22%	-0.65%
working_capital_proxy	-3.17%	1.25%	1.67%	-1.43%	-0.24%	1.56%	-0.89%	-1.24%
lottery_max_5_21d	2.34%	2.78%	0.44%	-1.07%	-1.50%	-2.61%	-3.10%	-2.71%
growth_acc_5y	-0.86%	0.59%	-0.75%	0.59%	-0.62%	2.24%	-2.37%	-1.19%
cash_earnings_ttm	-5.38%	-2.16%	0.16%	1.57%	0.70%	3.12%	-0.15%	-2.14%
gpoa	5.46%	0.07%	2.06%	-1.91%	-1.11%	-5.27%	-2.19%	-2.89%
debt_to_assets	5.45%	0.07%	-0.52%	-0.19%	-0.75%	-3.60%	-2.94%	-2.48%
skew_63d	-1.28%	-0.77%	0.21%	-1.17%	-0.70%	1.33%	0.82%	-1.56%
gp_to_equity	5.82%	0.16%	1.99%	-1.83%	-1.57%	-4.81%	-3.23%	-3.47%
equity_issuance_yoy	4.02%	0.32%	-1.16%	0.00%	-0.01%	-5.44%	-0.96%	-3.23%
rsi_14d	2.88%	-0.34%	-0.37%	-0.97%	0.68%	0.37%	-5.53%	-3.30%
ncav_to_mc	5.78%	0.79%	-0.56%	-1.06%	-0.48%	-4.49%	-3.41%	-3.42%
debt_to_equity	4.47%	0.73%	0.34%	-1.01%	-1.29%	-1.94%	-4.20%	-2.89%
capex_to_revenue	1.15%	-0.62%	1.36%	-1.81%	-0.55%	-1.10%	-0.99%	-2.56%
momentum_0_21	2.45%	-0.09%	0.08%	-1.12%	0.11%	0.95%	-6.62%	-4.26%
cash_ratio	5.44%	0.37%	0.20%	-0.38%	-0.46%	-4.69%	-4.57%	-4.09%
price_zscore_63d	3.23%	-0.14%	-0.17%	-2.85%	0.67%	0.45%	-6.29%	-5.10%
momentum_21_756	3.57%	-0.66%	-1.53%	-2.38%	0.80%	1.13%	-8.11%	-7.18%
ma_dist_63d	3.24%	-0.53%	0.13%	-2.99%	0.60%	0.32%	-6.92%	-6.13%
current_ratio	4.36%	1.99%	0.98%	-1.55%	-0.31%	-4.03%	-7.41%	-5.96%
is_negative_invested_capital	1.27%	4.22%	1.98%	-3.23%	1.40%	1.96%	-19.47%	-11.88%
value_proxy_1260d	2.60%	0.52%	-1.09%	-4.20%	0.51%	2.43%	-9.65%	-8.88%

1. share_turnover_21d

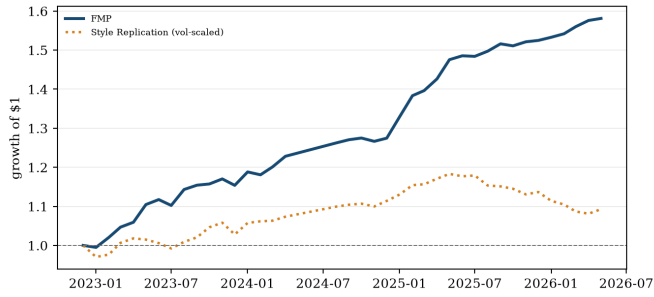


Figure: Common-window cumulative growth of \$1.

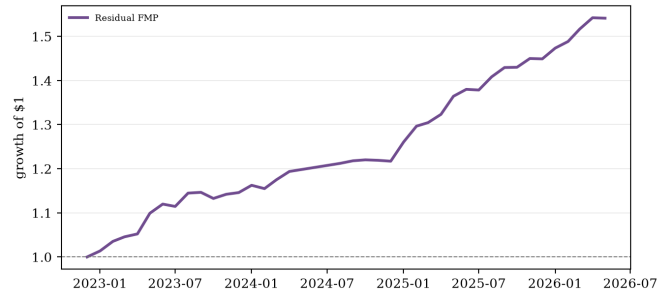


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
share_turnover_21d	14.68%	5.19%	2.83	5.22	-1.41%	0.329	13.00%	5.19%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
share_turnover_21d	-0.02 (-0.21)	-0.23 (-2.46)	-0.16 (-1.58)	-0.04 (-0.73)	0.08 (0.55)	0.05 (0.63)	13.83% (4.24)	0.337

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
share_turnover_21d	-0.36%	1.55%	0.97%	-0.41%	-0.29%	-0.60%	13.83%	14.68%

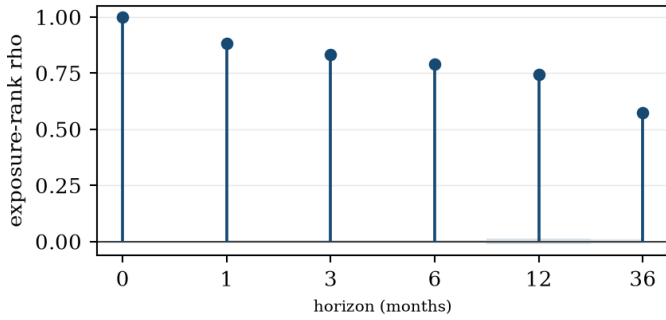


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

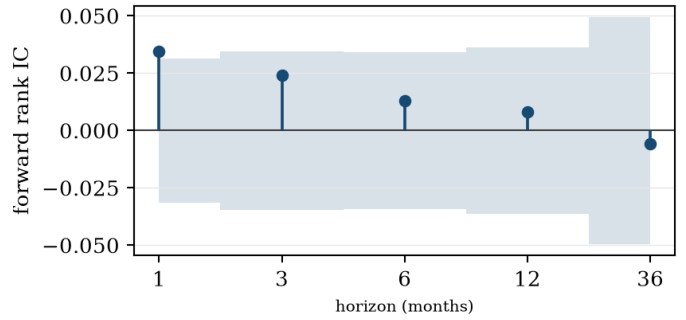


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	share_turnover_21d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	2.39
Bootstrap Sharpe	2.83 [2.33, 3.45]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.50 [0.07, 1.18]; max DD=-1.4% [-1.9%, -0.7%]

2. accruals_noa_proxy

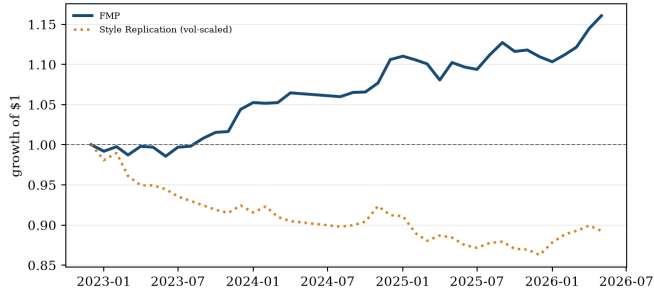


Figure: Common-window cumulative growth of \$1.

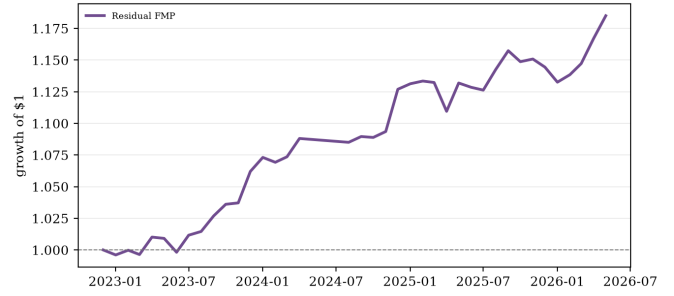


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_noa_proxy	4.78%	3.69%	1.30	2.41	-2.67%	0.067	5.69%	3.69%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_noa_proxy	-0.03 (-0.42)	0.09 (1.66)	-0.02 (-0.33)	0.05 (0.85)	0.06 (0.46)	-0.03 (-0.30)	5.43% (1.95)	0.069

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_noa_proxy	-0.73%	-0.64%	0.12%	0.50%	-0.22%	0.31%	5.43%	4.78%

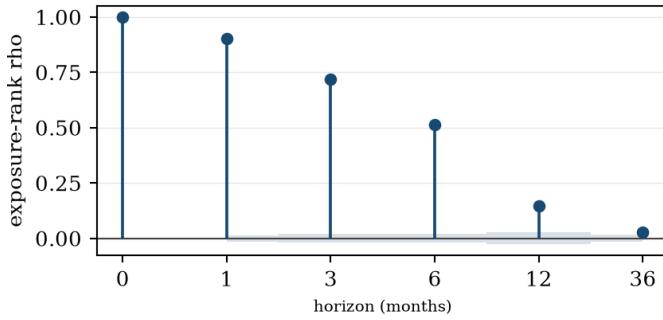


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

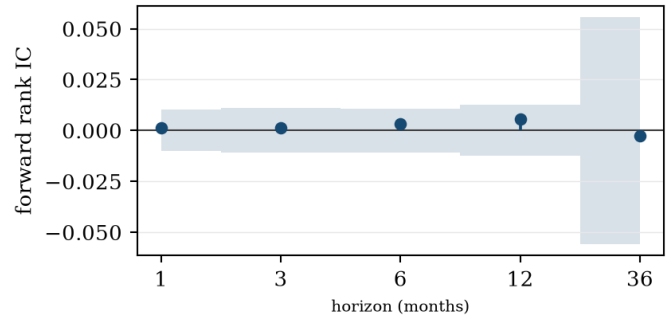


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	accruals_noa_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2034 / 2498 / 2685
Available-history Sharpe	1.38
Bootstrap Sharpe	1.30 [0.64, 2.08]; Pr(S<=0)=0.1%
Bootstrap tails	skew=0.28 [-0.04, 0.66]; max DD=-2.7% [-3.9%, -1.4%]

3. revenue_growth_yoy

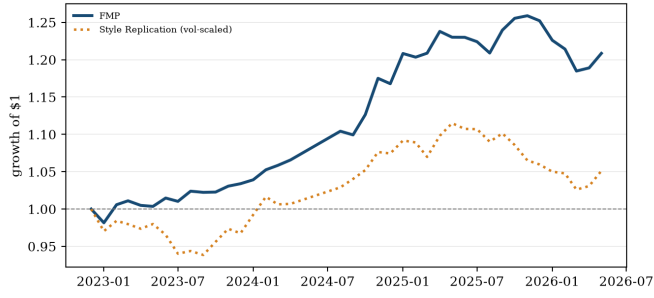


Figure: Common-window cumulative growth of \$1.

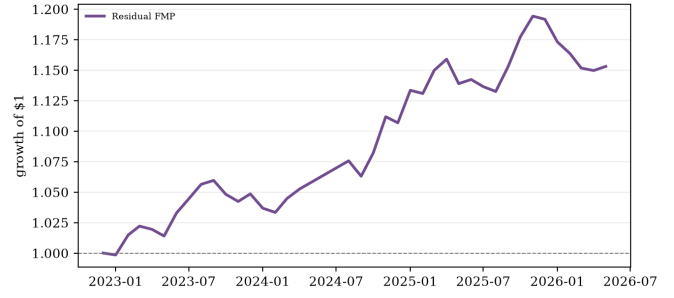


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
revenue_growth_yoy	6.13%	5.34%	1.15	1.88	-5.88%	0.393	5.06%	5.34%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
revenue_growth_yoy	-0.07 (-0.76)	-0.05 (-0.42)	-0.32 (-2.36)	0.01 (0.08)	0.35 (2.05)	-0.16 (-1.31)	4.59% (1.44)	0.396

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
revenue_growth_yoy	-1.44%	0.35%	2.01%	0.08%	-1.29%	1.83%	4.59%	6.13%

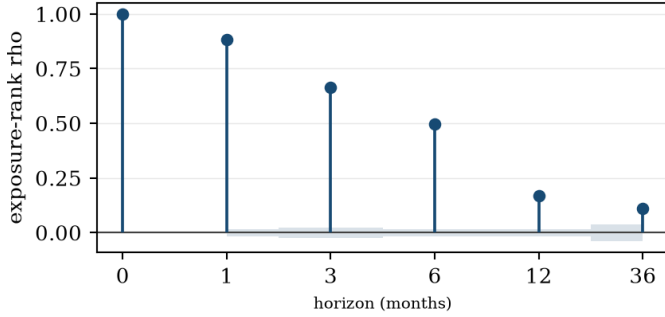


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

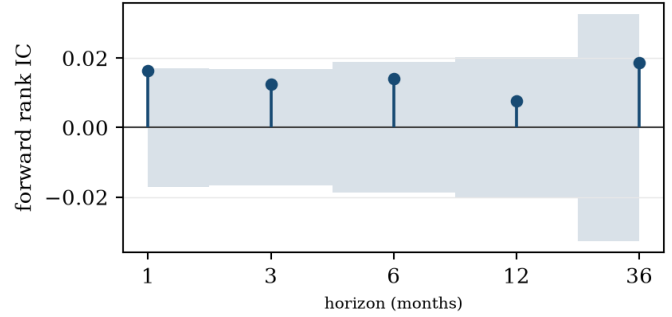


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	revenue_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2031 / 2418 / 2633
Available-history Sharpe	1.00
Bootstrap Sharpe	1.15 [0.04, 2.30]; Pr(S<=0)=2.1%
Bootstrap tails	skew=0.54 [-0.16, 1.11]; max DD=-5.9% [-9.2%, -0.9%]

4. sga_to_revenue

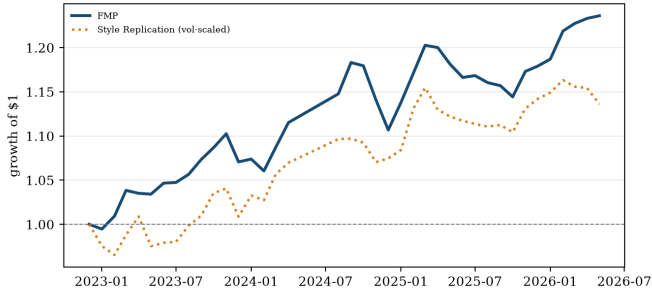


Figure: Common-window cumulative growth of \$1.

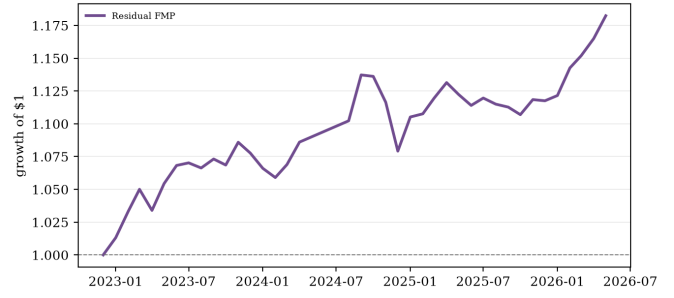


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
sga_to_revenue	6.89%	6.03%	1.14	2.02	-6.45%	0.425	4.14%	6.03%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
sga_to_revenue	0.08 (1.10)	-0.28 (-3.72)	0.11 (1.27)	-0.00 (-0.07)	0.16 (1.15)	0.06 (0.58)	5.40% (1.52)	0.437

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
sga_to_revenue	1.58%	1.90%	-0.68%	-0.05%	-0.58%	-0.69%	5.40%	6.89%

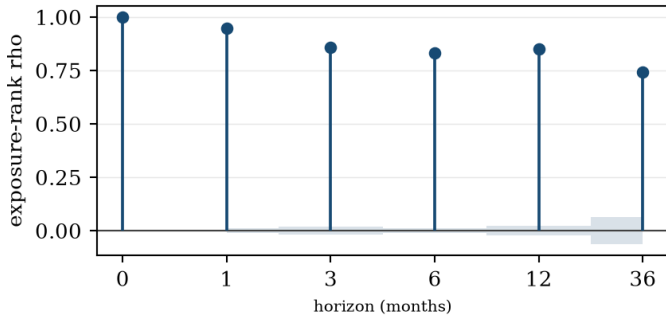


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

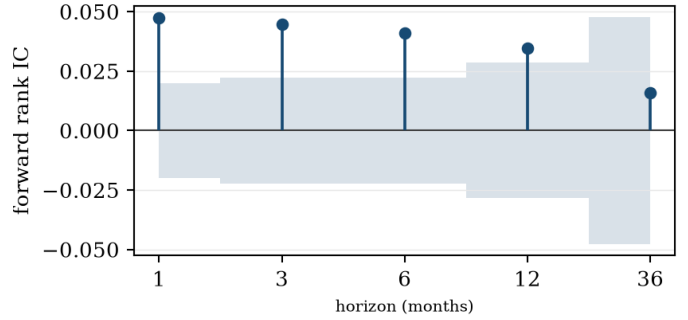


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	sga_to_revenue(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1970 / 2320 / 2505
Available-history Sharpe	1.31
Bootstrap Sharpe	1.14 [0.39, 2.20]; Pr(S<=0)=0.1%
Bootstrap tails	skew=-0.36 [-0.78, 0.46]; max DD=-6.4% [-9.8%, -3.1%]

5. momentum_21_126

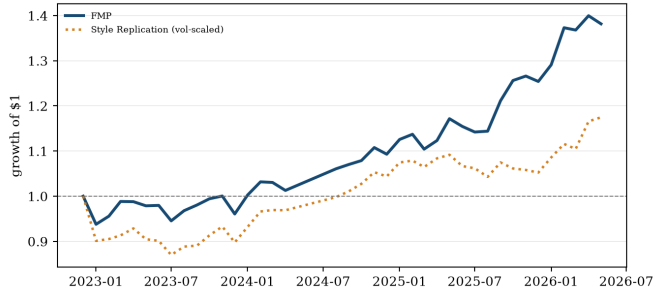


Figure: Common-window cumulative growth of \$1.

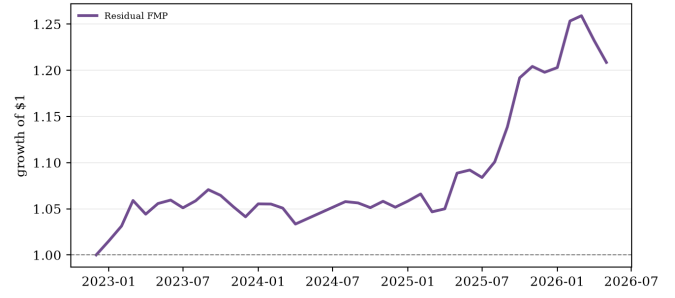


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_126	10.68%	9.44%	1.13	2.83	-4.34%	0.622	6.32%	9.44%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_126	-0.03 (-0.26)	-0.14 (-1.35)	-0.12 (-0.77)	0.42 (4.22)	0.43 (1.84)	-0.06 (-0.35)	6.16% (1.85)	0.622

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_126	-0.66%	0.97%	0.74%	4.36%	-1.60%	0.71%	6.16%	10.68%

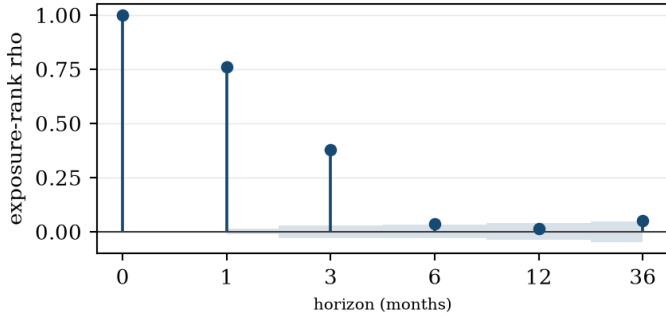


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

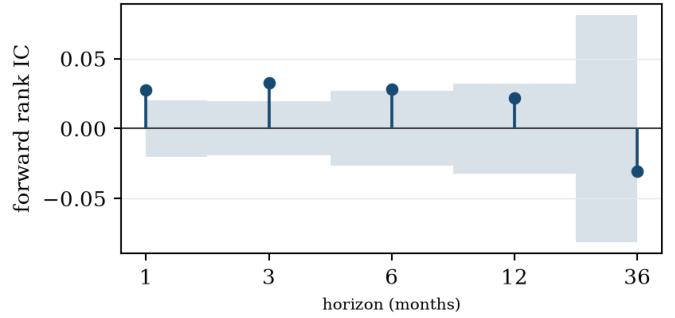


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	momentum_21_126(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.59
Bootstrap Sharpe	1.13 [0.44, 1.91]; Pr(S<=0)=0.2%
Bootstrap tails	skew=-0.26 [-0.68, 0.27]; max DD=-4.3% [-11.7%, -2.9%]

6. momentum_21_252_vol_63d

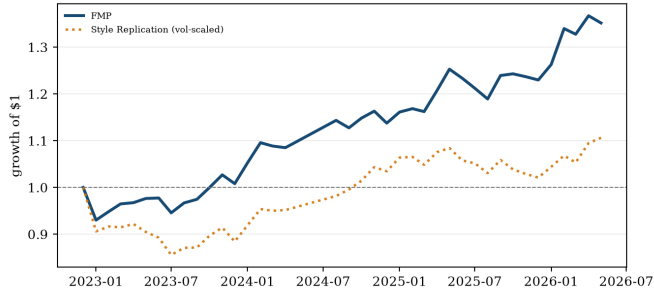


Figure: Common-window cumulative growth of \$1.

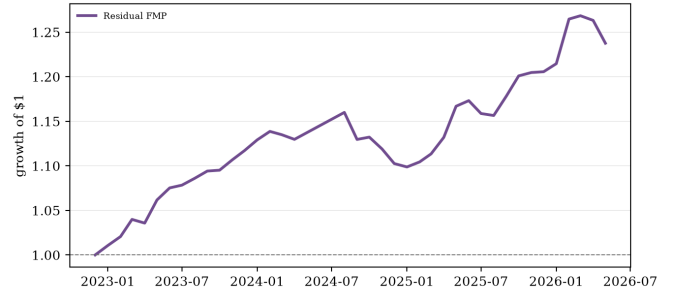


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_252_vol_63d	9.96%	9.17%	1.09	2.55	-5.09%	0.735	6.87%	9.17%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_252_vol_63d	-0.15	-0.12	-0.30	0.37	0.63	-0.19	6.86%	0.735
	(-2.32)	(-1.10)	(-3.19)	(6.82)	(4.38)	(-1.62)	(1.84)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_252_vol_63d	-3.19%	0.81%	1.88%	3.85%	-2.34%	2.09%	6.86%	9.96%

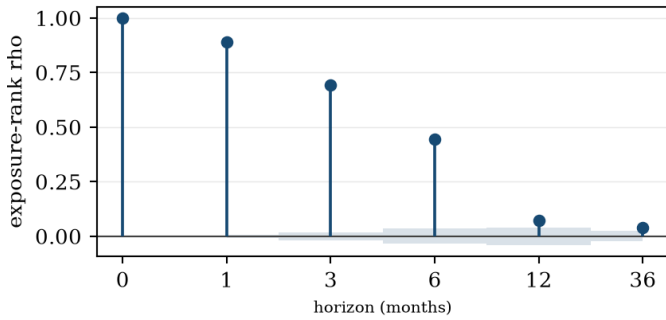


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

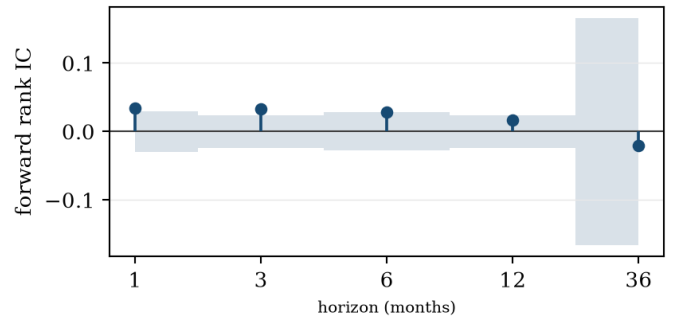


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	momentum_21_252_vol_63d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2811
Available-history Sharpe	0.66
Bootstrap Sharpe	1.09 [0.43, 1.85]; Pr(S<=0)=0.2%
Bootstrap tails	skew=-0.35 [-0.93, 0.47]; max DD=-5.1% [-12.7%, -3.2%]

7. rd_to_opex

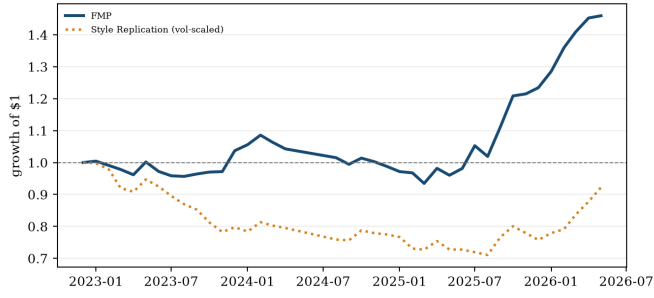


Figure: Common-window cumulative growth of \$1.

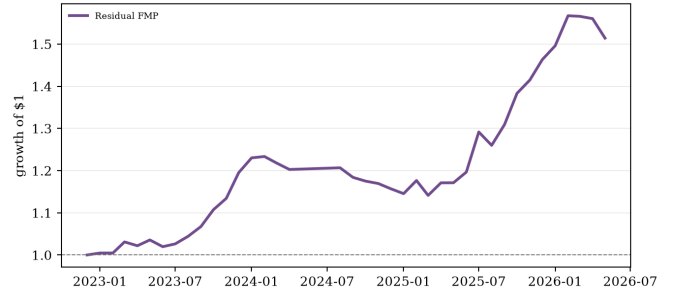


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
rd_to_opex	12.65%	11.73%	1.08	1.58	-13.92%	0.441	13.91%	11.73%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
rd_to_opex	-0.53 (-4.64)	0.12 (0.66)	0.25 (1.24)	0.37 (3.56)	-0.29 (-0.84)	-0.68 (-3.61)	13.54% (1.88)	0.441

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
rd_to_opex	-11.12%	-0.83%	-1.55%	3.90%	1.07%	7.64%	13.54%	12.65%

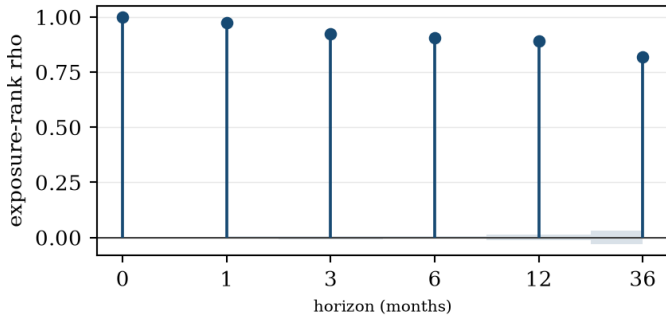


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

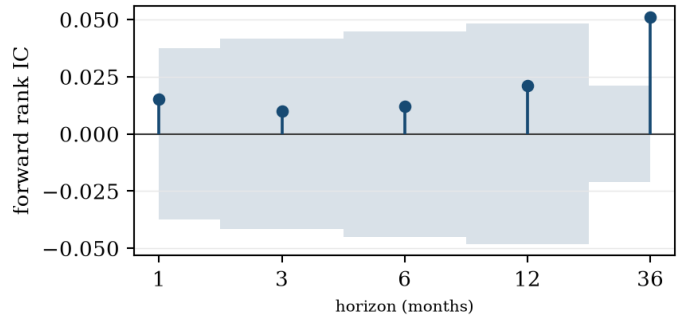


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	rd_to_opex(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	963 / 1306 / 1426
Available-history Sharpe	0.84
Bootstrap Sharpe	1.08 [-0.10, 2.20]; Pr(S<=0)=3.6%
Bootstrap tails	skew=0.84 [0.30, 1.29]; max DD=-13.9% [-19.9%, -4.8%]

8. resid_momentum_21_252

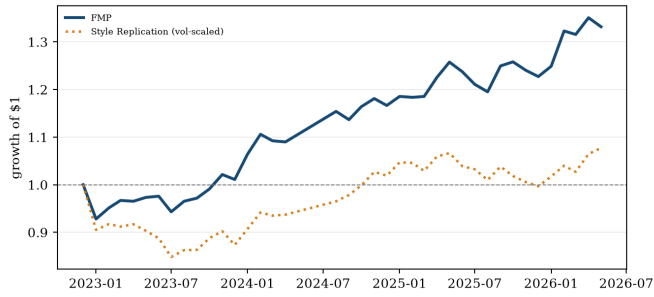


Figure: Common-window cumulative growth of \$1.

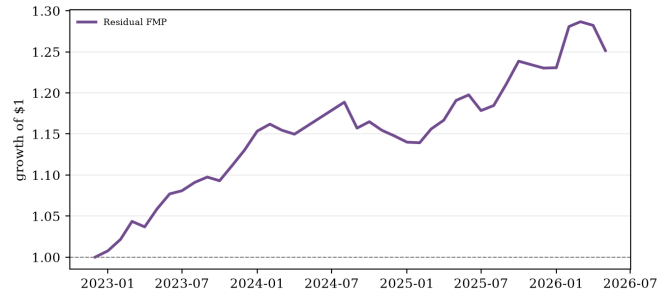


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
resid_momentum_21_252	9.50%	9.23%	1.03	2.41	-4.97%	0.703	7.17%	9.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
resid_momentum_21_252	-0.21	-0.10	-0.32	0.36	0.62	-0.22	7.23%	0.703
	(-3.06)	(-0.94)	(-3.62)	(6.99)	(3.69)	(-1.57)	(2.37)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
resid_momentum_21_252	-4.38%	0.71%	2.02%	3.76%	-2.30%	2.46%	7.23%	9.50%

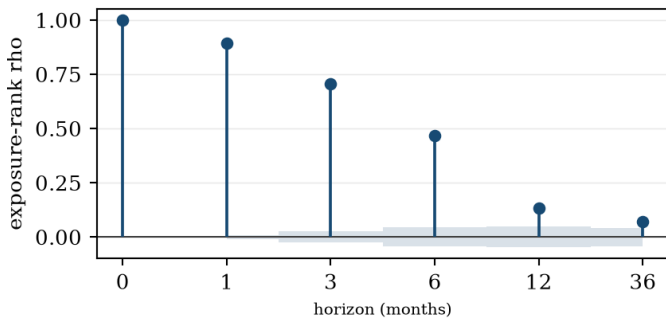


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

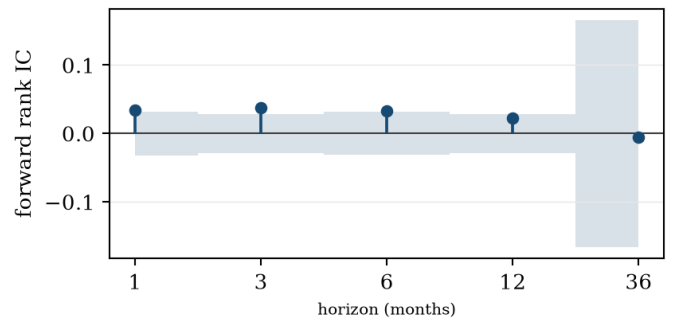


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	resid_momentum_21_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2001 / 2286 / 2734
Available-history Sharpe	0.57
Bootstrap Sharpe	1.03 [0.31, 1.88]; Pr(S<=0)=0.2%
Bootstrap tails	skew=-0.30 [-0.93, 0.65]; max DD=-5.0% [-13.0%, -3.4%]

9. ret_zscore_63d

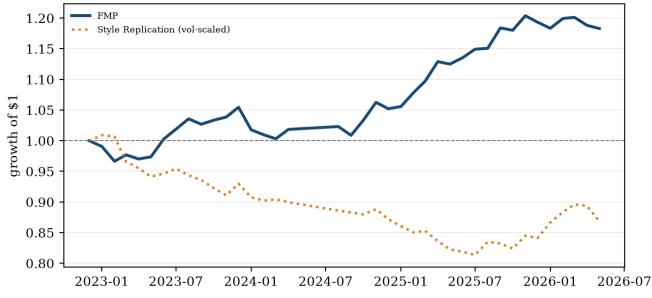


Figure: Common-window cumulative growth of \$1.

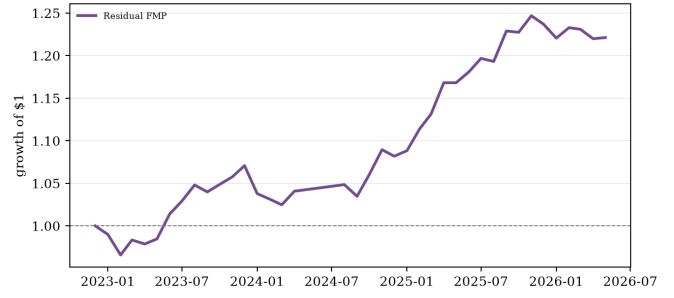


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ret_zscore_63d	5.46%	5.32%	1.03	1.84	-4.88%	0.029	6.19%	5.32%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ret_zscore_63d	-0.02 (-0.19)	0.07 (0.45)	0.04 (0.34)	0.02 (0.20)	0.03 (0.15)	-0.02 (-0.09)	6.46% (1.57)	0.030

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ret_zscore_63d	-0.49%	-0.50%	-0.27%	0.19%	-0.13%	0.20%	6.46%	5.46%

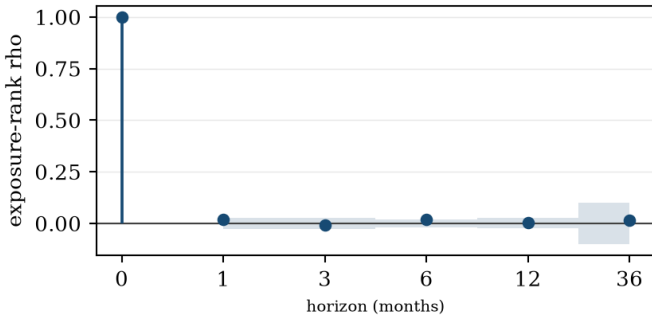


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

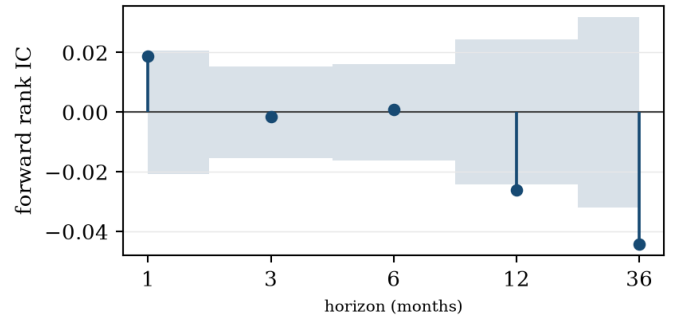


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	ret_zscore_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2811
Available-history Sharpe	1.02
Bootstrap Sharpe	1.03 [0.07, 2.13]; Pr(S<=0)=1.8%
Bootstrap tails	skew=-0.24 [-0.66, 0.32]; max DD=-4.9% [-9.2%, -1.8%]

10. dio

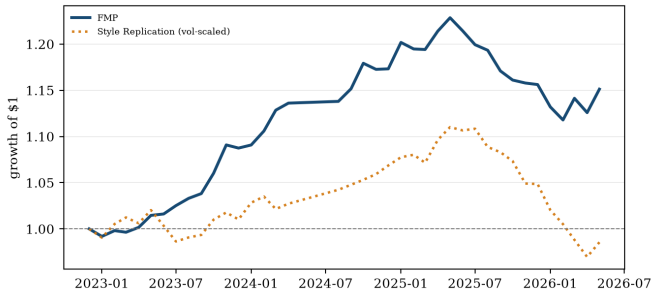


Figure: Common-window cumulative growth of \$1.

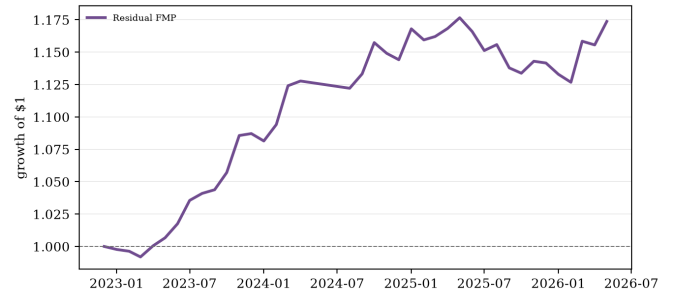


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dio	4.55%	4.47%	1.02	1.55	-9.00%	0.239	4.73%	4.47%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dio	-0.09 (-0.88)	-0.08 (-1.03)	-0.23 (-3.02)	-0.08 (-1.57)	0.12 (1.01)	-0.05 (-0.38)	5.14% (2.44)	0.243

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dio	-1.79%	0.52%	1.46%	-0.84%	-0.45%	0.51%	5.14%	4.55%

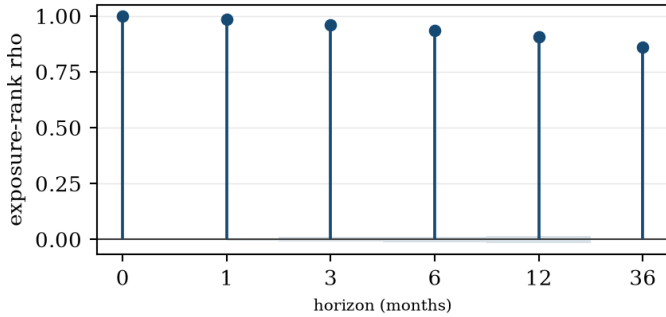


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

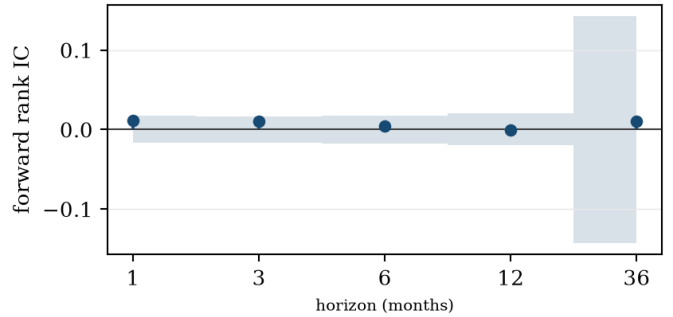


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	dio(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1792 / 1858 / 2001
Available-history Sharpe	1.00
Bootstrap Sharpe	1.02 [-0.54, 2.92]; Pr(S<=0)=8.4%
Bootstrap tails	skew=0.12 [-0.14, 0.65]; max DD=-9.0% [-14.1%, -0.8%]

11. momentum_21_252

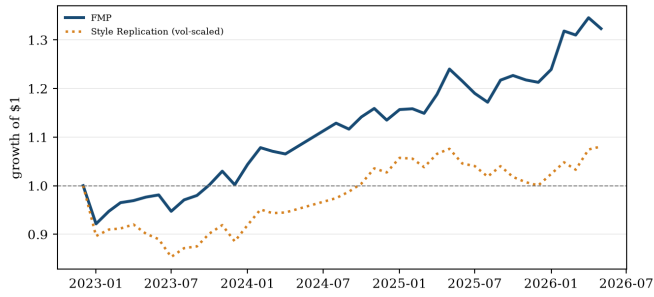


Figure: Common-window cumulative growth of \$1.

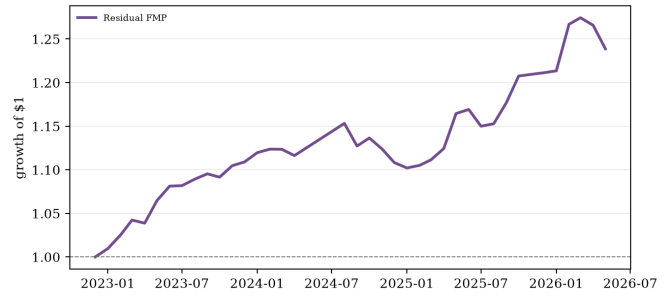


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_252	9.33%	9.62%	0.97	2.45	-5.50%	0.721	6.85%	9.62%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_252	-0.14 (-1.86)	-0.08 (-0.72)	-0.31 (-3.07)	0.39 (6.98)	0.58 (3.68)	-0.07 (-0.57)	6.90% (1.94)	0.721

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_252	-2.83%	0.56%	1.91%	4.13%	-2.15%	0.80%	6.90%	9.33%

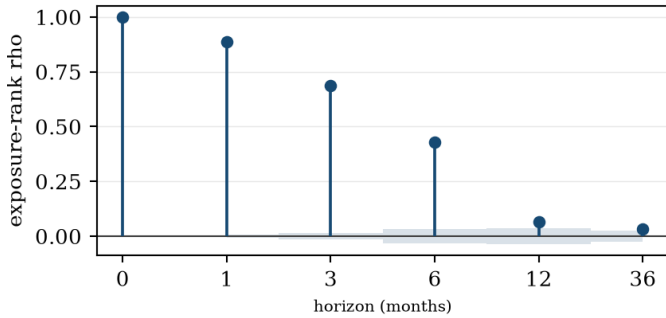


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

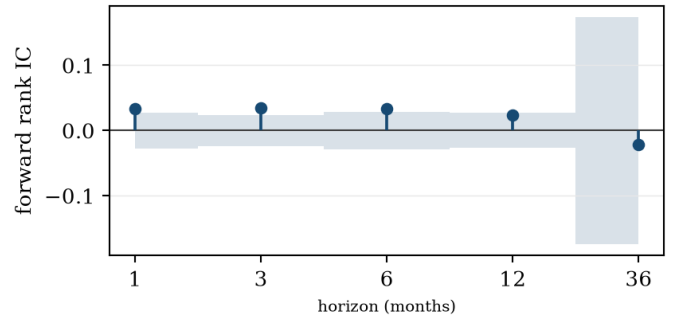


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	momentum_21_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.53
Bootstrap Sharpe	0.97 [0.35, 1.71]; Pr(S<=0)=0.2%
Bootstrap tails	skew=-0.50 [-1.12, 0.54]; max DD=-5.5% [-13.9%, -3.4%]

12. bab_252v_1260c_SPY

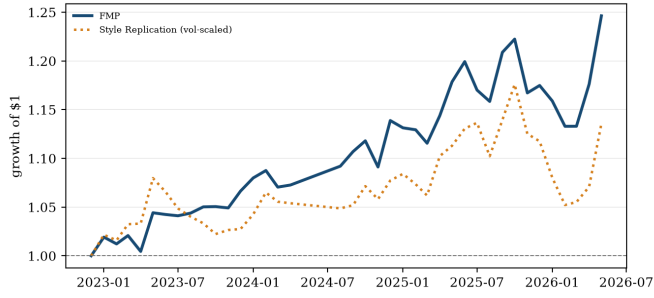


Figure: Common-window cumulative growth of \$1.

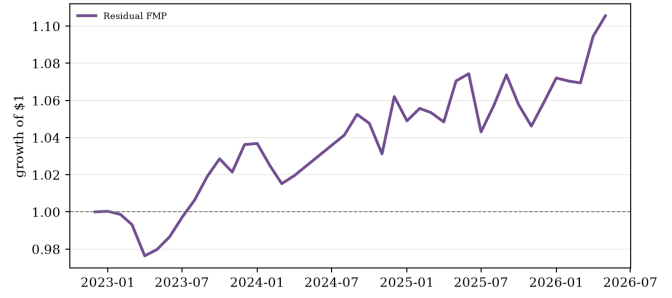


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_252v_1260c_SPY	7.24%	7.50%	0.97	2.20	-7.32%	0.651	3.78%	7.50%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_252v_1260c_SPY	-0.16 (-2.05)	-0.29 (-2.96)	-0.18 (-1.98)	-0.05 (-0.74)	-0.34 (-1.81)	-0.32 (-2.40)	3.27% (1.48)	0.653

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_252v_1260c_SPY	-3.39%	1.99%	1.10%	-0.53%	1.27%	3.54%	3.27%	7.24%

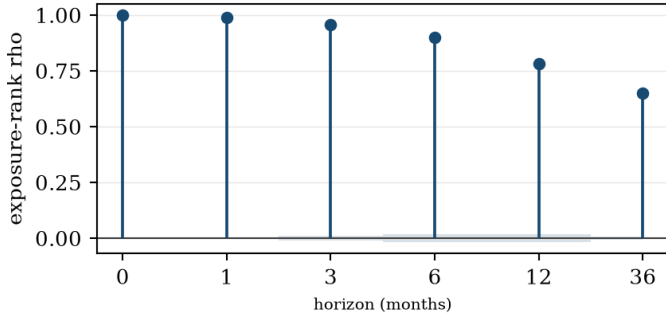


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

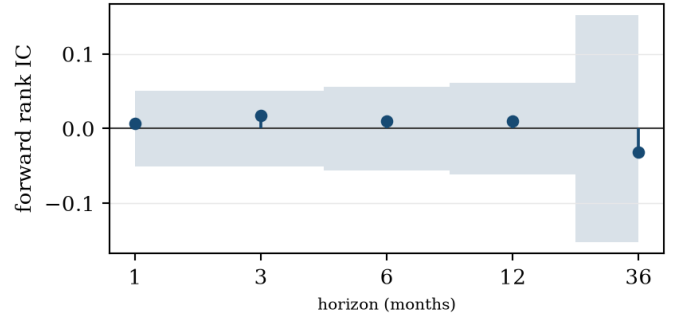


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	bab_252v_1260c_SPY(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.94
Bootstrap Sharpe	0.97 [0.38, 1.62]; Pr(S<=0)=0.2%
Bootstrap tails	skew=0.29 [-0.18, 1.04]; max DD=-7.3% [-10.5%, -2.4%]

13. bab_shrunk_252v_1260c_SPY

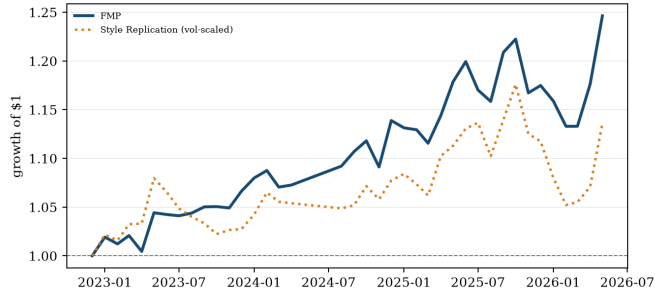


Figure: Common-window cumulative growth of \$1.

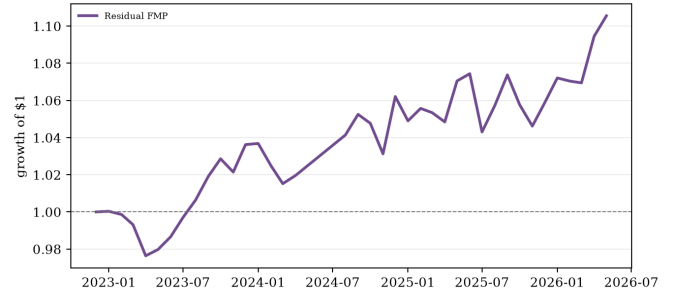


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_shrunk_252v_1260c_SPY	7.24%	7.50%	0.97	2.20	-7.32%	0.651	3.78%	7.50%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_shrunk_252v_1260c_SPY	-0.16	-0.29	-0.18	-0.05	-0.34	-0.32	3.27%	0.653
	(-2.05)	(-2.96)	(-1.98)	(-0.74)	(-1.81)	(-2.40)	(1.48)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_shrunk_252v_1260c_SPY	-3.39%	1.99%	1.10%	-0.53%	1.27%	3.54%	3.27%	7.24%

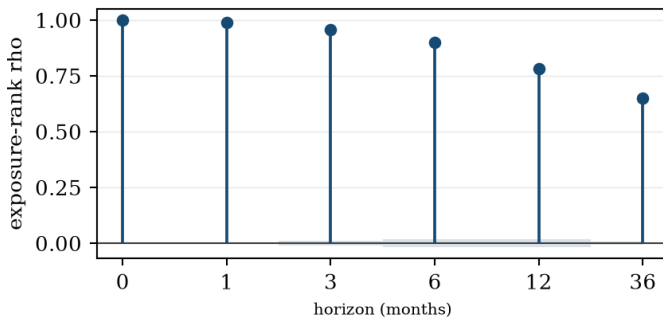


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

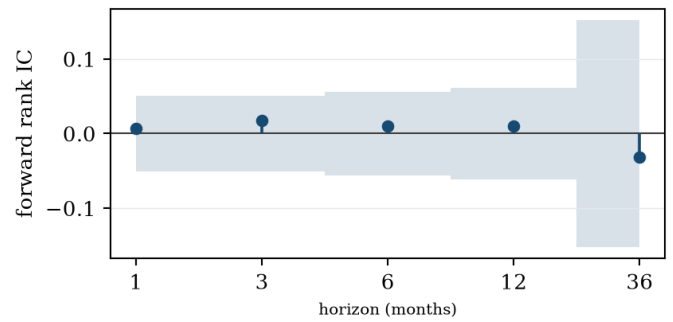


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	bab_shrunk_252v_1260c_SPY(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.94
Bootstrap Sharpe	0.97 [0.38, 1.62]; Pr(S<=0)=0.2%
Bootstrap tails	skew=0.29 [-0.18, 1.04]; max DD=-7.3% [-10.5%, -2.4%]

14. beta_252v_1260c_SPY

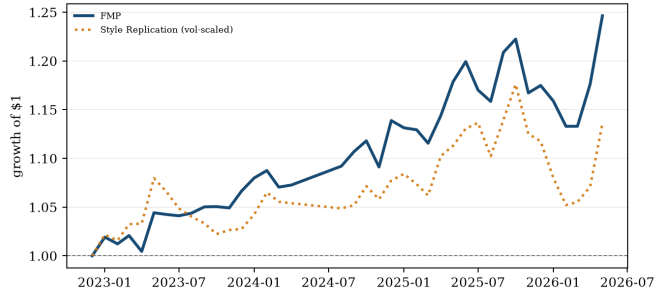


Figure: Common-window cumulative growth of \$1.

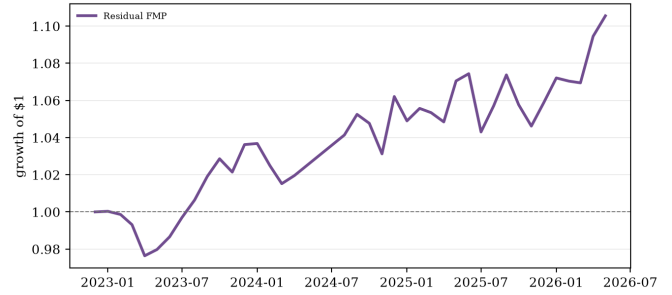


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_252v_1260c_SPY	7.24%	7.50%	0.97	2.20	-7.32%	0.651	3.78%	7.50%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_252v_1260c_SPY	-0.16 (-2.05)	-0.29 (-2.96)	-0.18 (-1.98)	-0.05 (-0.74)	-0.34 (-1.81)	-0.32 (-2.40)	3.27% (1.48)	0.653

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_252v_1260c_SPY	-3.39%	1.99%	1.10%	-0.53%	1.27%	3.54%	3.27%	7.24%

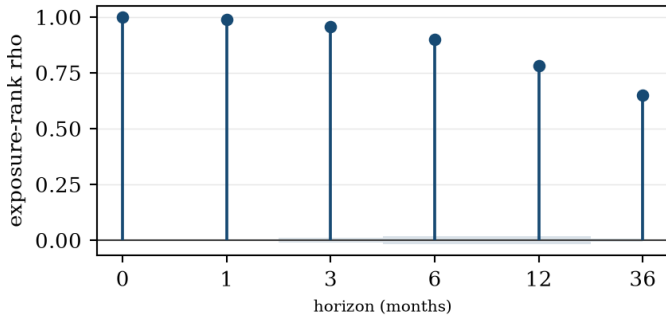


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

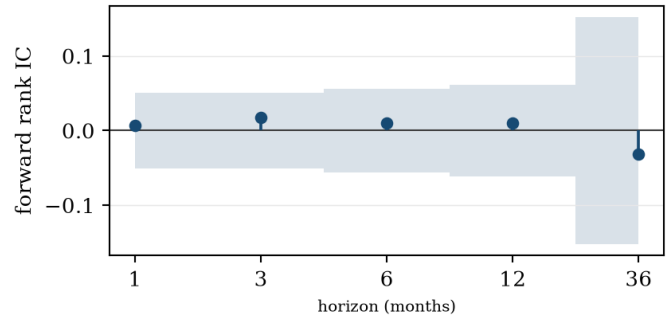


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	beta_252v_1260c_SPY(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.94
Bootstrap Sharpe	0.97 [0.38, 1.62]; Pr(S<=0)=0.2%
Bootstrap tails	skew=0.29 [-0.18, 1.04]; max DD=-7.3% [-10.5%, -2.4%]

15. beta_shrunk_252v_1260c_SPY

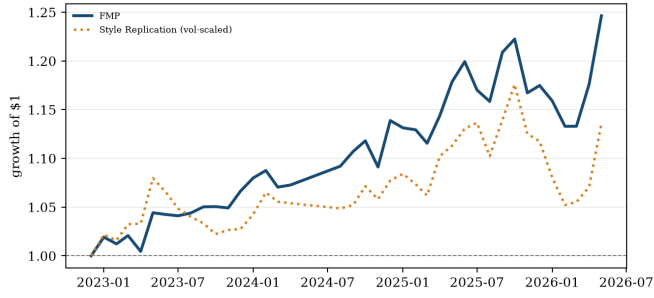


Figure: Common-window cumulative growth of \$1.

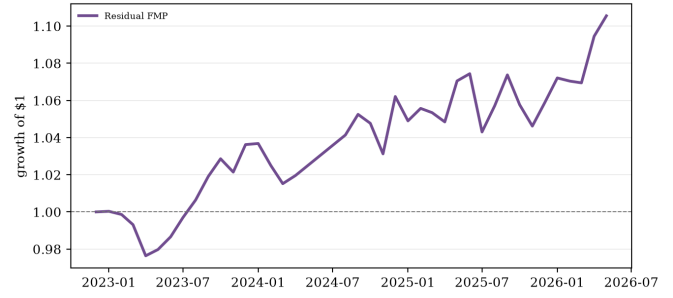


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_shrunk_252v_1260c_SPY	7.24%	7.50%	0.97	2.20	-7.32%	0.651	3.78%	7.50%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_shrunk_252v_1260c_SPY	-0.16	-0.29	-0.18	-0.05	-0.34	-0.32	3.27%	0.653
	(-2.05)	(-2.96)	(-1.98)	(-0.74)	(-1.81)	(-2.40)	(1.48)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_shrunk_252v_1260c_SPY	-3.39%	1.99%	1.10%	-0.53%	1.27%	3.54%	3.27%	7.24%

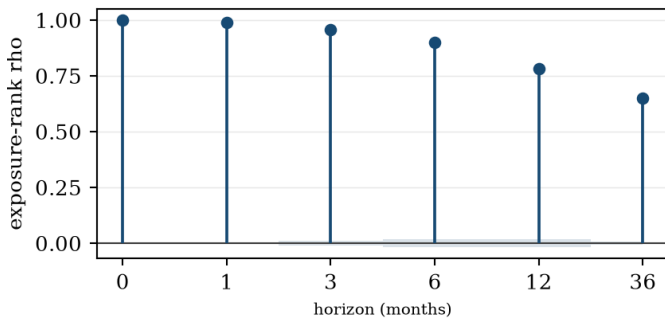


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

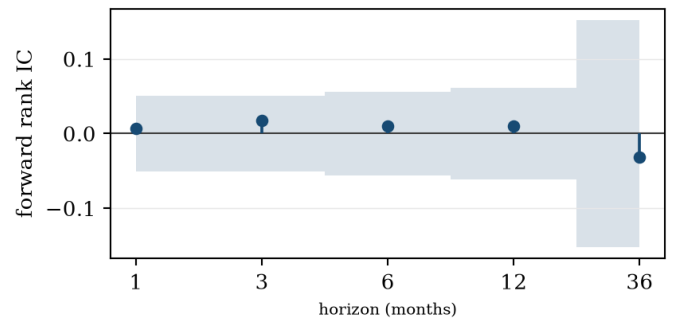


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	beta_shrunk_252v_1260c_SPY(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.94
Bootstrap Sharpe	0.97 [0.38, 1.62]; Pr(S<=0)=0.2%
Bootstrap tails	skew=0.29 [-0.18, 1.04]; max DD=-7.3% [-10.5%, -2.4%]

16. bab_252v_1260c_VTI

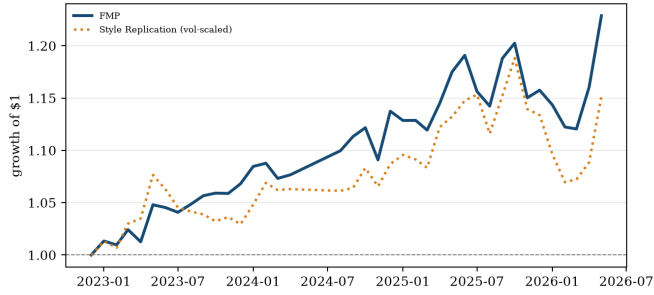


Figure: Common-window cumulative growth of \$1.

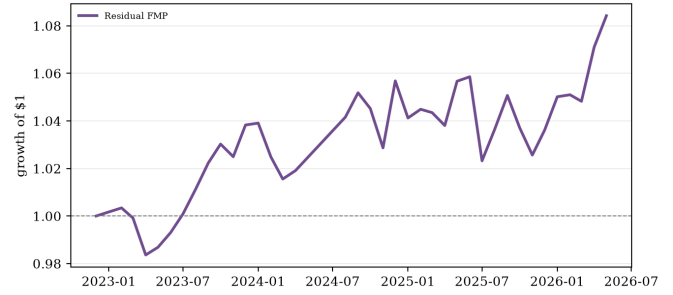


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_252v_1260c_VTI	6.77%	7.17%	0.94	2.19	-6.82%	0.630	3.04%	7.17%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_252v_1260c_VTI	-0.15 (-2.03)	-0.35 (-3.67)	-0.15 (-1.80)	-0.04 (-0.54)	-0.34 (-1.91)	-0.26 (-2.14)	2.65% (1.26)	0.631

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_252v_1260c_VTI	-3.02%	2.37%	0.95%	-0.38%	1.27%	2.93%	2.65%	6.77%

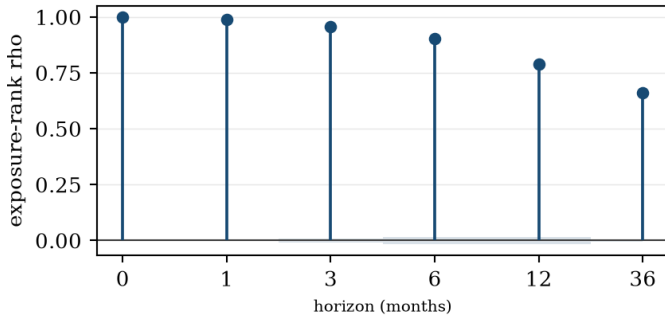


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

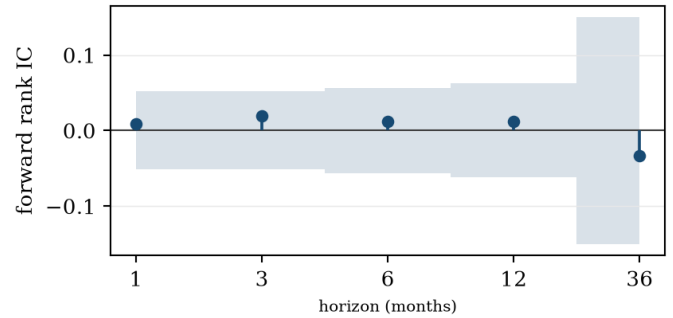


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	bab_252v_1260c_VTI(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.92
Bootstrap Sharpe	0.94 [0.33, 1.68]; Pr(S<=0)=0.3%
Bootstrap tails	skew=0.24 [-0.30, 1.08]; max DD=-6.8% [-9.8%, -2.7%]

17. bab_shrunk_252v_1260c_VTI

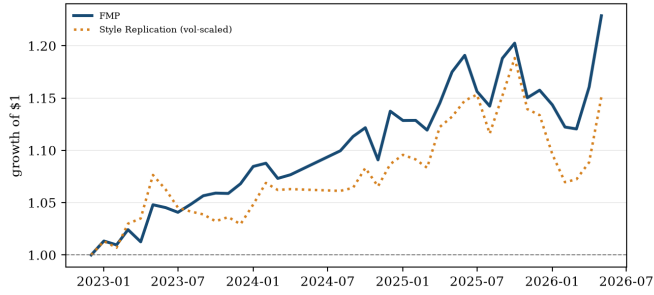


Figure: Common-window cumulative growth of \$1.

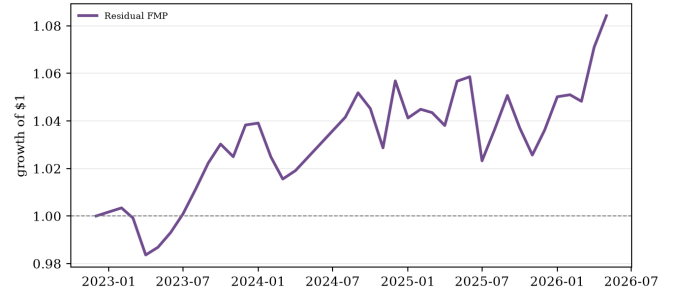


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_shrunk_252v_1260c_VTI	6.77%	7.17%	0.94	2.19	-6.82%	0.630	3.04%	7.17%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_shrunk_252v_1260c_VTI	-0.15	-0.35	-0.15	-0.04	-0.34	-0.26	2.65%	0.631
	(-2.03)	(-3.67)	(-1.80)	(-0.54)	(-1.91)	(-2.14)	(1.26)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_shrunk_252v_1260c_VTI	-3.02%	2.37%	0.95%	-0.38%	1.27%	2.93%	2.65%	6.77%

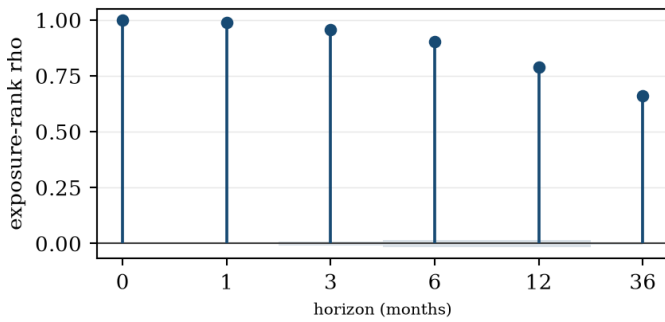


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

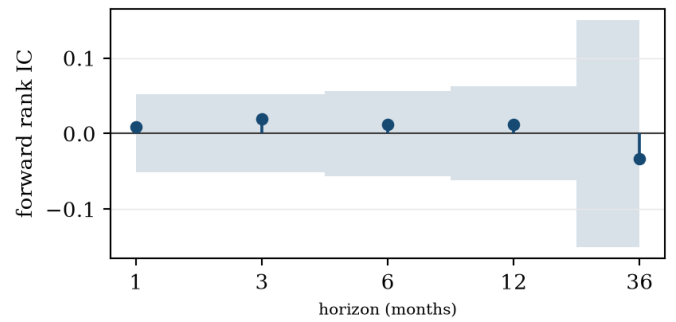


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	bab_shrunk_252v_1260c_VTI(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.92
Bootstrap Sharpe	0.94 [0.33, 1.68]; Pr(S<=0)=0.3%
Bootstrap tails	skew=0.24 [-0.30, 1.08]; max DD=-6.8% [-9.8%, -2.7%]

18. beta_252v_1260c_VTI

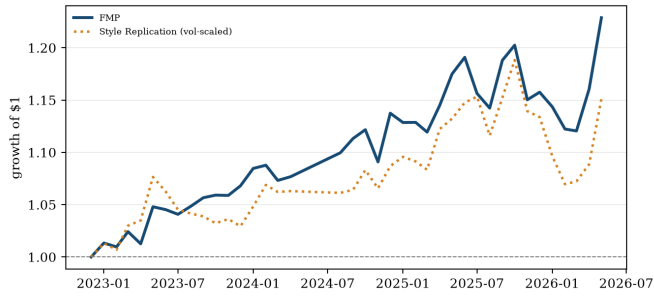


Figure: Common-window cumulative growth of \$1.

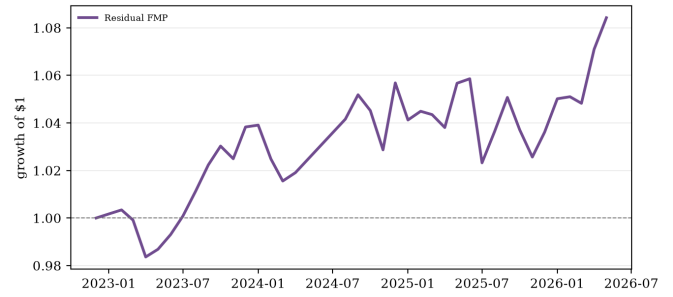


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_252v_1260c_VTI	6.77%	7.17%	0.94	2.19	-6.82%	0.630	3.04%	7.17%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_252v_1260c_VTI	-0.15 (-2.03)	-0.35 (-3.67)	-0.15 (-1.80)	-0.04 (-0.54)	-0.34 (-1.91)	-0.26 (-2.14)	2.65% (1.26)	0.631

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_252v_1260c_VTI	-3.02%	2.37%	0.95%	-0.38%	1.27%	2.93%	2.65%	6.77%

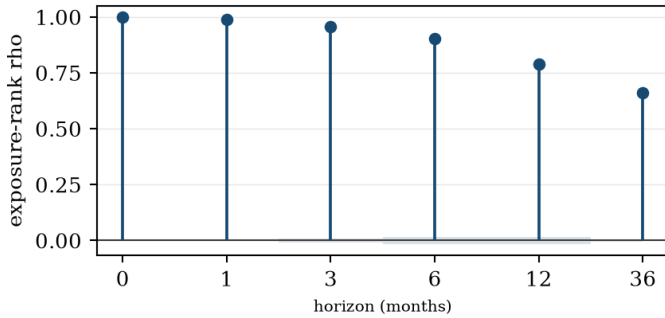


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

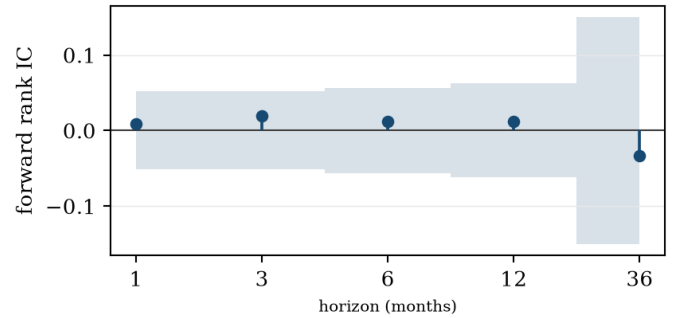


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	beta_252v_1260c_VTI(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.92
Bootstrap Sharpe	0.94 [0.33, 1.68]; Pr(S<=0)=0.3%
Bootstrap tails	skew=0.24 [-0.30, 1.08]; max DD=-6.8% [-9.8%, -2.7%]

19. beta_shrunk_252v_1260c_VTI

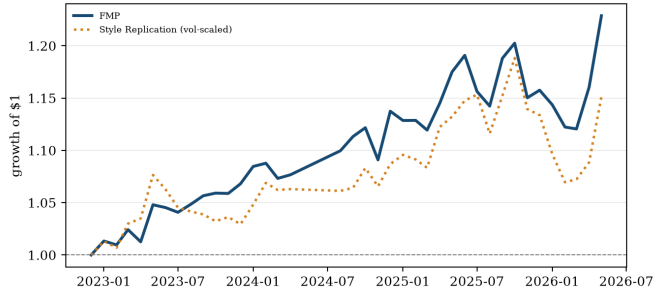


Figure: Common-window cumulative growth of \$1.

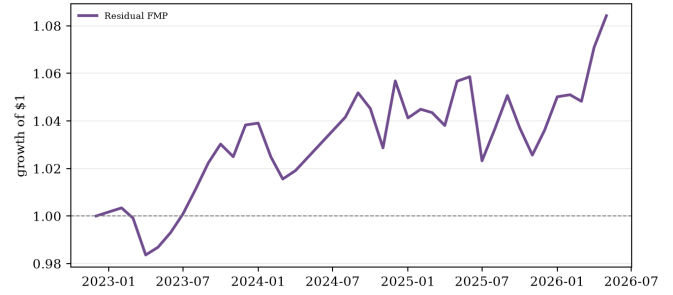


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_shrunk_252v_1260c_VTI	6.77%	7.17%	0.94	2.19	-6.82%	0.630	3.04%	7.17%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_shrunk_252v_1260c_VTI	-0.15	-0.35	-0.15	-0.04	-0.34	-0.26	2.65%	0.631
	(-2.03)	(-3.67)	(-1.80)	(-0.54)	(-1.91)	(-2.14)	(1.26)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_shrunk_252v_1260c_VTI	-3.02%	2.37%	0.95%	-0.38%	1.27%	2.93%	2.65%	6.77%

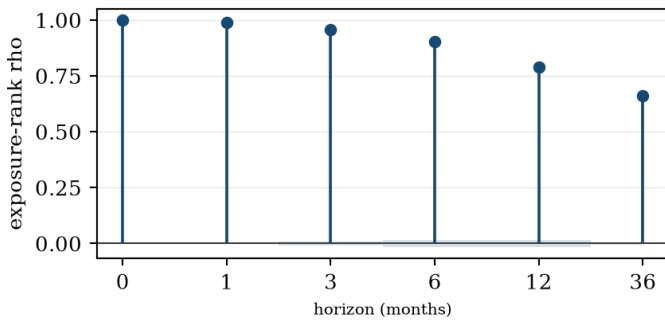


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

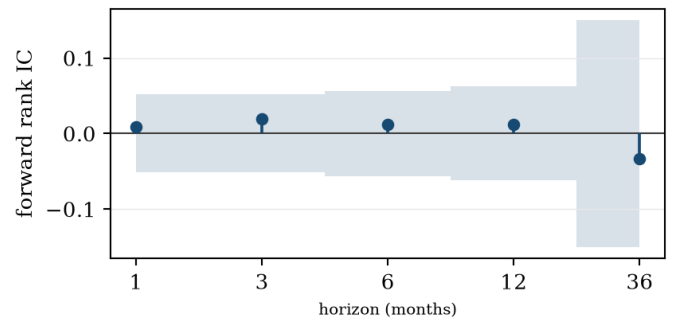


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	beta_shrunk_252v_1260c_VTI(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.92
Bootstrap Sharpe	0.94 [0.33, 1.68]; Pr(S<=0)=0.3%
Bootstrap tails	skew=0.24 [-0.30, 1.08]; max DD=-6.8% [-9.8%, -2.7%]

20. fcf_growth_yoy_ttm

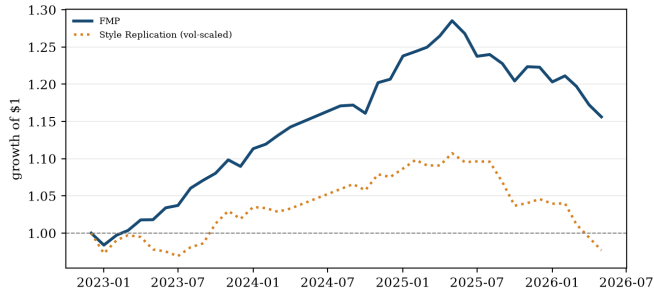


Figure: Common-window cumulative growth of \$1.

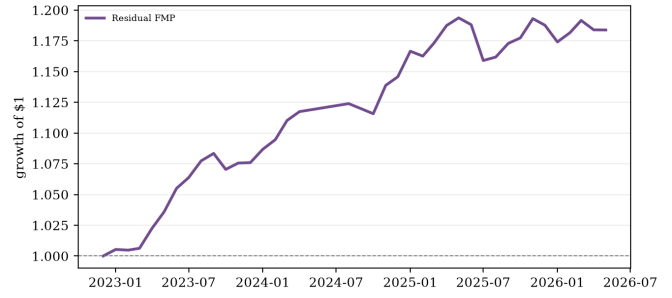


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf_growth_yoy_ttm	4.71%	5.00%	0.94	1.41	-10.05%	0.584	5.18%	5.00%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf_growth_yoy_ttm	0.09 (1.59)	0.01 (0.18)	-0.30 (-4.35)	-0.09 (-1.72)	0.43 (3.68)	0.16 (1.68)	5.39% (2.38)	0.584

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf_growth_yoy_ttm	1.88%	-0.10%	1.89%	-0.93%	-1.59%	-1.83%	5.39%	4.71%

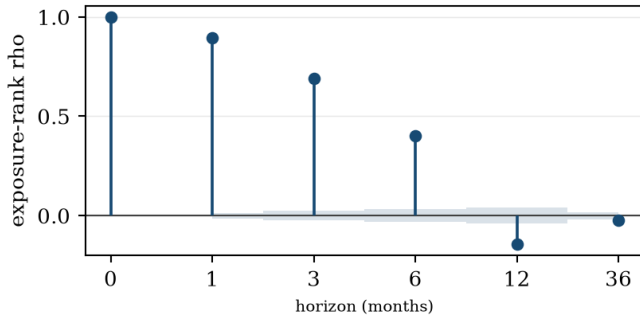


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

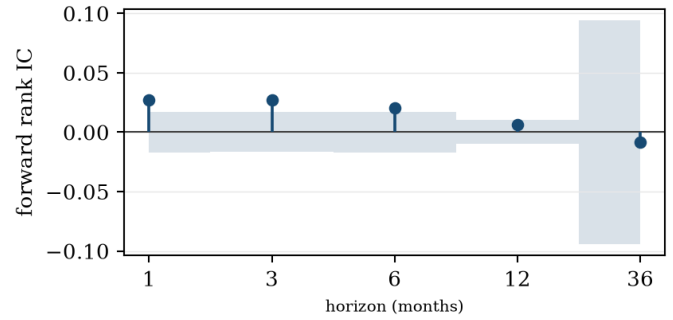


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	fcf_growth_yoy_ttm(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2051 / 2472 / 2642
Available-history Sharpe	1.05
Bootstrap Sharpe	0.94 [-0.54, 2.94]; Pr(S<=0)=10.3%
Bootstrap tails	skew=-0.12 [-0.60, 0.38]; max DD=-10.0% [-16.3%, -1.6%]

21. bab_252v_1260c_FRENCH_CRSP

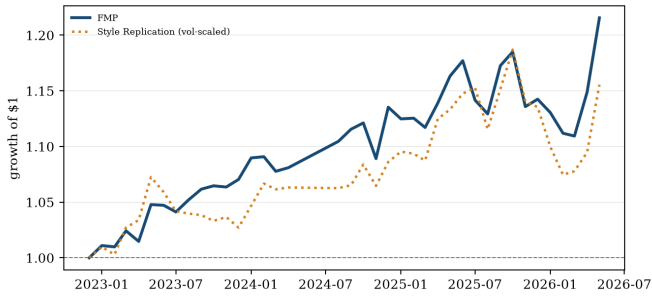


Figure: Common-window cumulative growth of \$1.

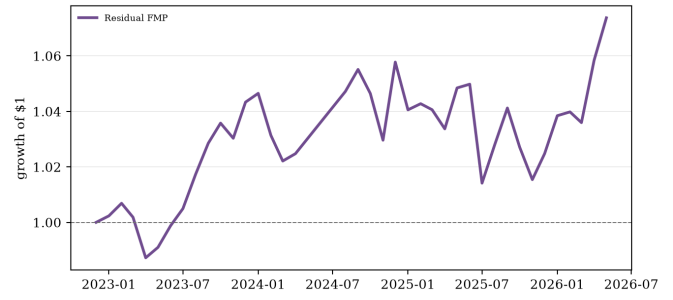


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_252v_1260c_FRENCH_CRSP	6.41%	6.94%	0.92	2.14	-6.38%	0.598	2.70%	6.94%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_252v_1260c_FRENCH_CRSP	-0.14	-0.35	-0.13	-0.02	-0.33	-0.24	2.34%	0.599
	(-1.87)	(-3.76)	(-1.52)	(-0.36)	(-1.83)	(-1.99)	(1.06)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_252v_1260c_FRENCH_CRSP	-2.82%	2.41%	0.81%	-0.25%	1.23%	2.69%	2.34%	6.41%

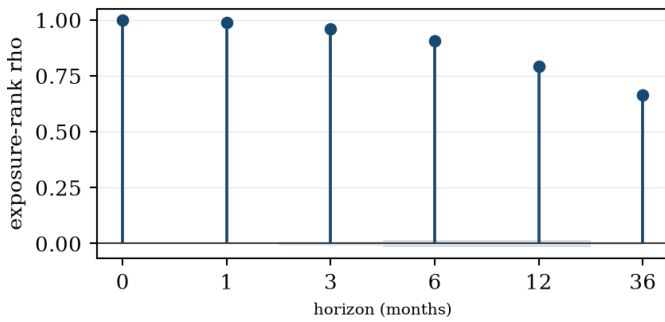


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

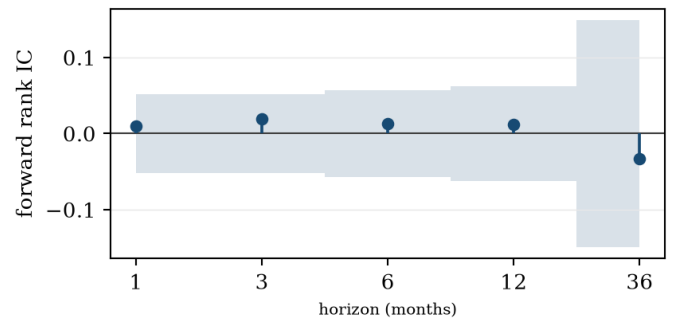


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	bab_252v_1260c_FRENCH_CRSP(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.91
Bootstrap Sharpe	0.92 [0.27, 1.72]; Pr(S<=0)=0.4%
Bootstrap tails	skew=0.27 [-0.30, 1.11]; max DD=-6.4% [-9.4%, -2.9%]

22. bab_shrunk_252v_1260c_FRENCH_CRSP

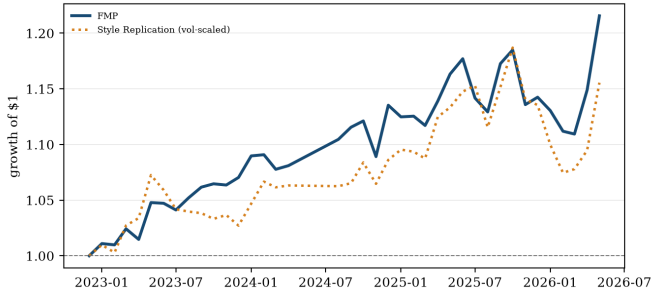


Figure: Common-window cumulative growth of \$1.

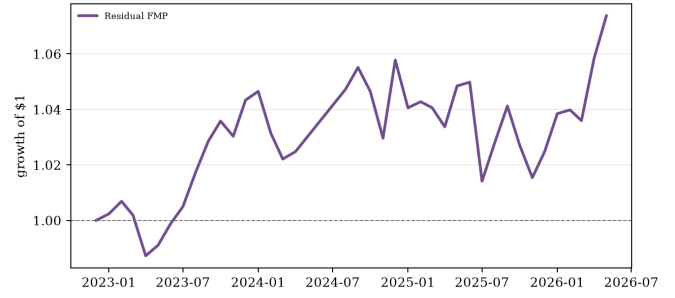


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_shrunk_252v_1260c_FRENCH_CRSP	6.41%	6.94%	0.92	2.14	-6.38%	0.598	2.70%	6.94%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_shrunk_252v_1260c_FRENCH_CRSP	-0.14 (-1.87)	-0.35 (-3.76)	-0.13 (-1.52)	-0.02 (-0.36)	-0.33 (-1.83)	-0.24 (-1.99)	2.34% (1.06)	0.599

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_shrunk_252v_1260c_FRENCH_CRSP	-2.82%	2.41%	0.81%	-0.25%	1.23%	2.69%	2.34%	6.41%

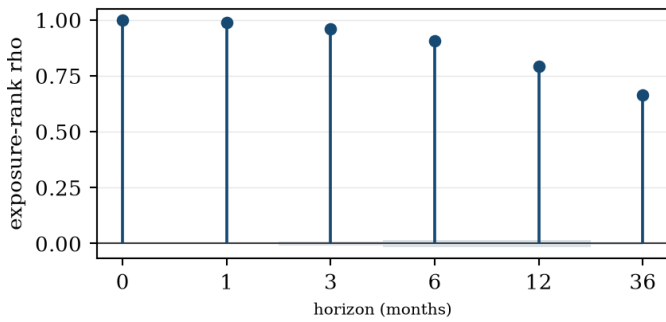


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

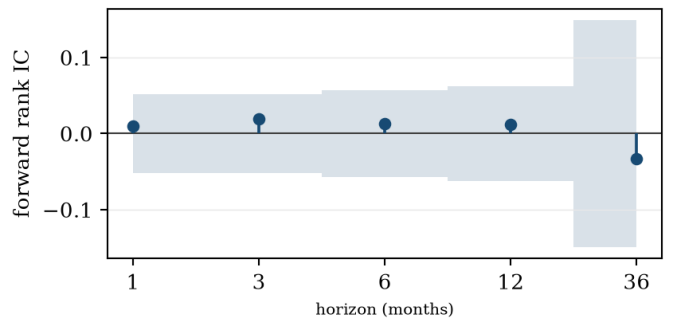


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	bab_shrunk_252v_1260c_FRENCH_CRSP(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.91
Bootstrap Sharpe	0.92 [0.27, 1.72]; Pr(S<=0)=0.4%
Bootstrap tails	skew=0.27 [-0.30, 1.11]; max DD=-6.4% [-9.4%, -2.9%]

23. beta_252v_1260c_FRENCH_CRSP

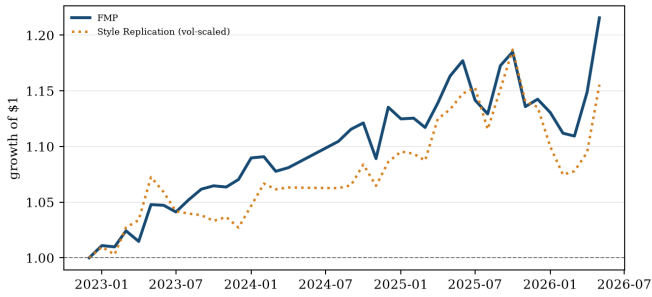


Figure: Common-window cumulative growth of \$1.

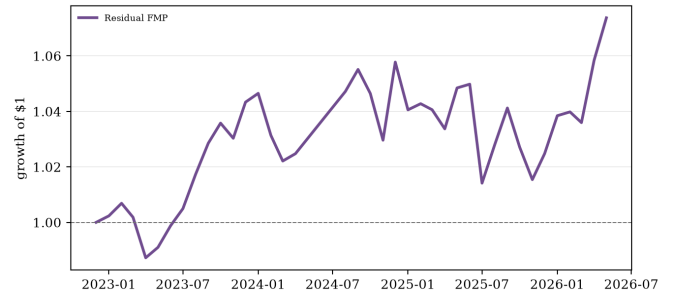


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_252v_1260c_FRENCH_CRSP	6.41%	6.94%	0.92	2.14	-6.38%	0.598	2.70%	6.94%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_252v_1260c_FRENCH_CRSP	-0.14	-0.35	-0.13	-0.02	-0.33	-0.24	2.34%	0.599
	(-1.87)	(-3.76)	(-1.52)	(-0.36)	(-1.83)	(-1.99)	(1.06)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_252v_1260c_FRENCH_CRSP	-2.82%	2.41%	0.81%	-0.25%	1.23%	2.69%	2.34%	6.41%

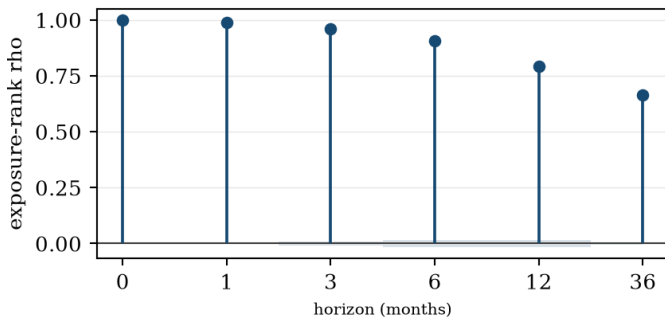


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

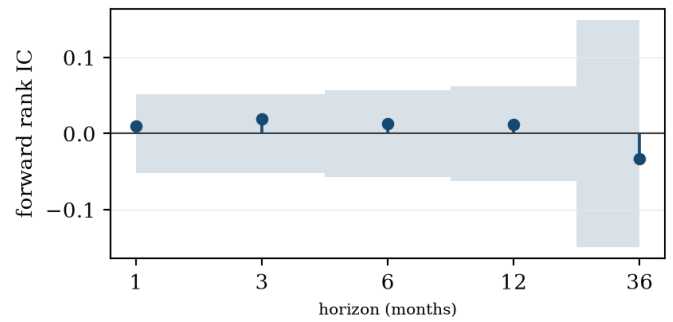


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	beta_252v_1260c_FRENCH_CRSP(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.91
Bootstrap Sharpe	0.92 [0.27, 1.72]; Pr(S<=0)=0.4%
Bootstrap tails	skew=0.27 [-0.30, 1.11]; max DD=-6.4% [-9.4%, -2.9%]

24. beta_shrunk_252v_1260c_FRENCH_CRSP

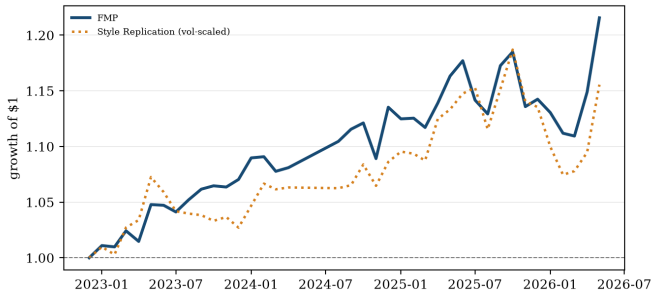


Figure: Common-window cumulative growth of \$1.

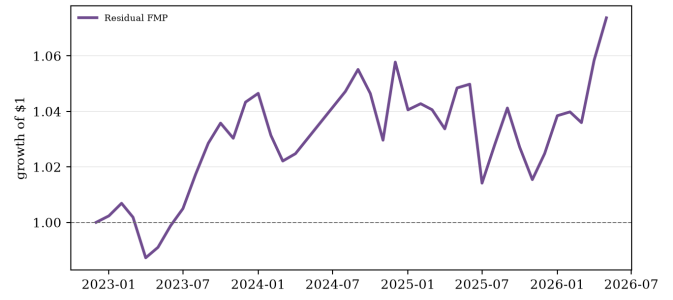


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_shrunk_252v_1260c_FRENCH_CRSP	6.41%	6.94%	0.92	2.14	-6.38%	0.598	2.70%	6.94%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_shrunk_252v_1260c_FRENCH_CRSP	-0.14 (-1.87)	-0.35 (-3.76)	-0.13 (-1.52)	-0.02 (-0.36)	-0.33 (-1.83)	-0.24 (-1.99)	2.34% (1.06)	0.599

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_shrunk_252v_1260c_FRENCH_CRSP	-2.82%	2.41%	0.81%	-0.25%	1.23%	2.69%	2.34%	6.41%

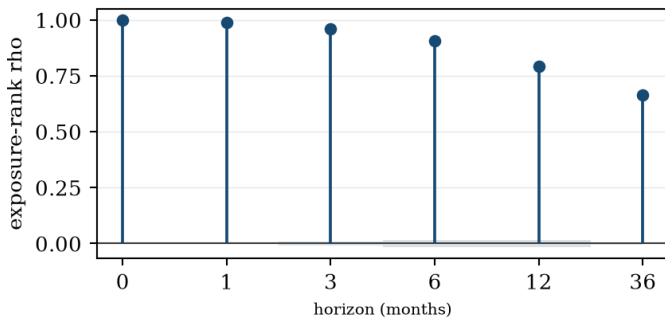


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

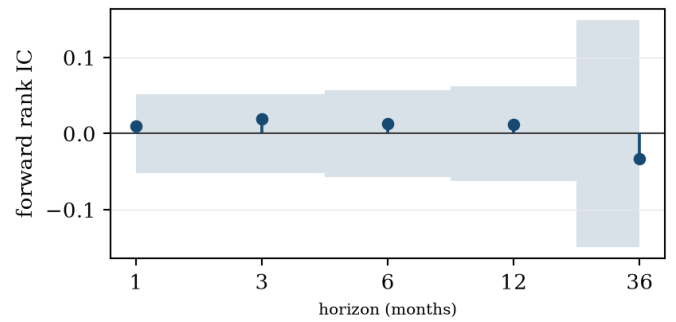


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	beta_shrunk_252v_1260c_FRENCH_CRSP(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.91
Bootstrap Sharpe	0.92 [0.27, 1.72]; Pr(S<=0)=0.4%
Bootstrap tails	skew=0.27 [-0.30, 1.11]; max DD=-6.4% [-9.4%, -2.9%]

25. gross_profit_margin_ttm

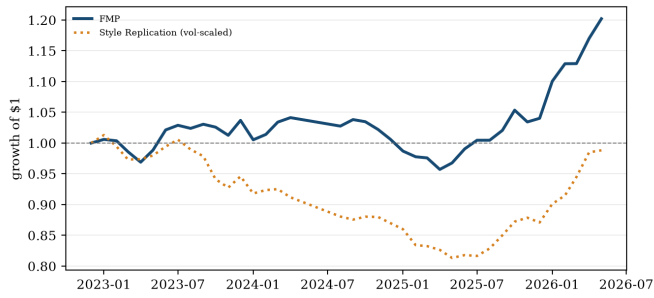


Figure: Common-window cumulative growth of \$1.

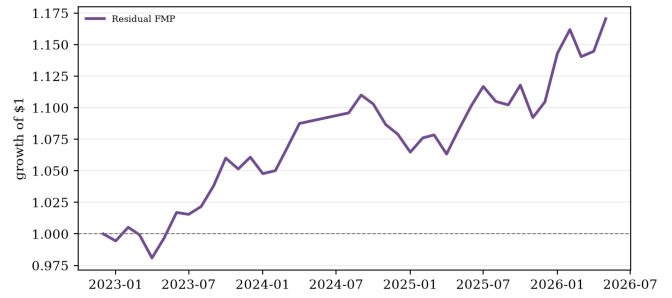


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gross_profit_margin_ttm	6.04%	6.65%	0.91	1.42	-8.08%	0.452	6.15%	6.65%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gross_profit_margin_ttm	0.01	0.22	0.37	0.23	-0.36	-0.07	5.10%	0.461
	(0.09)	(1.80)	(3.52)	(2.96)	(-2.02)	(-0.60)	(1.55)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gross_profit_margin_ttm	0.21%	-1.48%	-2.31%	2.46%	1.33%	0.73%	5.10%	6.04%

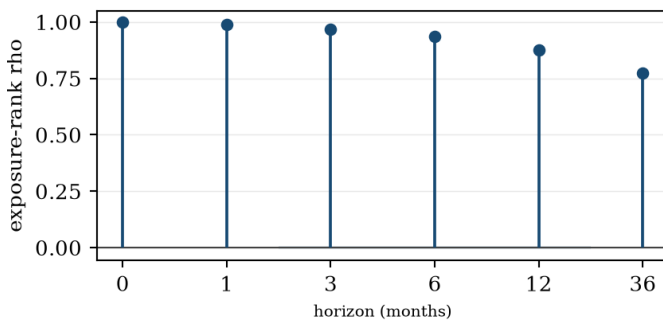


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

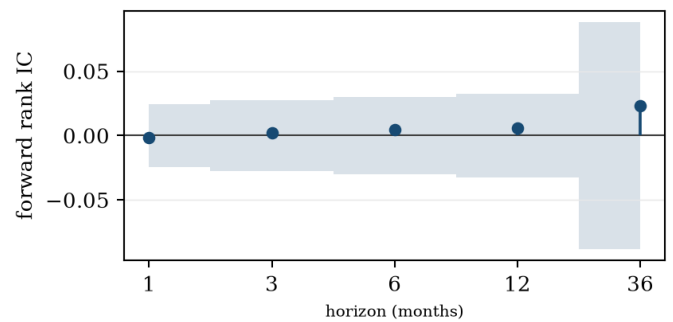


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	gross_profit_margin_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2046 / 2481 / 2633
Available-history Sharpe	0.62
Bootstrap Sharpe	0.91 [-0.31, 1.97]; Pr(S<=0)=7.4%
Bootstrap tails	skew=0.50 [-0.07, 0.89]; max DD=-8.1% [-13.8%, -3.6%]

26. ebit_growth_yoy_ttm

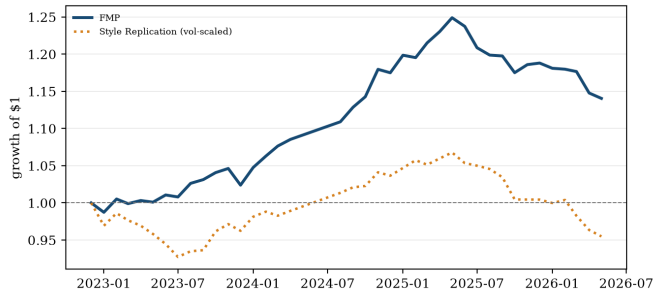


Figure: Common-window cumulative growth of \$1.

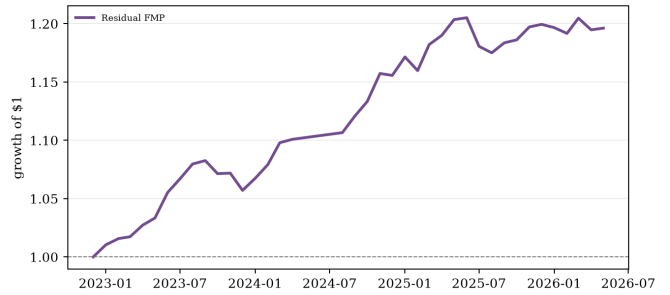


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ebit_growth_yoy_ttm	4.27%	4.75%	0.90	1.30	-8.69%	0.515	5.23%	4.75%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ebit_growth_yoy_ttm	-0.08 (-1.38)	-0.01 (-0.08)	-0.32 (-4.10)	-0.06 (-0.94)	0.51 (4.03)	-0.07 (-0.71)	5.72% (2.17)	0.519

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ebit_growth_yoy_ttm	-1.74%	0.05%	1.96%	-0.58%	-1.90%	0.75%	5.72%	4.27%

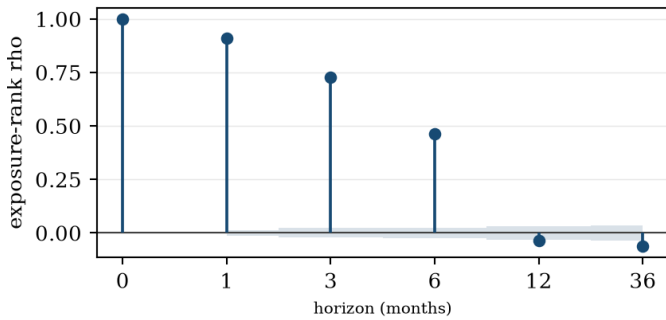


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

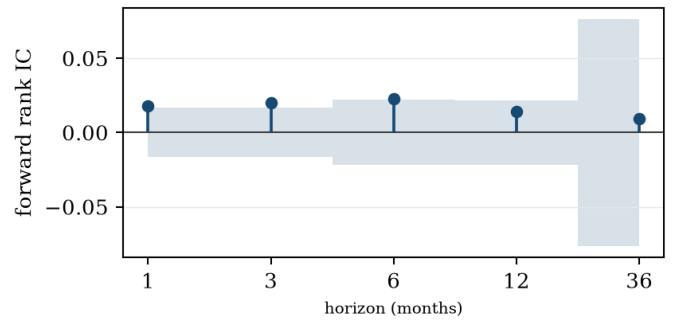


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	ebit_growth_yoy_ttm(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2086 / 2478 / 2574
Available-history Sharpe	0.83
Bootstrap Sharpe	0.90 [-0.71, 2.61]; Pr(S<=0)=12.6%
Bootstrap tails	skew=-0.24 [-0.80, 0.20]; max DD=-8.7% [-14.6%, -2.1%]

27. eps_growth_yoy

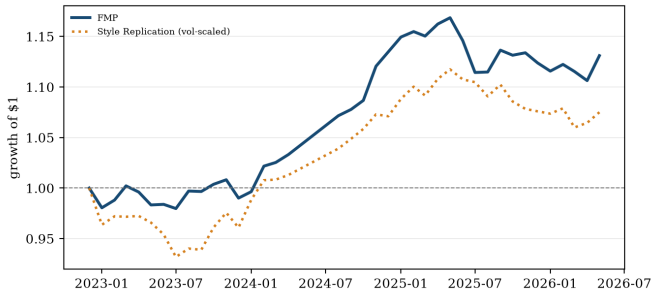


Figure: Common-window cumulative growth of \$1.

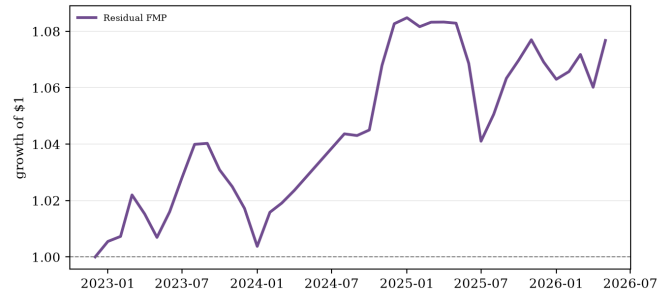


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
eps_growth_yoy	4.01%	4.82%	0.83	1.35	-5.32%	0.470	2.36%	4.82%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
eps_growth_yoy	-0.06 (-0.81)	-0.13 (-1.70)	-0.25 (-2.67)	0.03 (0.47)	0.40 (2.72)	-0.15 (-1.20)	2.40% (1.02)	0.470

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
eps_growth_yoy	-1.32%	0.89%	1.56%	0.28%	-1.50%	1.70%	2.40%	4.01%

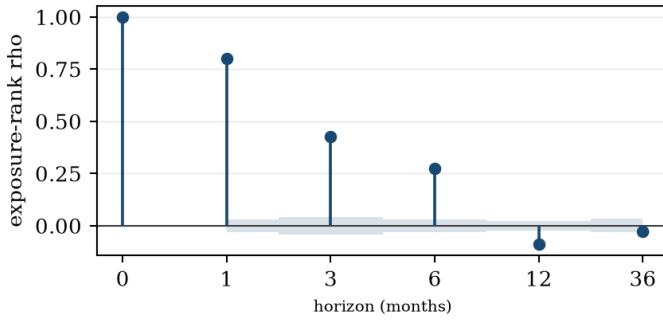


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

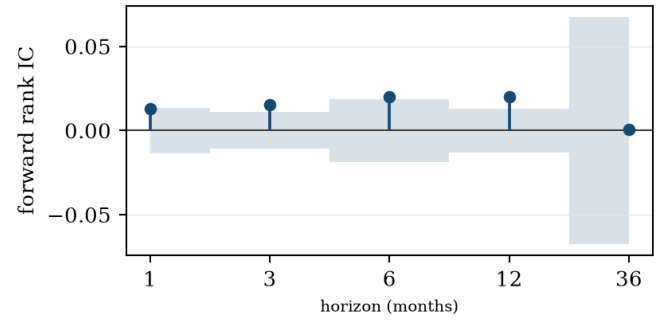


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	eps_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2061 / 2512 / 2709
Available-history Sharpe	0.93
Bootstrap Sharpe	0.83 [-0.29, 2.06]; Pr(S<=0)=9.0%
Bootstrap tails	skew=0.15 [-0.29, 0.57]; max DD=-5.3% [-9.3%, -2.2%]

28. resid_ret_252v_1260c_SPY

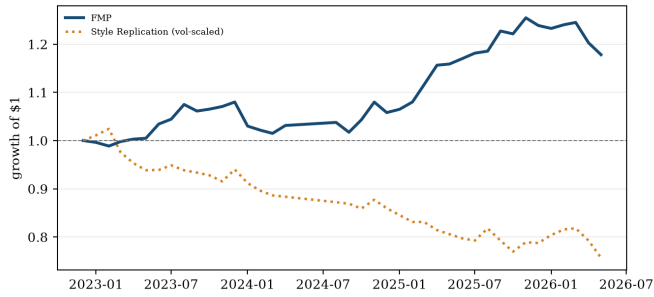


Figure: Common-window cumulative growth of \$1.

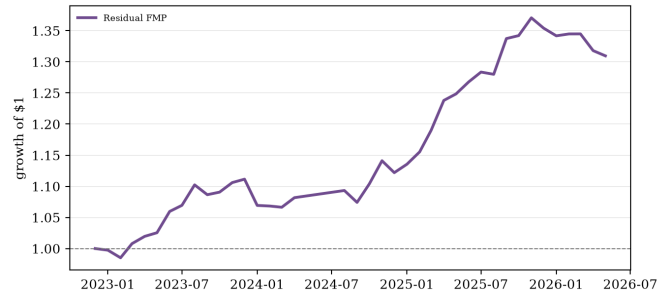


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
resid_ret_252v_1260c_SPY	5.40%	6.59%	0.82	1.39	-6.09%	0.068	7.63%	6.59%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
resid_ret_252v_1260c_SPY	-0.08	0.16	0.00	-0.01	0.10	0.02	8.74%	0.083
	(-0.49)	(0.88)	(0.01)	(-0.11)	(0.37)	(0.06)	(1.95)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
resid_ret_252v_1260c_SPY	-1.58%	-1.06%	-0.01%	-0.13%	-0.38%	-0.17%	8.74%	5.40%

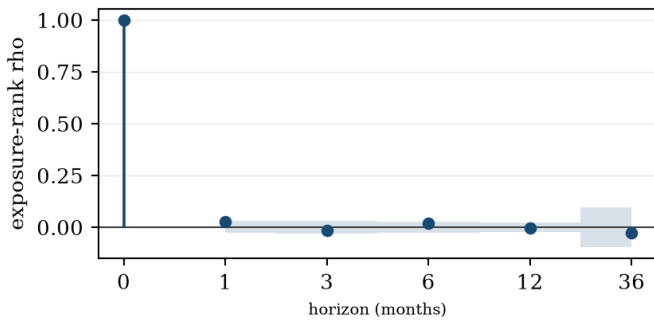


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

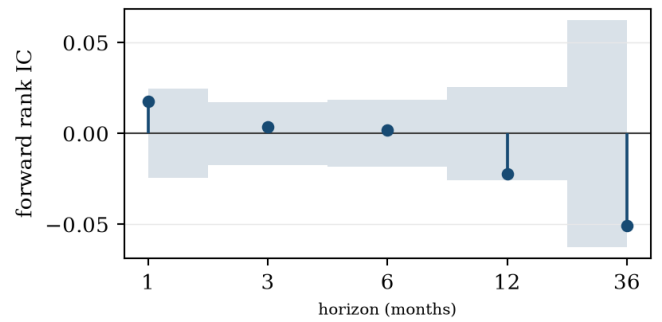


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	resid_ret_252v_1260c_SPY(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.92
Bootstrap Sharpe	0.82 [-0.18, 1.94]; Pr(S<=0)=5.7%
Bootstrap tails	skew=-0.34 [-0.73, 0.12]; max DD=-6.1% [-13.2%, -3.2%]

29. net_income_growth_yoy

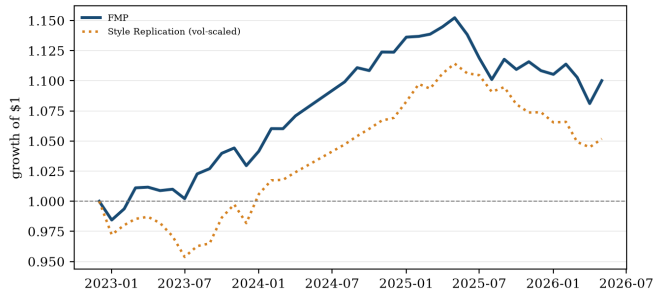


Figure: Common-window cumulative growth of \$1.

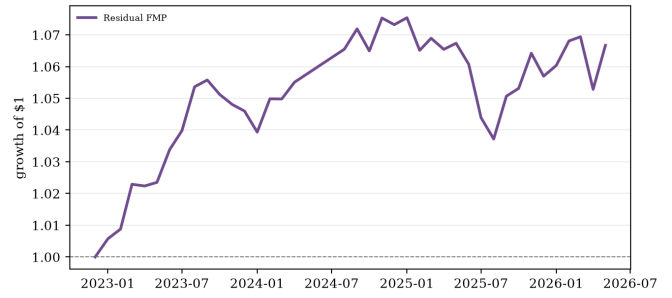


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_growth_yoy	3.09%	4.00%	0.77	1.40	-6.18%	0.558	1.84%	4.00%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_growth_yoy	-0.05	-0.16	-0.24	-0.02	0.33	-0.09	2.07%	0.559
	(-0.77)	(-2.27)	(-3.45)	(-0.34)	(2.70)	(-0.74)	(1.31)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_growth_yoy	-1.12%	1.07%	1.48%	-0.21%	-1.22%	1.01%	2.07%	3.09%

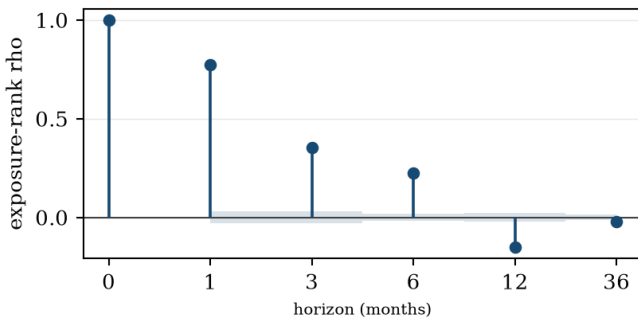


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

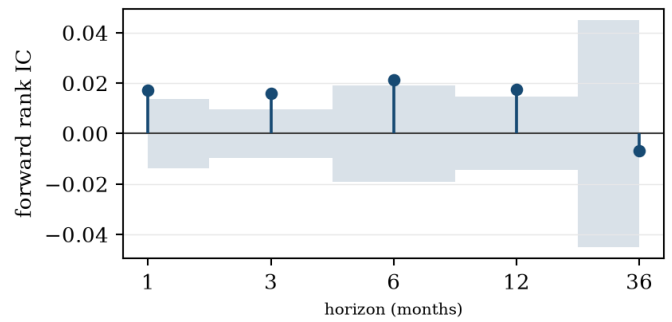


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	net_income_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2088 / 2567 / 2779
Available-history Sharpe	1.03
Bootstrap Sharpe	0.77 [-0.36, 2.11]; Pr(S<=0)=8.7%
Bootstrap tails	skew=-0.13 [-0.37, 0.16]; max DD=-6.2% [-9.3%, -1.4%]

30. evs_ttm

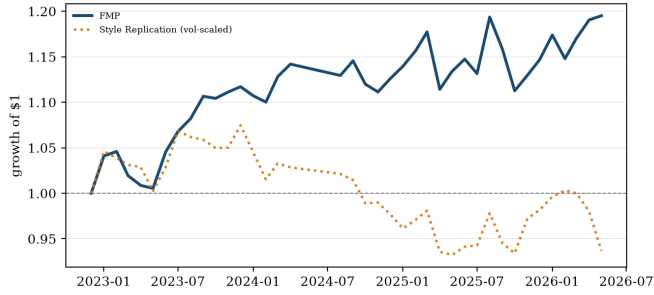


Figure: Common-window cumulative growth of \$1.

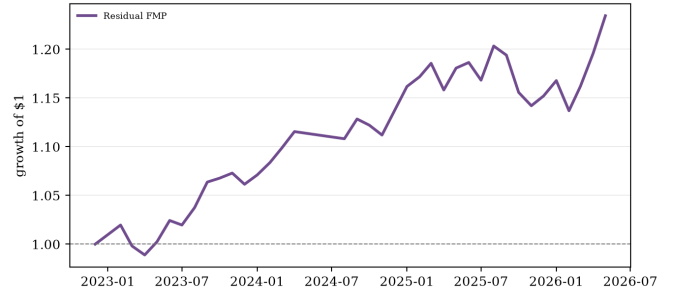


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
evs_ttm	5.93%	7.68%	0.77	1.88	-6.79%	0.436	7.10%	7.68%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
evs_ttm	0.29 (2.40)	0.20 (1.43)	0.23 (1.68)	-0.14 (-1.66)	0.01 (0.02)	0.24 (1.30)	6.83% (2.28)	0.437

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
evs_ttm	6.11%	-1.39%	-1.46%	-1.47%	-0.02%	-2.67%	6.83%	5.93%

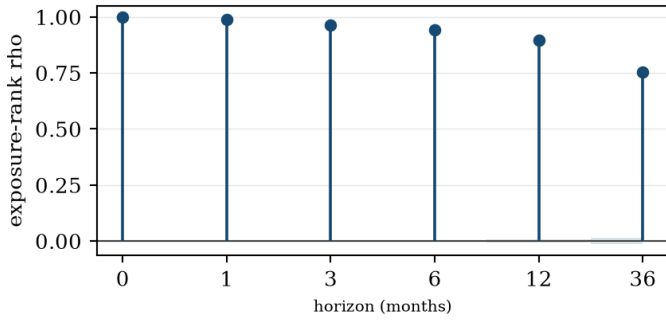


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

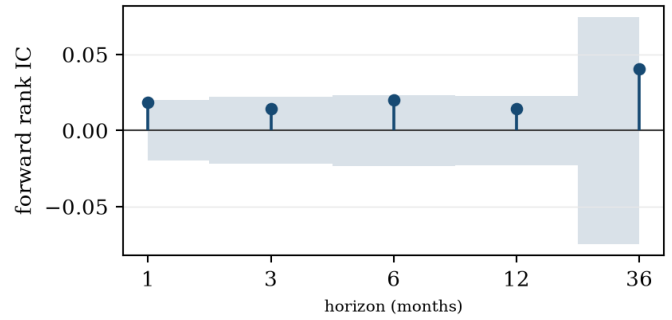


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	evs_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1989 / 2404 / 2553
Available-history Sharpe	1.07
Bootstrap Sharpe	0.77 [0.14, 1.66]; Pr(S<=0)=0.4%
Bootstrap tails	skew=-0.43 [-1.08, 0.19]; max DD=-6.8% [-10.4%, -3.9%]

31. ev_to_assets

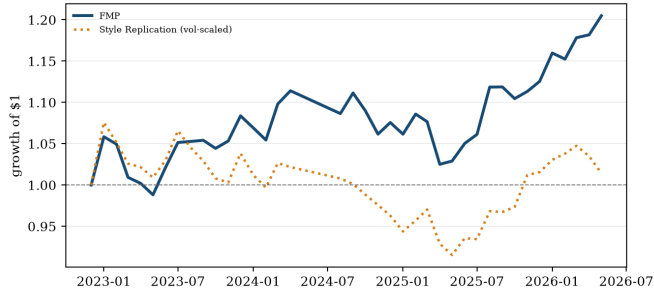


Figure: Common-window cumulative growth of \$1.

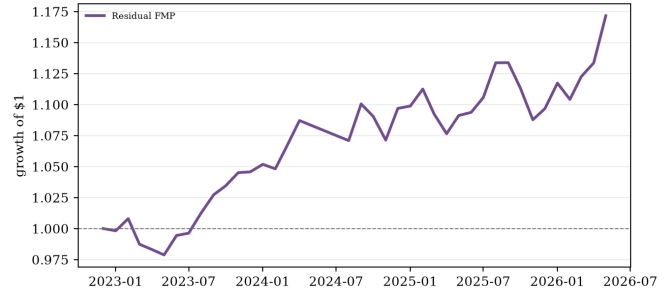


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ev_to_assets	6.22%	8.17%	0.76	1.64	-7.99%	0.610	5.62%	8.17%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ev_to_assets	0.22 (1.93)	0.11 (0.86)	0.38 (3.54)	-0.18 (-3.59)	-0.01 (-0.03)	-0.13 (-0.87)	5.14% (1.61)	0.612

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ev_to_assets	4.59%	-0.77%	-2.39%	-1.87%	0.02%	1.50%	5.14%	6.22%

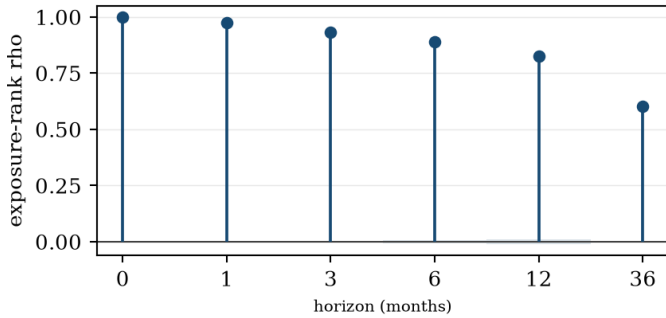


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

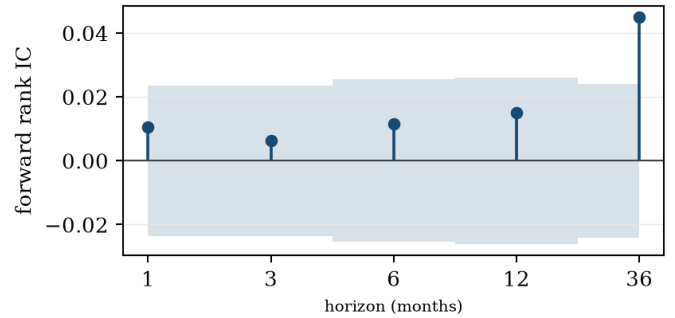


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	ev_to_assets(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2059 / 2529 / 2728
Available-history Sharpe	0.79
Bootstrap Sharpe	0.76 [0.05, 1.49]; Pr(S<=0)=1.9%
Bootstrap tails	skew=0.13 [-0.21, 0.48]; max DD=-8.0% [-13.1%, -4.8%]

32. pb

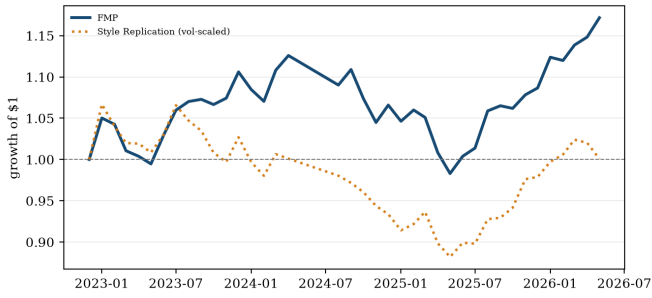


Figure: Common-window cumulative growth of \$1.

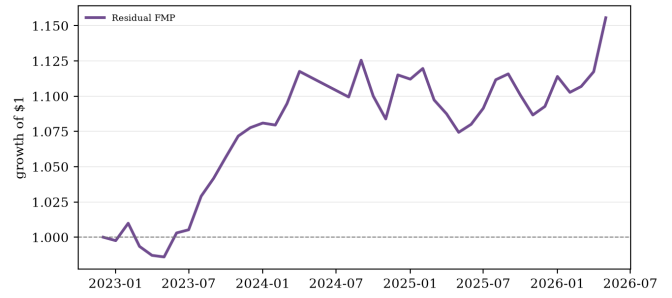


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
pb	5.32%	7.89%	0.67	1.26	-12.69%	0.600	5.07%	7.89%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
pb	0.19 (2.02)	0.12 (1.05)	0.47 (4.61)	-0.09 (-2.19)	-0.18 (-0.90)	-0.07 (-0.47)	4.70% (1.42)	0.601

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
pb	3.89%	-0.83%	-2.94%	-0.97%	0.68%	0.78%	4.70%	5.32%

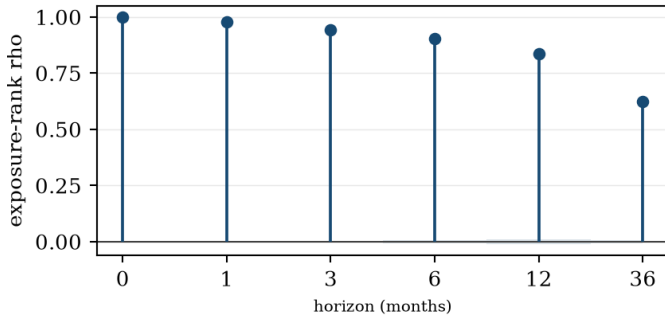


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

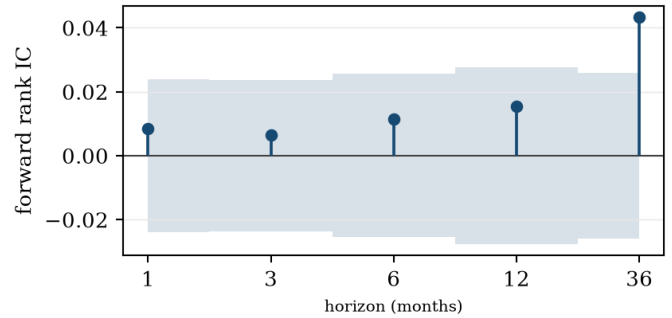


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	pb(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1982 / 2418 / 2590
Available-history Sharpe	0.67
Bootstrap Sharpe	0.67 [-0.27, 1.77]; Pr(S<=0)=8.9%
Bootstrap tails	skew=-0.07 [-0.30, 0.23]; max DD=-12.7% [-18.8%, -5.3%]

33. accruals_wc_proxy

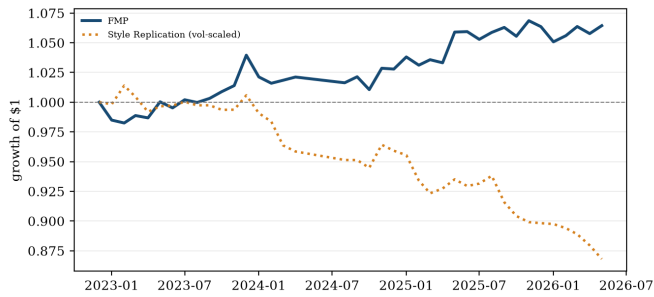


Figure: Common-window cumulative growth of \$1.

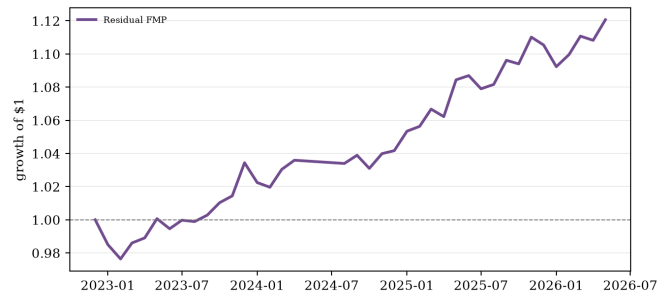


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_wc_proxy	2.03%	3.30%	0.61	1.65	-2.79%	0.176	3.87%	3.30%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_wc_proxy	0.01 (0.16)	0.16 (2.05)	-0.08 (-1.41)	-0.01 (-0.15)	0.01 (0.10)	0.10 (1.00)	3.65% (1.98)	0.178

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_wc_proxy	0.20%	-1.08%	0.48%	-0.10%	-0.03%	-1.09%	3.65%	2.03%

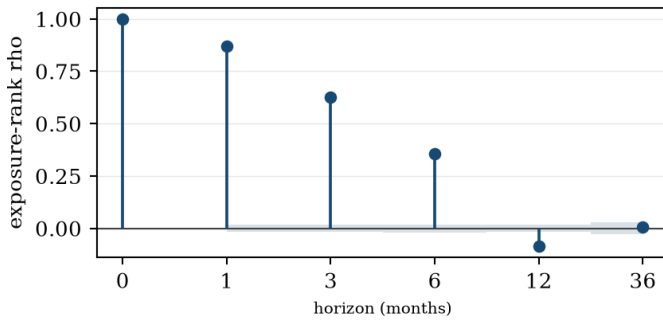


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

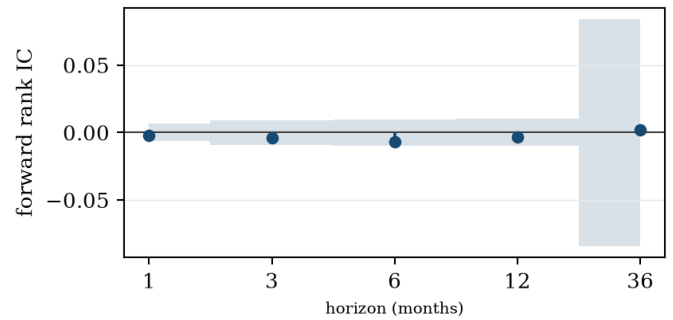


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	accruals_wc_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2086 / 2568 / 2782
Available-history Sharpe	0.54
Bootstrap Sharpe	0.61 [0.09, 1.11]; Pr(S<=0)=0.9%
Bootstrap tails	skew=0.47 [-0.04, 0.81]; max DD=-2.8% [-4.4%, -1.7%]

34. roic_nopat

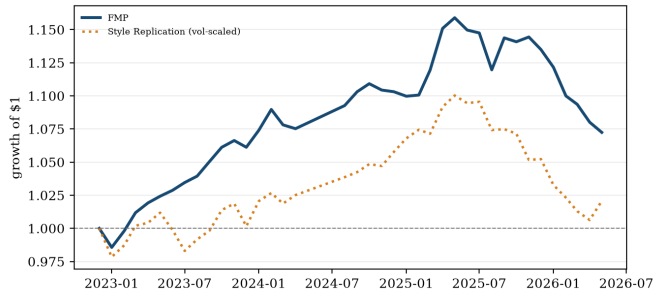


Figure: Common-window cumulative growth of \$1.

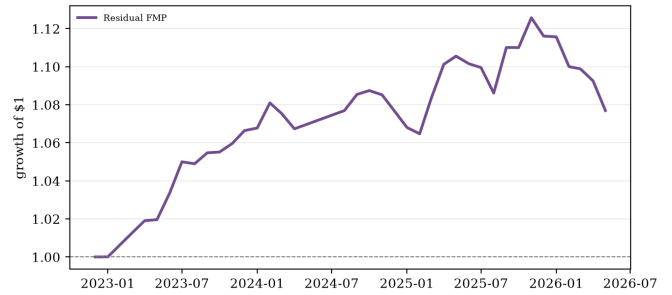


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
roic_nopat	2.29%	3.95%	0.58	0.90	-7.46%	0.375	1.84%	3.95%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
roic_nopat	-0.09 (-1.05)	-0.14 (-1.86)	-0.18 (-2.75)	-0.03 (-0.58)	0.06 (0.60)	-0.01 (-0.08)	2.39% (1.00)	0.382

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
roic_nopat	-1.80%	0.94%	1.14%	-0.29%	-0.22%	0.13%	2.39%	2.29%

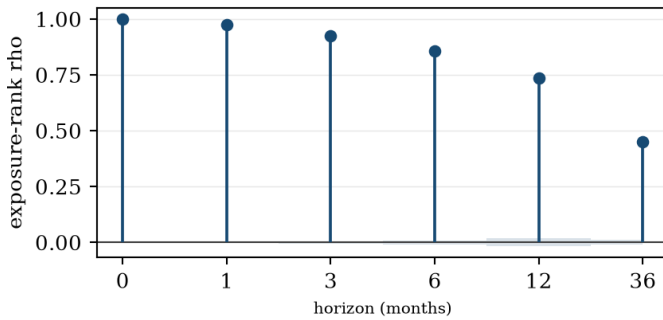


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

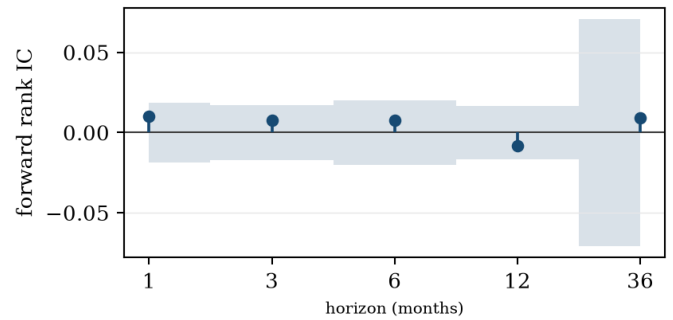


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	roic_nopat(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1310 / 1358 / 1404
Available-history Sharpe	0.43
Bootstrap Sharpe	0.58 [-0.58, 2.16]; Pr(S<=0)=15.7%
Bootstrap tails	skew=-0.06 [-0.75, 0.52]; max DD=-7.5% [-13.6%, -1.3%]

35. ps_ttm

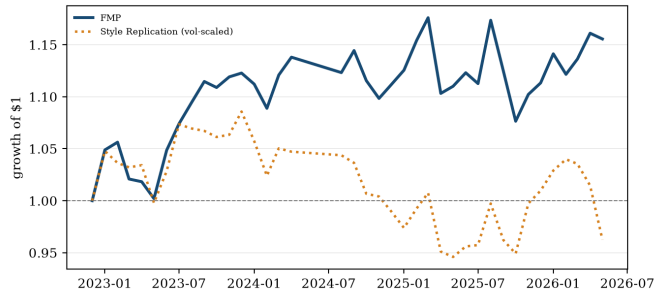


Figure: Common-window cumulative growth of \$1.

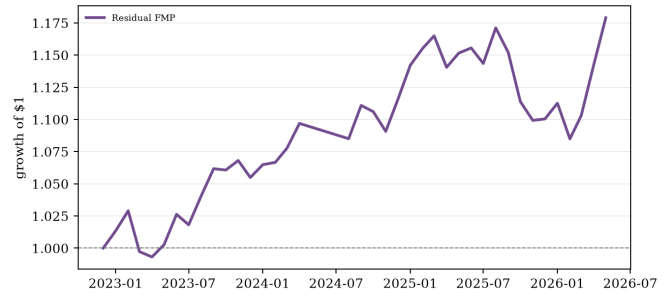


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ps_ttm	4.95%	8.82%	0.56	1.34	-8.46%	0.532	5.56%	8.82%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ps_ttm	0.39 (2.83)	0.19 (1.14)	0.33 (2.09)	-0.17 (-1.59)	0.04 (0.13)	0.30 (1.37)	5.39% (1.65)	0.532

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ps_ttm	8.09%	-1.28%	-2.03%	-1.75%	-0.15%	-3.32%	5.39%	4.95%

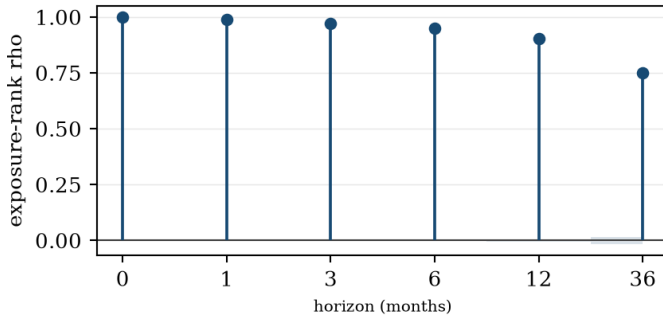


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

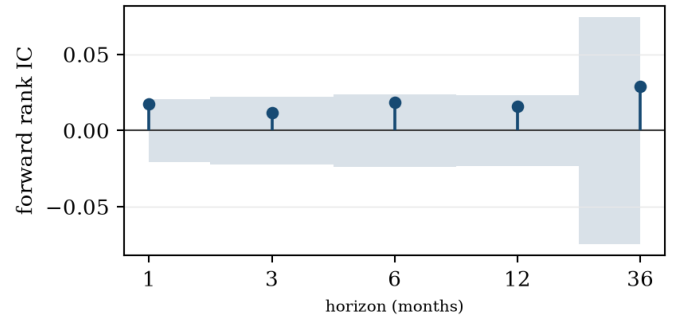


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	ps_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2041 / 2478 / 2642
Available-history Sharpe	0.87
Bootstrap Sharpe	0.56 [-0.04, 1.35]; Pr(S<=0)=3.4%
Bootstrap tails	skew=-0.43 [-0.94, 0.17]; max DD=-8.5% [-13.2%, -5.1%]

36. dpo

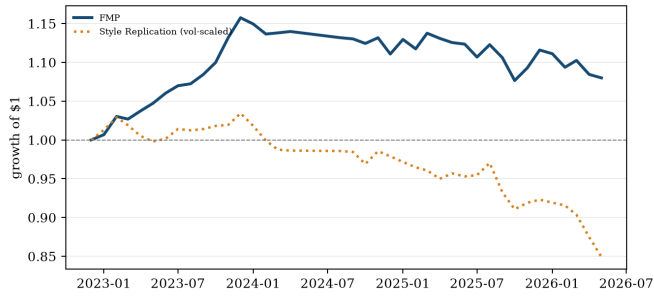


Figure: Common-window cumulative growth of \$1.

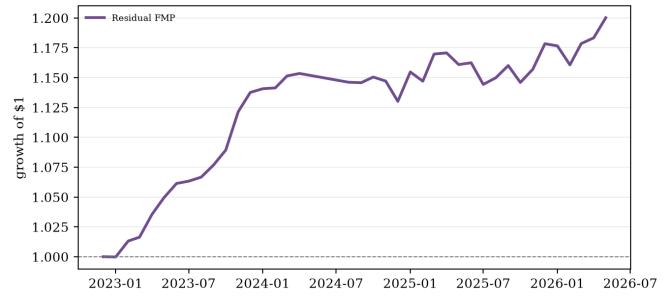


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dpo	2.53%	4.64%	0.55	0.94	-7.00%	0.367	5.59%	4.64%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dpo	0.06 (1.00)	0.22 (4.36)	-0.11 (-1.42)	-0.12 (-2.37)	0.13 (0.89)	0.18 (2.53)	5.84% (2.45)	0.368

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dpo	1.31%	-1.48%	0.70%	-1.29%	-0.49%	-2.06%	5.84%	2.53%

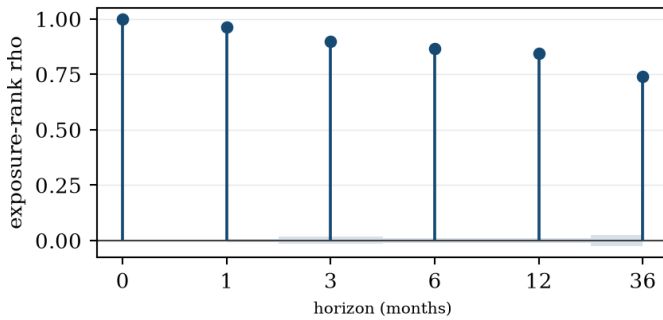


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

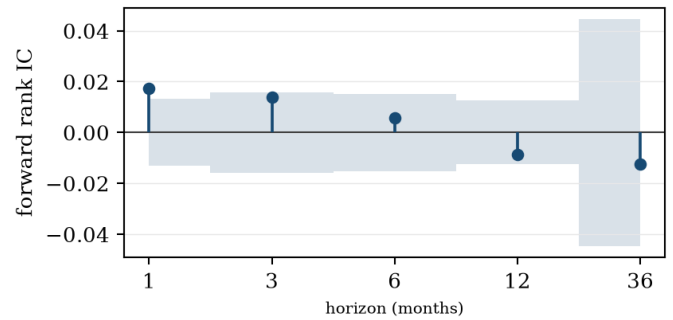


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	dpo(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2017 / 2440 / 2524
Available-history Sharpe	0.96
Bootstrap Sharpe	0.55 [-0.54, 1.92]; Pr(S<=0)=16.8%
Bootstrap tails	skew=-0.01 [-0.46, 0.48]; max DD=-7.0% [-10.5%, -3.2%]

37. corr_1260d_FRENCH_CRSP

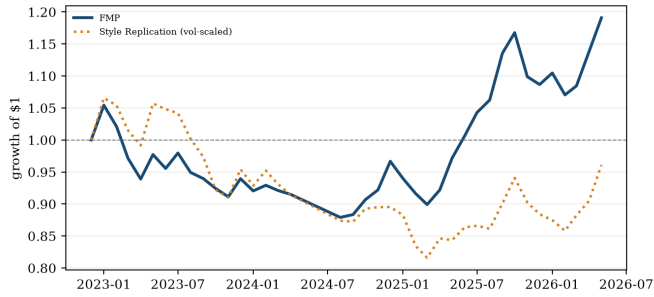


Figure: Common-window cumulative growth of \$1.

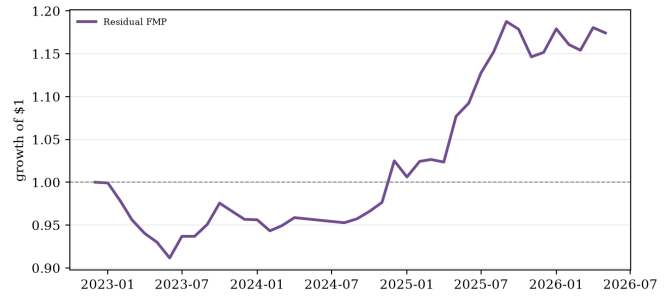


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
corr_1260d_FRENCH_CRSP	6.17%	11.47%	0.54	0.91	-16.62%	0.645	6.67%	11.47%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
corr_1260d_FRENCH_CRSP	-0.30	0.22	0.03	0.05	-0.50	-0.56	5.30%	0.650
	(-1.90)	(2.54)	(0.25)	(0.64)	(-2.04)	(-2.18)	(1.02)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
corr_1260d_FRENCH_CRSP	-6.17%	-1.50%	-0.17%	0.53%	1.87%	6.31%	5.30%	6.17%

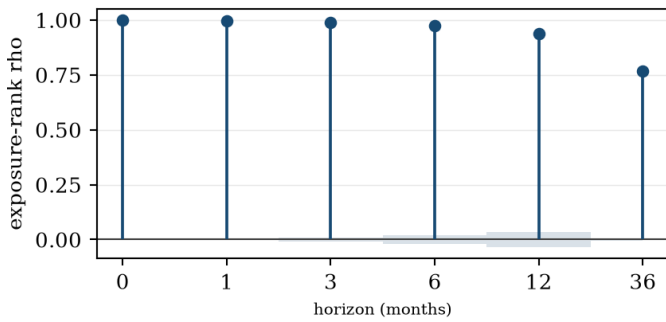


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

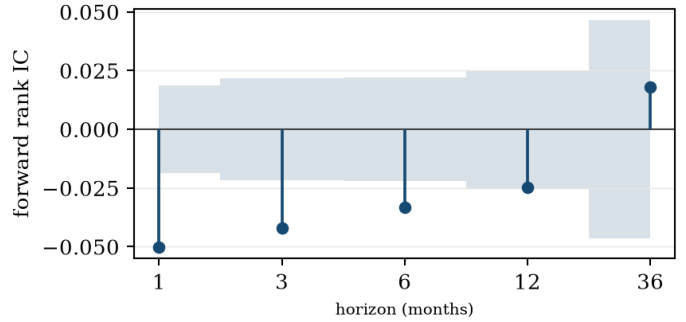


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	corr_1260d_FRENCH_CRSP(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.20
Bootstrap Sharpe	0.54 [-0.42, 1.52]; Pr(S<=0)=13.2%
Bootstrap tails	skew=0.03 [-0.47, 0.51]; max DD=-16.6% [-23.9%, -8.3%]

38. corr_1260d_VTI

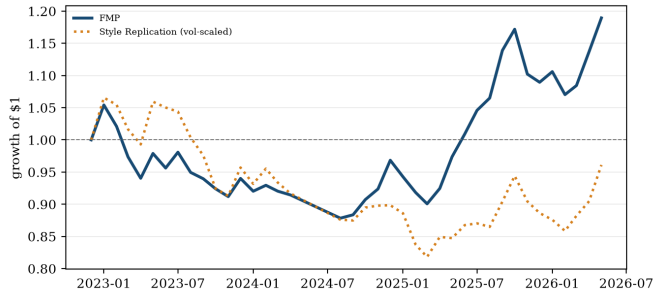


Figure: Common-window cumulative growth of \$1.

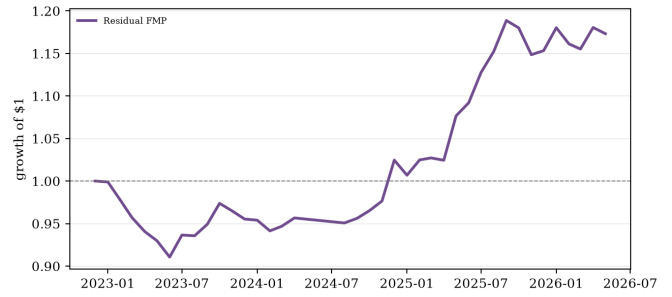


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
corr_1260d_VTI	6.13%	11.51%	0.53	0.90	-16.68%	0.651	6.63%	11.51%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
corr_1260d_VTI	-0.30 (-1.95)	0.22 (2.55)	0.01 (0.14)	0.05 (0.60)	-0.51 (-2.12)	-0.56 (-2.19)	5.27% (1.00)	0.656

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
corr_1260d_VTI	-6.19%	-1.50%	-0.09%	0.49%	1.90%	6.26%	5.27%	6.13%

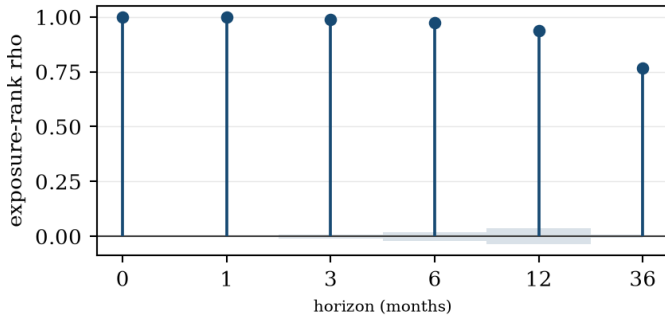


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

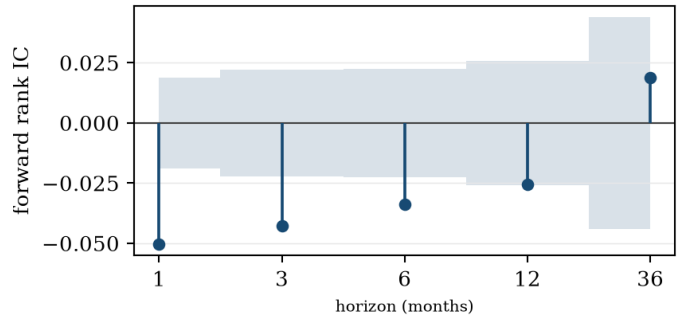


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	corr_1260d_VTI(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.19
Bootstrap Sharpe	0.53 [-0.42, 1.52]; Pr(S<=0)=13.2%
Bootstrap tails	skew=0.04 [-0.46, 0.52]; max DD=-16.7% [-24.0%, -8.5%]

39. corr_1260d_SPY

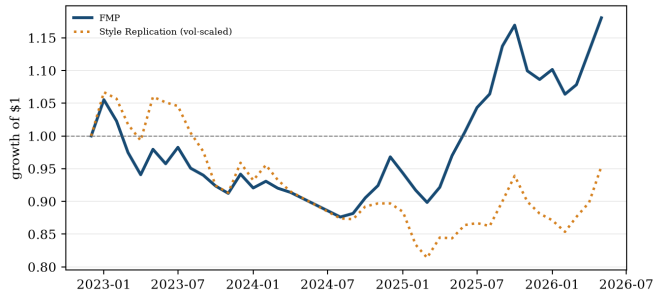


Figure: Common-window cumulative growth of \$1.

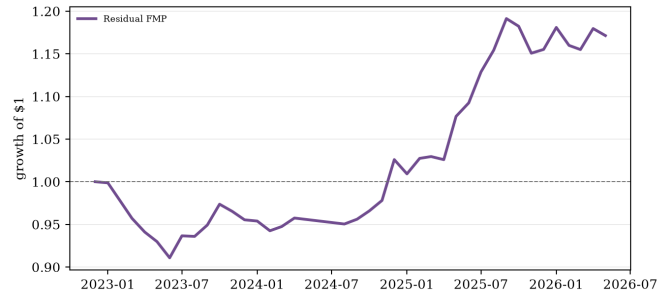


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
corr_1260d_SPY	5.90%	11.60%	0.51	0.86	-16.98%	0.660	6.59%	11.60%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
corr_1260d_SPY	-0.29 (-1.94)	0.25 (2.83)	0.01 (0.05)	0.04 (0.56)	-0.51 (-2.14)	-0.55 (-2.18)	5.22% (0.98)	0.665

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
corr_1260d_SPY	-6.11%	-1.69%	-0.03%	0.44%	1.90%	6.17%	5.22%	5.90%

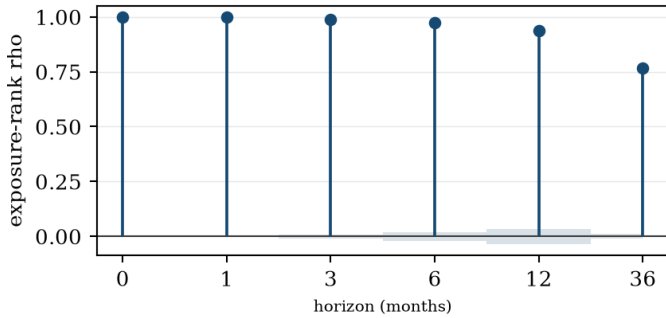


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

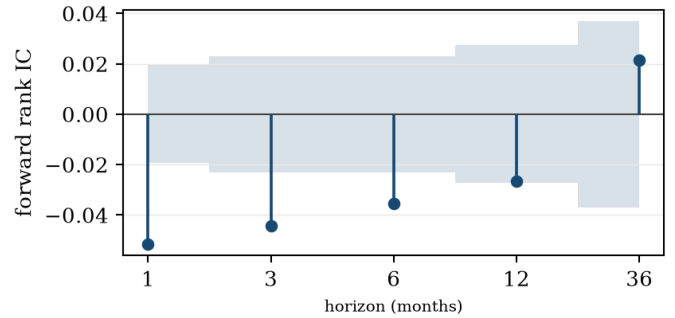


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	corr_1260d_SPY(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.18
Bootstrap Sharpe	0.51 [-0.44, 1.49]; Pr(S<=0)=14.2%
Bootstrap tails	skew=0.02 [-0.49, 0.50]; max DD=-17.0% [-24.3%, -8.7%]

40. assets_to_mc

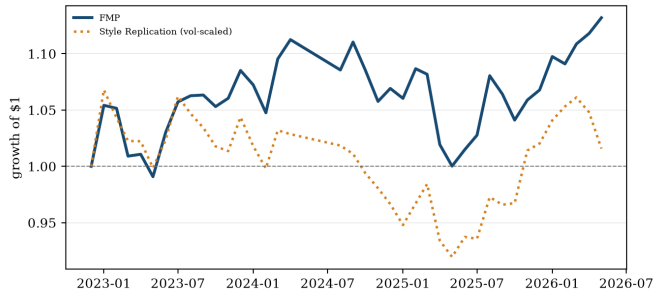


Figure: Common-window cumulative growth of \$1.

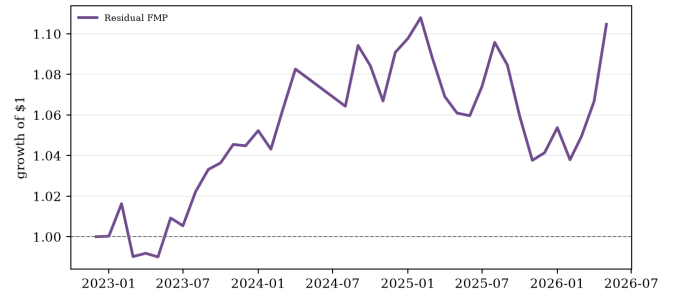


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
assets_to_mc	4.27%	8.56%	0.50	1.12	-10.09%	0.611	3.61%	8.56%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
assets_to_mc	0.28 (2.48)	0.11 (0.76)	0.44 (3.51)	-0.16 (-2.94)	0.03 (0.12)	-0.04 (-0.23)	3.28% (0.99)	0.612

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
assets_to_mc	5.85%	-0.74%	-2.71%	-1.73%	-0.10%	0.42%	3.28%	4.27%

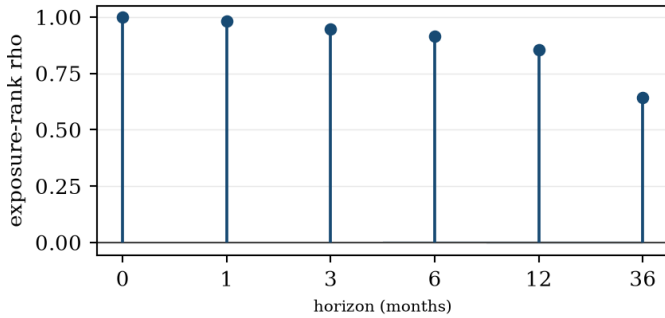


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

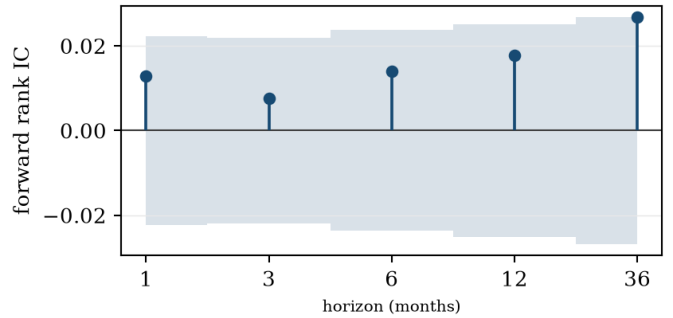


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	assets_to_mc(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2105 / 2582 / 2811
Available-history Sharpe	0.65
Bootstrap Sharpe	0.50 [-0.21, 1.26]; Pr(S<=0)=8.8%
Bootstrap tails	skew=-0.06 [-0.39, 0.34]; max DD=-10.1% [-16.0%, -5.7%]

41. dso

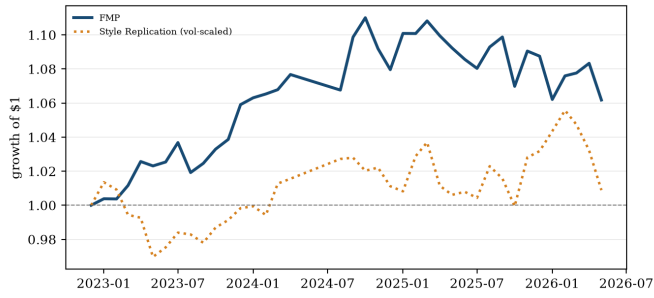


Figure: Common-window cumulative growth of \$1.

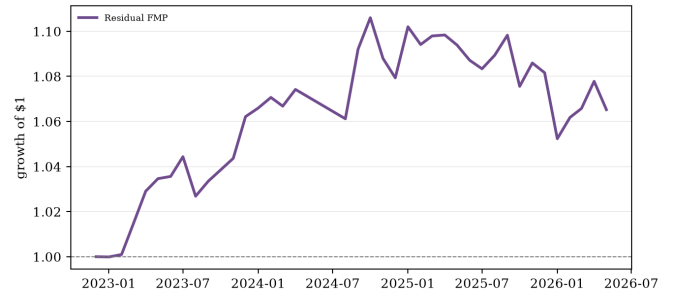


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dso	1.98%	4.26%	0.47	1.14	-4.35%	0.116	1.86%	4.26%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dso	0.04 (0.64)	0.01 (0.09)	0.03 (0.30)	-0.04 (-0.63)	0.16 (0.98)	-0.03 (-0.27)	2.07% (0.94)	0.118

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dso	0.83%	-0.05%	-0.16%	-0.46%	-0.60%	0.34%	2.07%	1.98%

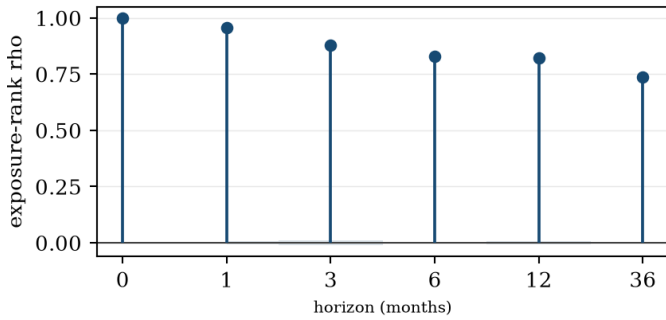


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

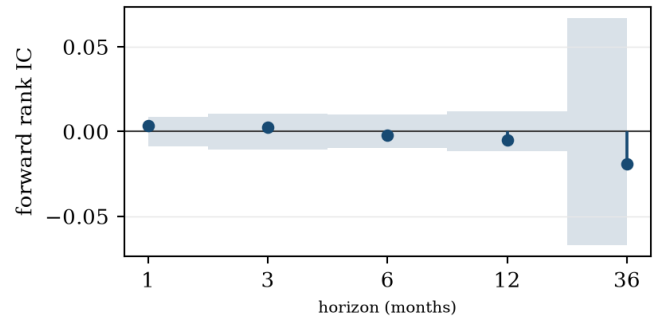


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	dso(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1999 / 2400 / 2539
Available-history Sharpe	0.66
Bootstrap Sharpe	0.47 [-0.26, 1.36]; Pr(S<=0)=10.9%
Bootstrap tails	skew=-0.31 [-0.97, 0.32]; max DD=-4.3% [-7.6%, -2.6%]

42. sales_to_price

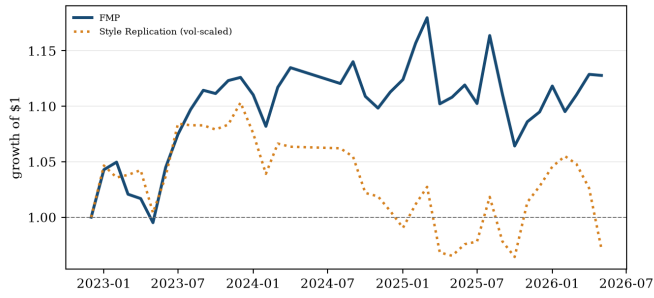


Figure: Common-window cumulative growth of \$1.

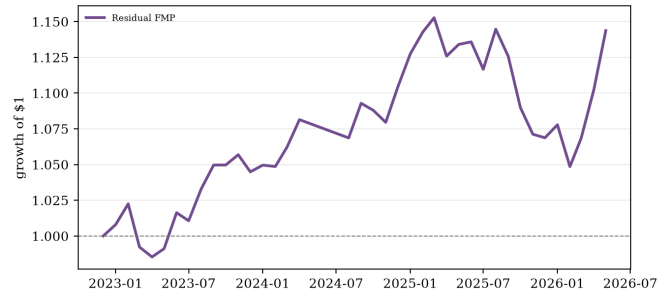


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
sales_to_price	4.20%	9.04%	0.47	1.08	-9.80%	0.544	4.58%	9.04%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
sales_to_price	0.43 (3.31)	0.17 (1.07)	0.33 (2.15)	-0.18 (-1.95)	0.02 (0.06)	0.36 (1.77)	4.43% (1.29)	0.545

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
sales_to_price	9.03%	-1.17%	-2.03%	-1.91%	-0.06%	-4.08%	4.43%	4.20%

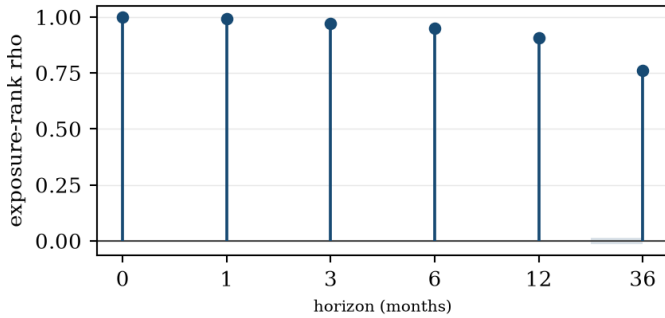


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

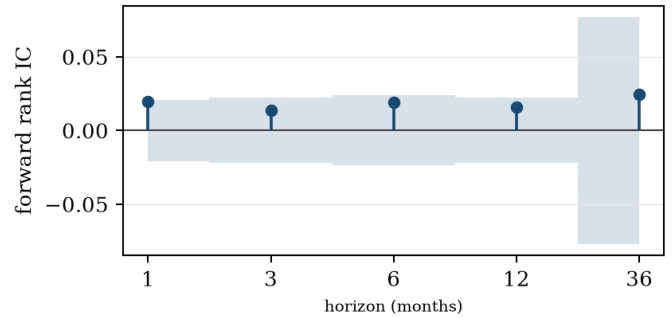


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	sales_to_price(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2086 / 2534 / 2728
Available-history Sharpe	0.81
Bootstrap Sharpe	0.47 [-0.15, 1.30]; Pr(S<=0)=7.6%
Bootstrap tails	skew=-0.46 [-0.99, 0.18]; max DD=-9.8% [-15.3%, -5.2%]

43. altman_z

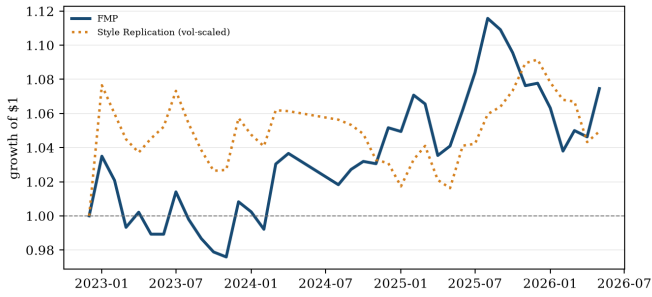


Figure: Common-window cumulative growth of \$1.

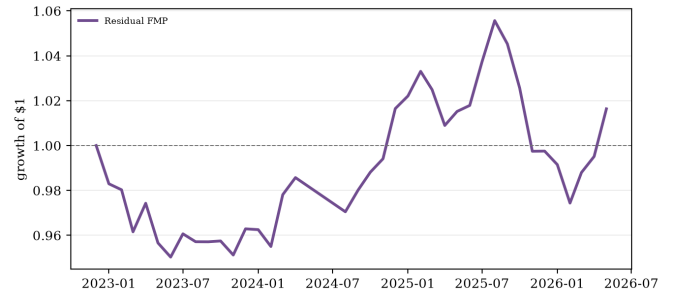


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
altman_z	2.46%	6.23%	0.40	0.81	-6.97%	0.444	1.32%	6.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
altman_z	0.11 (1.17)	-0.05 (-0.51)	0.09 (0.83)	-0.24 (-3.28)	0.03 (0.17)	-0.22 (-1.86)	0.62% (0.18)	0.450

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
altman_z	2.22%	0.35%	-0.54%	-2.52%	-0.10%	2.45%	0.62%	2.46%

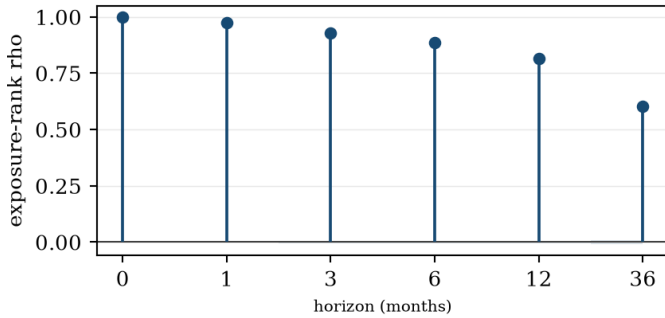


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

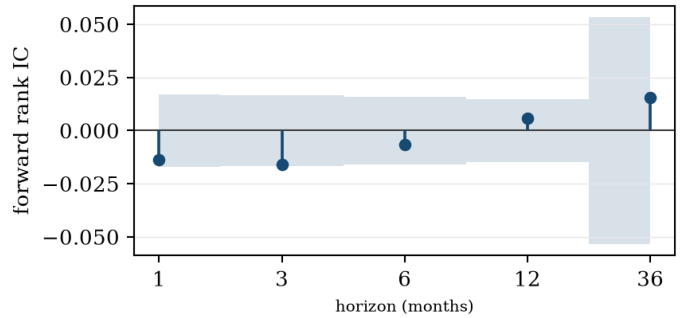


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	altman_z(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2053 / 2424 / 2545
Available-history Sharpe	0.30
Bootstrap Sharpe	0.40 [-0.24, 1.07]; Pr(S<=0)=10.3%
Bootstrap tails	skew=0.38 [-0.08, 0.82]; max DD=-7.0% [-12.0%, -3.8%]

44. capital_intensity

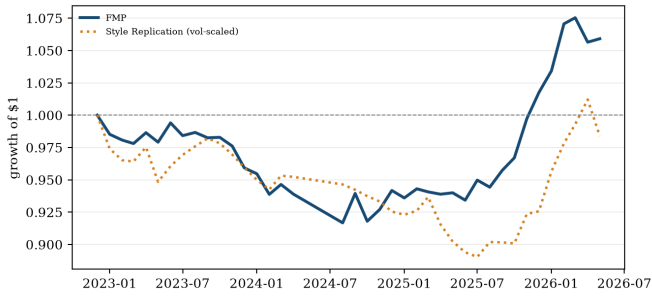


Figure: Common-window cumulative growth of \$1.

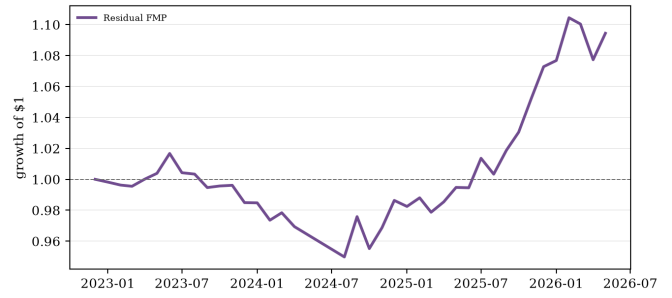


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
capital_intensity	1.93%	4.91%	0.39	0.58	-7.78%	0.178	2.10%	4.91%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
capital_intensity	0.01 (0.10)	0.05 (0.54)	0.18 (1.69)	0.11 (1.53)	-0.05 (-0.28)	0.10 (0.70)	2.95% (0.95)	0.190

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
capital_intensity	0.15%	-0.31%	-1.12%	1.17%	0.17%	-1.08%	2.95%	1.93%

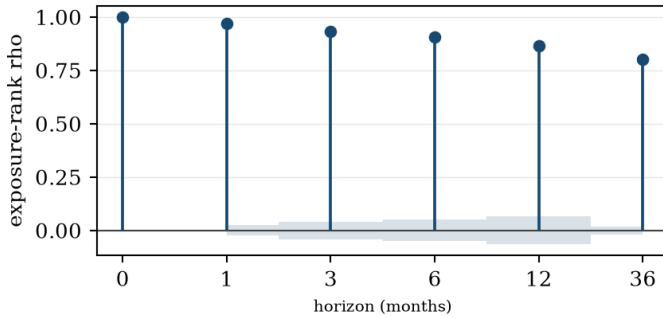


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

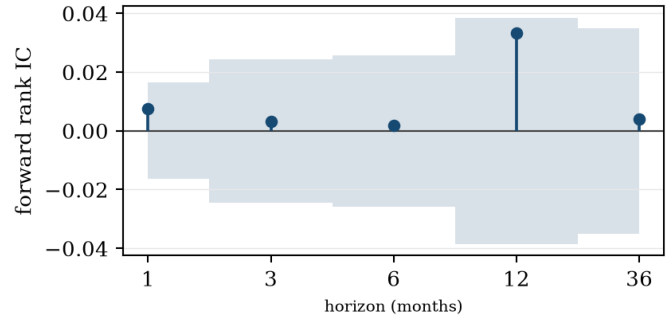


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	capital_intensity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	41 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	2 / 3
Stocks, min / median / max	22 / 842 / 2049
Available-history Sharpe	-0.26
Bootstrap Sharpe	0.39 [-0.94, 1.61]; Pr(S<=0)=27.4%
Bootstrap tails	skew=0.38 [-0.08, 0.90]; max DD=-7.8% [-14.0%, -3.1%]

45. asset_turnover

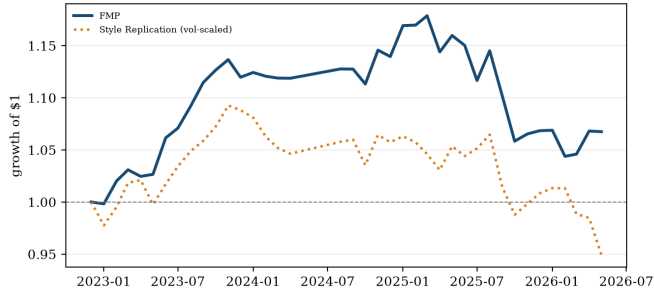


Figure: Common-window cumulative growth of \$1.

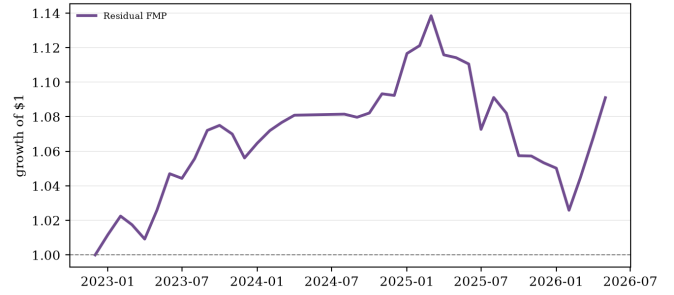


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
asset_turnover	2.25%	6.12%	0.37	0.64	-11.44%	0.417	3.19%	6.12%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
asset_turnover	0.29 (4.75)	0.20 (1.97)	-0.10 (-0.85)	-0.02 (-0.29)	0.05 (0.24)	0.50 (5.23)	2.86% (0.92)	0.418

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
asset_turnover	6.14%	-1.34%	0.59%	-0.23%	-0.19%	-5.60%	2.86%	2.25%

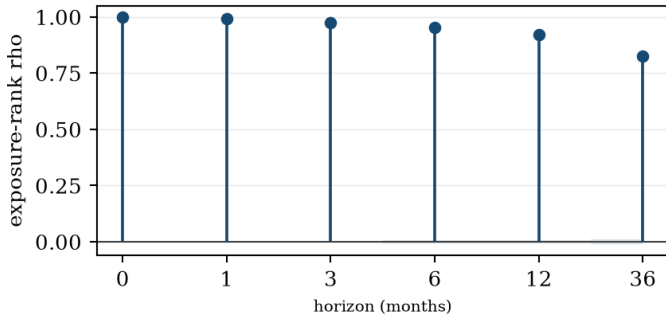


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

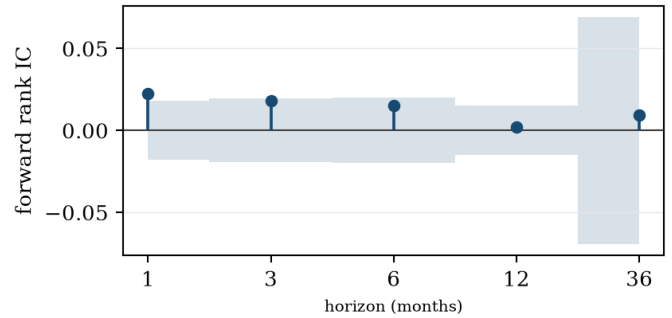


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	asset_turnover(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2086 / 2532 / 2727
Available-history Sharpe	0.81
Bootstrap Sharpe	0.37 [-0.70, 1.94]; Pr(S<=0)=24.0%
Bootstrap tails	skew=-0.55 [-0.81, 0.35]; max DD=-11.4% [-19.7%, -2.1%]

46. equity_growth_yoy

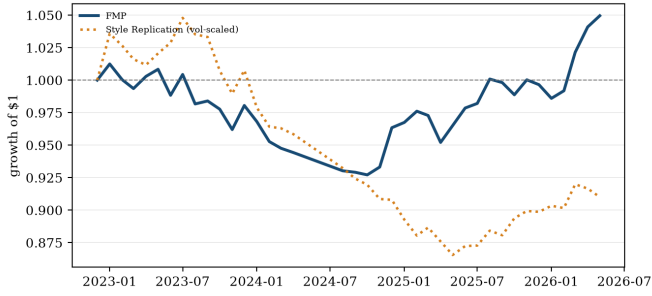


Figure: Common-window cumulative growth of \$1.

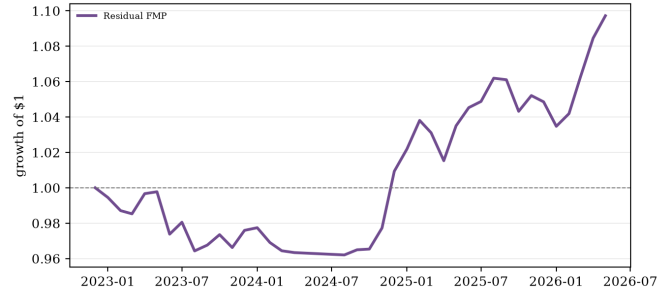


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
equity_growth_yoy	1.64%	4.78%	0.34	0.59	-8.43%	0.247	3.06%	4.78%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
equity_growth_yoy	-0.00 (-0.00)	0.11 (1.42)	0.17 (1.85)	-0.01 (-0.06)	-0.24 (-1.47)	0.04 (0.33)	3.02% (0.99)	0.247

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
equity_growth_yoy	-0.00%	-0.75%	-1.08%	-0.05%	0.91%	-0.39%	3.02%	1.64%

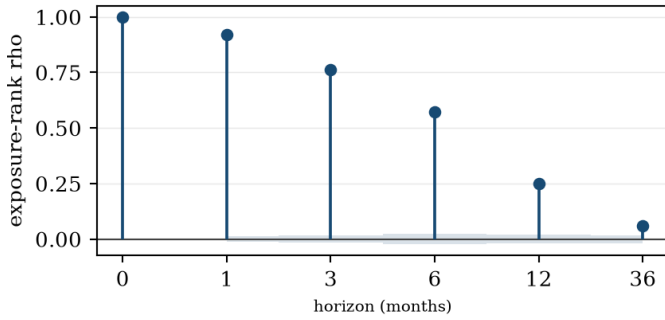


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

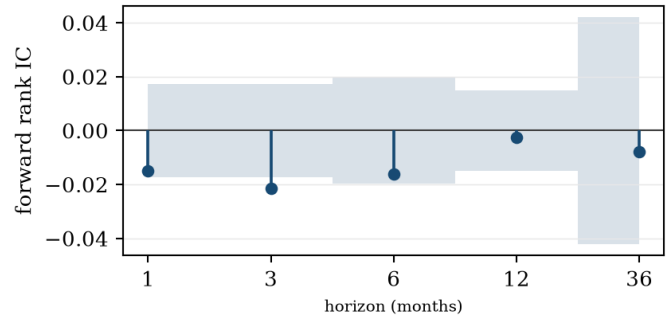


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	equity_growth_yoy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2092 / 2572 / 2795
Available-history Sharpe	0.43
Bootstrap Sharpe	0.34 [-0.67, 1.36]; Pr(S<=0)=23.9%
Bootstrap tails	skew=0.23 [-0.17, 0.70]; max DD=-8.4% [-13.2%, -2.5%]

47. debt_issuance_yoy

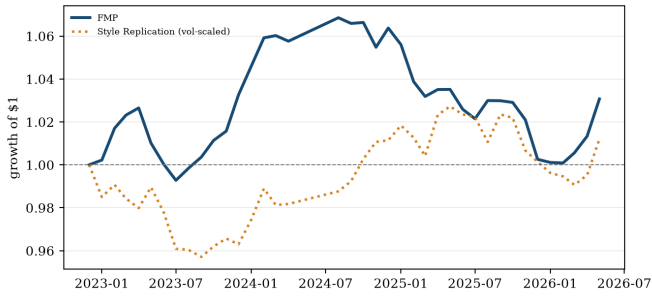


Figure: Common-window cumulative growth of \$1.

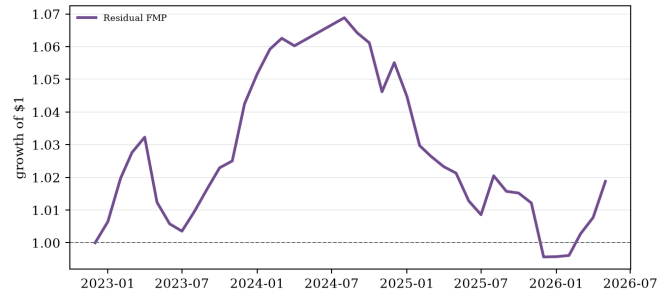


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_issuance_yoy	1.00%	3.12%	0.32	0.45	-6.33%	0.110	0.86%	3.12%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_issuance_yoy	-0.05 (-1.11)	-0.02 (-0.21)	-0.08 (-1.34)	0.02 (0.38)	0.07 (0.53)	-0.08 (-0.90)	0.63% (0.25)	0.113

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_issuance_yoy	-1.03%	0.12%	0.50%	0.16%	-0.25%	0.87%	0.63%	1.00%

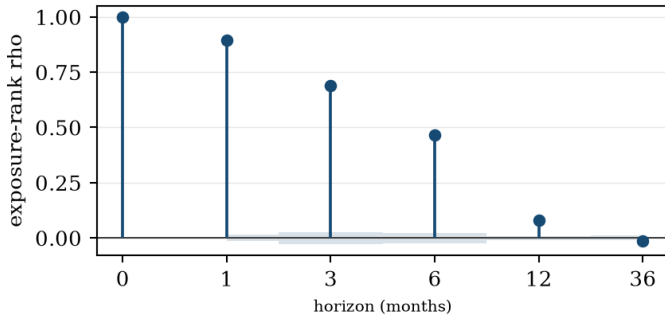


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

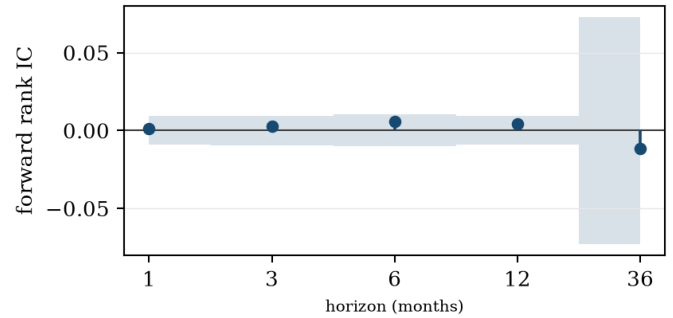


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	debt_issuance_yoy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2031 / 2488 / 2680
Available-history Sharpe	0.23
Bootstrap Sharpe	0.32 [-0.93, 1.63]; Pr(S<=0)=29.6%
Bootstrap tails	skew=-0.21 [-0.58, 0.09]; max DD=-6.3% [-10.3%, -3.3%]

48. dollar_volume_21d

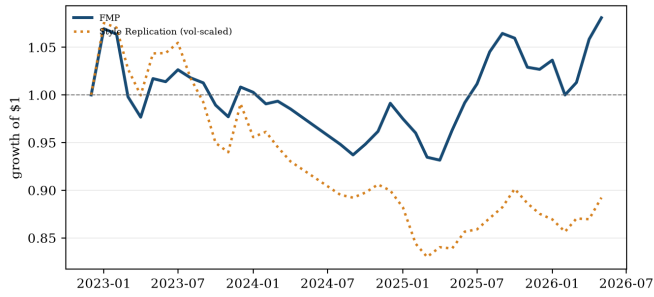


Figure: Common-window cumulative growth of \$1.

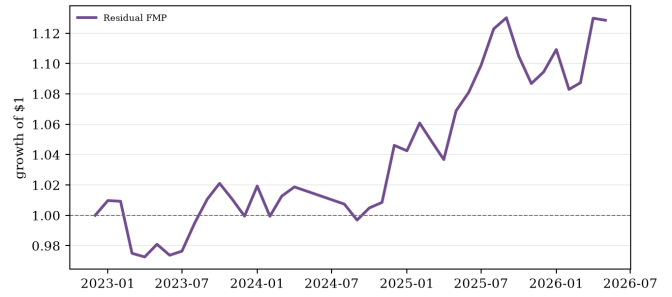


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dollar_volume_21d	2.86%	9.11%	0.31	0.60	-12.89%	0.596	5.33%	9.11%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dollar_volume_21d	-0.11 (-0.63)	0.34 (2.89)	0.02 (0.17)	-0.09 (-0.82)	-0.27 (-0.92)	-0.32 (-1.30)	3.99% (1.18)	0.603

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dollar_volume_21d	-2.29%	-2.35%	-0.15%	-0.92%	1.01%	3.58%	3.99%	2.86%

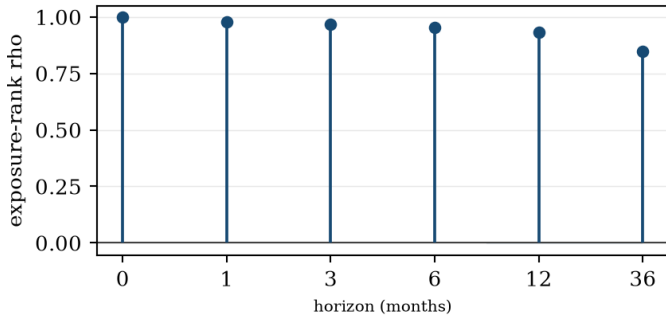


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

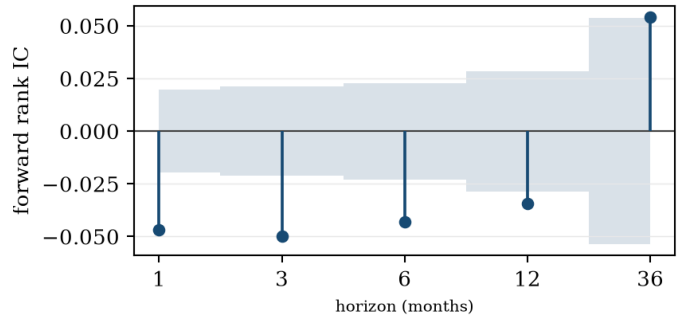


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	dollar_volume_21d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.22
Bootstrap Sharpe	0.31 [-0.51, 1.10]; Pr(S<=0)=20.2%
Bootstrap tails	skew=0.15 [-0.32, 0.60]; max DD=-12.9% [-19.8%, -6.2%]

49. growth_gmar_5y

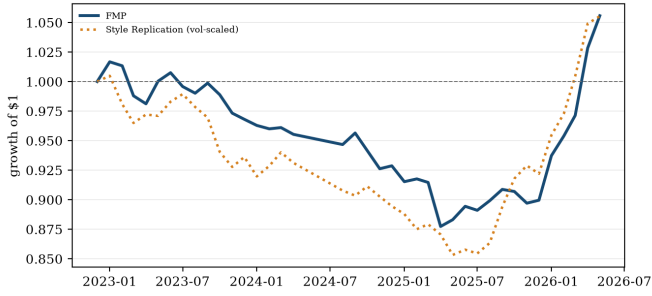


Figure: Common-window cumulative growth of \$1.

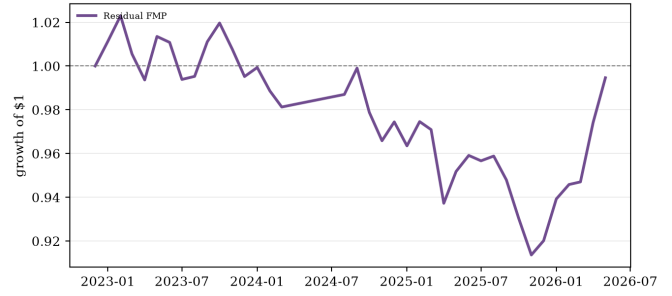


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_gmar_5y	1.90%	6.16%	0.31	0.43	-13.71%	0.360	0.77%	6.16%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_gmar_5y	0.00 (0.02)	0.07 (0.58)	0.36 (2.47)	0.22 (1.79)	-0.25 (-1.49)	-0.12 (-0.80)	-0.05% (-0.02)	0.367

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_gmar_5y	0.07%	-0.48%	-2.23%	2.28%	0.95%	1.36%	-0.05%	1.90%

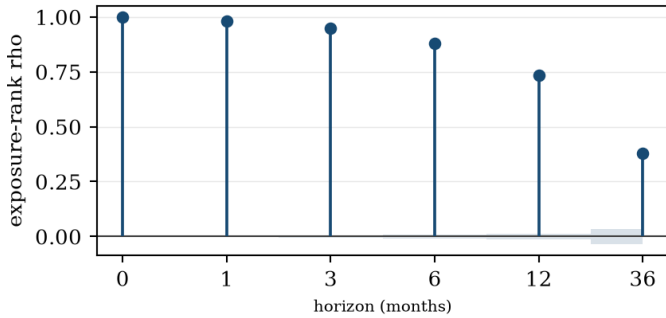


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

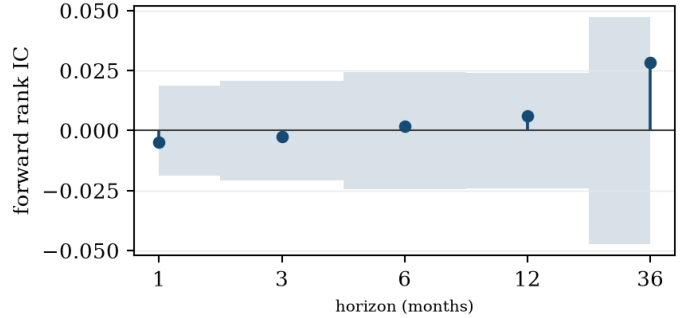


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	growth_gmar_5y(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1897 / 2080 / 2382
Available-history Sharpe	0.16
Bootstrap Sharpe	0.31 [-1.52, 1.50]; Pr(S<=0)=31.4%
Bootstrap tails	skew=0.84 [-1.09, 1.60]; max DD=-13.7% [-19.2%, -4.4%]

50. log_market_cap

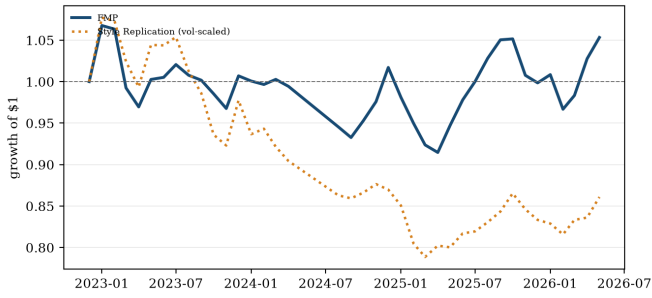


Figure: Common-window cumulative growth of \$1.

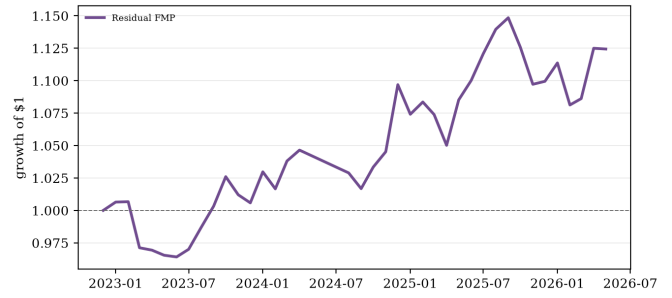


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
log_market_cap	2.16%	10.30%	0.21	0.39	-14.34%	0.591	5.39%	10.30%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
log_market_cap	-0.15 (-0.84)	0.42 (3.61)	0.04 (0.27)	-0.05 (-0.52)	-0.36 (-1.17)	-0.33 (-1.37)	3.92% (0.88)	0.597

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
log_market_cap	-3.22%	-2.85%	-0.22%	-0.52%	1.34%	3.71%	3.92%	2.16%

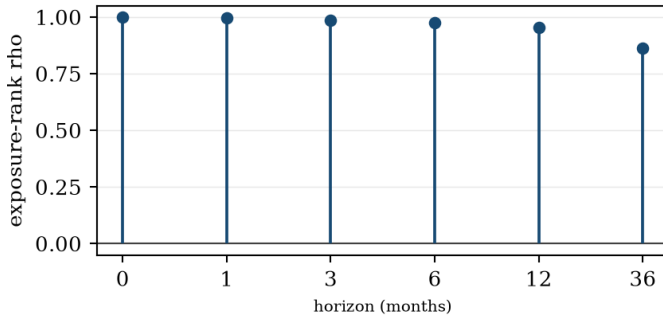


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

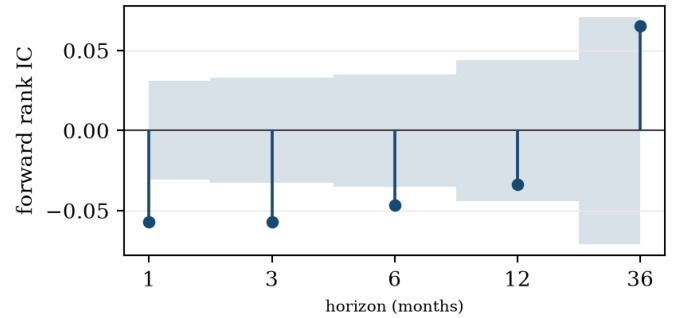


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	log_market_cap(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.14
Bootstrap Sharpe	0.21 [-0.49, 0.91]; Pr(S<=0)=25.8%
Bootstrap tails	skew=-0.11 [-0.50, 0.26]; max DD=-14.3% [-22.4%, -9.2%]

51. market_cap

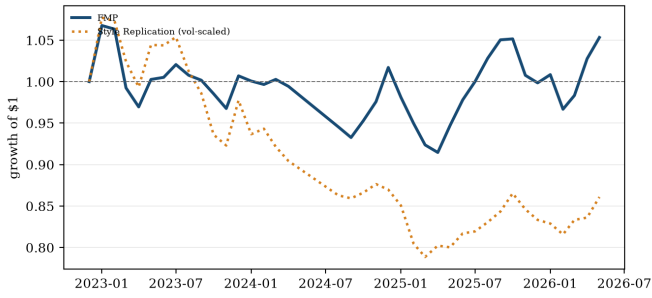


Figure: Common-window cumulative growth of \$1.

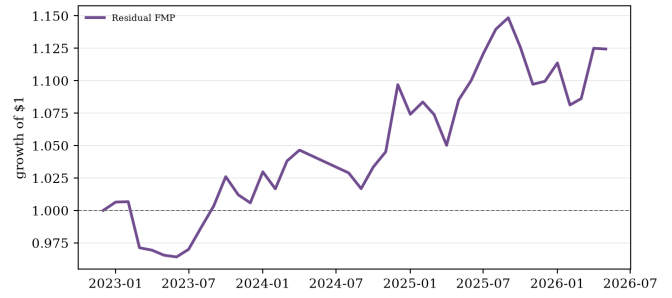


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
market_cap	2.16%	10.30%	0.21	0.39	-14.34%	0.591	5.39%	10.30%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
market_cap	-0.15 (-0.84)	0.42 (3.61)	0.04 (0.27)	-0.05 (-0.52)	-0.36 (-1.17)	-0.33 (-1.37)	3.92% (0.88)	0.597

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
market_cap	-3.22%	-2.85%	-0.22%	-0.52%	1.34%	3.71%	3.92%	2.16%

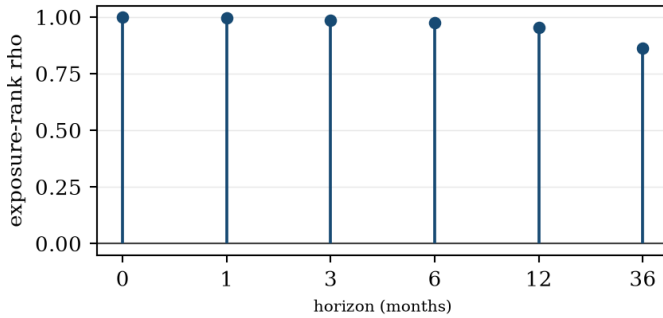


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

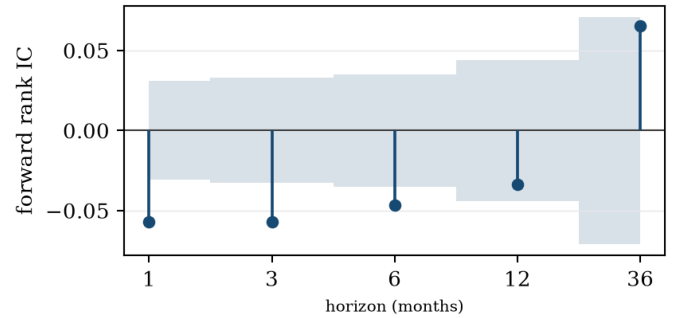


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	market_cap(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.14
Bootstrap Sharpe	0.21 [-0.49, 0.91]; Pr(S<=0)=25.8%
Bootstrap tails	skew=-0.11 [-0.50, 0.26]; max DD=-14.3% [-22.4%, -9.2%]

52. surprise

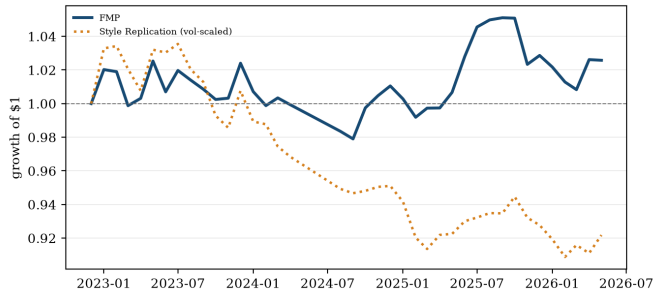


Figure: Common-window cumulative growth of \$1.

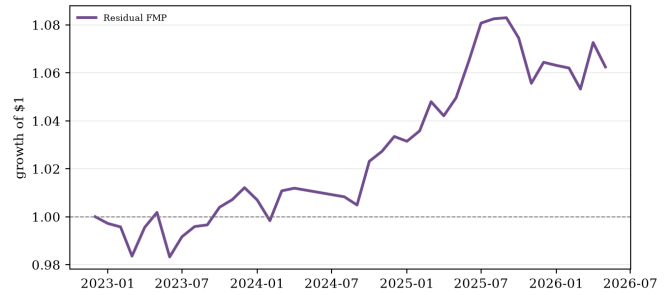


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
surprise	0.89%	4.26%	0.21	0.43	-4.52%	0.450	2.56%	4.26%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
surprise	-0.05 (-0.55)	0.14 (3.68)	-0.02 (-0.21)	-0.05 (-0.97)	-0.18 (-2.01)	-0.06 (-0.62)	1.96% (1.05)	0.456

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
surprise	-1.09%	-0.97%	0.10%	-0.51%	0.68%	0.72%	1.96%	0.89%

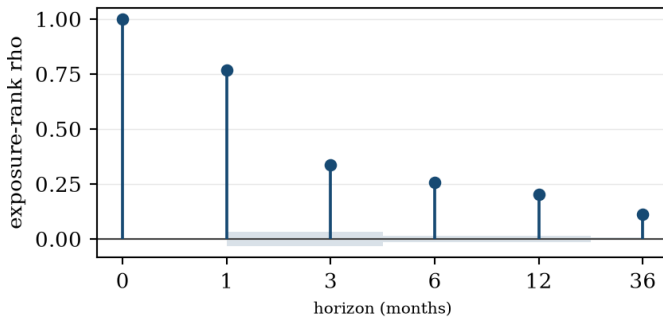


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

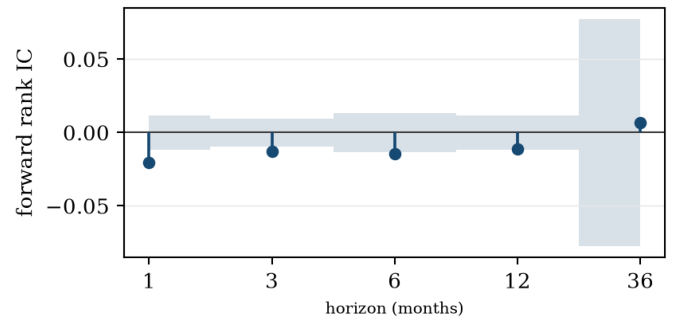


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	surprise(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2014 / 2390 / 2668
Available-history Sharpe	-0.02
Bootstrap Sharpe	0.21 [-0.46, 0.90]; Pr(S<=0)=23.8%
Bootstrap tails	skew=0.04 [-0.39, 0.46]; max DD=-4.5% [-8.9%, -2.8%]

53. fcf_margin

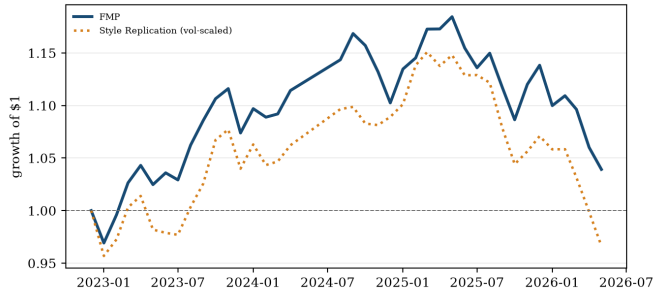


Figure: Common-window cumulative growth of \$1.

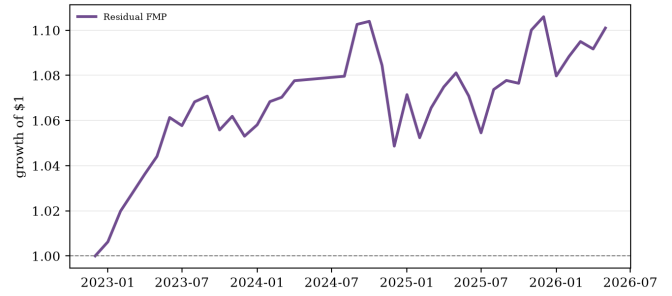


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf_margin	1.50%	7.71%	0.20	0.36	-12.26%	0.675	2.16%	7.71%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf_margin	0.13 (2.53)	-0.19 (-2.04)	-0.22 (-3.15)	-0.13 (-1.77)	0.33 (2.75)	0.40 (4.84)	3.13% (1.01)	0.681

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf_margin	2.81%	1.28%	1.34%	-1.33%	-1.23%	-4.51%	3.13%	1.50%

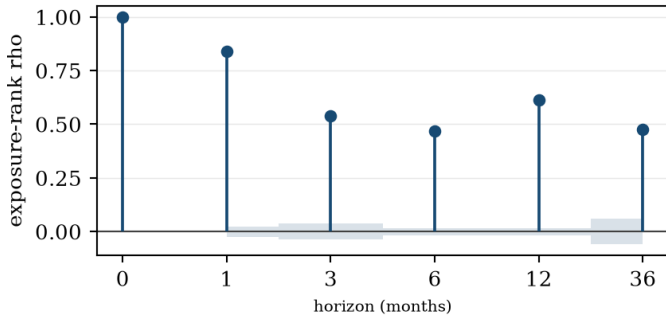


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

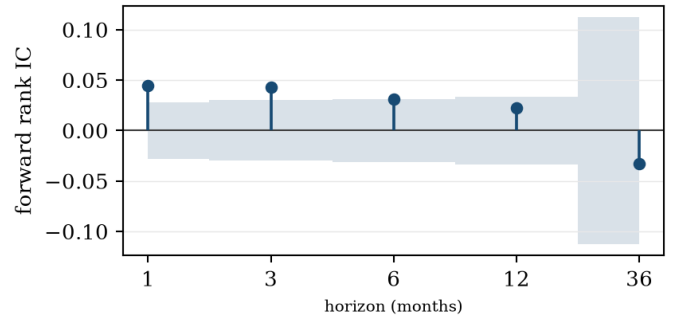


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	fcf_margin(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2022 / 2382 / 2565
Available-history Sharpe	0.51
Bootstrap Sharpe	0.20 [-0.68, 1.15]; Pr(S<=0)=30.4%
Bootstrap tails	skew=-0.27 [-0.75, 0.18]; max DD=-12.3% [-22.4%, -5.6%]

54. accruals

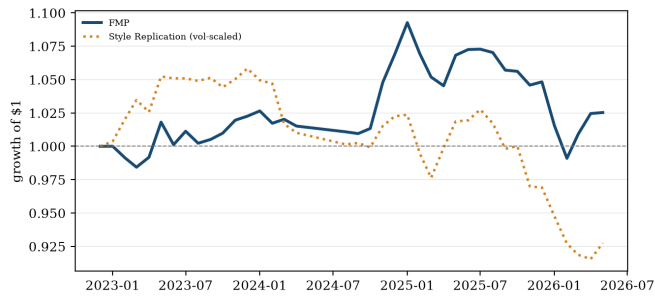


Figure: Common-window cumulative growth of \$1.

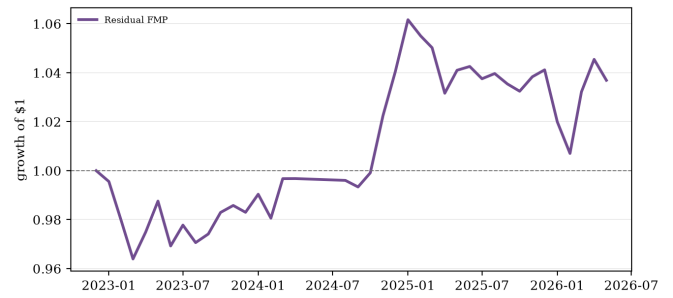


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals	0.90%	4.87%	0.19	0.34	-9.30%	0.293	2.13%	4.87%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals	0.04 (0.38)	0.10 (1.06)	-0.17 (-1.56)	-0.04 (-0.52)	-0.17 (-1.13)	0.15 (1.18)	1.22% (0.38)	0.304

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals	0.76%	-0.70%	1.05%	-0.40%	0.64%	-1.67%	1.22%	0.90%

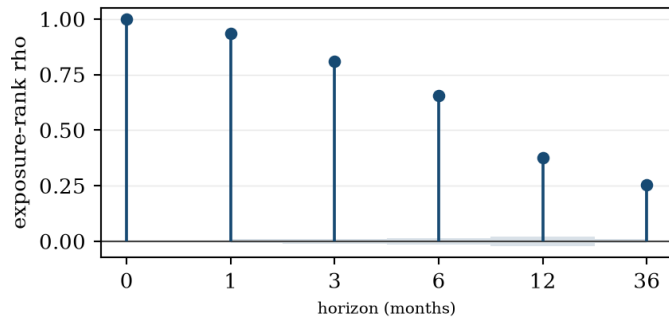


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

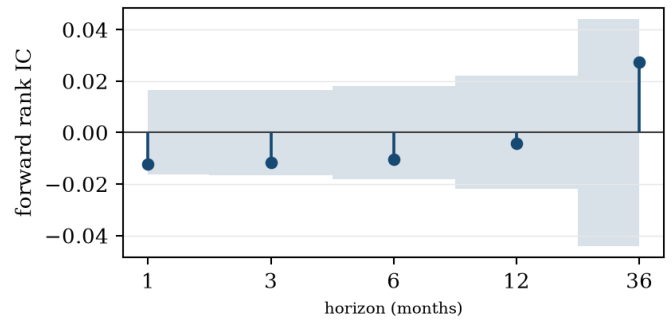


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	accruals(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2059 / 2484 / 2672
Available-history Sharpe	0.18
Bootstrap Sharpe	0.19 [-0.64, 1.02]; Pr(S<=0)=31.5%
Bootstrap tails	skew=0.15 [-0.38, 0.81]; max DD=-9.3% [-14.0%, -2.5%]

55. ebit_margin_ttm

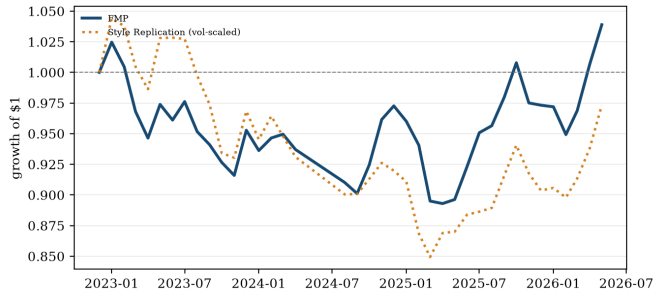


Figure: Common-window cumulative growth of \$1.

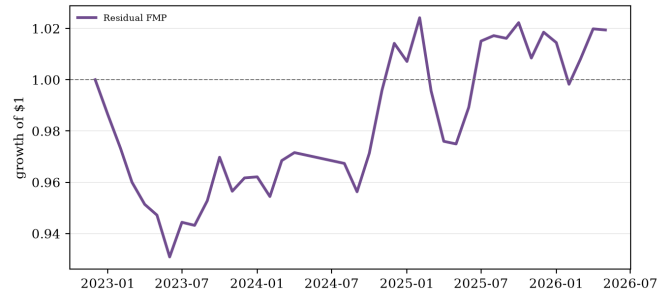


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ebit_margin_ttm	1.54%	8.34%	0.19	0.31	-12.85%	0.682	1.97%	8.34%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ebit_margin_ttm	-0.14 (-1.30)	0.29 (3.49)	0.03 (0.36)	0.08 (1.21)	-0.31 (-1.86)	-0.35 (-3.07)	0.71% (0.17)	0.689

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ebit_margin_ttm	-2.99%	-1.96%	-0.19%	0.87%	1.15%	3.96%	0.71%	1.54%

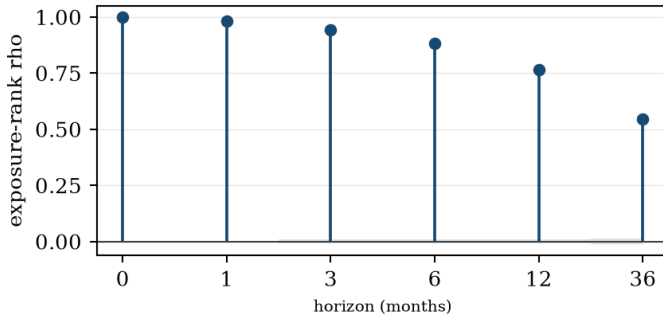


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

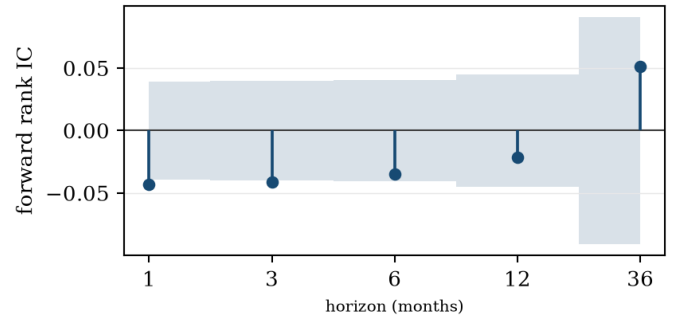


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	ebit_margin_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2045 / 2435 / 2548
Available-history Sharpe	-0.01
Bootstrap Sharpe	0.19 [-0.58, 0.96]; Pr(S<=0)=30.3%
Bootstrap tails	skew=0.02 [-0.49, 0.59]; max DD=-12.9% [-19.8%, -7.7%]

56. amihud_illiq_21d

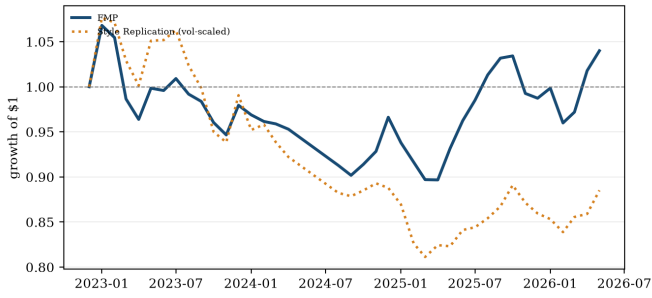


Figure: Common-window cumulative growth of \$1.

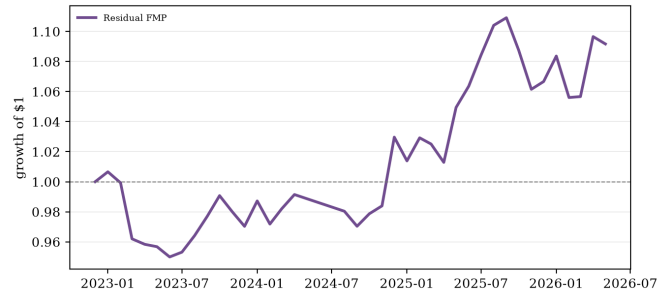


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
amihud_illiq_21d	1.71%	9.82%	0.17	0.32	-16.04%	0.608	4.34%	9.82%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
amihud_illiq_21d	-0.13 (-0.75)	0.37 (3.40)	0.04 (0.35)	-0.06 (-0.58)	-0.38 (-1.28)	-0.31 (-1.21)	2.95% (0.78)	0.614

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
amihud_illiq_21d	-2.71%	-2.52%	-0.28%	-0.64%	1.43%	3.48%	2.95%	1.71%

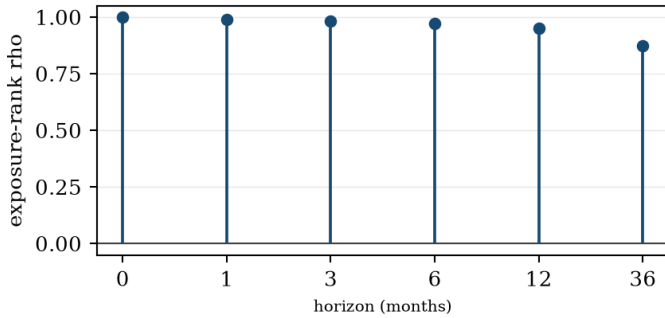


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

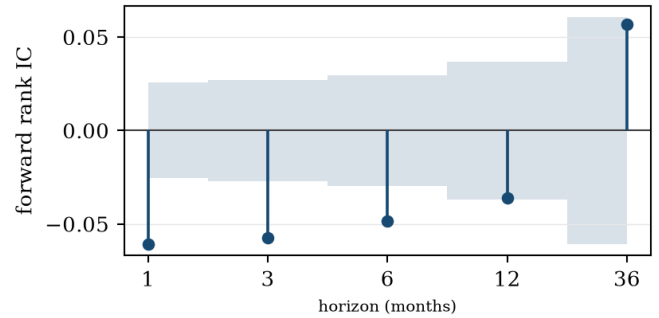


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	amihud_illiq_21d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2093 / 2574 / 2801
Available-history Sharpe	0.04
Bootstrap Sharpe	0.17 [-0.65, 0.98]; Pr(S<=0)=31.2%
Bootstrap tails	skew=0.07 [-0.39, 0.55]; max DD=-16.0% [-24.0%, -7.4%]

57. growth_gpoa_5y

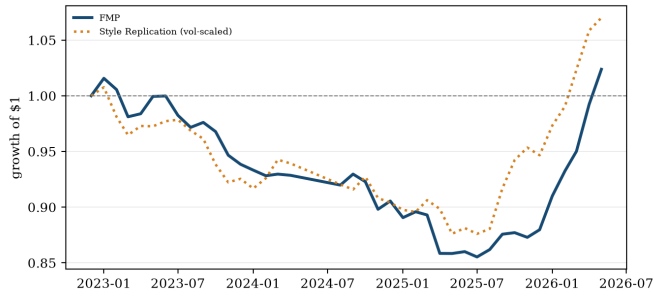


Figure: Common-window cumulative growth of \$1.

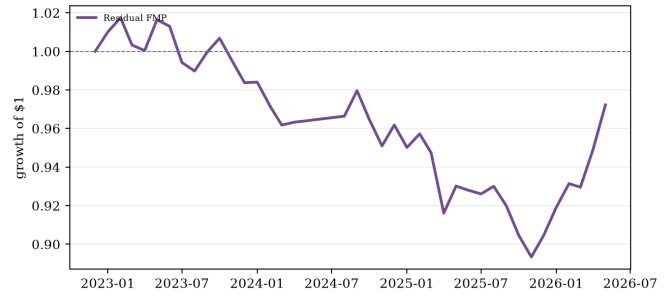


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_gpoa_5y	0.91%	5.90%	0.15	0.20	-15.79%	0.416	-0.58%	5.90%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_gpoa_5y	-0.08 (-0.63)	-0.04 (-0.39)	0.38 (3.29)	0.19 (1.96)	-0.22 (-1.60)	-0.24 (-1.72)	-0.79% (-0.25)	0.417

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_gpoa_5y	-1.76%	0.28%	-2.34%	2.04%	0.82%	2.68%	-0.79%	0.91%

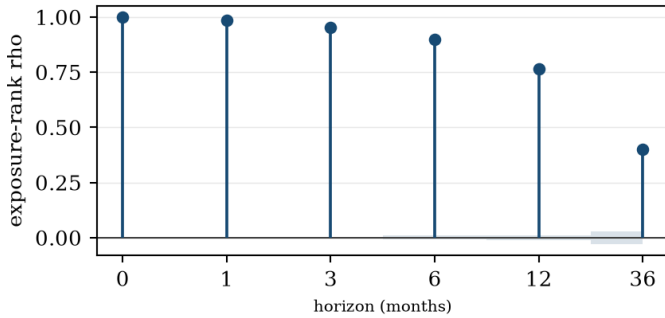


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

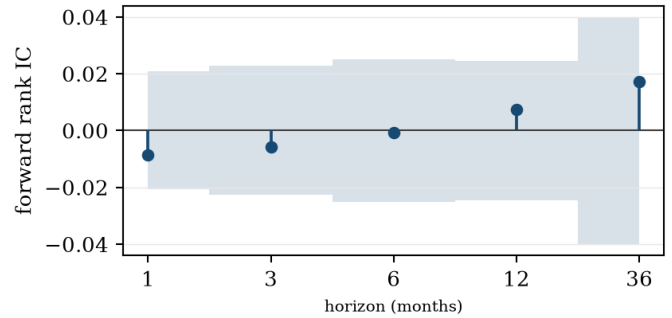


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	growth_gpoa_5y(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1905 / 2104 / 2397
Available-history Sharpe	-0.09
Bootstrap Sharpe	0.15 [-1.71, 1.51]; Pr(S<=0)=39.1%
Bootstrap tails	skew=0.32 [-0.98, 0.85]; max DD=-15.8% [-21.6%, -5.0%]

58. ev

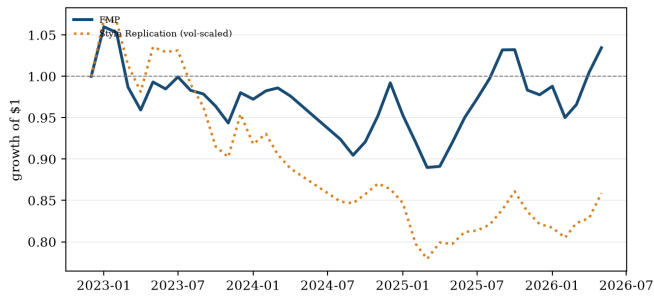


Figure: Common-window cumulative growth of \$1.

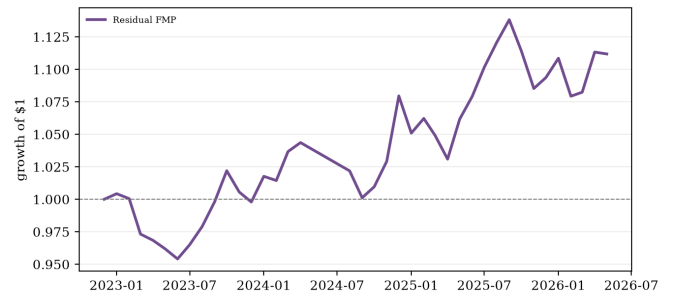


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ev	1.61%	10.51%	0.15	0.28	-16.05%	0.611	4.95%	10.51%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ev	-0.24 (-1.49)	0.42 (3.90)	-0.02 (-0.16)	-0.01 (-0.15)	-0.32 (-1.16)	-0.42 (-1.91)	3.56% (0.78)	0.616

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ev	-4.93%	-2.88%	0.12%	-0.13%	1.19%	4.68%	3.56%	1.61%

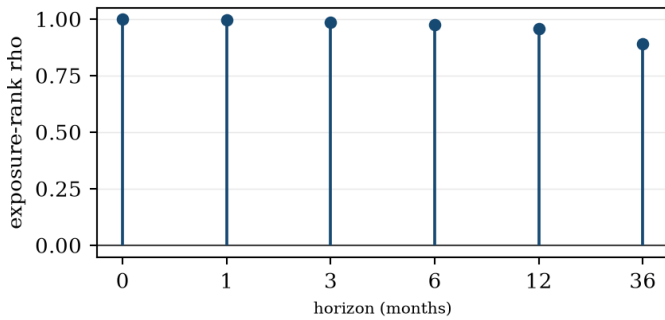


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

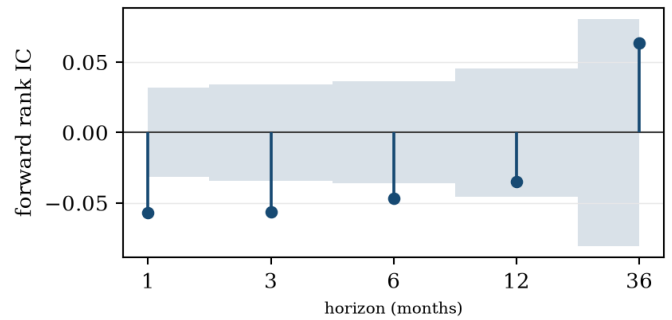


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	ev(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2059 / 2529 / 2728
Available-history Sharpe	0.07
Bootstrap Sharpe	0.15 [-0.56, 0.90]; Pr(S<=0)=32.1%
Bootstrap tails	skew=-0.21 [-0.61, 0.15]; max DD=-16.0% [-24.6%, -9.5%]

59. ret_volume_corr_63d

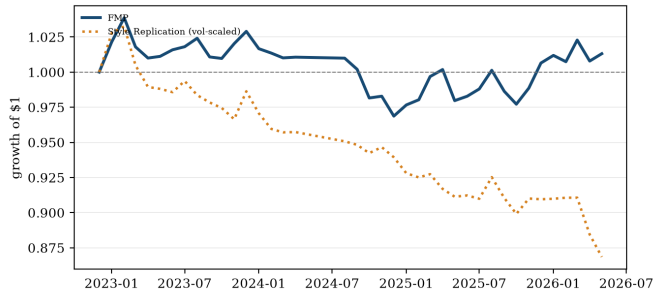


Figure: Common-window cumulative growth of \$1.

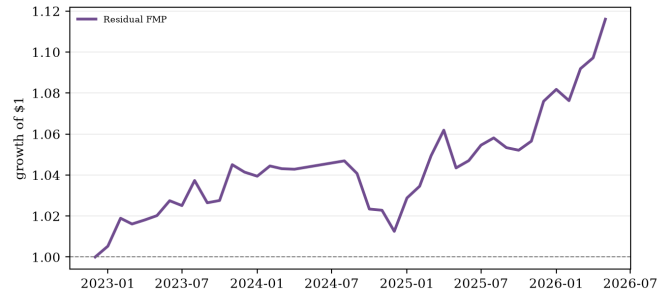


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ret_volume_corr_63d	0.49%	3.97%	0.12	0.28	-6.74%	0.391	3.22%	3.97%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ret_volume_corr_63d	-0.05 (-0.52)	0.16 (3.19)	-0.02 (-0.42)	-0.11 (-4.10)	0.14 (1.62)	-0.05 (-0.60)	3.52% (1.38)	0.394

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ret_volume_corr_63d	-0.95%	-1.13%	0.14%	-1.15%	-0.54%	0.59%	3.52%	0.49%

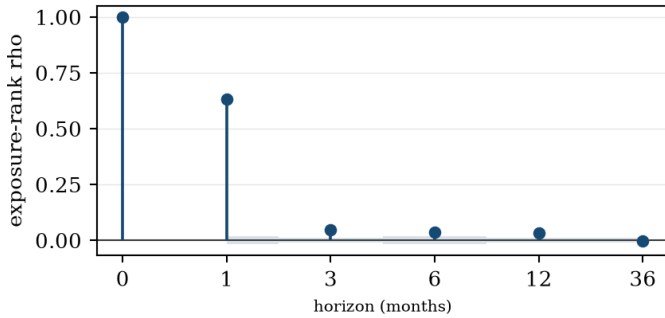


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

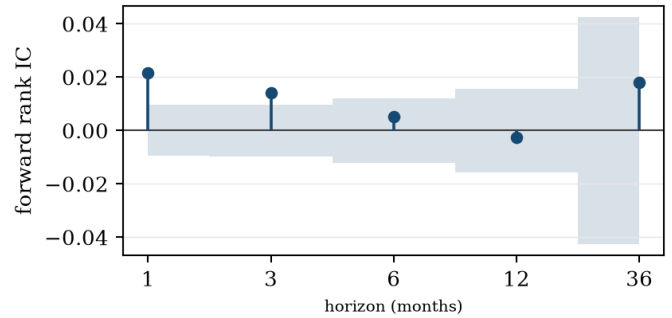


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	ret_volume_corr_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2811
Available-history Sharpe	0.48
Bootstrap Sharpe	0.12 [-0.74, 0.78]; Pr(S<=0)=36.1%
Bootstrap tails	skew=-0.23 [-0.57, 0.05]; max DD=-6.7% [-10.5%, -2.8%]

60. gp_ev

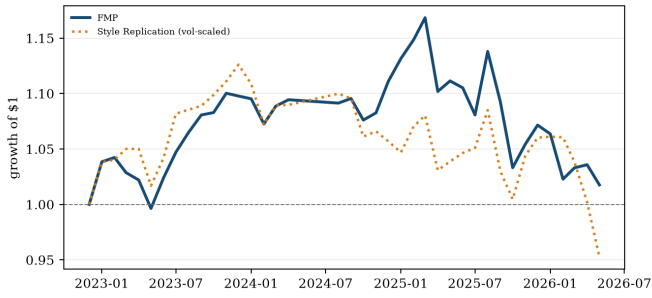


Figure: Common-window cumulative growth of \$1.

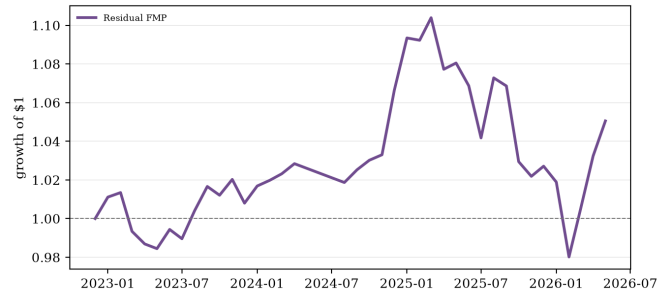


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gp_ev	0.88%	8.21%	0.11	0.24	-12.90%	0.510	1.74%	8.21%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gp_ev	0.40 (4.60)	0.13 (1.17)	0.08 (0.59)	-0.26 (-2.67)	0.14 (0.47)	0.39 (2.67)	1.72% (0.47)	0.510

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gp_ev	8.27%	-0.91%	-0.50%	-2.75%	-0.53%	-4.41%	1.72%	0.88%

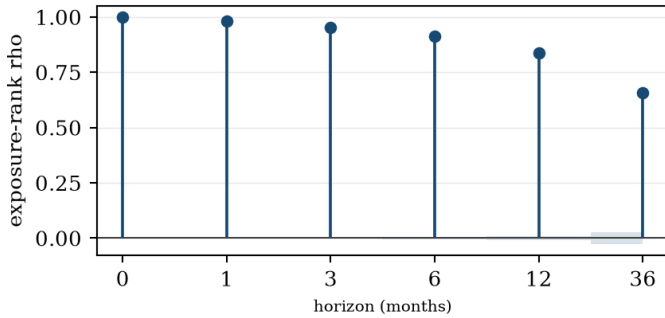


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

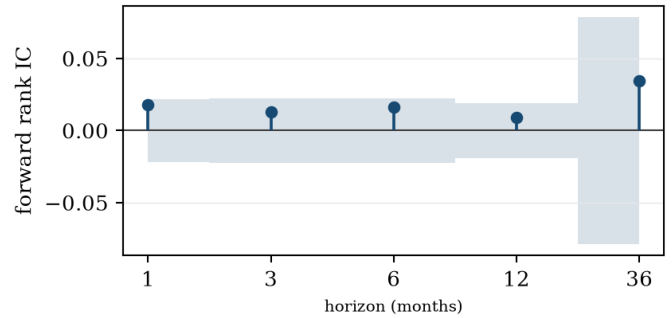


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	gp_ev(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2025 / 2444 / 2585
Available-history Sharpe	0.55
Bootstrap Sharpe	0.11 [-0.51, 1.10]; Pr(S<=0)=35.5%
Bootstrap tails	skew=-0.55 [-1.27, 0.03]; max DD=-12.9% [-21.3%, -4.4%]

61. gross_profit_to_ev

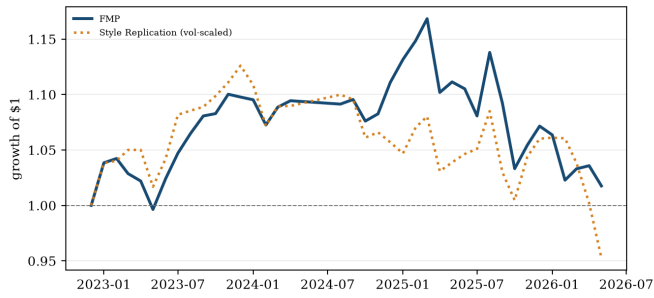


Figure: Common-window cumulative growth of \$1.

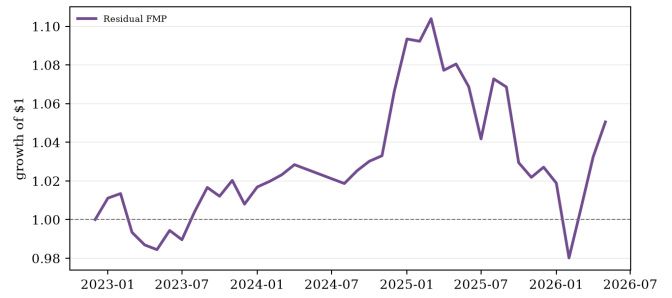


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gross_profit_to_ev	0.88%	8.21%	0.11	0.24	-12.90%	0.510	1.74%	8.21%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gross_profit_to_ev	0.40 (4.60)	0.13 (1.17)	0.08 (0.59)	-0.26 (-2.67)	0.14 (0.47)	0.39 (2.67)	1.72% (0.47)	0.510

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gross_profit_to_ev	8.27%	-0.91%	-0.50%	-2.75%	-0.53%	-4.41%	1.72%	0.88%

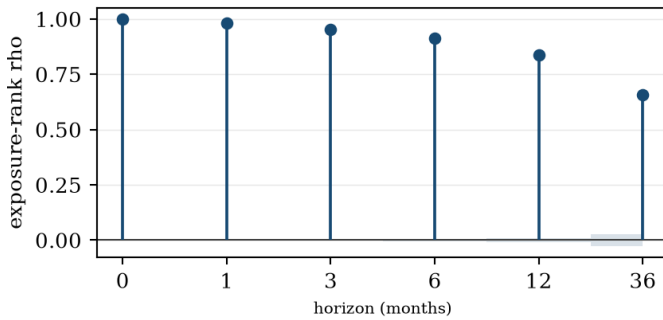


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

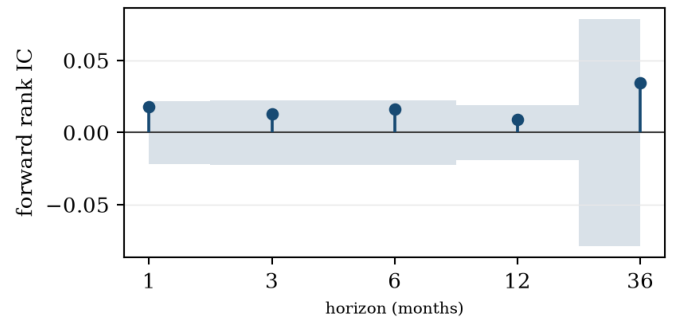


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	gross_profit_to_ev(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2025 / 2444 / 2585
Available-history Sharpe	0.55
Bootstrap Sharpe	0.11 [-0.51, 1.10]; Pr(S<=0)=35.5%
Bootstrap tails	skew=-0.55 [-1.27, 0.03]; max DD=-12.9% [-21.3%, -4.4%]

62. stoch_k_14d

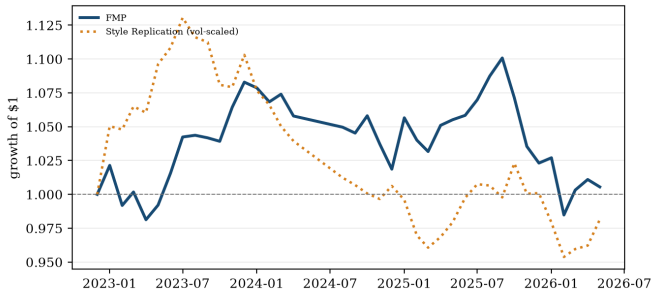


Figure: Common-window cumulative growth of \$1.

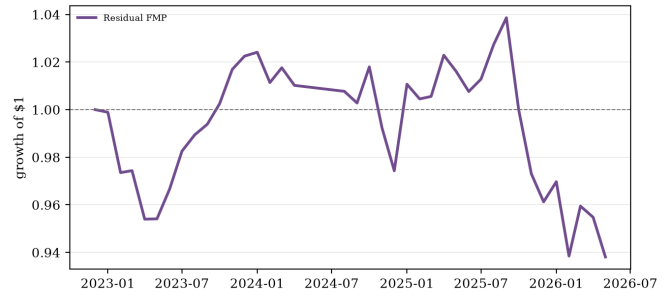


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
stoch_k_14d	0.37%	6.27%	0.06	0.11	-10.54%	0.151	0.52%	6.27%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
stoch_k_14d	0.14 (1.48)	0.04 (0.41)	0.02 (0.16)	-0.04 (-1.02)	-0.25 (-1.21)	0.07 (0.50)	-1.86% (-0.46)	0.201

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
stoch_k_14d	2.98%	-0.30%	-0.13%	-0.45%	0.92%	-0.79%	-1.86%	0.37%

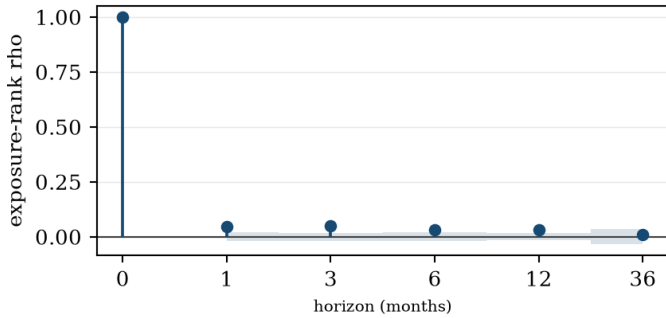


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

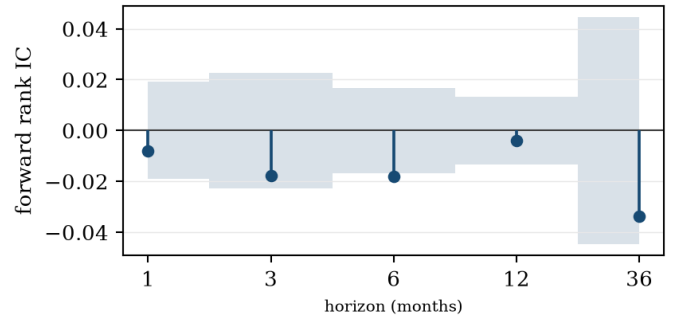


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	stoch_k_14d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2105 / 2583 / 2810
Available-history Sharpe	0.32
Bootstrap Sharpe	0.06 [-0.68, 0.90]; Pr(S<=0)=41.1%
Bootstrap tails	skew=-0.27 [-0.75, 0.54]; max DD=-10.5% [-18.1%, -4.2%]

63. earnings_yield

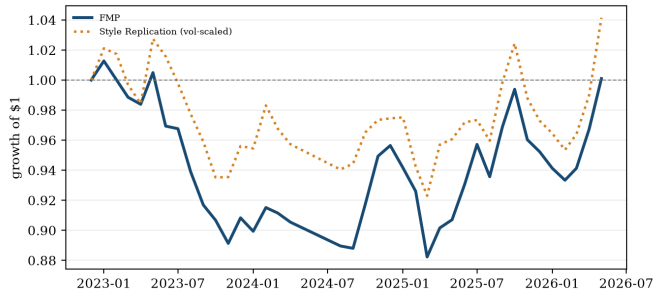


Figure: Common-window cumulative growth of \$1.

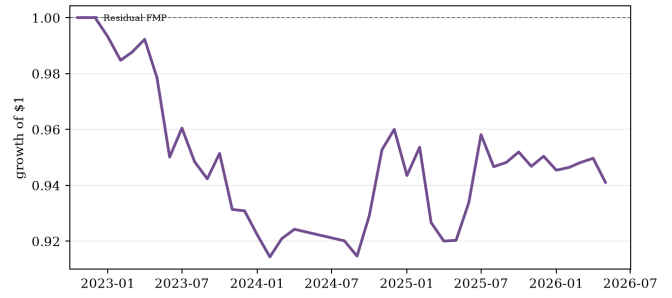


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
earnings_yield	0.32%	7.68%	0.04	0.07	-12.88%	0.683	-0.98%	7.68%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
earnings_yield	-0.25 (-2.91)	0.05 (0.52)	-0.13 (-1.62)	0.09 (2.11)	-0.18 (-1.51)	-0.47 (-6.12)	-1.83% (-0.56)	0.688

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
earnings_yield	-5.13%	-0.33%	0.78%	0.91%	0.67%	5.23%	-1.83%	0.32%

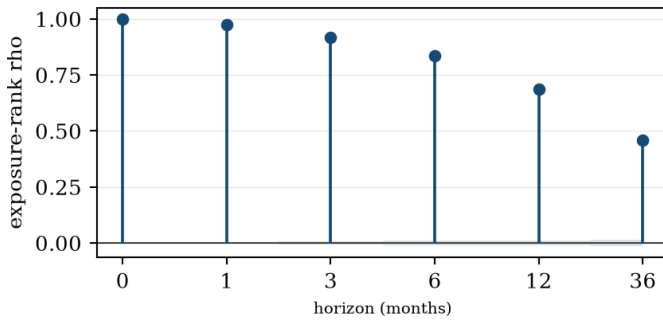


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

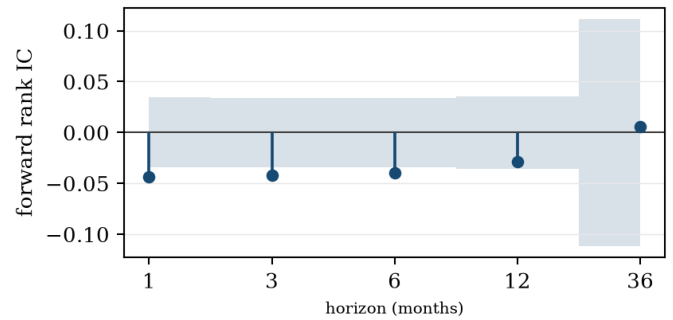


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	earnings_yield(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2036 / 2434 / 2543
Available-history Sharpe	-0.24
Bootstrap Sharpe	0.04 [-0.91, 0.86]; Pr(S<=0)=44.2%
Bootstrap tails	skew=-0.00 [-0.43, 0.47]; max DD=-12.9% [-23.1%, -7.5%]

64. ebit_to_ev

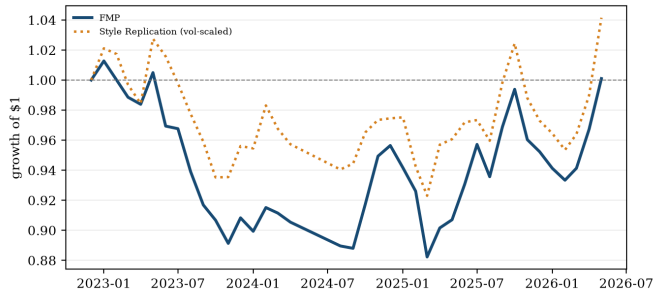


Figure: Common-window cumulative growth of \$1.

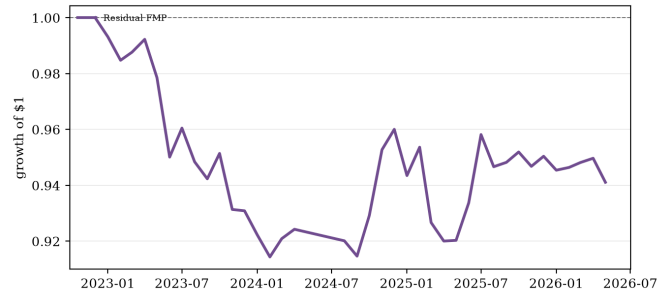


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ebit_to_ev	0.32%	7.68%	0.04	0.07	-12.88%	0.683	-0.98%	7.68%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ebit_to_ev	-0.25 (-2.91)	0.05 (0.52)	-0.13 (-1.62)	0.09 (2.11)	-0.18 (-1.51)	-0.47 (-6.12)	-1.83% (-0.56)	0.688

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ebit_to_ev	-5.13%	-0.33%	0.78%	0.91%	0.67%	5.23%	-1.83%	0.32%

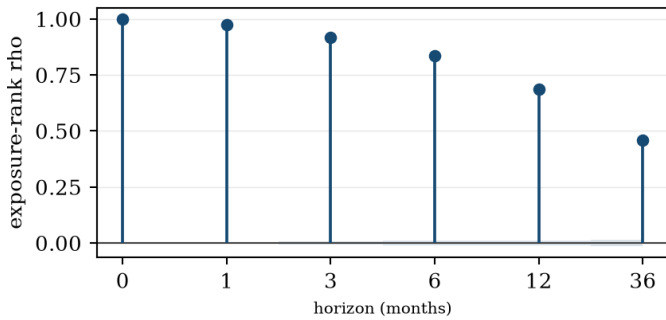


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

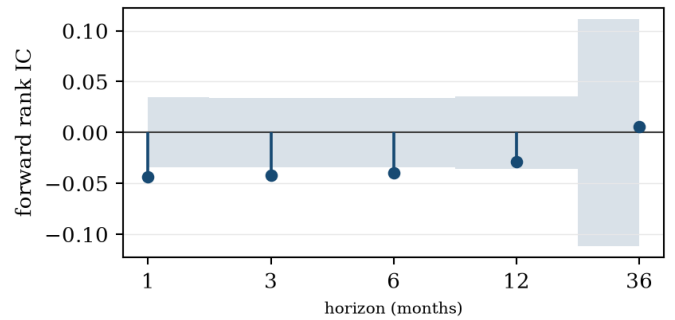


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	ebit_to_ev(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2036 / 2434 / 2543
Available-history Sharpe	-0.24
Bootstrap Sharpe	0.04 [-0.91, 0.86]; Pr(S<=0)=44.2%
Bootstrap tails	skew=-0.00 [-0.43, 0.47]; max DD=-12.9% [-23.1%, -7.5%]

65. is_positive_revenue

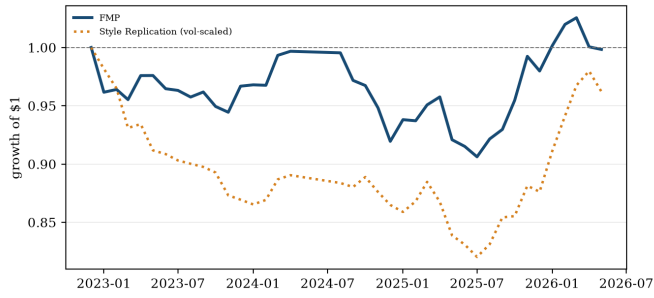


Figure: Common-window cumulative growth of \$1.

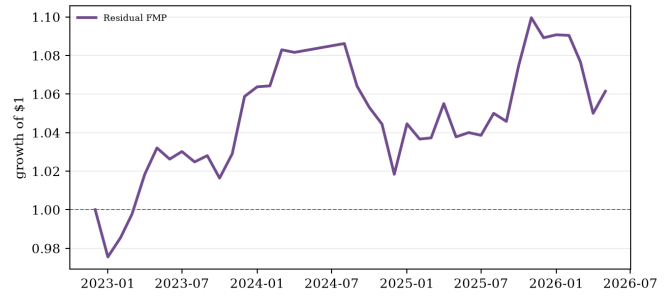


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
is_positive_revenue	0.14%	6.27%	0.02	0.04	-9.08%	0.340	0.74%	6.27%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
is_positive_revenue	-0.18 (-2.03)	0.02 (0.18)	0.26 (1.92)	0.16 (1.24)	0.13 (0.67)	-0.21 (-1.41)	2.01% (0.56)	0.356

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
is_positive_revenue	-3.70%	-0.11%	-1.65%	1.70%	-0.47%	2.36%	2.01%	0.14%

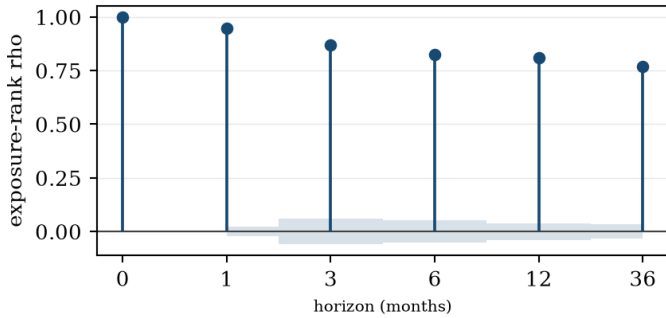


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

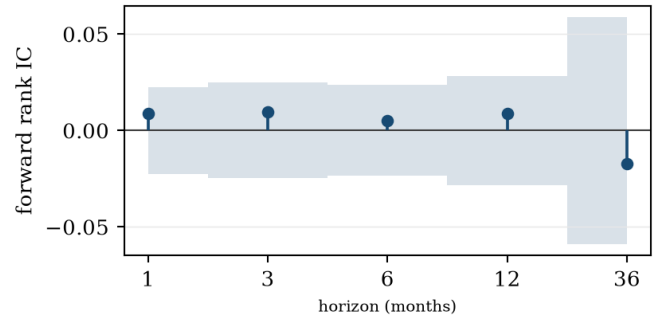


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	is_positive_revenue(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2103 / 2577 / 2805
Available-history Sharpe	0.05
Bootstrap Sharpe	0.02 [-0.89, 0.85]; Pr(S<=0)=46.2%
Bootstrap tails	skew=-0.15 [-0.48, 0.21]; max DD=-9.1% [-18.2%, -6.0%]

66. net_income_margin_ttm

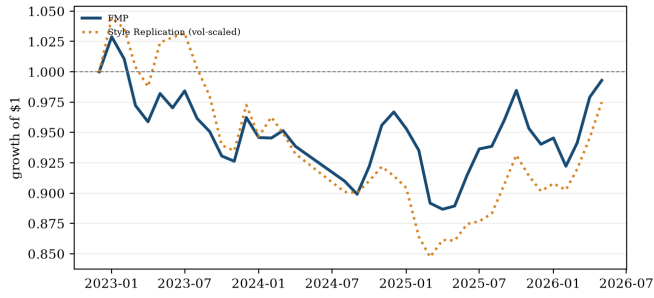


Figure: Common-window cumulative growth of \$1.

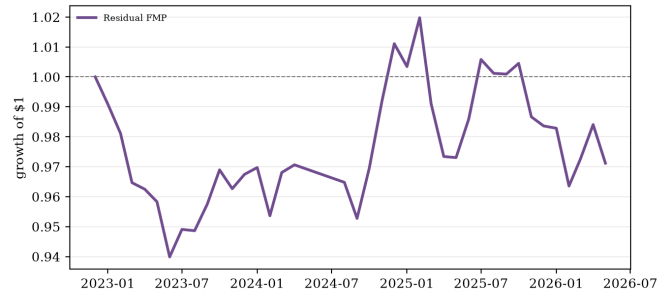


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_margin_ttm	0.08%	7.95%	0.01	0.02	-13.82%	0.667	0.48%	7.95%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_margin_ttm	-0.09	0.30	0.10	0.09	-0.33	-0.29	-0.82%	0.676
	(-0.88)	(3.57)	(1.10)	(1.50)	(-1.99)	(-2.43)	(-0.21)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_margin_ttm	-1.96%	-2.02%	-0.61%	0.98%	1.22%	3.30%	-0.82%	0.08%

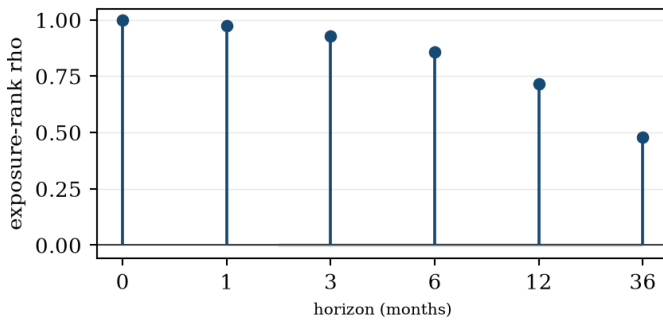


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

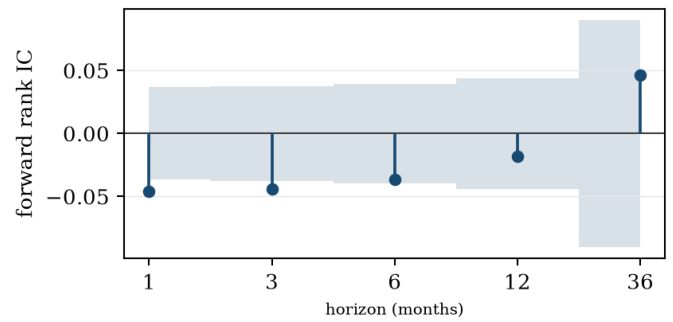


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	net_income_margin_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2046 / 2486 / 2650
Available-history Sharpe	-0.20
Bootstrap Sharpe	0.01 [-0.73, 0.73]; Pr(S<=0)=48.4%
Bootstrap tails	skew=0.05 [-0.43, 0.58]; max DD=-13.8% [-20.8%, -7.6%]

67. roe_volatility

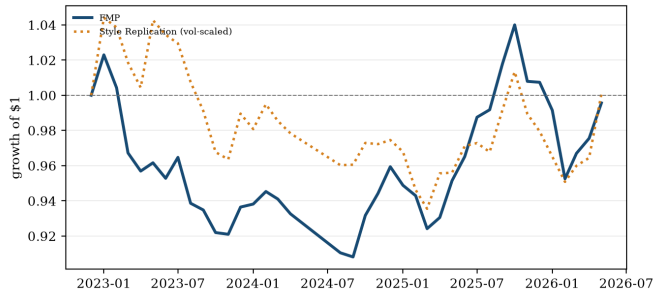


Figure: Common-window cumulative growth of \$1.

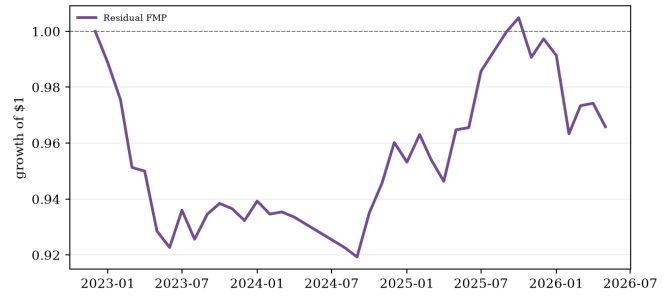


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
roe_volatility	0.05%	6.29%	0.01	0.01	-11.23%	0.555	-0.11%	6.29%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
roe_volatility	-0.14 (-1.48)	0.04 (0.59)	-0.06 (-0.91)	-0.05 (-0.96)	-0.17 (-0.90)	-0.34 (-2.32)	-1.01% (-0.29)	0.563

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
roe_volatility	-2.90%	-0.29%	0.37%	-0.58%	0.63%	3.83%	-1.01%	0.05%

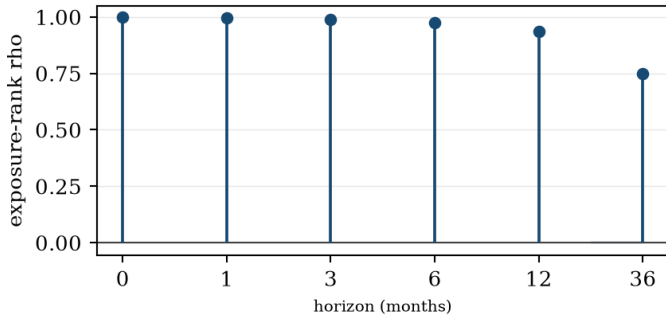


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

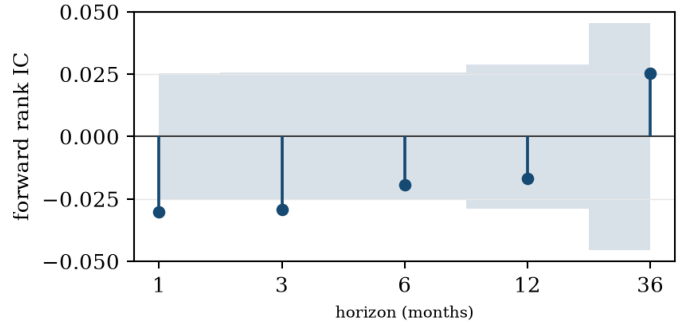


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	roe_volatility(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1949 / 2348 / 2499
Available-history Sharpe	-0.35
Bootstrap Sharpe	0.01 [-0.85, 0.97]; Pr(S<=0)=48.4%
Bootstrap tails	skew=-0.42 [-0.78, -0.05]; max DD=-11.2% [-19.6%, -5.1%]

68. accruals_total

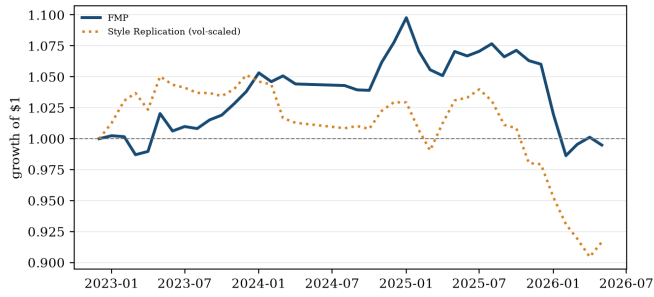


Figure: Common-window cumulative growth of \$1.

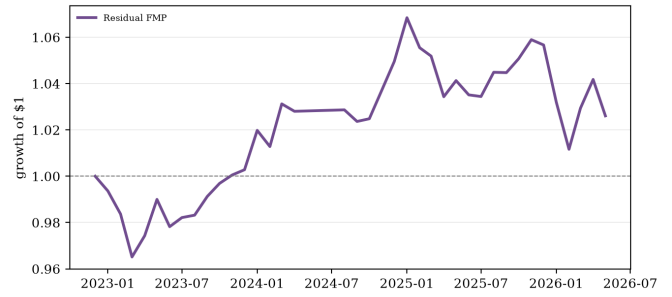


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_total	-0.05%	4.80%	-0.01	-0.02	-10.15%	0.326	1.46%	4.80%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_total	-0.01 (-0.06)	0.09 (0.86)	-0.23 (-2.61)	-0.10 (-1.32)	-0.08 (-0.50)	0.08 (0.58)	0.89% (0.32)	0.331

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_total	-0.11%	-0.61%	1.44%	-1.04%	0.28%	-0.89%	0.89%	-0.05%

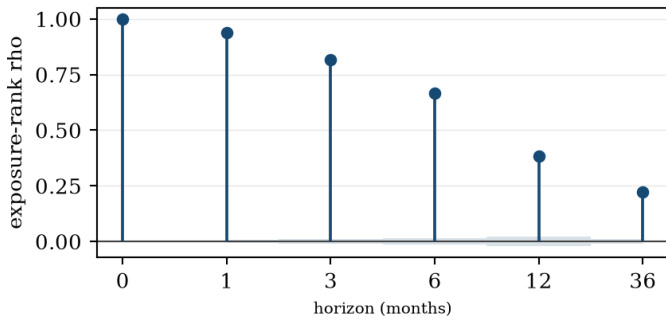


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

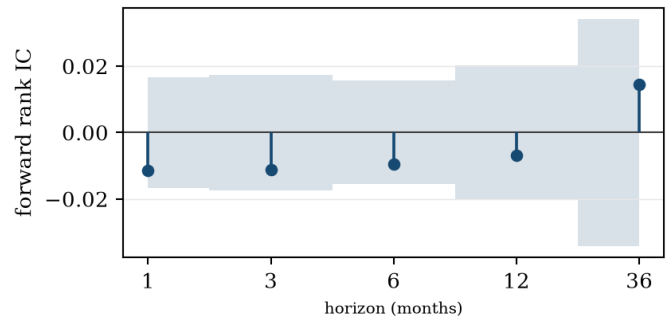


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	accruals_total(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2059 / 2484 / 2672
Available-history Sharpe	0.16
Bootstrap Sharpe	-0.01 [-0.90, 1.12]; Pr(S<=0)=52.6%
Bootstrap tails	skew=-0.57 [-1.09, 0.56]; max DD=-10.1% [-17.3%, -2.5%]

69. fcf_to_market_cap

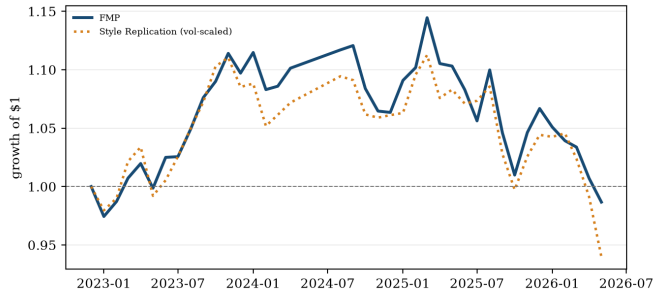


Figure: Common-window cumulative growth of \$1.

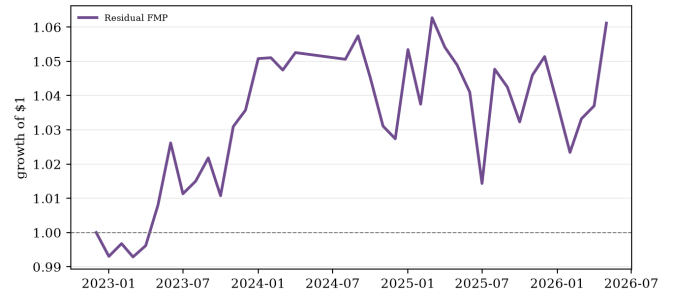


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf_to_market_cap	-0.10%	8.11%	-0.01	-0.03	-13.79%	0.673	1.24%	8.11%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf_to_market_cap	0.33 (4.98)	-0.03 (-0.34)	0.01 (0.08)	-0.15 (-2.51)	0.15 (1.05)	0.62 (8.04)	1.98% (0.83)	0.677

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf_to_market_cap	6.79%	0.24%	-0.04%	-1.61%	-0.55%	-6.91%	1.98%	-0.10%

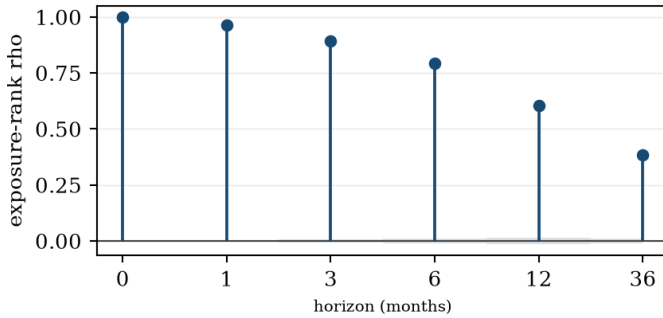


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

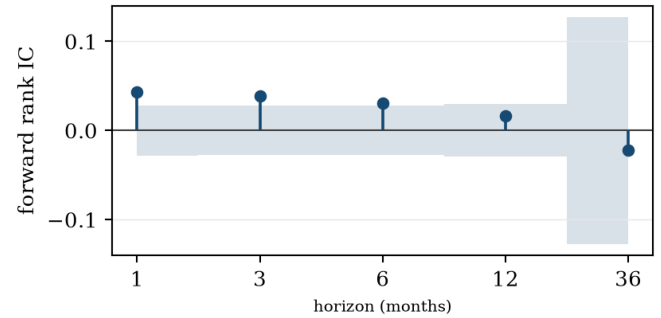


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	fcf_to_market_cap(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2068 / 2485 / 2673
Available-history Sharpe	0.37
Bootstrap Sharpe	-0.01 [-0.76, 1.00]; Pr(S<=0)=53.8%
Bootstrap tails	skew=-0.12 [-0.65, 0.35]; max DD=-13.8% [-24.1%, -5.4%]

70. pe_ttm

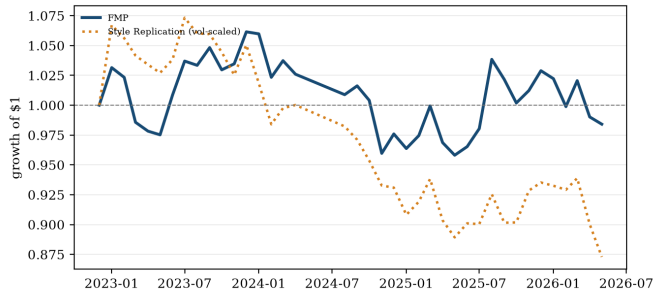


Figure: Common-window cumulative growth of \$1.

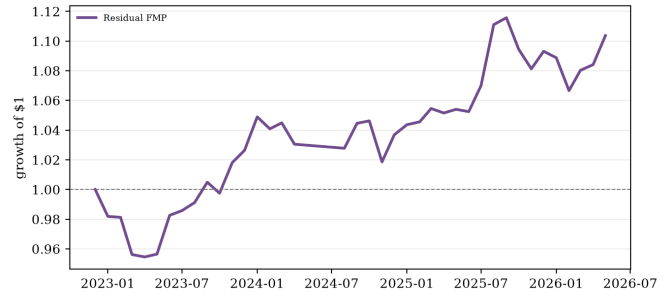


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
pe_ttm	-0.21%	7.78%	-0.03	-0.06	-9.74%	0.583	2.84%	7.78%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
pe_ttm	0.10 (0.98)	0.12 (0.82)	0.27 (2.66)	-0.25 (-3.27)	-0.15 (-0.65)	0.09 (0.56)	3.24% (1.09)	0.585

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
pe_ttm	2.16%	-0.84%	-1.70%	-2.67%	0.56%	-0.96%	3.24%	-0.21%

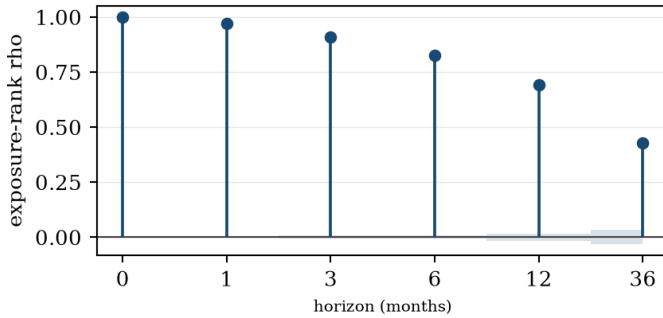


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

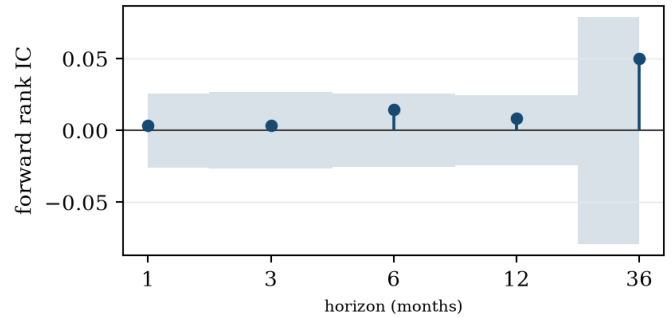


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	pe_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1595 / 1784 / 1874
Available-history Sharpe	0.18
Bootstrap Sharpe	-0.03 [-0.67, 0.58]; Pr(S<=0)=55.8%
Bootstrap tails	skew=0.26 [-0.29, 0.67]; max DD=-9.7% [-18.5%, -5.8%]

71. surprisePercentage

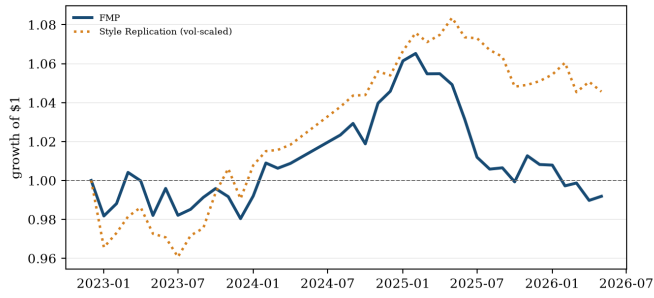


Figure: Common-window cumulative growth of \$1.

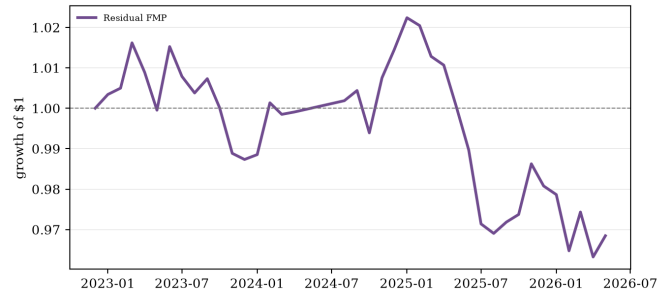


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
surprisePercentage	-0.19%	3.74%	-0.05	-0.09	-7.08%	0.384	-1.11%	3.74%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
surprisePercentage	0.03 (0.39)	-0.05 (-1.07)	-0.12 (-1.90)	0.04 (0.91)	0.21 (2.23)	0.05 (0.51)	-0.97% (-0.53)	0.384

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
surprisePercentage	0.65%	0.36%	0.75%	0.39%	-0.80%	-0.58%	-0.97%	-0.19%

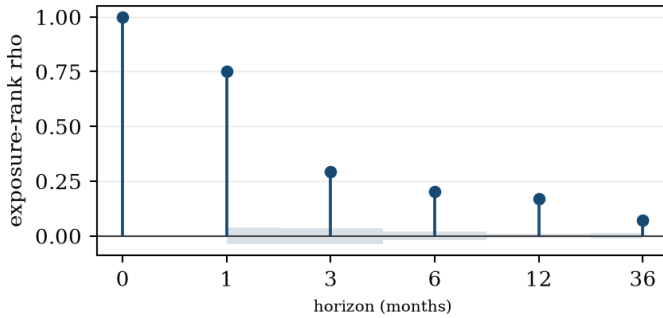


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

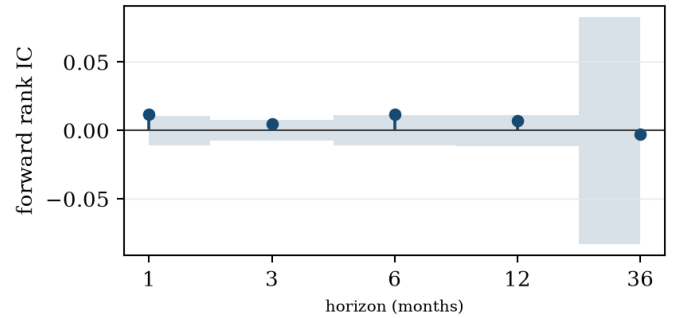


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	surprisePercentage(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1974 / 2376 / 2558
Available-history Sharpe	0.17
Bootstrap Sharpe	-0.05 [-1.16, 1.02]; Pr(S<=0)=55.5%
Bootstrap tails	skew=0.04 [-0.27, 0.34]; max DD=-7.1% [-13.7%, -2.4%]

72. fcf_to_equity

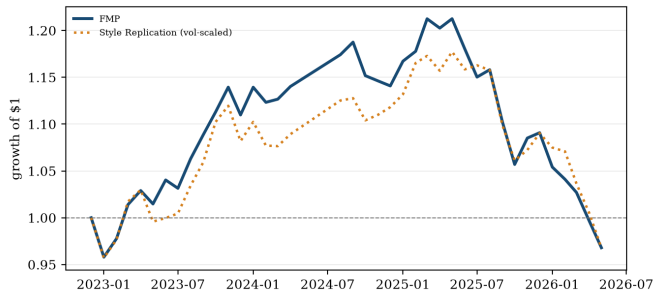


Figure: Common-window cumulative growth of \$1.

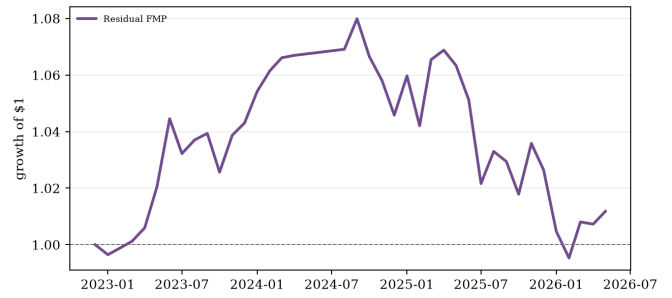


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf_to_equity	-0.68%	8.43%	-0.08	-0.13	-20.17%	0.759	-0.08%	8.43%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf_to_equity	0.29 (4.19)	-0.14 (-2.01)	-0.27 (-4.87)	-0.16 (-2.61)	0.28 (1.87)	0.64 (7.91)	0.45% (0.17)	0.760

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf_to_equity	6.10%	0.98%	1.65%	-1.68%	-1.04%	-7.15%	0.45%	-0.68%

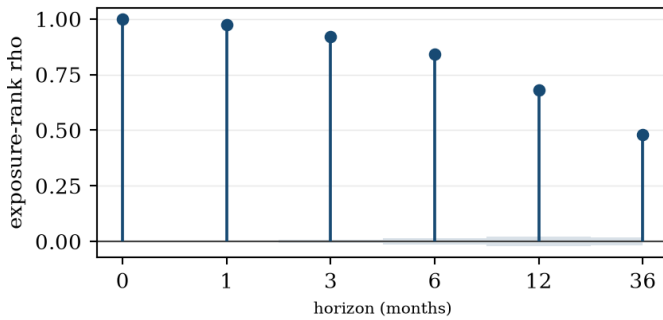


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

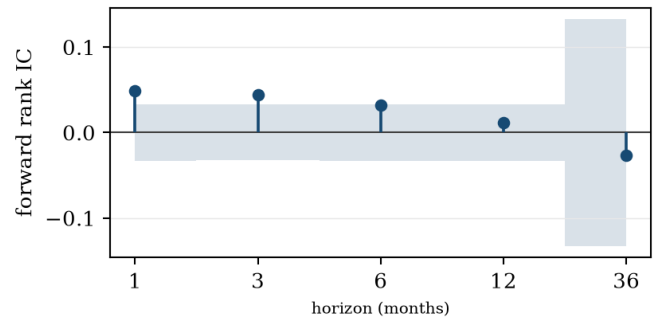


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	fcf_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1951 / 2332 / 2470
Available-history Sharpe	0.25
Bootstrap Sharpe	-0.08 [-1.35, 1.35]; Pr(S<=0)=56.5%
Bootstrap tails	skew=-0.28 [-0.72, 0.19]; max DD=-20.2% [-35.2%, -4.2%]

73. bb_pctb_20d

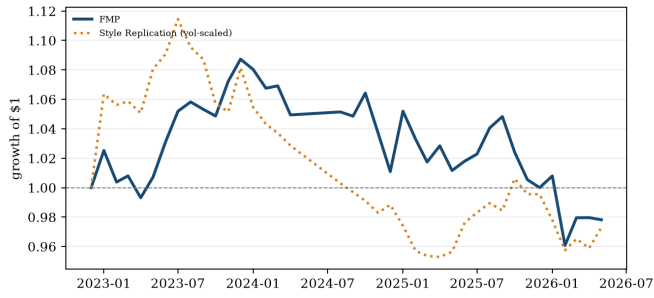


Figure: Common-window cumulative growth of \$1.

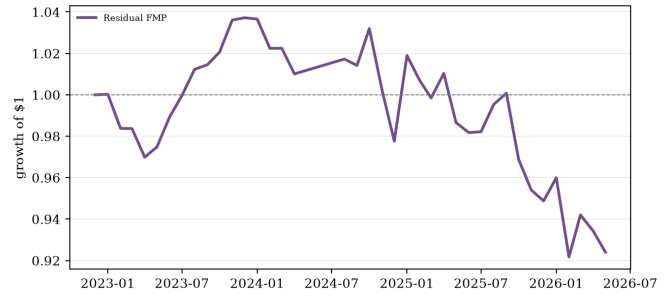


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bb_pctb_20d	-0.51%	6.20%	-0.08	-0.19	-11.64%	0.129	-0.26%	6.20%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bb_pctb_20d	0.12 (1.24)	0.04 (0.39)	0.04 (0.37)	-0.06 (-1.30)	-0.18 (-0.92)	0.01 (0.09)	-2.34% (-0.74)	0.168

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bb_pctb_20d	2.46%	-0.28%	-0.24%	-0.64%	0.68%	-0.15%	-2.34%	-0.51%

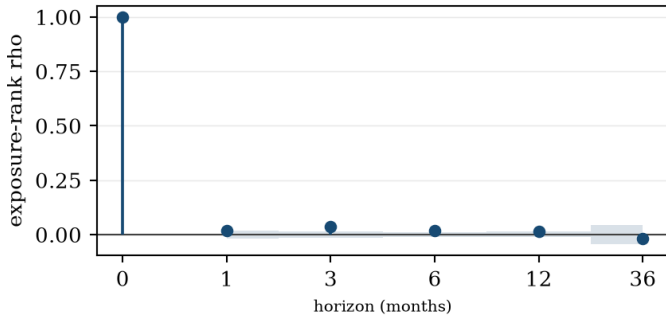


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

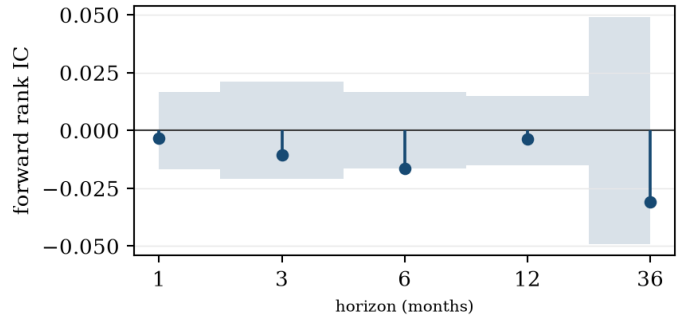


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	bb_pctb_20d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2105 / 2584 / 2811
Available-history Sharpe	0.26
Bootstrap Sharpe	-0.08 [-0.75, 0.73]; Pr(S<=0)=61.9%
Bootstrap tails	skew=-0.16 [-0.88, 0.61]; max DD=-11.6% [-17.2%, -4.9%]

74. accruals_percent

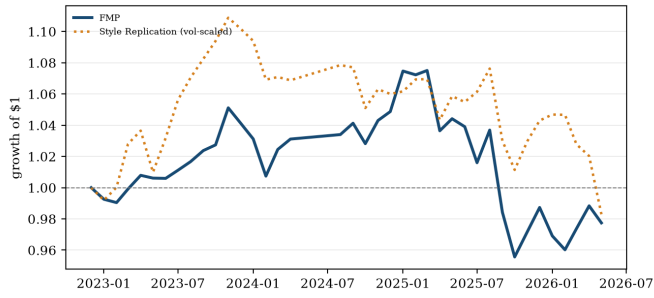


Figure: Common-window cumulative growth of \$1.

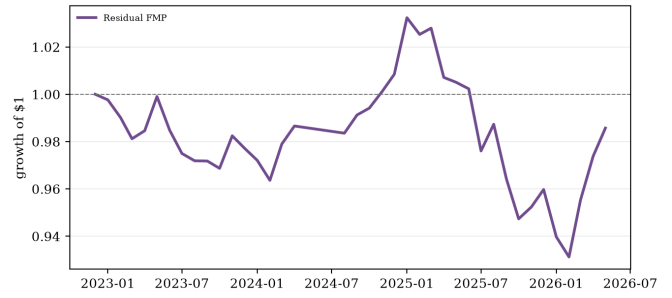


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_percent	-0.56%	5.75%	-0.10	-0.21	-11.11%	0.409	-0.31%	5.75%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_percent	0.31 (2.95)	0.10 (1.13)	0.02 (0.21)	-0.06 (-0.62)	-0.03 (-0.18)	0.47 (3.55)	-0.36% (-0.13)	0.409

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_percent	6.36%	-0.66%	-0.15%	-0.59%	0.12%	-5.28%	-0.36%	-0.56%

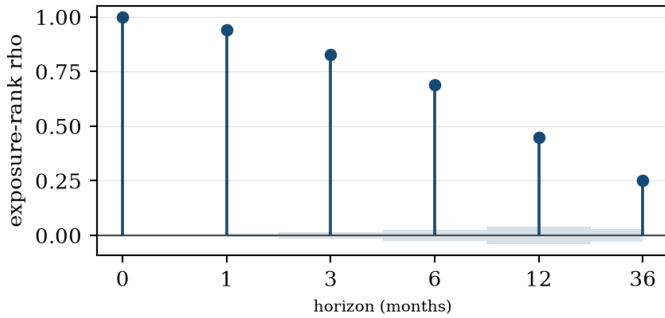


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

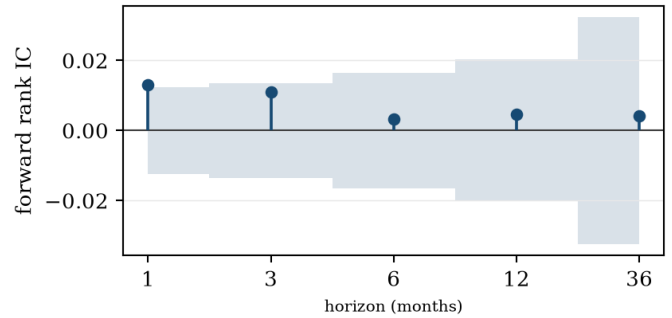


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	accruals_percent(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2019 / 2428 / 2614
Available-history Sharpe	0.23
Bootstrap Sharpe	-0.10 [-0.79, 0.97]; Pr(S<=0)=61.3%
Bootstrap tails	skew=-1.00 [-1.34, -0.16]; max DD=-11.1% [-20.4%, -4.2%]

75. net_operating_assets_proxy

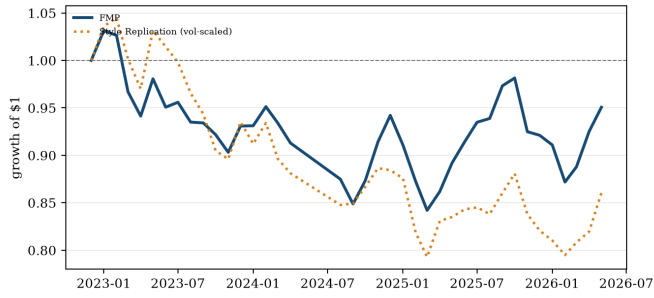


Figure: Common-window cumulative growth of \$1.

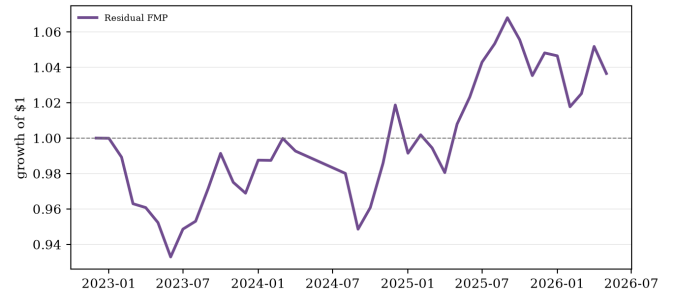


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_operating_assets_proxy	-1.05%	10.54%	-0.10	-0.18	-18.37%	0.661	2.36%	10.54%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_operating_assets_proxy	-0.38	0.38	-0.23	0.06	-0.29	-0.44	1.31%	0.664
	(-2.61)	(3.55)	(-2.01)	(0.77)	(-1.20)	(-2.04)	(0.30)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_operating_assets_proxy	-7.88%	-2.62%	1.44%	0.68%	1.09%	4.93%	1.31%	-1.05%

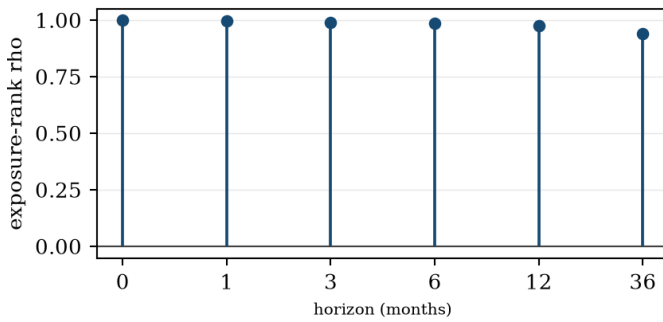


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

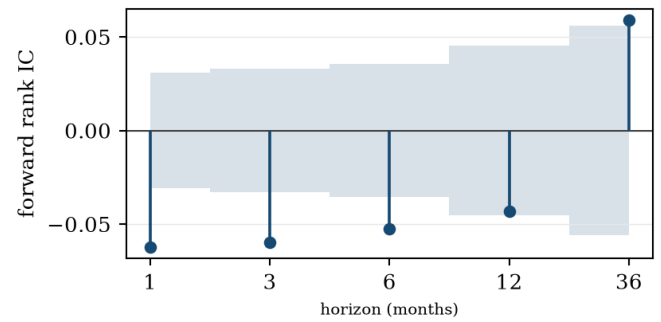


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	net_operating_assets_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2061 / 2529 / 2728
Available-history Sharpe	-0.24
Bootstrap Sharpe	-0.10 [-0.80, 0.63]; Pr(S<=0)=65.0%
Bootstrap tails	skew=-0.19 [-0.67, 0.29]; max DD=-18.4% [-28.6%, -11.2%]

76. net_income_growth_yoy_sign_flip

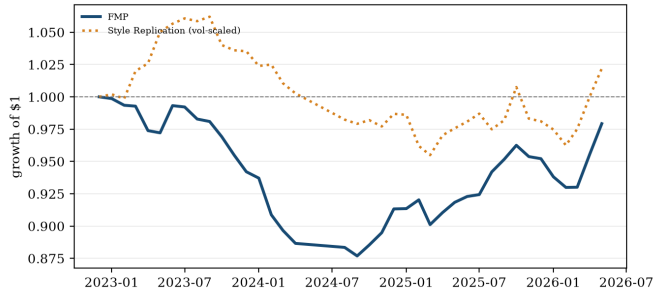


Figure: Common-window cumulative growth of \$1.

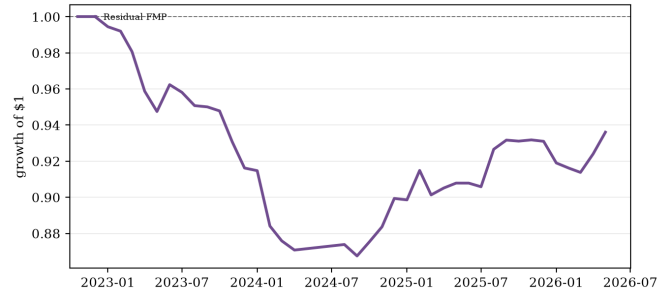


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_growth_yoy_sign_flip	-0.56%	4.58%	-0.12	-0.17	-12.19%	0.204	-0.92%	4.58%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_growth_yoy_sign_flip	0.05	0.01	0.06	0.06	-0.32	0.09	-2.01%	0.226
	(0.73)	(0.11)	(0.72)	(1.22)	(-1.90)	(0.67)	(-0.72)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_growth_yoy_sign_flip	1.02%	-0.07%	-0.35%	0.66%	1.17%	-0.99%	-2.01%	-0.56%

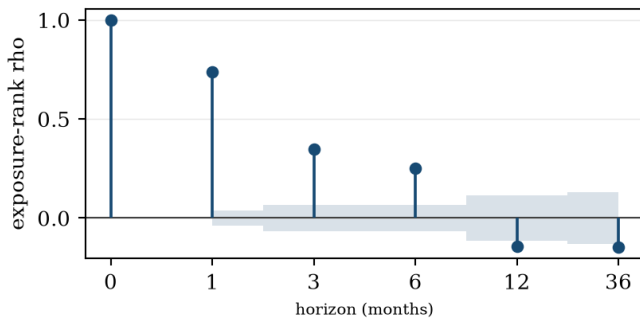


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

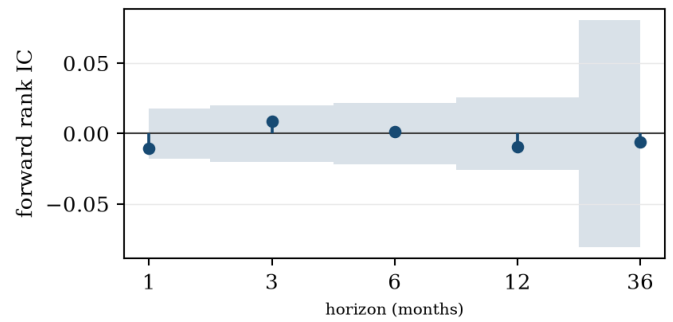


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	net_income_growth_yoy_sign_flip(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2094 / 2574 / 2796
Available-history Sharpe	-0.11
Bootstrap Sharpe	-0.12 [-1.56, 1.14]; Pr(S<=0)=59.2%
Bootstrap tails	skew=0.20 [-0.49, 0.83]; max DD=-12.2% [-21.0%, -3.4%]

77. interest_to_debt

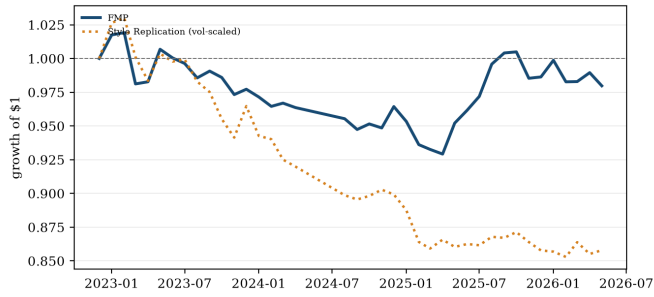


Figure: Common-window cumulative growth of \$1.

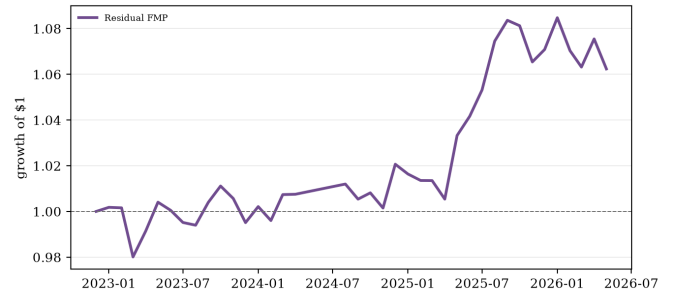


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
interest_to_debt	-0.55%	4.47%	-0.12	-0.24	-8.83%	0.330	2.17%	4.47%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
interest_to_debt	-0.12 (-1.04)	0.20 (2.42)	-0.00 (-0.07)	-0.01 (-0.39)	-0.09 (-0.66)	-0.10 (-0.66)	1.98% (0.89)	0.331

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
interest_to_debt	-2.48%	-1.33%	0.03%	-0.14%	0.32%	1.08%	1.98%	-0.55%

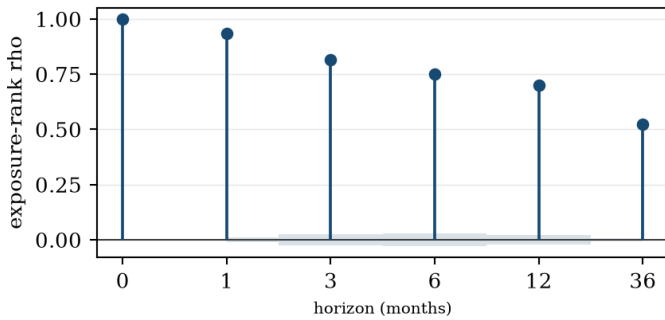


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

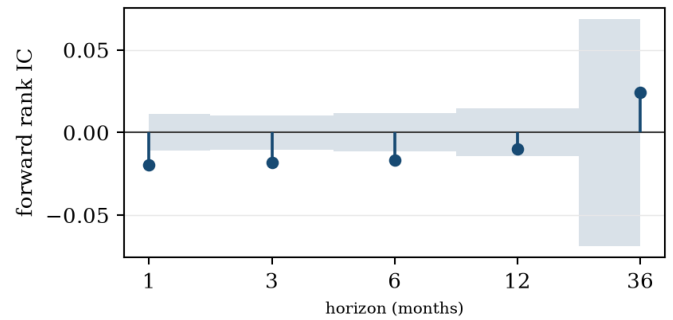


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	interest_to_debt(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1971 / 2170 / 2298
Available-history Sharpe	-0.14
Bootstrap Sharpe	-0.12 [-1.05, 0.76]; Pr(S<=0)=65.0%
Bootstrap tails	skew=-0.12 [-0.65, 0.91]; max DD=-8.8% [-13.0%, -3.9%]

78. net_income_to_market_cap

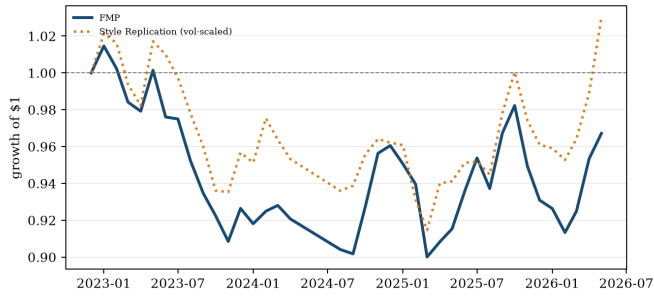


Figure: Common-window cumulative growth of \$1.

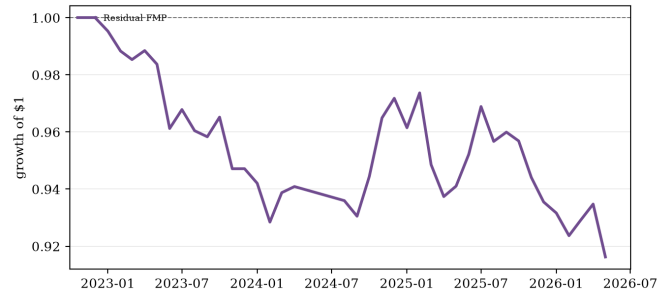


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_to_market_cap	-0.84%	6.62%	-0.13	-0.22	-11.29%	0.647	-1.76%	6.62%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_to_market_cap	-0.18	0.09	-0.05	0.09	-0.15	-0.39	-2.69%	0.654
	(-2.05)	(1.01)	(-0.61)	(1.90)	(-1.41)	(-4.78)	(-0.91)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_to_market_cap	-3.78%	-0.61%	0.31%	0.96%	0.57%	4.38%	-2.69%	-0.84%

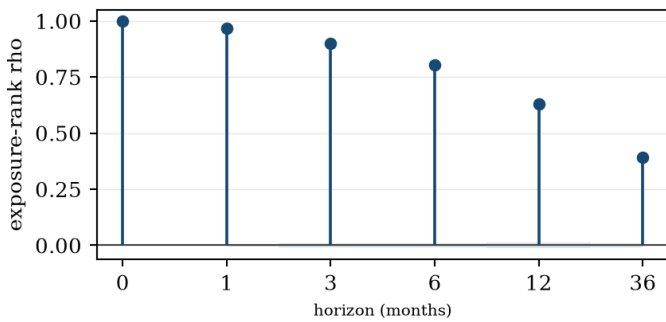


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

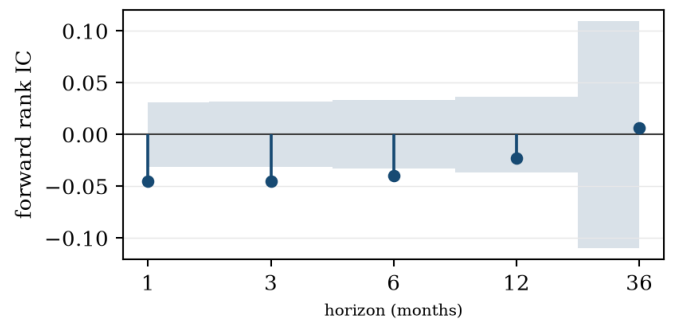


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	net_income_to_market_cap(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2090 / 2544 / 2736
Available-history Sharpe	-0.35
Bootstrap Sharpe	-0.13 [-1.00, 0.61]; Pr(S<=0)=65.2%
Bootstrap tails	skew=0.04 [-0.43, 0.60]; max DD=-11.3% [-20.6%, -6.3%]

79. asset_growth_yoy

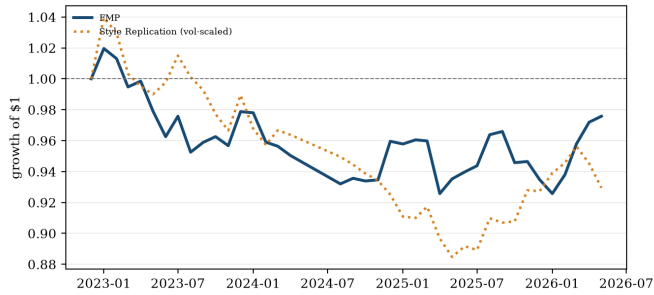


Figure: Common-window cumulative growth of \$1.

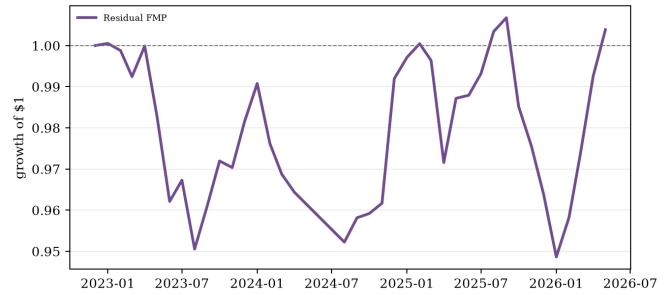


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
asset_growth_yoy	-0.65%	5.01%	-0.13	-0.24	-9.21%	0.222	0.38%	5.01%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
asset_growth_yoy	0.02 (0.26)	0.11 (1.72)	0.13 (1.29)	-0.04 (-0.49)	0.00 (0.00)	-0.06 (-0.55)	0.22% (0.06)	0.223

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
asset_growth_yoy	0.50%	-0.74%	-0.84%	-0.42%	-0.00%	0.63%	0.22%	-0.65%

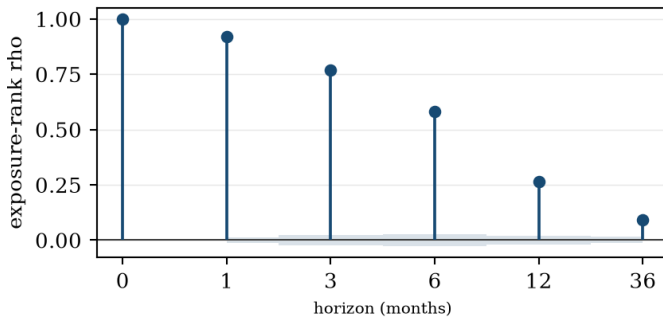


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

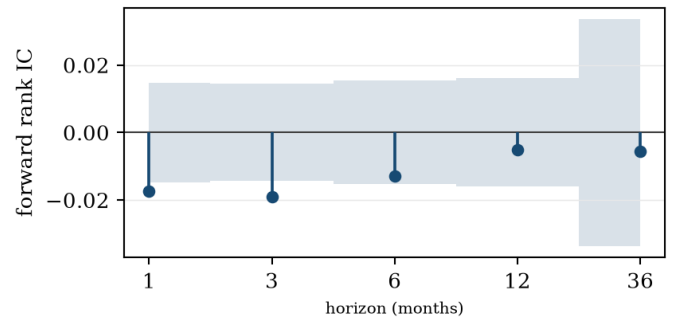


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	asset_growth_yoy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2095 / 2571 / 2791
Available-history Sharpe	0.07
Bootstrap Sharpe	-0.13 [-0.82, 0.58]; Pr(S<=0)=68.5%
Bootstrap tails	skew=-0.22 [-0.68, 0.27]; max DD=-9.2% [-14.4%, -4.2%]

80. working_capital_proxy

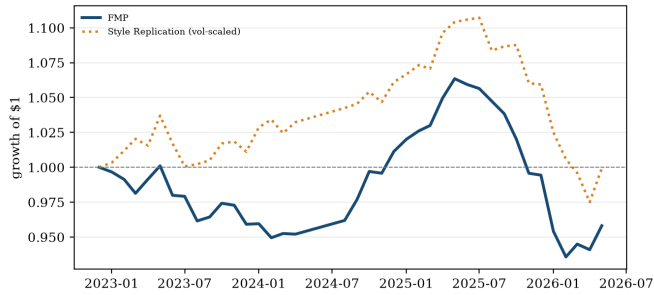


Figure: Common-window cumulative growth of \$1.

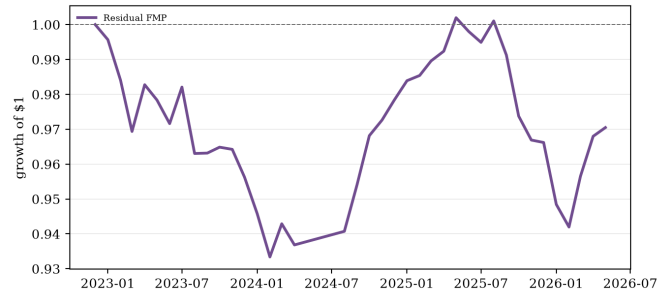


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
working_capital_proxy	-1.24%	4.66%	-0.27	-0.36	-12.01%	0.466	-1.30%	4.66%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
working_capital_proxy	-0.15 (-2.69)	-0.18 (-3.14)	-0.27 (-2.71)	-0.14 (-2.13)	0.06 (0.42)	-0.14 (-1.35)	-0.89% (-0.34)	0.469

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
working_capital_proxy	-3.17%	1.25%	1.67%	-1.43%	-0.24%	1.56%	-0.89%	-1.24%

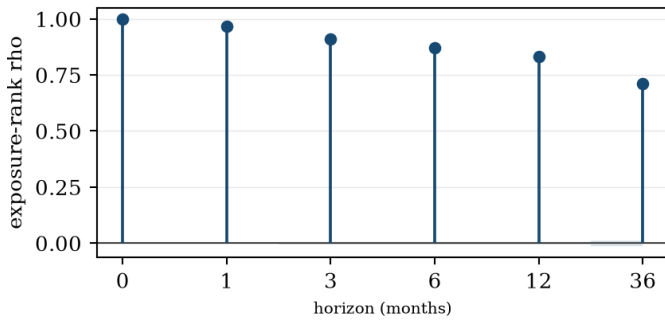


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

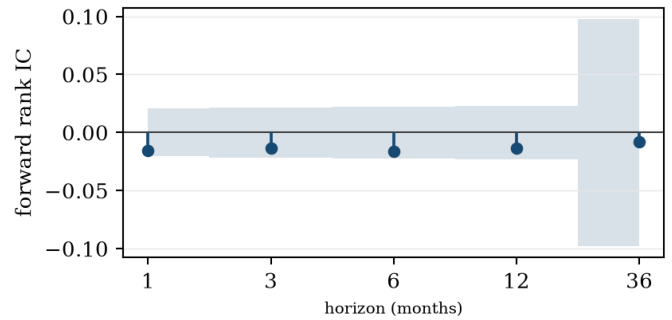


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	working_capital_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2095 / 2579 / 2803
Available-history Sharpe	-0.39
Bootstrap Sharpe	-0.27 [-1.49, 1.42]; Pr(S<=0)=65.1%
Bootstrap tails	skew=-0.66 [-1.03, -0.03]; max DD=-12.0% [-23.3%, -3.7%]

81. lottery_max_5_21d

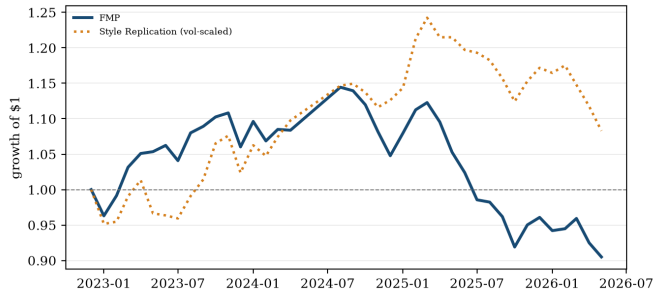


Figure: Common-window cumulative growth of \$1.

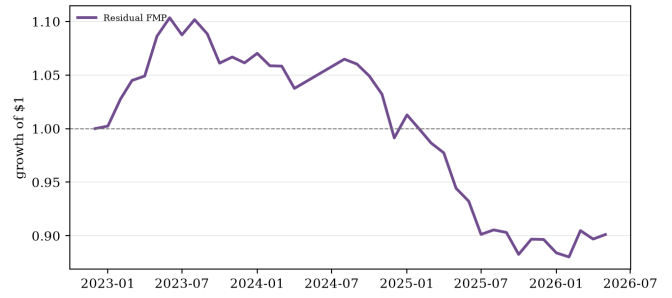


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
lottery_max_5_21d	-2.71%	9.52%	-0.28	-0.51	-20.92%	0.566	-4.93%	9.52%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
lottery_max_5_21d	0.11 (0.95)	-0.41 (-4.57)	-0.07 (-0.66)	-0.10 (-1.54)	0.40 (1.56)	0.23 (1.47)	-3.10% (-0.53)	0.577

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
lottery_max_5_21d	2.34%	2.78%	0.44%	-1.07%	-1.50%	-2.61%	-3.10%	-2.71%

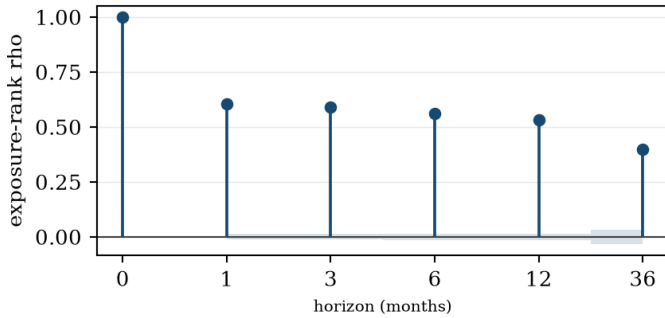


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

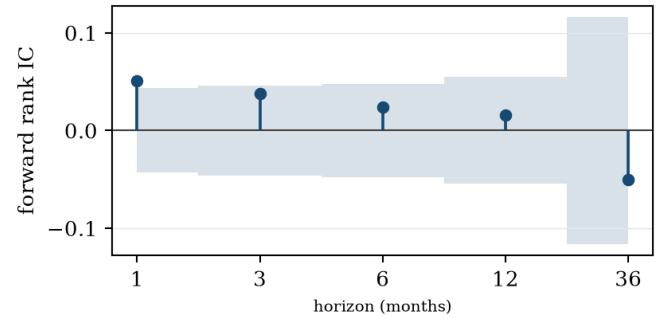


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	lottery_max_5_21d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	0.07
Bootstrap Sharpe	-0.28 [-1.30, 0.65]; Pr(S<=0)=73.1%
Bootstrap tails	skew=0.17 [-0.28, 0.65]; max DD=-20.9% [-35.4%, -8.4%]

82. growth_acc_5y

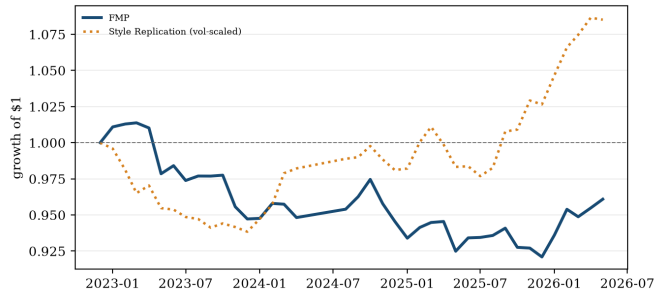


Figure: Common-window cumulative growth of \$1.

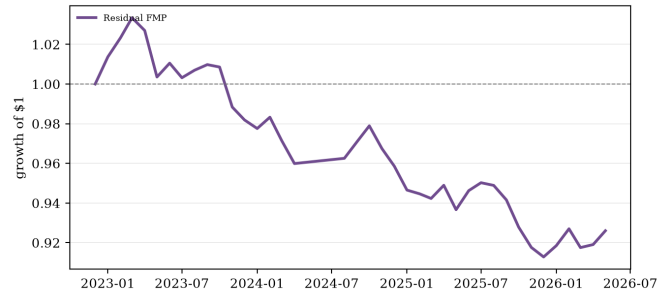


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_acc_5y	-1.19%	3.85%	-0.31	-0.63	-9.16%	0.280	-2.59%	3.85%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_acc_5y	-0.04 (-0.49)	-0.09 (-1.69)	0.12 (1.85)	0.06 (0.90)	0.17 (1.92)	-0.20 (-2.16)	-2.37% (-1.15)	0.282

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_acc_5y	-0.86%	0.59%	-0.75%	0.59%	-0.62%	2.24%	-2.37%	-1.19%

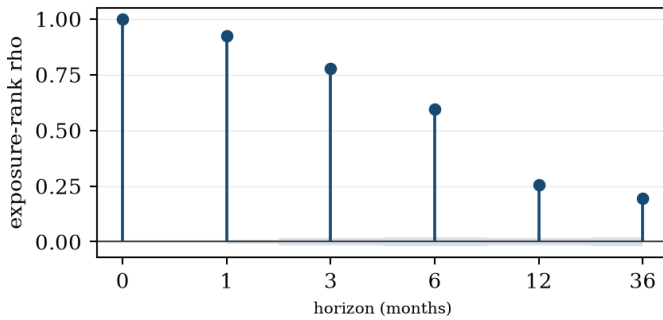


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

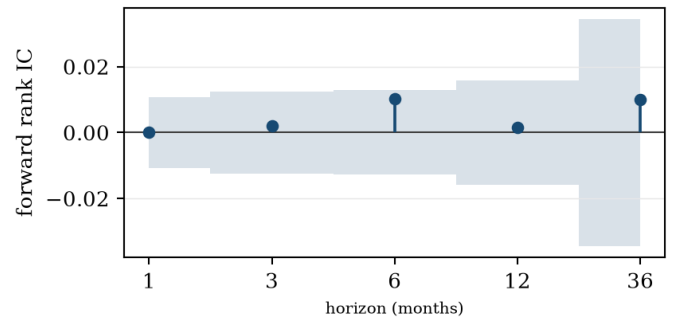


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	growth_acc_5y(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1894 / 2121 / 2499
Available-history Sharpe	-0.29
Bootstrap Sharpe	-0.31 [-1.07, 0.50]; Pr(S<=0)=79.1%
Bootstrap tails	skew=-0.71 [-1.22, -0.21]; max DD=-9.2% [-13.8%, -3.9%]

83. cash_earnings_ttm

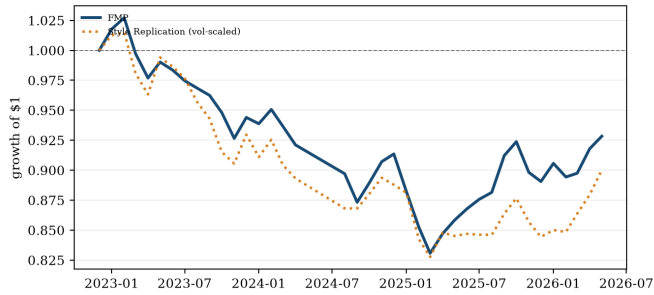


Figure: Common-window cumulative growth of \$1.

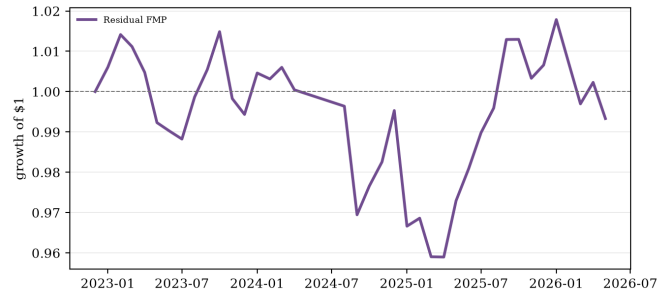


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
cash_earnings_ttm	-2.14%	6.49%	-0.33	-0.58	-19.13%	0.677	0.42%	6.49%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
cash_earnings_ttm	-0.26 (-3.53)	0.32 (5.98)	-0.03 (-0.49)	0.15 (3.96)	-0.19 (-2.20)	-0.28 (-3.06)	-0.15% (-0.06)	0.679

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
cash_earnings_ttm	-5.38%	-2.16%	0.16%	1.57%	0.70%	3.12%	-0.15%	-2.14%

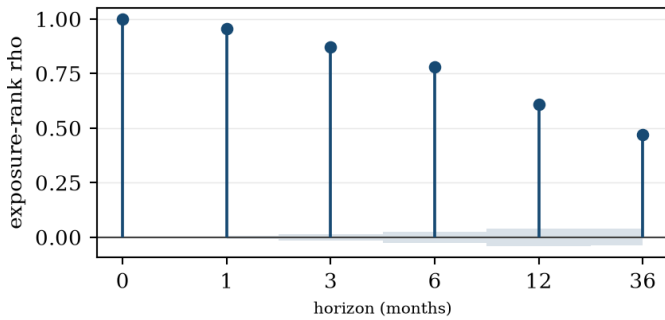


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

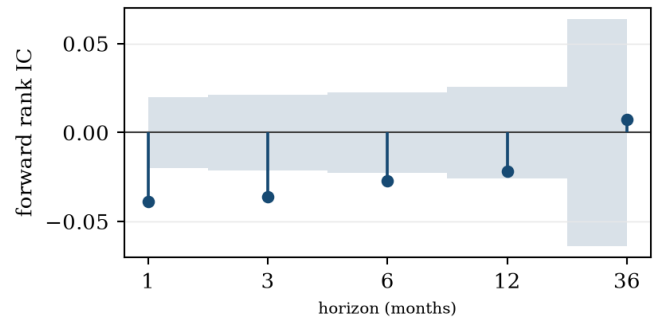


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	cash_earnings_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2059 / 2484 / 2672
Available-history Sharpe	-0.38
Bootstrap Sharpe	-0.33 [-1.35, 0.67]; Pr(S<=0)=76.8%
Bootstrap tails	skew=-0.16 [-0.61, 0.32]; max DD=-19.1% [-25.6%, -7.1%]

84. gpoa

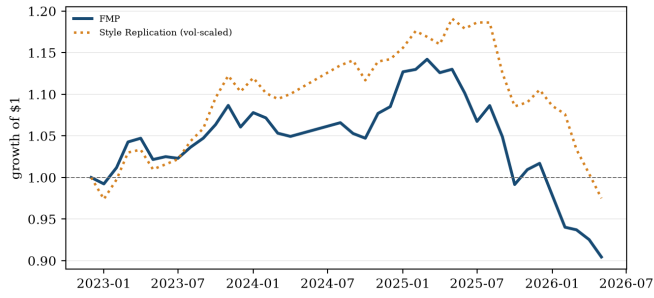


Figure: Common-window cumulative growth of \$1.

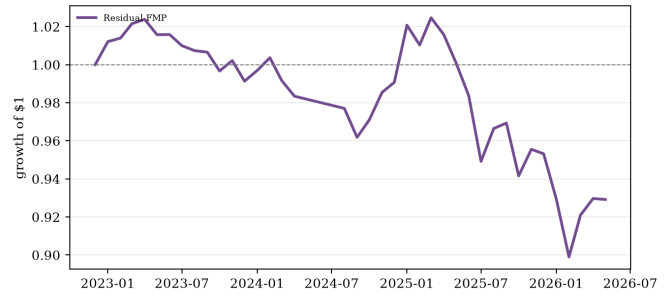


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gpoa	-2.89%	7.47%	-0.39	-0.62	-20.80%	0.525	-2.51%	7.47%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gpoa	0.26 (3.86)	-0.01 (-0.11)	-0.33 (-3.16)	-0.18 (-1.98)	0.30 (1.23)	0.47 (5.00)	-2.19% (-0.67)	0.526

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gpoa	5.46%	0.07%	2.06%	-1.91%	-1.11%	-5.27%	-2.19%	-2.89%

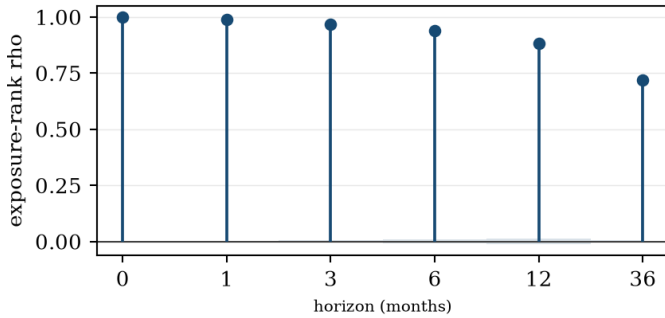


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

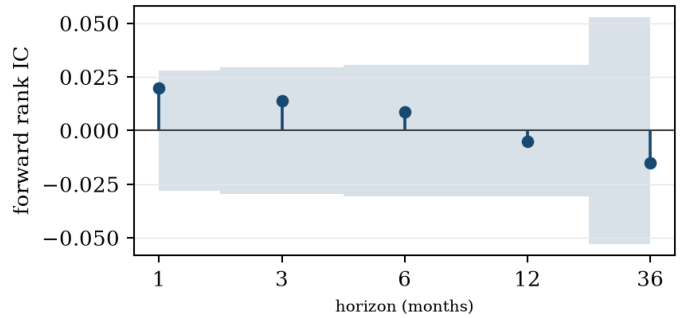


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	gpoa(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2081 / 2514 / 2675
Available-history Sharpe	0.16
Bootstrap Sharpe	-0.39 [-1.42, 0.88]; Pr(S<=0)=73.8%
Bootstrap tails	skew=-0.39 [-0.69, 0.14]; max DD=-20.8% [-34.6%, -3.8%]

85. debt_to_assets

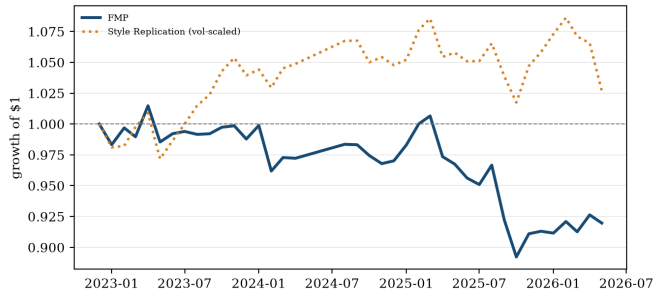


Figure: Common-window cumulative growth of \$1.

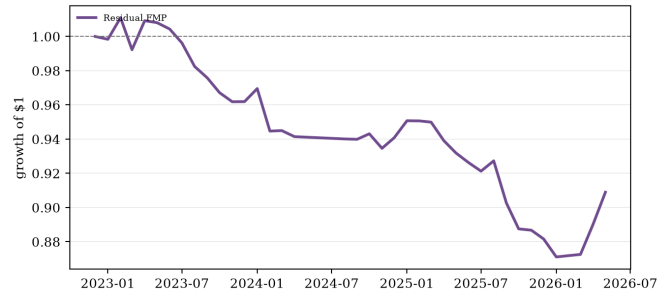


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_to_assets	-2.48%	5.72%	-0.43	-1.04	-12.08%	0.574	-3.24%	5.72%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_to_assets	0.26 (3.59)	-0.01 (-0.10)	0.08 (1.01)	-0.02 (-0.25)	0.20 (1.52)	0.32 (2.37)	-2.94% (-1.38)	0.575

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_to_assets	5.45%	0.07%	-0.52%	-0.19%	-0.75%	-3.60%	-2.94%	-2.48%

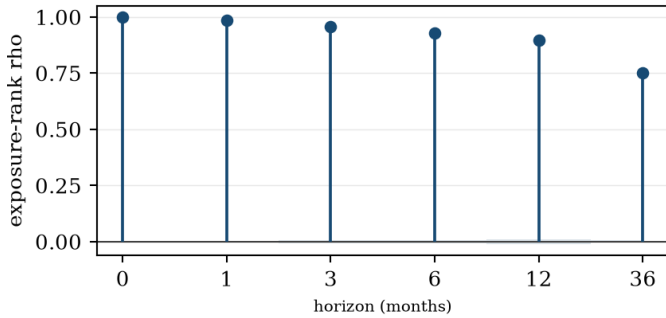


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

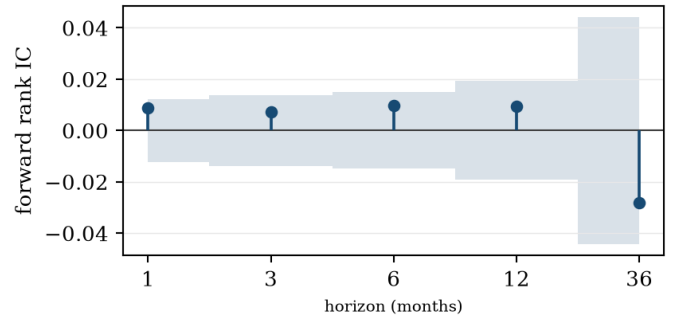


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	debt_to_assets(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2063 / 2532 / 2731
Available-history Sharpe	-0.12
Bootstrap Sharpe	-0.43 [-0.94, 0.15]; Pr(S<=0)=94.0%
Bootstrap tails	skew=-0.88 [-1.26, -0.45]; max DD=-12.1% [-20.5%, -4.7%]

86. skew_63d

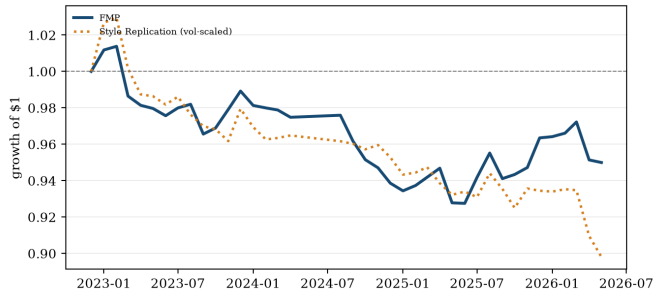


Figure: Common-window cumulative growth of \$1.

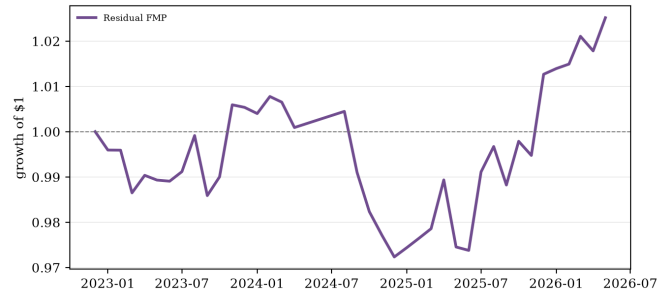


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
skew_63d	-1.56%	3.54%	-0.44	-0.91	-8.51%	0.400	0.55%	3.54%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
skew_63d	-0.06 (-0.86)	0.11 (2.72)	-0.03 (-0.78)	-0.11 (-4.42)	0.19 (1.81)	-0.12 (-1.28)	0.82% (0.39)	0.403

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
skew_63d	-1.28%	-0.77%	0.21%	-1.17%	-0.70%	1.33%	0.82%	-1.56%

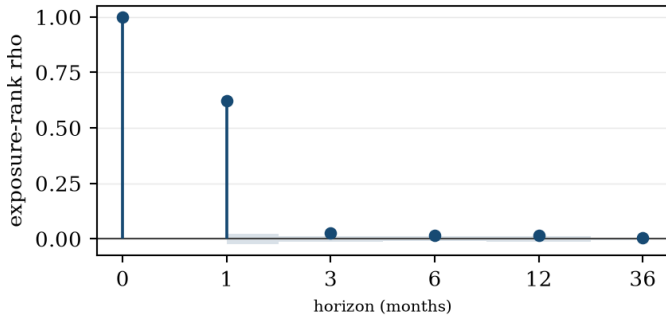


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

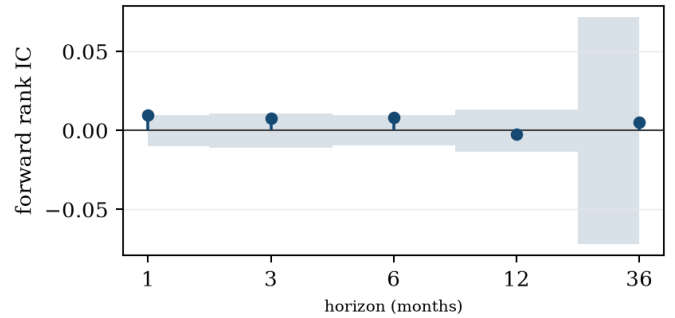


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	skew_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2811
Available-history Sharpe	-0.01
Bootstrap Sharpe	-0.44 [-1.32, 0.34]; Pr(S<=0)=87.5%
Bootstrap tails	skew=-0.54 [-0.93, -0.01]; max DD=-8.5% [-13.3%, -4.0%]

87. gp_to_equity

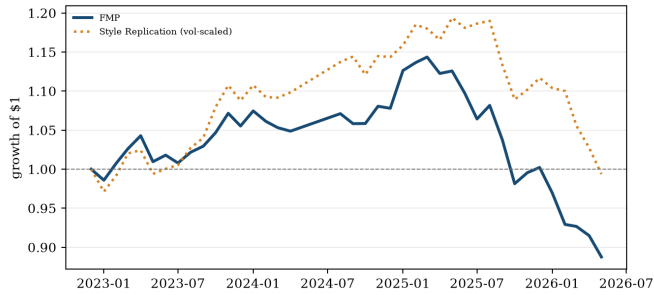


Figure: Common-window cumulative growth of \$1.

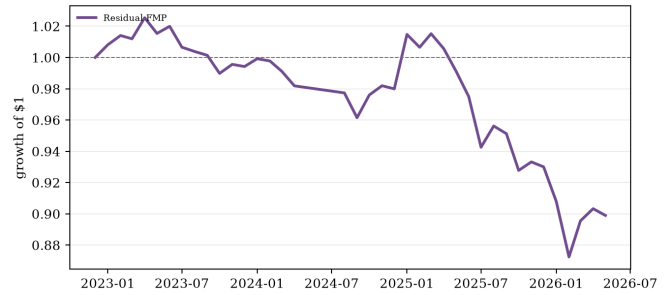


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gp_to_equity	-3.47%	7.65%	-0.45	-0.73	-22.38%	0.554	-3.54%	7.65%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gp_to_equity	0.28 (4.38)	-0.02 (-0.30)	-0.32 (-3.06)	-0.17 (-1.99)	0.42 (1.90)	0.43 (5.55)	-3.23% (-1.02)	0.555

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gp_to_equity	5.82%	0.16%	1.99%	-1.83%	-1.57%	-4.81%	-3.23%	-3.47%

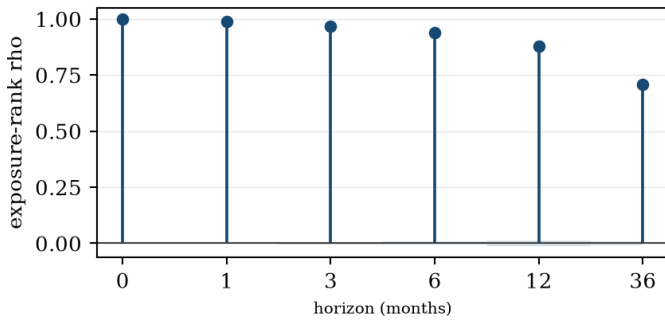


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

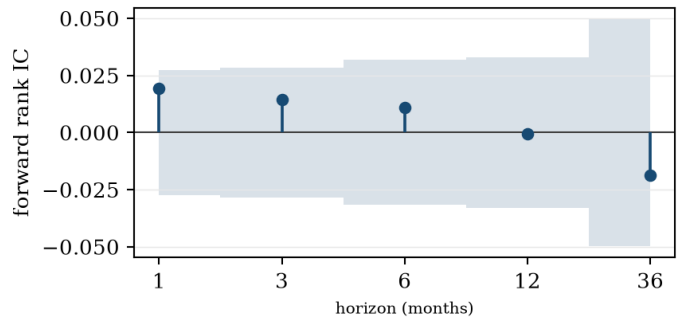


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	gp_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1965 / 2360 / 2472
Available-history Sharpe	0.03
Bootstrap Sharpe	-0.45 [-1.62, 0.89]; Pr(S<=0)=75.8%
Bootstrap tails	skew=-0.34 [-0.69, 0.23]; max DD=-22.4% [-37.2%, -4.0%]

88. equity_issuance_yoy

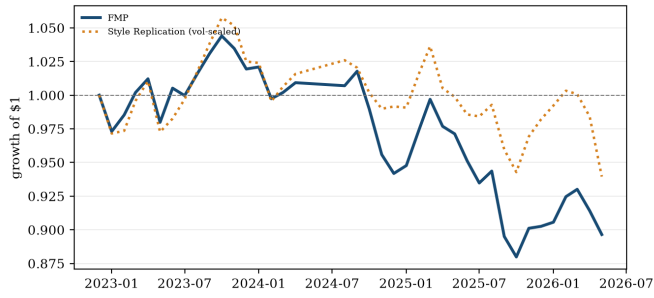


Figure: Common-window cumulative growth of \$1.

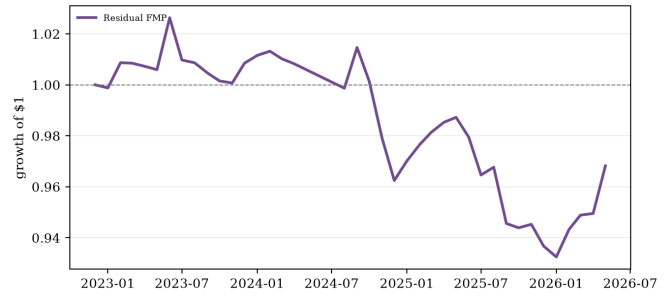


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
equity_issuance_yoy	-3.23%	6.70%	-0.48	-0.90	-15.73%	0.720	-1.74%	6.70%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
equity_issuance_yoy	0.19 (3.02)	-0.05 (-0.52)	0.19 (2.96)	0.00 (0.01)	0.00 (0.02)	0.48 (4.09)	-0.96% (-0.39)	0.725

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
equity_issuance_yoy	4.02%	0.32%	-1.16%	0.00%	-0.01%	-5.44%	-0.96%	-3.23%

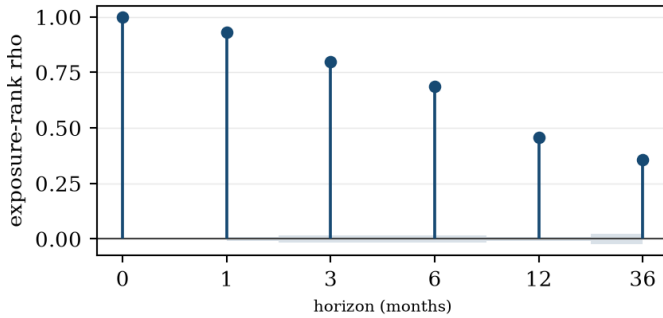


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

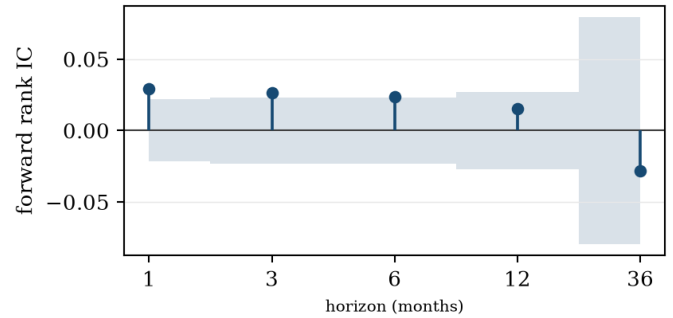


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	equity_issuance_yoy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2095 / 2576 / 2796
Available-history Sharpe	-0.09
Bootstrap Sharpe	-0.48 [-1.20, 0.23]; Pr(S<=0)=91.5%
Bootstrap tails	skew=-0.40 [-0.75, 0.02]; max DD=-15.7% [-25.4%, -6.9%]

89. rsi_14d

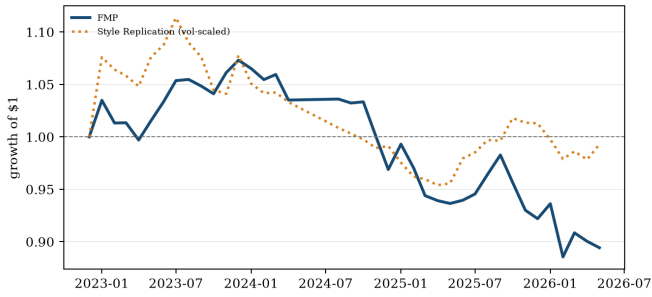


Figure: Common-window cumulative growth of \$1.

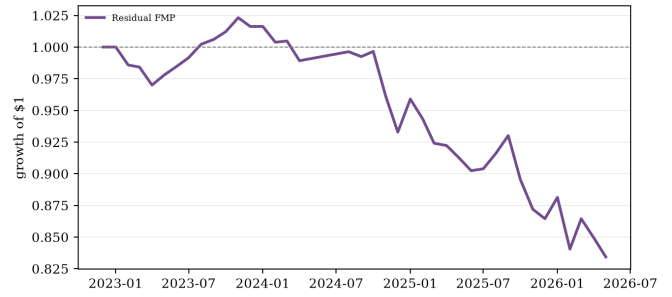


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
rsi_14d	-3.30%	6.80%	-0.49	-1.01	-17.48%	0.193	-3.29%	6.80%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
rsi_14d	0.14 (1.58)	0.05 (0.41)	0.06 (0.53)	-0.09 (-1.67)	-0.18 (-0.83)	-0.03 (-0.23)	-5.53% (-1.41)	0.232

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
rsi_14d	2.88%	-0.34%	-0.37%	-0.97%	0.68%	0.37%	-5.53%	-3.30%

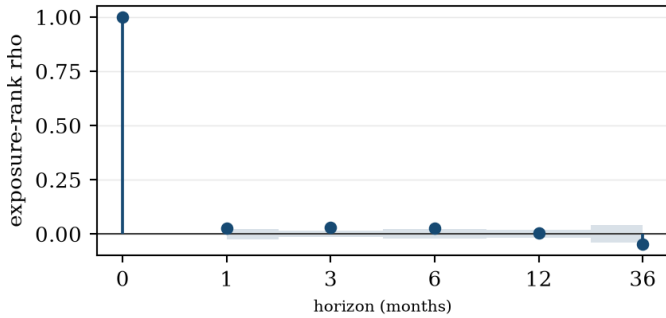


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

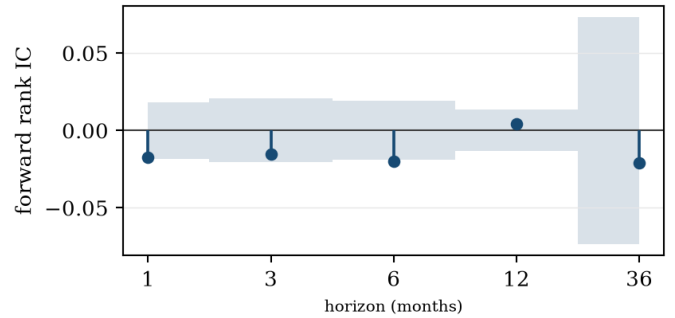


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	rsi_14d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2105 / 2584 / 2810
Available-history Sharpe	-0.01
Bootstrap Sharpe	-0.49 [-1.30, 0.34]; Pr(S<=0)=87.8%
Bootstrap tails	skew=-0.32 [-0.72, 0.22]; max DD=-17.5% [-25.8%, -6.7%]

90. ncav_to_mc

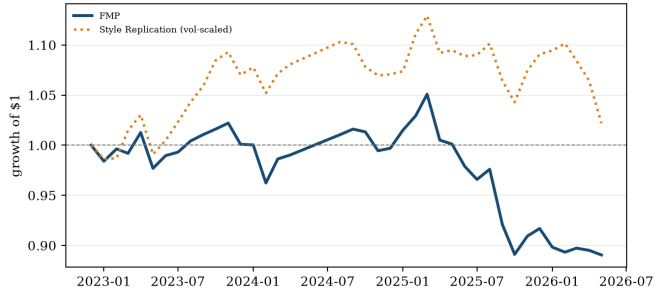


Figure: Common-window cumulative growth of \$1.

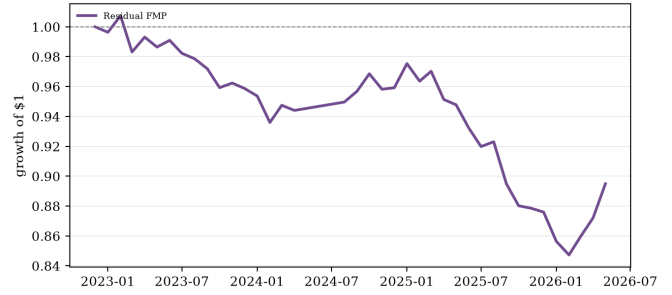


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ncav_to_mc	-3.42%	6.84%	-0.50	-1.03	-15.26%	0.575	-4.12%	6.84%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ncav_to_mc	0.28 (2.94)	-0.12 (-1.04)	0.09 (0.89)	-0.10 (-1.32)	0.13 (0.71)	0.40 (2.49)	-3.41% (-1.38)	0.579

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ncav_to_mc	5.78%	0.79%	-0.56%	-1.06%	-0.48%	-4.49%	-3.41%	-3.42%

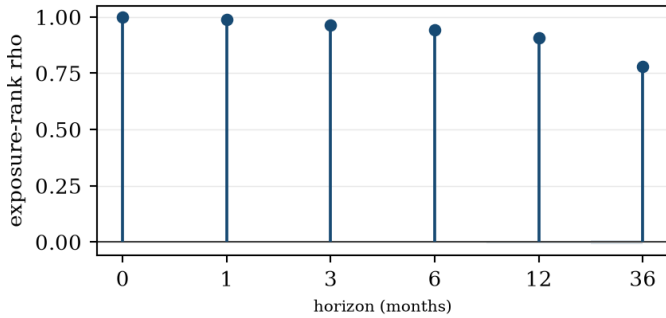


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

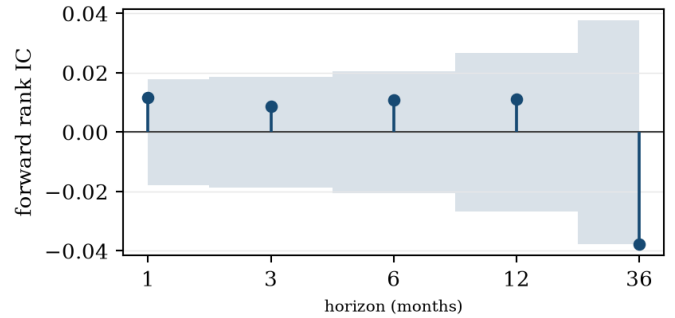


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	ncav_to_mc(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2100 / 2583 / 2811
Available-history Sharpe	-0.19
Bootstrap Sharpe	-0.50 [-1.22, 0.33]; Pr(S<=0)=89.0%
Bootstrap tails	skew=-0.91 [-1.19, -0.59]; max DD=-15.3% [-28.9%, -5.8%]

91. debt_to_equity

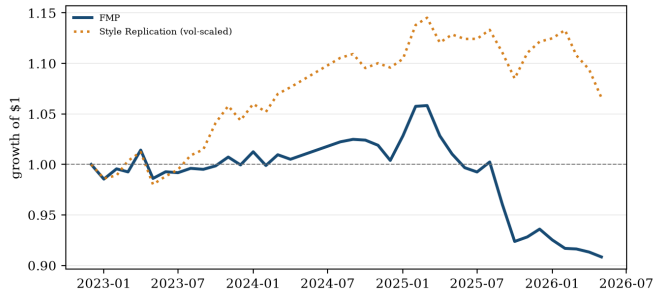


Figure: Common-window cumulative growth of \$1.

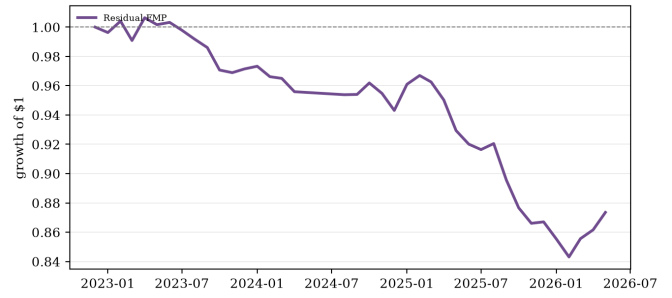


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_to_equity	-2.89%	5.39%	-0.54	-1.04	-14.16%	0.529	-4.46%	5.39%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_to_equity	0.21 (3.37)	-0.11 (-1.18)	-0.06 (-0.57)	-0.10 (-1.59)	0.35 (2.49)	0.17 (1.40)	-4.20% (-1.99)	0.530

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_to_equity	4.47%	0.73%	0.34%	-1.01%	-1.29%	-1.94%	-4.20%	-2.89%

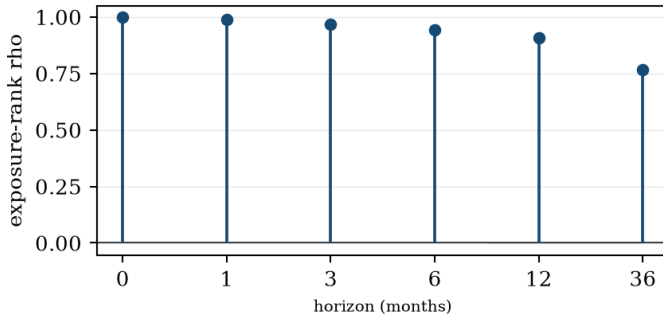


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

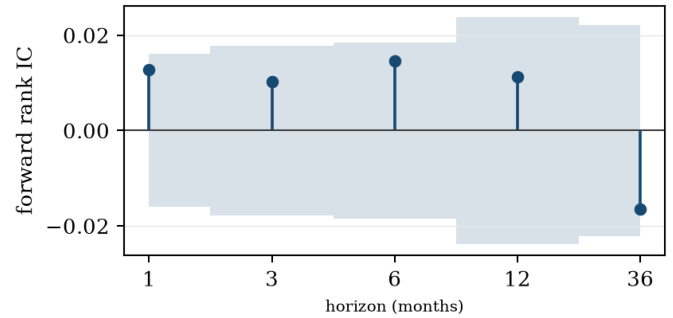


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	debt_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1982 / 2417 / 2593
Available-history Sharpe	-0.15
Bootstrap Sharpe	-0.54 [-1.49, 0.48]; Pr(S<=0)=85.0%
Bootstrap tails	skew=-0.51 [-1.10, 0.19]; max DD=-14.2% [-26.1%, -2.8%]

92. capex_to_revenue

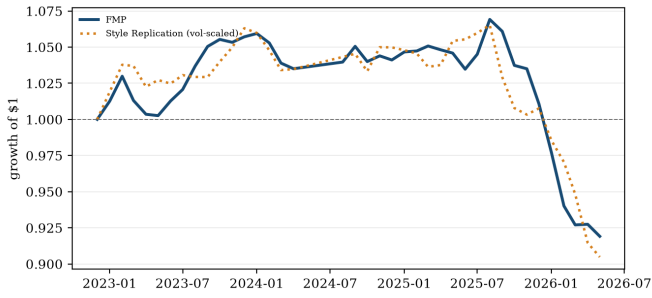


Figure: Common-window cumulative growth of \$1.

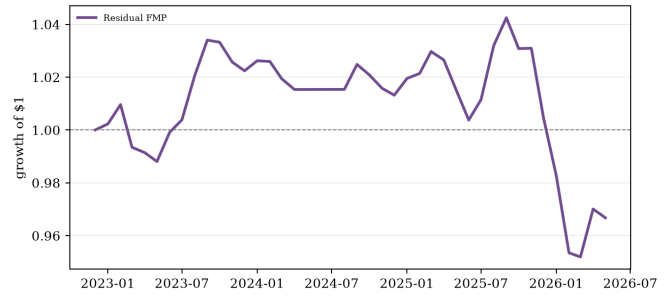


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
capex_to_revenue	-2.56%	4.58%	-0.56	-0.73	-14.04%	0.284	-0.93%	4.58%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
capex_to_revenue	0.06 (0.65)	0.09 (0.95)	-0.22 (-2.54)	-0.17 (-2.33)	0.15 (0.90)	0.10 (0.73)	-0.99% (-0.25)	0.284

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
capex_to_revenue	1.15%	-0.62%	1.36%	-1.81%	-0.55%	-1.10%	-0.99%	-2.56%

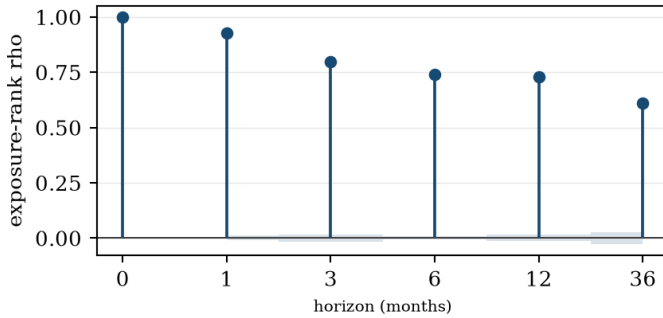


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

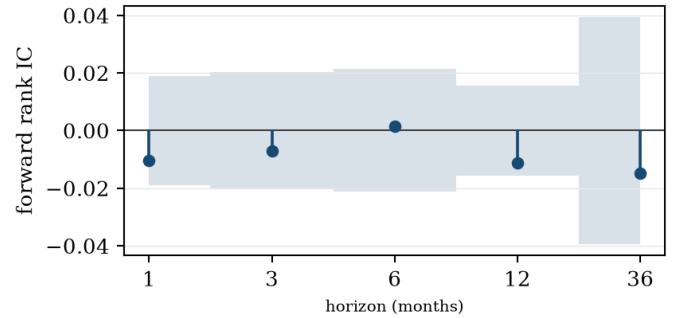


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	capex_to_revenue(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2040 / 2434 / 2619
Available-history Sharpe	-0.18
Bootstrap Sharpe	-0.56 [-1.61, 0.86]; Pr(S<=0)=78.8%
Bootstrap tails	skew=-0.71 [-1.17, 0.23]; max DD=-14.0% [-25.5%, -2.6%]

93. momentum_0_21

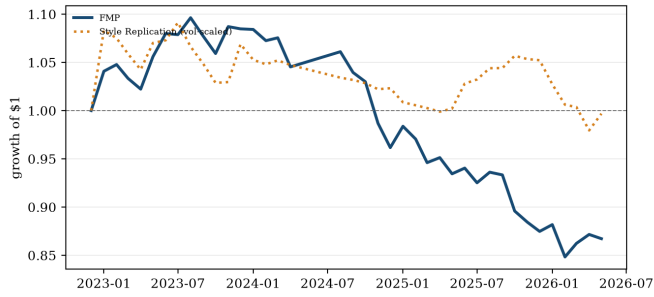


Figure: Common-window cumulative growth of \$1.

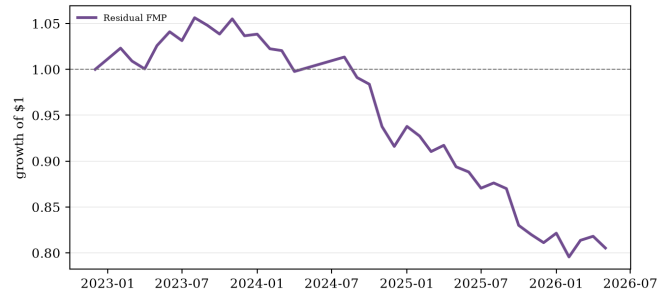


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_0_21	-4.26%	6.90%	-0.62	-1.20	-22.62%	0.109	-4.30%	6.90%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_0_21	0.12 (1.08)	0.01 (0.08)	-0.01 (-0.07)	-0.11 (-1.10)	-0.03 (-0.11)	-0.08 (-0.68)	-6.62% (-1.33)	0.151

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_0_21	2.45%	-0.09%	0.08%	-1.12%	0.11%	0.95%	-6.62%	-4.26%

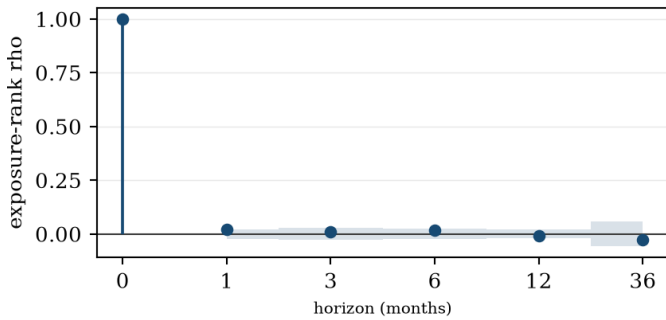


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

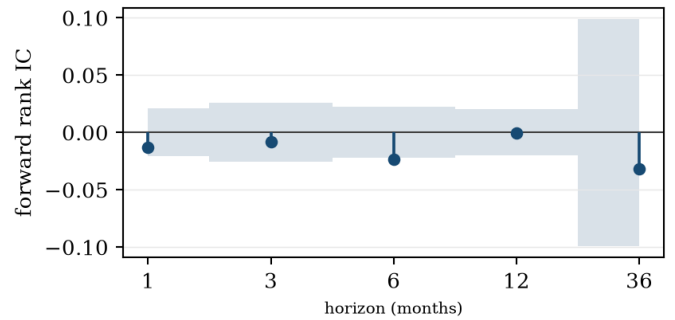


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	momentum_0_21(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	-0.18
Bootstrap Sharpe	-0.62 [-1.77, 0.46]; Pr(S<=0)=86.9%
Bootstrap tails	skew=0.12 [-0.17, 0.37]; max DD=-22.6% [-31.5%, -6.8%]

94. cash_ratio

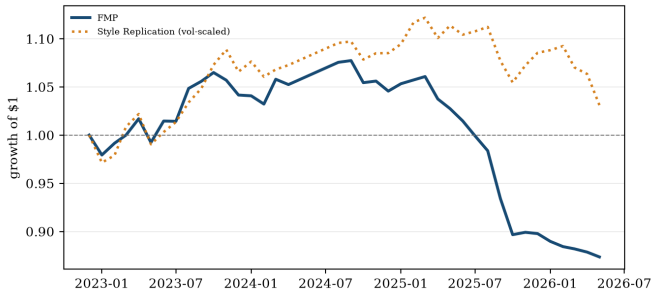


Figure: Common-window cumulative growth of \$1.

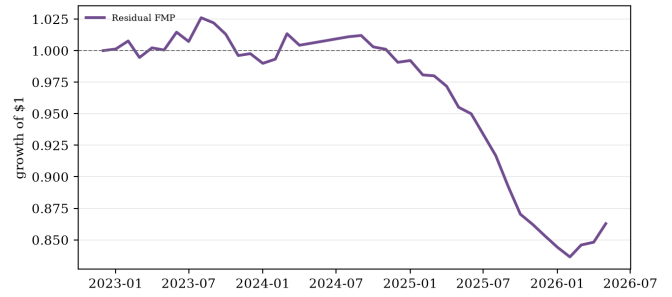


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
cash_ratio	-4.09%	5.83%	-0.70	-1.02	-18.91%	0.541	-4.93%	5.83%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
cash_ratio	0.26 (3.44)	-0.05 (-0.64)	-0.03 (-0.32)	-0.04 (-0.63)	0.12 (0.85)	0.42 (3.03)	-4.57% (-1.65)	0.542

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
cash_ratio	5.44%	0.37%	0.20%	-0.38%	-0.46%	-4.69%	-4.57%	-4.09%

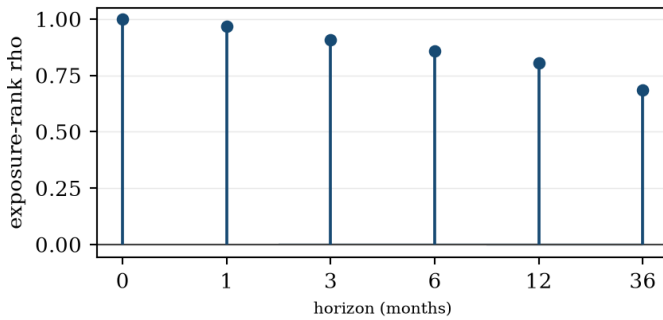


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

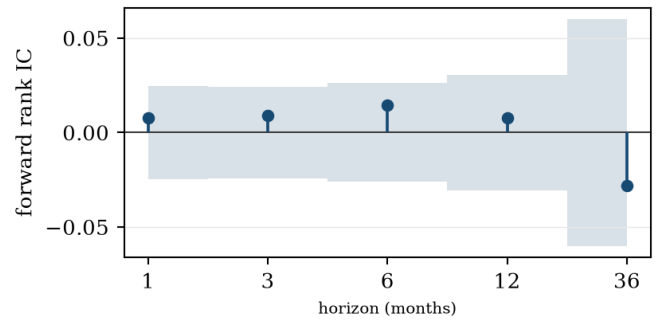


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	cash_ratio(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2091 / 2576 / 2803
Available-history Sharpe	-0.22
Bootstrap Sharpe	-0.70 [-2.24, 0.68]; Pr(S<=0)=82.0%
Bootstrap tails	skew=-0.35 [-1.00, 0.57]; max DD=-18.9% [-34.4%, -3.8%]

95. price_zscore_63d

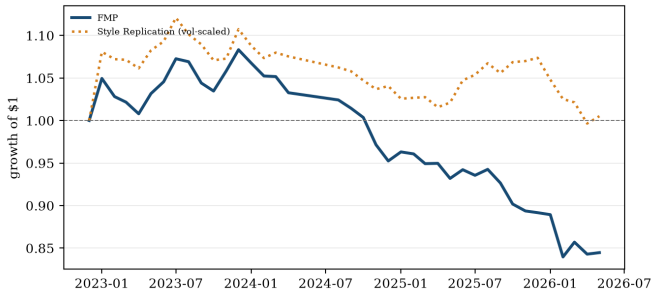


Figure: Common-window cumulative growth of \$1.

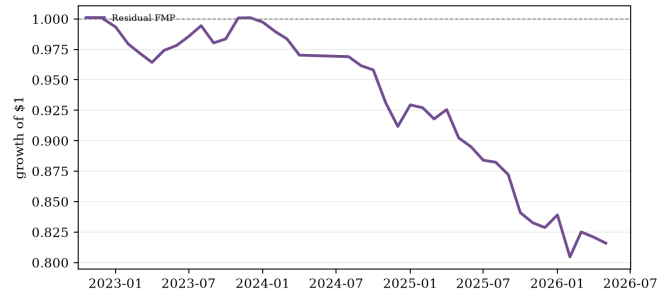


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
price_zscore_63d	-5.10%	6.67%	-0.77	-1.57	-22.49%	0.469	-5.36%	6.67%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
price_zscore_63d	0.16 (1.67)	0.02 (0.21)	0.03 (0.24)	-0.27 (-4.95)	-0.18 (-0.82)	-0.04 (-0.41)	-6.29% (-2.07)	0.476

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
price_zscore_63d	3.23%	-0.14%	-0.17%	-2.85%	0.67%	0.45%	-6.29%	-5.10%

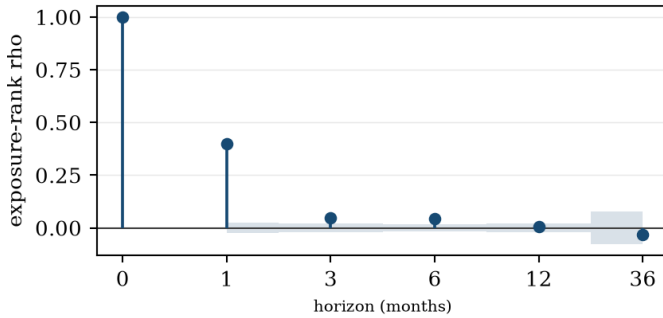


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

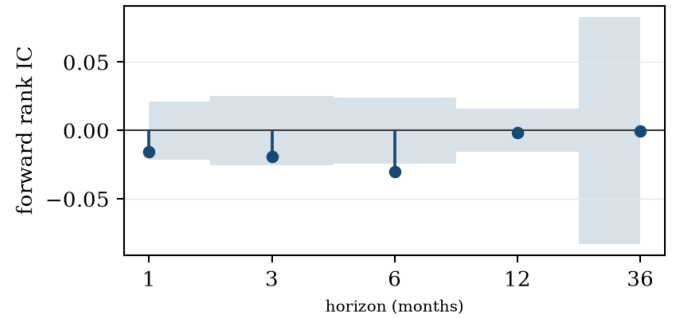


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	price_zscore_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2811
Available-history Sharpe	-0.26
Bootstrap Sharpe	-0.77 [-2.08, 0.15]; Pr(S<=0)=95.3%
Bootstrap tails	skew=0.31 [-0.60, 1.12]; max DD=-22.5% [-30.8%, -8.3%]

96. momentum_21_756

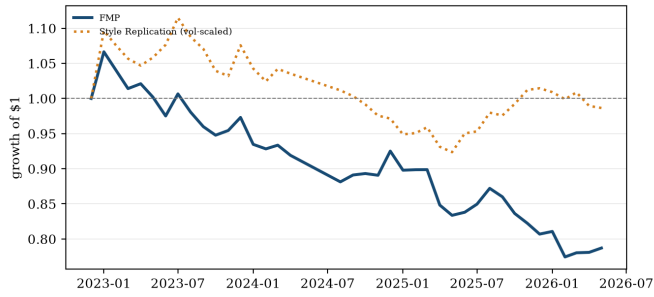


Figure: Common-window cumulative growth of \$1.

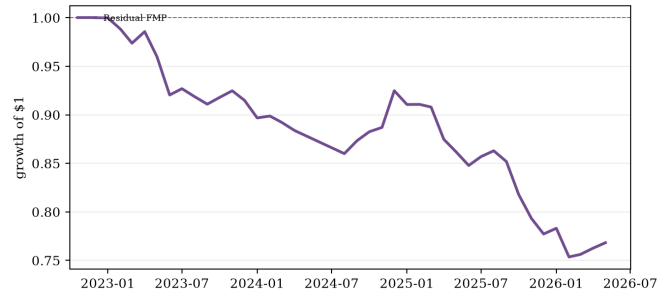


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_756	-7.18%	8.50%	-0.85	-1.99	-27.38%	0.459	-7.13%	8.50%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_756	0.17 (1.72)	0.10 (0.93)	0.25 (1.79)	-0.23 (-3.28)	-0.21 (-1.04)	-0.10 (-1.03)	-8.11% (-1.64)	0.464

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_756	3.57%	-0.66%	-1.53%	-2.38%	0.80%	1.13%	-8.11%	-7.18%

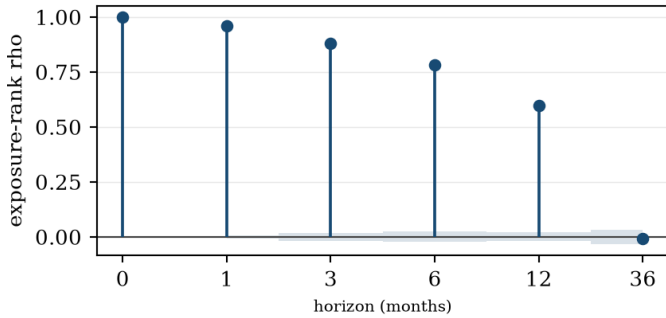


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

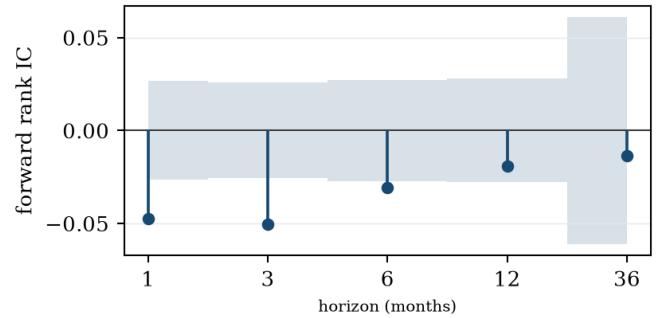


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	momentum_21_756(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	-0.68
Bootstrap Sharpe	-0.85 [-1.46, -0.29]; Pr(S<=0)=99.7%
Bootstrap tails	skew=0.53 [-0.33, 1.14]; max DD=-27.4% [-32.0%, -15.4%]

97. ma_dist_63d

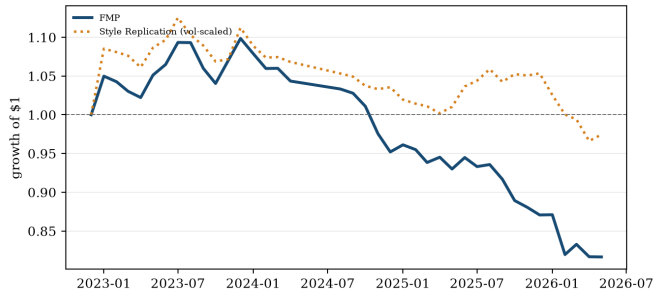


Figure: Common-window cumulative growth of \$1.

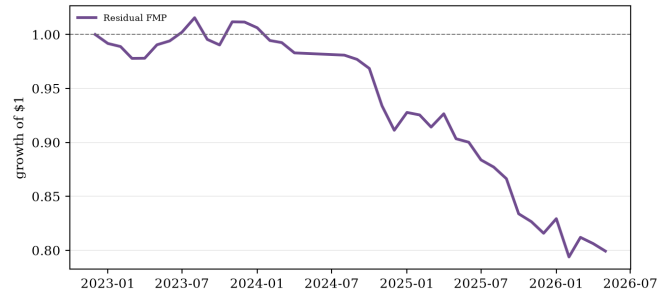


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ma_dist_63d	-6.13%	7.22%	-0.85	-1.60	-25.66%	0.449	-5.75%	7.22%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ma_dist_63d	0.16 (1.44)	0.08 (0.69)	-0.02 (-0.18)	-0.28 (-5.13)	-0.16 (-0.73)	-0.03 (-0.27)	-6.92% (-1.93)	0.459

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ma_dist_63d	3.24%	-0.53%	0.13%	-2.99%	0.60%	0.32%	-6.92%	-6.13%

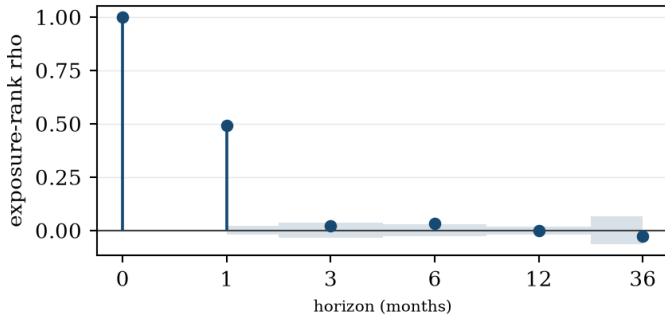


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

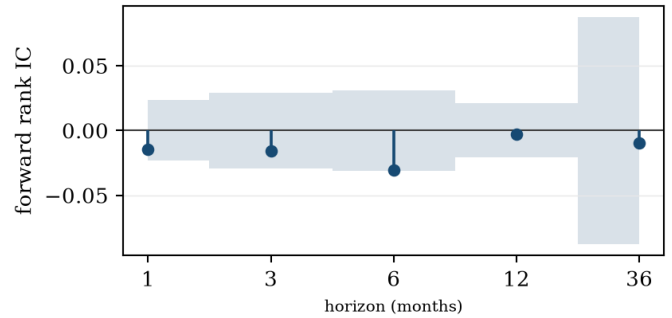


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	ma_dist_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2106 / 2585 / 2812
Available-history Sharpe	-0.28
Bootstrap Sharpe	-0.85 [-2.20, 0.14]; Pr(S<=0)=95.5%
Bootstrap tails	skew=0.30 [-0.44, 0.96]; max DD=-25.7% [-34.9%, -9.6%]

98. current_ratio

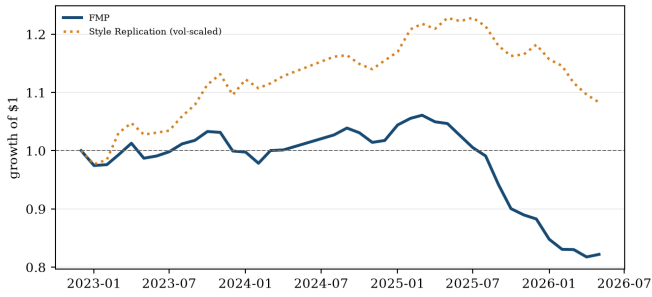


Figure: Common-window cumulative growth of \$1.

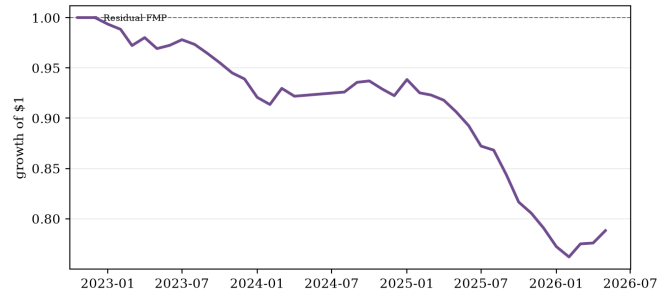


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
current_ratio	-5.96%	6.57%	-0.91	-1.23	-22.93%	0.573	-8.03%	6.57%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
current_ratio	0.21 (2.49)	-0.29 (-3.39)	-0.16 (-1.33)	-0.15 (-2.25)	0.08 (0.46)	0.36 (2.77)	-7.41% (-2.72)	0.576

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
current_ratio	4.36%	1.99%	0.98%	-1.55%	-0.31%	-4.03%	-7.41%	-5.96%

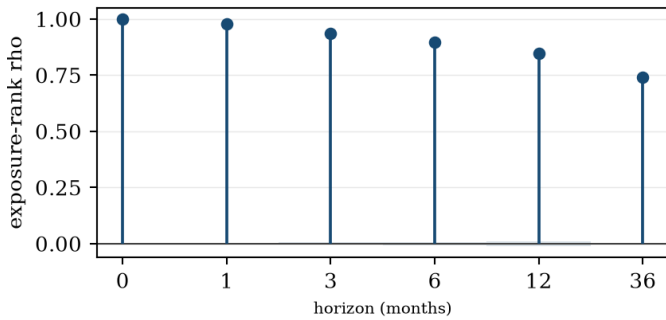


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

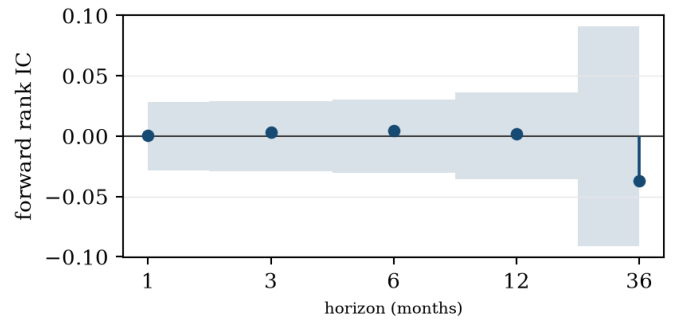


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	current_ratio(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	2098 / 2578 / 2806
Available-history Sharpe	-0.62
Bootstrap Sharpe	-0.91 [-2.51, 0.49]; Pr(S<=0)=86.6%
Bootstrap tails	skew=-0.44 [-0.73, -0.09]; max DD=-22.9% [-42.1%, -5.3%]

99. is_negative_invested_capital

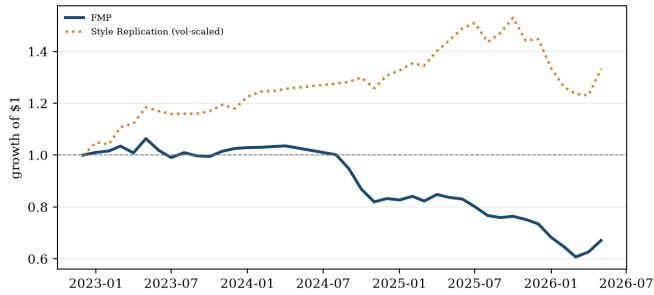


Figure: Common-window cumulative growth of \$1.

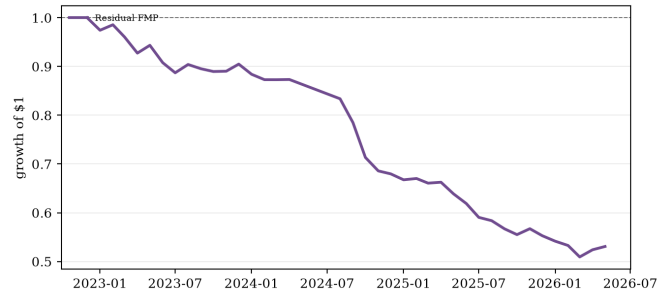


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
is_negative_invested_capital	-11.88%	11.66%	-1.02	-1.56	-42.91%	0.454	-18.46%	11.66%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
is_negative_invested_capital	0.06	-0.62	-0.32	-0.31	-0.38	-0.17	-19.47%	0.458
	(0.25)	(-3.48)	(-1.50)	(-2.61)	(-1.22)	(-0.72)	(-2.54)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
is_negative_invested_capital	1.27%	4.22%	1.98%	-3.23%	1.40%	1.96%	-19.47%	-11.88%

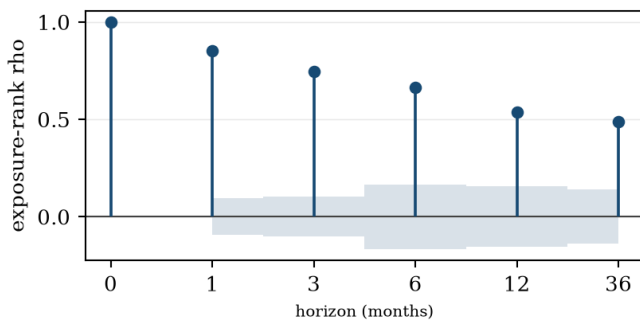


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

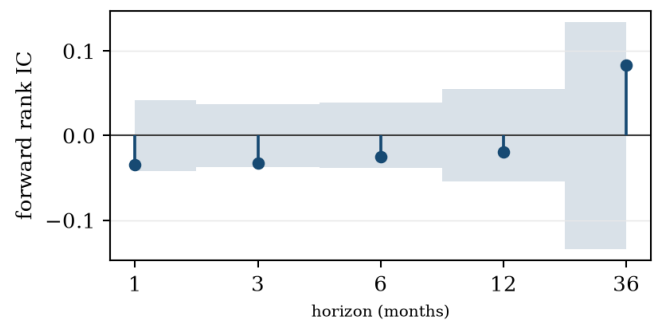


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	is_negative_invested_capital(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	40 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	2 / 2
Stocks, min / median / max	22 / 840 / 2048
Available-history Sharpe	-1.04
Bootstrap Sharpe	-1.02 [-2.15, 0.01]; Pr(S<=0)=97.4%
Bootstrap tails	skew=-0.04 [-0.86, 0.42]; max DD=-42.9% [-55.3%, -18.2%]

100. value_proxy_1260d

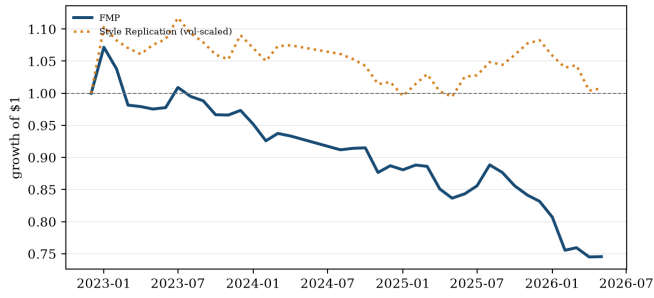


Figure: Common-window cumulative growth of \$1.

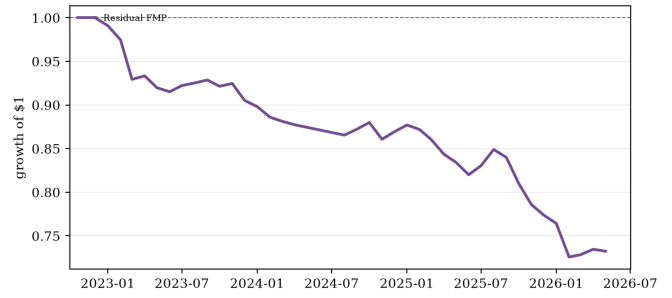


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
value_proxy_1260d	-8.88%	8.62%	-1.03	-2.18	-30.47%	0.602	-9.34%	8.62%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
value_proxy_1260d	0.12 (1.04)	-0.08 (-0.61)	0.17 (1.63)	-0.40 (-6.58)	-0.14 (-0.51)	-0.22 (-0.99)	-9.65% (-2.32)	0.603

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
value_proxy_1260d	2.60%	0.52%	-1.09%	-4.20%	0.51%	2.43%	-9.65%	-8.88%

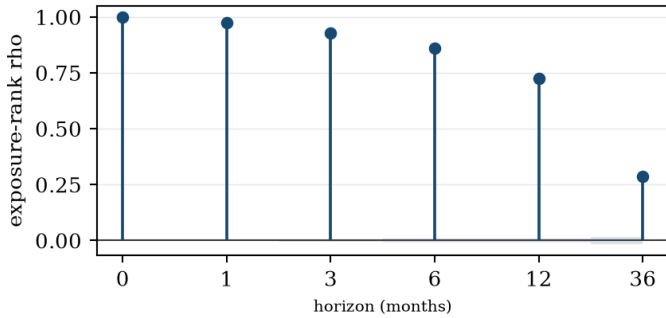


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

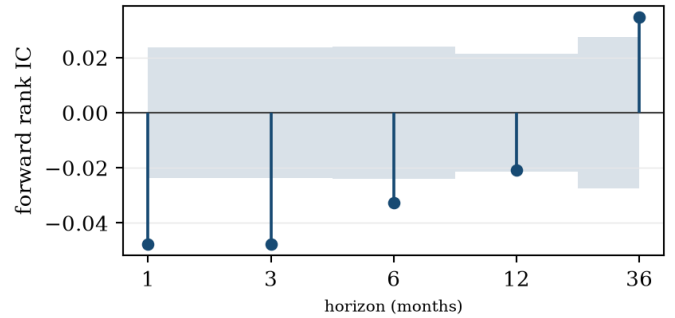


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	38
Signals	value_proxy_1260d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp; skip-infeasible=True
Components required	1 of 1
Available / comparison months	43 / 38
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 5
Stocks, min / median / max	1889 / 2159 / 2506
Available-history Sharpe	-0.84
Bootstrap Sharpe	-1.03 [-1.79, -0.48]; Pr(S<=0)=99.9%
Bootstrap tails	skew=0.52 [-0.53, 1.00]; max DD=-30.5% [-36.7%, -17.1%]

Reproducibility

Run ID: 20260906T164702Z_no-clustering-bootstrap-screen-sharpe-0-oos-2023_e771371fdc.

Gold snapshot: version=20260829T051952Z.

Configuration hash: e771371fdc.

Symbols requested / loaded: 0 / 3595.

Seed: 0. Comparison months: 38, 2023-01-01 to 2026-05-01.

Command:

```
C:/Users/User/Desktop/alpha/.venv/Scripts/python.exeC:/Users/User/Desktop/alpha/src/experiments/factor_model_runner/run.py--configdata/outputs/automatic_factor_grouping/signal_screening/20260906T064949Z_e2fdcf7479/factor_runner_handoffs/20260906T164336Z_sharpe_gt_0_9c2d37bba4/factor_model_oos.py--no-default-liquidity-screen.
```

Direction is frozen upstream; this report performs no return-based sign selection. The reported direction is applied consistently before every downstream regression, style calculation, plot, table, PCA coordinate, and index assignment. The run directory contains the source configuration, normalized configuration, manifest, canonical tables, monthly returns, PCA coordinates, plots, report source, and PDF.